

Countable Information, Uncountably Many Boxes

Sharp measurable prediction bounds, exact decision-tree enumeration, and an all-orders asymptotic expansion

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Abstract

We study the distinction between choosing an unread coordinate and assigning a measurable event to a correct prediction. For a nonempty standard Borel set of boxes, a finite alphabet of size q , and m deterministic players whose output maps are measurable for the cylinder σ -algebra, we prove that the optimal guaranteed number of correct guesses is exactly $\lfloor m/q \rfloor$. The proof does not assume that success events are measurable. A simultaneous measure-extension construction supplies fair success probabilities for any countable collection of such players. Conversely, measurability of a blind player's success event forces its target range to be countable, pointwise or outside a cylinder-null set according to the chosen completion.

The finite problem leads to a separate structural result. Blind binary output maps correspond to labeled perfect matchings of the hypercube, whereas executable maps correspond precisely to recursively sliceable matchings. We derive an exact inclusion–exclusion recurrence for the latter, prove that the two classes first differ at four boxes, and establish

$$F_n = \frac{e^{\gamma 2^n}}{n} \left(1 - \frac{3}{2n} + \frac{17}{4n^2} - \frac{12}{n^3} + O(n^{-4}) \right),$$

with a rational algorithm for every further asymptotic coefficient. We also prove an explicit doubly exponential upper bound on the fraction of blind maps that are executable. Finite checks, an oracle counterexample-extraction theorem, and a Busy Beaver obstruction connect the analysis to the certificate-oriented architecture of ProveIt.

Scope and status. All stated theorems have proofs in this note. The finite computations were executed independently in Python. The infinite results have not been checked in Lean or Rocq, and historical priority is not established. The restricted measurable problem is settled here; Glazer's unrestricted question about what ZF proves is not settled here. Neither authorship nor endorsement by Elliot Glazer is implied.

[write] Since 4 October 2026 the report has four Parts: Part **II** (from Section 17; batch 90, labels `mbg:hat:`) treats surplus in hat guessing and settles Q9 for one family; Part **III** (from Section 30; batch 92, labels `mbg:ext:`) classifies the extensions of the product measure to adaptive success events and answers Q2 in part and Q5; Part **IV** (from Section 45; batch 92, two merged manuscripts, labels `mbg:qry:` and `mbg:insp:`) determines the inspection cost of divergent surplus in hat guessing and settles Q9 for a larger family. Part **I**'s numbering is unchanged.

Contents

I Sharp measurable prediction bounds, exact decision-tree enumeration, and an all-orders asymptotic expansion	5
1 Research question and contribution map	5
2 Configurations, blindness, and measurable output	8
3 The sharp measurable-output bound	9
4 A simultaneous fair extension of probability	11
5 Measurable success forces a countable target set	12
6 A continuous selector with maximally nonmeasurable success	13
7 Finite output maps and executable decision trees	14
8 Exact enumeration of executable maps	15
9 All-orders asymptotics for sliceable matchings	17
10 Executable blind maps are doubly exponentially rare	20
11 How much does a coordinate budget reveal?	21
12 Effective finite certificates and the Busy Beaver barrier	22
13 Executed checks and reproducibility	23
14 Foundations and a concrete ProveIt integration plan	24
15 Further research questions and topics	25
16 Conclusion	28
A Assumption and claim ledger	28

B	Source and priority audit	29
II	No universal sublinear bound for measurable hat guessing	31
17	The research intersection and the precise questions	31
18	The game and its elementary constraints	35
19	A transfer from partial predictions to compulsory guesses	36
20	The finite syndrome block	37
21	A bound at every prefix	39
22	Concatenation and a deterministic growth profile	40
23	An exact logarithmic answer and explicit faster examples	42
24	No deterministic sublinear universal bound	43
25	The linear boundary remains unavoidable	45
26	Probability, Baire category, and the exact success set	46
27	Effective construction and finite information costs	47
28	Verification and a concrete route into ProveIt	48
29	Further research questions	50
C	The exact prefix constant	52
D	A self-contained completeness fact for FIN	53
E	Reproducibility and source notes	54
III	Adaptive box games beyond cylinder measurability	55
30	Research setting and the questions answered	55
31	Configurations, output maps, and the base probability space	59
32	Classical support reduction, with the required exceptional-set control	60

33 Target atoms and measurable internal success	62
34 The finite feasible-pattern object	63
35 The envelope lemma	64
36 Classification of all finite-team extensions	65
37 The exact atomic-mass law for one player	66
38 The joint-law polytope and exact inequalities	68
39 Full Borel and universal minimax	70
40 Examples: what the extension freedom means	71
41 A complete classification for countably many players	73
42 The Baire-property clause: a reduction, not a solution	75
43 Foundations, verification, and a route into ProveIt	76
44 Further research questions and topics	78
F Assumption and result ledger	80
G Sources, attribution, and reproducibility	81
 IV Two queries in expectation: sharp information thresholds and the exact inspection cost of infinite hat guessing	 83
45 The research intersection and the results	84
46 The game, its algorithms, and three notions of cost	89
47 From conditional obstructions to probability zero	90
48 Finite colors obstruct both measure and category success	91
49 Quantitative bounds from Boolean degree	93
50 Stronger finite bounds from sparse supports	95
51 Expected query cost: the lower half of the exact threshold	97

52 The efficient implementation of a known block strategy	98
53 Infinite schedules, exact rates, and the sharp threshold	100
54 Prescribed budgets: count and distance are different	103
55 Free private randomness changes the expected-cost problem	106
56 Finite certificates, reproducibility, and formalization	107
57 Further research questions	109
58 Conclusions and contribution ledger	110
59 The question and the research intersection	111
60 The model and the main statements	113
61 Bounded inspection cannot produce a divergent surplus	115
62 The geometric lower frontier for inspection costs	118
63 An optimal inspection implementation of Eldredge's blocks	119
64 Inspection, dependence, and the density of correct guesses	125
65 Sharp inspection cost for finite first-moment extremizers	128
66 A concrete route to formalization in ProveIt	133
67 Finite verification and reproducibility	136
68 Further research questions	137
69 What has been resolved	139

Part I

Sharp measurable prediction bounds, exact decision-tree enumeration, and an all-orders asymptotic expansion

1 Research question and contribution map

Glazer studies prediction games in which players inspect boxes and then guess an unopened value. In Section 5 of [1], he asks whether ZF proves that three players with continuum many binary boxes can guarantee at least two correct guesses. His work on weak choice principles [2] and the finite and infinite puzzles described in his research profile [3] make regularity, executability, and explicit axiomatic boundaries natural questions to investigate.

The relevant ProveIt materials are its Busy Beaver machine interface, finite-list coding in arithmetic, and explicit first-order set-theoretic developments [6, 7, 8, 9]. Rather than importing an unverified mathematical claim from those projects, we use their architecture: distinguish a semantic statement from an executable certificate, and distinguish both from the theorem that validates the certificate. The repository snapshot used throughout is 6fef5383b126be343cccc9baf47081ef37ab5afe.

The central issue is not a numerical estimate. On an uncountable product, the following statements are different:

- (i) the rule selecting a box and a guess is measurable;
- (ii) the event that the selected box contains the guess is measurable;
- (iii) the rule can actually be implemented without inspecting its final target.

None of these distinctions can safely be suppressed. We obtain two complementary lines of results.

1.1 The continuum line

Theorem 3.1 gives the exact measurable-output minimax value, including the three-player binary case. Theorem 4.1 constructs a simultaneous fair extension of the product probability for countably many players. Theorem 5.1 shows that measurable success is a much stronger restriction on the range of targets than measurable output. The diagonal example in Section 6 makes the difference explicit: its output rule is continuous, but its success event has inner measure zero and outer measure one. We classify every extension obtained by adjoining that event.

1.2 The finite line

Theorem 7.3 characterizes executable binary output maps by a recursive property of hypercube matchings. The sequence F_n counting the underlying executable matchings begins

$$1, 2, 9, 232, 206065, 212181312096, \dots$$

It is *not* the sequence of all perfect matchings, OEIS A005271 [5]; that sequence already differs at $n = 4$. We prove an exact recurrence, a full asymptotic expansion, and an explicit rarity bound. These results are

not asserted to be absent from the literature or the OEIS: exact-sequence searches did not yield a usable classification during this investigation.

1.3 What is, and is not, claimed as a contribution

The countable-support lemma, finite-coordinate averaging, Radon–Nikodym theorem, compactness of a finite-alphabet sequence space, and halting-time diagonalization are standard ingredients. The contribution developed here is their precise application to the output/success distinction, together with the decision-tree classification, enumeration, asymptotic calculation, and executable finite checks. The measure-extension classification is a special-case application of the classical theory represented by Łoś and Marczewski [4], not a claim to have invented measure extension.

These statements constitute a research note with complete proofs and testable finite consequences. They are not presented as a verified historical breakthrough or as a resolution of an unrestricted choiceless-set-theory problem.

1.4 Provenance and place in the collection

[write] Added 3 October 2026, batch 87. This report is a single manuscript, printed in full: manuscript 02 of batch 87 of ProveIt’s incoming-reports intake, delivered as the archive `glazer_box_games_research.zip` (inner directory `glazer_box_games/`, main file `article.tex`, a 24-page US Letter PDF) in the arrival commit `fa0a0576e`, and placed in commit `d151b39ca` in the research-report collection as `ordinals-and-order-types/measurable-box-games`. The author line reads “Prepared for Vladimir Reshetnikov” and carries no AI wording; like the other deliveries of the intake, the report is AI-assisted, unrefereed and not formalized. Its five sister manuscripts of batch 87 concern Glazer’s *other* question paper, *A Topological Tennenbaum Theorem* (arXiv:2311.13699), and became Parts VI–IX of the surreal-collection report `Algebra/SurrealNumbers/docs/foundations-and-computation/polis-h-models-of-omnific-arithmetic`; this manuscript shares no theorem, question or notation with them. Being a single source, the report needed no merge choices. The write prefixed all 61 delivered labels with `mbg:` and updated every reference; added the four labels `mbg:sec:provenance`, `mbg:sec:notation`, `mbg:app:ledger` and `mbg:app:audit`; added these dated **[write]** notes, the title-page line, the `pageanchor` switch around the title page (a delivered build gave a duplicate `page.1` destination) and a dated addition to the bibliography entry [1]. No statement, proof, number or non-claim of the manuscript was changed.

Pin and repository claims. The pin `6fef5383b126be343cccc9baf47081ef37ab5afe` (`6fef5383b`, quoted throughout) is the ProveIt commit the manuscript was written against; it is an ancestor of the placement. The three repository files it read, `Computability/BusyBeaver/Lean/BusyBeaver/Core.lean`, `Logic/PeanoArithmetic/ListCoding/README.md` and `SetTheory/ClosureAxiomatization/README.md`, were byte-identical at the pin and at the write (blobs `4961655e`, `74164ac6`, `fb0ab511`), and the descriptions of them in Sections 12 and 14 are accurate: the Busy Beaver core’s opening comment separates the blank-tape machines and the maximum property of the score function from a separately stated compiler fact; the list-coding README documents validity guards, unique decoding and round trips; the closure-axiomatization README separates its semantic directions from the deductive-equivalence layer. At placement the intake searched the whole tracked tree for `box`, `hat`, guessing and prediction games, for `2211.10474` and for `A005271`, and found nothing: no report or formal project in ProveIt treats this question, and the report continues no repository material. Its place in the collection confers no formal status; no statement of it is formalized in Lean or Rocq.

Glazer’s paper. The arXiv record of [1], read on 3 October 2026, lists a single version, v1 of 15 November 2022 (6 pages; math.LO, math.CO; DOI `10.48550/arXiv.2211.10474`); the date line “January 9, 2023” that the manuscript’s “inspected PDF” carries is printed in arXiv’s PDF of that same v1, not a revision. The question of Section 5 (“More box games”) of that paper reads “Does ZF prove that $G([3], \mathbb{R}, \{0, 1\})$ is winnable?”, where winning means that at most one of the three players guesses wrong; it is Research question Q1 of Section 15. **It is not the “Glazer’s Question 1” of the surreal report above**, which is Question 1 of the Topological Tennenbaum paper, about the reverse-mathematical strength of Glazer’s Theorem 2 there.

Review in the Hilbert’s-tenth research tree. Before placement, another session reviewed all six batch-87 archives at their arrival revision (commit f15962acd): Computability/HilbertTenthProblem/Papers/research-h-wip/native-stream-queue/review_baire_polish_arithmetic_intake.md, with an independent cross-reading in review_baire_polish_arithmetic_intake_independent.md beside it and a receipt in review_baire_polish_arithmetic_intake.json. For this manuscript it read the delivered lines 64–73, 82–110, 386–428, 776–820 and 927–955 (the abstract; Section 1 through Section 1.3; Section 7 from its start through Lemma 7.4; Section 12; the conclusion and the ledger of Appendix A). It confirms the statements as scoped here: legal finite binary decision trees correspond to recursively sliceable matchings and blind output tables need not be executable; the extraction of Theorem 12.1 is a partial procedure for which the totality and legality promise is essential; no computable horizon bound from program length exists, and the Busy Beaver comparison is conditional on an encoding and an additive-overhead compiler. It records that these results give no fixed-size Diophantine history certificate and no paid arithmetic schedule, so they leave the Hilbert’s-tenth programme’s 84-operation universal polynomial unchanged, and that the probability, matching-enumeration and asymptotic claims were *not* reviewed there. Nothing in the review required a change of scope here.

OEIS, checked at intake. On 3 October 2026 the intake searched the OEIS for 1, 2, 9, 232, 206065, for 9, 232, 206065 and for the ternary 2, 21, 33976: none matches an entry. The entry A005271, “Number of perfect matchings in n -cube”, begins 1, 2, 9, 272, 589185, 16332454526976, ..., agreeing with $M_1, \dots, M_5$ of Section 8. This is a check of the database on that date, not of the literature, so the manuscript’s own reservation in Section 1.2 stands. Nothing was submitted to the OEIS, by the manuscript’s author or by the intake; any submission is a human editor’s decision after review. An independent program at intake reproduced $F_1, \dots, F_6$, the ternary $F_1^{(3)}, \dots, F_5^{(3)}$, D_6 , the integer comparison of Theorem 10.1, the 232 sliceable and 40 nonsliceable matchings of Q_4 by brute force, and the two decimal enclosures of γ .

Neighbouring reports. None shares a theorem. The games of ordinals-and-order-types/games-on-ordinals/ are different objects: in open-query-membership-games a Seeker asks open-set queries about a hidden point on a topological space (its transcripts and finite-budget certificates are adaptive query strategies, but nobody guesses an unread coordinate); point-separating-game-values and ordinal-chomp-transition-at-two concern point-separating games and ordinal Chomp. ordinals-and-order-types/non-baire-translation-invariant-ideal is the nearest in spirit: it builds, in ZFC from a free ultrafilter, an ideal without the Baire property, the kind of choice-built irregular object whose absence Research question Q2 asks about for the Baire-property version of Theorem 3.1.

[write] Added 4 October 2026, batch 90. Since 4 October 2026 this report has two Parts. Part I is the batch-87 manuscript described in the preceding note, printed as before: its sections, statements, equations, labels and bibliography numbers are unchanged (checked against a build of the batch-87 text), so the preceding note’s “single manuscript, printed in full” now describes Part I. Part II is manuscript 02 of batch 90, *No Universal Sublinear Bound for Measurable Hat Guessing*, pinned at commit 7c0f2d9f9; its provenance is Section 17.3. It treats Eldredge’s infinite binary hat game, in which every player’s target is its own hat, and answers the two growth questions of his Remark 6.8 [10]; its Corollary 20.3 settles Research question Q9 below for the parameters $(m, q, n) = (2a(2^r - 1), 2, 2a(2^r - 1))$ and the threshold $a2^r$ (see the note after the questions of Section 15). The statement above that nothing in ProveIt treats box, hat, guessing or prediction games was true at the batch-87 placement; Part II is now such material, and the batch-90 placement (commit 12076b2e8) found no other. The neighbouring reports named above share no theorem with Part II either.

[write] Added 4 October 2026, batch 92. Since the batch-92 write (4 October 2026) this report has four Parts; Parts I and II are printed as before (their sections, statements, equations, labels and bibliography numbers are unchanged). Part III is manuscript 03 of batch 92, *Adaptive Box Games Beyond Cylinder Measurability*, pinned at commit 1afa38bf9 (provenance: Section 30.3). It answers Research question Q2 for full product-topology Borel and for universally measurable outputs (Theorem 39.2), reduces its Baire-property clause without settling it (Proposition 42.1), and answers Q5 (Theorems 36.1 and 38.1). Part IV merges manuscripts 06 (the base) and 04 of batch 92, both pinned at commit 9fe62865d, which independently determine the inspection cost of divergent surplus in Eldredge’s hat game, answering Part II’s Research question 29.2; manuscript 04 also settles Q9 for the parameters $(m(2^r - 1), 2, m(2^r - 1))$ and the threshold $m2^{r-1}$, for every $m \geq 1$ (provenance and the merge: Sections 45.5–45.6). The notes after the questions of Section 15 record what is now answered. The neighbouring reports named above share no theorem with Parts III–IV; non-baire-translation-invariant-ideal remains the nearest in spirit to Q2’s open Baire-property clause.

1.5 Notation used in this report

[write] **Added 3 October 2026, batch 87.** The manuscript reuses several letters with local meanings. None was renamed; this table fixes the readings, with the tempting false one where confusion is likely.

Symbol	Meaning (and what it is not)
γ	the growth constant $\lim 2^{-n} \log F_n \approx 0.438158$ of (11); <i>not</i> Euler's constant $0.5772 \dots$
F_n, M_n	sliceable (executable) matchings of Q_n ; all perfect matchings (A005271). They first differ at $n = 4$ ($232 < 272$).
B_n, L_n	blind and legal binary output maps, $2^{2^{n-1}} M_n$ and $2^{2^{n-1}} F_n$; B_n has nothing to do with a Busy Beaver function
$\text{BB}_{\text{time}}, H$	a program-length running-time Busy Beaver function and the maximal horizon (Section 12); <i>not</i> Rado's n -state score function formalized in Computability/BusyBeaver. In Section 11, H is entropy.
I	the set of boxes; in Section 11 also mutual information $I(Z; X_K)$
$\mathbb{E}, E_\tau, E$	expectation; the success set of τ ; the diagonal success set of Section 6
C	a group of players sharing an external target (proof of Theorem 3.1); Cantor space $2^{\mathbb{N}}$ (Section 6); the formal series $C(z)$ of (23)
D	the integer score $D(u)$ (proof of Theorem 3.1); the count D_6 of (25); an event in the proof of Theorem 11.1
r	list size (Corollary 3.3); the ratio $r_n = F_n/F_{n-1}^2$; the label $r = x \upharpoonright \mathbb{N}$ in Section 6
$R, \mathbb{R}$	the formal series $R(z)$ of (21); the real line
T, L	the replacement map $T(Y, U)$ (Theorem 4.1); the time and prefix-length bounds T, L in the proof of Theorem 12.1 (L is not L_n)
Q, Q_n, q	the alphabet $\{0, \dots, q-1\}$; the cube graph; the alphabet size (not a q -series variable)
Σ_I	the cylinder σ -algebra, <i>not</i> the Borel σ -algebra of the product topology and not its completion
Q1–Q12	Research questions of Section 15; Q1 is Glazer's box-game question, not the "Question 1" of his Topological Tennenbaum paper

2 Configurations, blindness, and measurable output

Fix $q \geq 2$, put $Q = \{0, \dots, q-1\}$, and let I be a nonempty set. A *configuration* is a function $x : I \rightarrow Q$. Write

$$X_I = Q^I, \quad \Sigma_I = \sigma\{x : x_i = a \mid i \in I, a \in Q\}.$$

The probability measure μ_I on Σ_I is the product measure under which every fixed finite family of coordinates is independent and uniform on Q .

Remark 2.1 (Which σ -algebra?). The cylinder σ -algebra Σ_I is not, by definition, the full Borel σ -algebra of the uncountable product topology. Nor is it automatically its own completion. Every measurability assertion below names the relevant σ -algebra. This is essential, not a technical convention that can be changed without affecting the theorems.

For $i \in I$ and $v \in Q$, let $x^{i \leftarrow v}$ denote the configuration obtained by replacing x_i by v and leaving all other coordinates unchanged.

Definition 2.2 (Blind output map). An output map is a total function

$$\tau = (b, a) : X_I \longrightarrow I \times Q.$$

It is *blind* if, whenever $\tau(x) = (i, c)$,

$$\tau(x^{i \leftarrow v}) = (i, c) \quad \text{for every } v \in Q. \quad (1)$$

Its success set is

$$E_\tau = \{x : x_{b(x)} = a(x)\}.$$

For a team $\tau_1, \dots, \tau_m$, the score is $S(x) = \sum_{p=1}^m \mathbf{1}_{E_{\tau_p}}(x)$.

Every total deterministic legal probing strategy induces a blind output map. Indeed, altering only the value of its ultimately unopened box changes none of the observations along the original run. Induction on the successive moves gives the same history and the same final output. This remains a necessary condition when a move reveals an entire set of boxes or histories are transfinite. We do not identify blindness with legal executability; Section 7 gives a finite counterexample to that identification.

From now on, for the continuum theorems, I carries a standard Borel structure. Measurability of τ means measurability from (X_I, Σ_I) to $I \times Q$ with its product Borel structure. In particular, singleton targets, equality of two target maps, and membership in any countable set of targets are measurable.

Definition 2.3 (Coordinate support). A set $A \subseteq X_I$ is *supported by* $J \subseteq I$ if membership in A is unchanged when coordinates outside J change. A function is supported by J if two configurations agreeing on J have the same function value.

Lemma 2.4 (Countable-coordinate dependence). *Every member of Σ_I has a countable support. Every Σ_I -measurable map to a countably separated measurable space has a countable support. Consequently a finite or countable family of measurable maps to standard Borel spaces has a common countable support.*

Proof. The sets possessing a countable support form a σ -algebra: complements keep the same support, and a countable union is supported by the union of its chosen supports. This σ -algebra contains each coordinate cylinder, proving the first assertion.

For a measurable function f , choose a countable family (C_k) of measurable sets separating points in its codomain. Choose a countable support J_k for $f^{-1}(C_k)$ and put $J = \bigcup_k J_k$. If $x \upharpoonright J = y \upharpoonright J$, then $f(x)$ and $f(y)$ belong to exactly the same C_k , hence are equal. A further countable union gives the last assertion.

The factorization is literal, not merely almost everywhere. Fix the symbol 0 outside J and let $s_J : Q^J \rightarrow Q^I$ be the corresponding section. Then s_J is measurable, $f_J = f \circ s_J$ is measurable, and $f = f_J \circ \pi_J$. $\square$

The lemma is proved here in the ordinary classical setting. The selection of countably many supports is among the steps that need a separate audit before attributing this proof to a weak choiceless base theory.

3 The sharp measurable-output bound

Theorem 3.1 (Exact minimax value). *Let I be a nonempty standard Borel space and let $m \geq 1$. For every team of m blind cylinder-measurable output maps there is a configuration x such that*

$$S(x) \leq \lfloor m/q \rfloor. \quad (2)$$

There is a legal team of constant output maps guaranteeing at least $\lfloor m/q \rfloor$ correct guesses on every configuration. Thus the optimum worst-case score in the measurable-output class is exactly $\lfloor m/q \rfloor$.

Proof. Choose a common countable support J for the outputs. On Q^J write the factored outputs as $(b_p(u), a_p(u))$. We separate targets inside J from targets outside J .

For $i \in J$ and $c \in Q$, the measurable fiber

$$A_{p,i,c} = \{u : b_p(u) = i, a_p(u) = c\}$$

is invariant under changing u_i , by blindness. A product integration in that one coordinate gives

$$\mu_J(A_{p,i,c} \cap \{u : u_i = c\}) = \frac{1}{q} \mu_J(A_{p,i,c}). \quad (3)$$

Therefore the internal score

$$S_{\text{in}}(u) = \sum_{\substack{1 \leq p \leq m \\ b_p(u) \in J}} \mathbf{1}_{\{u_{b_p(u)} = a_p(u)\}}$$

is measurable and satisfies

$$\mathbb{E}S_{\text{in}} = \frac{1}{q} \sum_{p=1}^m \mu_J(b_p \in J). \quad (4)$$

For each fixed u , group the remaining players according to their common target outside J . If C is one such group, let $n_{C,v}(u) = |\{p \in C : a_p(u) = v\}|$. Give the shared target the least symbol minimizing $n_{C,v}(u)$. Its contribution to the score is then

$$\min_{v \in Q} n_{C,v}(u) \leq |C|/q.$$

Define the integer-valued function

$$D(u) = S_{\text{in}}(u) + \sum_C \min_{v \in Q} n_{C,v}(u).$$

There are only finitely many players. The grouping is determined by finitely many measurable equality tests among their targets, so D is measurable. From (4),

$$\mathbb{E}D \leq \frac{1}{q} \sum_p \mu_J(b_p \in J) + \frac{1}{q} \sum_p \mu_J(b_p \notin J) = \frac{m}{q}.$$

An integer-valued function everywhere larger than $\lfloor m/q \rfloor$ would have expectation strictly larger than m/q . Choose u with $D(u) \leq \lfloor m/q \rfloor$. Complete u using the selected minimizing symbols at the finitely many external targets and the default symbol elsewhere. Since all outputs depend only on J , their values are unchanged. The resulting score is exactly $D(u)$.

For the reverse bound, fix a single box and divide the players' constant guesses as evenly as possible among the q symbols. Every symbol receives at least $\lfloor m/q \rfloor$ guesses. The players need not open any box, so this is a legal strategy. $\square$

Corollary 3.2 (The regular three-player continuum problem). *Three binary players with continuum many boxes and blind cylinder-measurable output maps cannot guarantee two correct guesses. They can guarantee one.*

This is a negative answer for a specified regularity class. It is not a statement that ZF disproves the existence of unrestricted winning strategies.

Corollary 3.3 (Lists of guesses). *Suppose every player supplies a set of exactly r possible symbols, where $1 \leq r < q$, and success means that the target value belongs to the list. With the corresponding blindness condition on the entire output (b, A) , the exact optimum guaranteed score is*

$$\lfloor mr/q \rfloor.$$

Proof. The internal fiber average in (3) is r/q . For an external target with t players, summing the number of containing lists over all symbols gives tr , so some symbol is in at most tr/q lists. The previous proof applies.

For sharpness, have all players choose the same box and let player p , indexed from zero, use

$$A_p = \{pr, pr + 1, \dots, pr + r - 1\} \pmod{q}.$$

Each list has r distinct symbols, and the mr consecutive residues distribute as evenly as possible among Q . $\square$

4 A simultaneous fair extension of probability

The preceding proof carefully avoided integrating the possibly nonmeasurable score on X_I . There is also a constructive way to enlarge the measurable space so that every player in a countable team has the expected fair marginal.

Theorem 4.1 (Independent replacement at external targets). *Let $(\tau_p)_{p \in \mathbb{N}}$ be a countable family of blind cylinder-measurable output maps on Q^I , with I standard Borel. There is a countably additive probability measure ν on*

$$\sigma(\Sigma_I, E_{\tau_0}, E_{\tau_1}, \dots)$$

such that $\nu \upharpoonright \Sigma_I = \mu_I$ and

$$\nu(E_{\tau_p}) = 1/q \quad (p \in \mathbb{N}).$$

For a finite team the same conclusion holds with just its success events adjoined.

Proof. Choose a common countable support J . Work on the auxiliary product space carrying an independent pair

$$Y \sim \mu_I, \quad U = (U_0, U_1, \dots) \sim \mu_{\mathbb{N}}.$$

All output pairs $(b_p(Y), a_p(Y))$ depend only on $Y \upharpoonright J$.

For each external target that occurs, use the first player selecting it as its representative. More precisely, when $b_p(Y) \notin J$, let

$$\rho_p(Y) = \min\{r \leq p : b_r(Y) = b_p(Y)\}.$$

Define a new configuration $T(Y, U)$ by leaving the coordinates of J unchanged, assigning the value $U_{\rho_p(Y)}$ to each selected external target $b_p(Y)$, and leaving all other coordinates equal to those of Y . Repeated targets receive the same replacement value, so the definition is consistent.

For each fixed coordinate i , T_i is measurable. Indeed, whether i is a selected target and which player is its first representative are countable combinations of measurable target-equality tests. The default case is also measurable. Hence T is measurable into (X_I, Σ_I) .

We next check the fixed-coordinate law. Conditional on $Y \upharpoonright J$, any fixed finite collection of coordinates outside J is transformed by replacing some of its entries with distinct independent uniform U coordinates. The selection is determined by $Y \upharpoonright J$, not by those outside values. Untouched entries remain independent

uniform values of Y , independent also of U . Thus every fixed finite coordinate law of T is the original product law. Uniqueness of the probability measure on the generated cylinder σ -algebra gives

$$\mathbb{P}(T \in A) = \mu_I(A) \quad (A \in \Sigma_I).$$

This argument can equivalently be written by partitioning Q^J into the countably many possible representative patterns for the specified finite set, avoiding any disintegration on an uncountable product.

All outputs are unchanged by T , since it fixes J . If player p selects an external target, its pulled-back success event is

$$\{b_p(Y) \notin J, U_{\rho_p(Y)} = a_p(Y)\},$$

which is measurable and has conditional probability $1/q$ on that case. If it selects an internal target, its success probability is the sum in (3). Altogether,

$$\mathbb{P}(T \in E_{\tau_p}) = \frac{1}{q}\mathbb{P}(b_p \notin J) + \frac{1}{q}\mathbb{P}(b_p \in J) = \frac{1}{q}.$$

Finally, the collection of all subsets $A \subseteq X_I$ for which $T^{-1}(A)$ is measurable is a σ -algebra containing both Σ_I and every success event. The formula $\nu(A) = \mathbb{P}(T \in A)$ defines a countably additive probability on this σ -algebra, and restriction gives the required extension. $\square$

Remark 4.2. The theorem establishes existence, not uniqueness. It does not assert that success events already belong to the original product σ -algebra, or that every extension must give probability $1/q$. Section 6 exhibits the strongest possible nonuniqueness for one player. It also does not assert that the successes of different players are independent.

5 Measurable success forces a countable target set

Theorem 5.1 (Pointwise and completed target collapse). *Let I be any set and let $\tau = (b, a)$ be a blind output map on Q^I . No measurability of τ is assumed.*

- (a) *If $E_\tau \in \Sigma_I$, there is a countable $J \subseteq I$ such that $b(x) \in J$ for every x .*
- (b) *If E_τ belongs to the μ_I -completion of Σ_I , there are a countable $J \subseteq I$ and a set $N \in \Sigma_I$ with $\mu_I(N) = 0$ such that $b(x) \in J$ whenever $x \notin N$.*

Proof. For (a), let J support E_τ . Suppose $b(x) = i \notin J$ and $a(x) = c$. Choose $d \neq c$. Blindness gives the same output (i, c) on $x^{i \leftarrow c}$ and $x^{i \leftarrow d}$. Exactly one of these configurations succeeds. But they agree on J , contradicting support of the success event.

For (b), choose $E_0, N \in \Sigma_I$ with $\mu_I(N) = 0$ and $E_\tau \triangle E_0 \subseteq N$. Let J support both E_0 and N . If $x \notin N$ and $b(x) = i \notin J$, the two replacements from the preceding argument also lie outside N . They have equal membership in E_0 , hence in E_τ , again a contradiction. $\square$

Remark 5.2 (Two nonconverses). A map supported by countably many input coordinates can have uncountable output range. Conversely, countable range of targets alone does not establish measurability of its target/guess fibers. To deduce fair probabilities by summing (3), the fibers must be measurable as well. The theorem should not be paraphrased as “measurable success alone implies probability $1/q$.”

6 A continuous selector with maximally nonmeasurable success

In this section take $q = 2$. Let $C = 2^{\mathbb{N}}$ be Cantor space and take the tagged disjoint union

$$I = \mathbb{N} \sqcup C.$$

The distinction between the two tags is part of the definition. For a configuration x , read its natural-number coordinates and form $r = x \upharpoonright \mathbb{N} \in C$. Select the box on the *other* side indexed by r , and guess zero:

$$b(x) = r \in C, \quad a(x) = 0. \quad (5)$$

This is a legal strategy when countably many boxes may be opened in one move. It is blind. Its output map is cylinder-measurable; with the product topology on the domain and the usual disjoint-union topology on I , it is even continuous.

Its success set is

$$E = \{x : x_{x \upharpoonright \mathbb{N}} = 0\},$$

where the outer evaluation uses the C -tagged coordinate.

Theorem 6.1 (Inner measure zero, outer measure one). *For the diagonal strategy,*

$$\mu_*(E) = 0, \quad \mu^*(E) = 1.$$

In particular, E belongs neither to Σ_I nor to its completion.

Proof. Let $A \in \Sigma_I$ with $A \subseteq E$, and let K be a countable support for A . For any $x \in A$, its selected label $r = x \upharpoonright \mathbb{N}$ must belong to $K \cap C$. Otherwise changing only the r -tagged coordinate from zero to one leaves A unchanged but destroys success. Thus

$$A \subseteq \{x : x \upharpoonright \mathbb{N} \in K \cap C\}.$$

A countable set of fair infinite binary sequences has probability zero, so $\mu_I(A) = 0$. This proves the inner-measure assertion. Repeating the argument for a measurable subset of E^c , now changing the selected bit from one to zero, shows $\mu_*(E^c) = 0$. The outer-measure assertion follows by complementation. $\square$

6.1 Every probability extension after adjoining success

Write $\Sigma_I[E] = \sigma(\Sigma_I, E)$. Its members have the form

$$(A \cap E) \cup (B \cap E^c), \quad A, B \in \Sigma_I. \quad (6)$$

This is a σ -algebra directly: take complements and countable unions in the two components.

Theorem 6.2 (Classification by a measurable density). *For every measurable $h : X_I \rightarrow [0, 1]$, the formula*

$$\nu_h((A \cap E) \cup (B \cap E^c)) = \int_A h \, d\mu_I + \int_B (1 - h) \, d\mu_I \quad (7)$$

defines a countably additive probability extending μ_I . Every countably additive extension to $\Sigma_I[E]$ is obtained this way, with h unique up to μ_I -almost-everywhere equality.

Proof. Suppose two pairs (A, B) and (A', B') represent the same set. Then $A \triangle A'$ is a measurable subset of E^c , and $B \triangle B'$ is a measurable subset of E . Both are null by Theorem 6.1, so the formula is well-defined.

For a disjoint sequence of represented sets, the corresponding A components are pairwise disjoint modulo null sets, as are the B components. Integration of h and $1 - h$, or monotone convergence after removal of those countably many null overlaps, proves countable additivity. Taking $A = B = X_I$ gives total mass one. Taking $A = B = D$ shows $\nu_h(D) = \mu_I(D)$ for $D \in \Sigma_I$.

Conversely, let ν be an extension and define $\lambda(A) = \nu(A \cap E)$ on Σ_I . This is a measure and $0 \leq \lambda \leq \mu_I$. The Radon–Nikodym theorem yields a measurable density $0 \leq h \leq 1$ with $\lambda(A) = \int_A h d\mu_I$. Since $\nu(B \cap E^c) = \mu_I(B) - \lambda(B)$, formula (7) follows. Uniqueness is the usual almost-everywhere uniqueness of that density. $\square$

For constant $h = \theta$, the extension assigns $\nu_h(E) = \theta$ for any $\theta \in [0, 1]$. Every fixed finite collection of boxes still consists of independent fair bits. Thus those fixed-coordinate laws alone do not determine the probability of a dynamically indexed success event. The fair extension of Theorem 4.1 supplies one consistent answer; it does not turn that answer into a consequence of the original cylinder law alone.

7 Finite output maps and executable decision trees

Now let $I = \{0, \dots, n-1\}$ and first take $Q = \{0, 1\}$. The cube graph Q_n has vertex set Q^n and edges joining configurations differing in one coordinate. A perfect matching partitions its vertices into edges.

Proposition 7.1 (Blind maps are labeled matchings). *Blind binary output maps on n boxes are in bijection with perfect matchings of Q_n whose edges each carry an arbitrary label in $\{0, 1\}$. If M_n is the number of perfect matchings, the number of blind output maps is*

$$B_n = 2^{2^{n-1}} M_n.$$

Every such map succeeds at exactly 2^{n-1} configurations.

Proof. If $\tau(x) = (i, a)$, blindness pairs x with the configuration obtained by flipping its i th coordinate, giving both vertices the same target direction i and guess label a . This pairing is an involution without fixed points, hence a perfect matching. Conversely, a matching and its labels specify a blind output map by using the edge direction as target and its label as guess. Each edge has exactly one endpoint whose target bit equals its label. There are 2^{n-1} edges. $\square$

This is a statement about extensional output maps, not about descriptions of programs. Different decision trees can compute the same map.

Definition 7.2 (Recursively sliceable matching). The unique matching of a one-dimensional cube is *sliceable*. For $n \geq 2$, a matching is sliceable if there is a coordinate k such that it has no edge in direction k and the induced matchings in both faces $x_k = 0$ and $x_k = 1$ are sliceable. Coordinates retain their labels during restriction.

Theorem 7.3 (Exact executability criterion). *A blind binary output map on finitely many boxes is realizable by a deterministic decision tree that never queries its eventual target if and only if its underlying matching is sliceable. The choice of guess labels does not affect realizability.*

Proof. A sliceable matching supplies a decision tree recursively: query a slicing coordinate and continue in the observed face. At a final one-dimensional face, its unique coordinate is still unread; output that coordinate and the label of its edge. This works for every assignment of labels.

Conversely, start with a legal decision tree. Repeated queries can be deleted. If a leaf is reached with more than one unread coordinate, its target is one of them. Refine that leaf by querying all the other unread coordinates and leave the target unread. Its prescribed output remains unchanged. The resulting tree ends at one-dimensional faces. Its first query k can be no eventual target, so the matching has no edge in direction k . The two restricted trees realize the induced face matchings. Induction on dimension proves sliceability.

A tree that initially makes no query is handled by the same refinement: its constant target is left unread while the other coordinates are queried. Thus the argument covers early stopping as well as full-depth trees. $\square$

Lemma 7.4 (Restriction lemma). *If a sliceable matching has no edges in any of a specified set K of coordinate directions, its restriction to every face obtained by fixing the coordinates of K is sliceable, provided that face has positive dimension.*

Proof. Use its legal decision tree and substitute the fixed values for queries in K , deleting those query nodes. Since none of these coordinates is ever a target, the resulting restricted tree is still legal. Apply Theorem 7.3. $\square$

7.1 The first failure of the blindness converse

For $n \leq 3$, every cube matching is sliceable. For $n = 2, 3$, the number of matching edges crossing each coordinate cut is even: each half contains 2^{n-1} vertices, and internal matching edges cover them in pairs. A matching of Q_2 has two edges and one of Q_3 has four. Consequently they cannot use every coordinate direction. Split along a missing direction and proceed inductively.

At dimension four consider the matching below. Vertices are shown as integers; coordinate i is bit i , with bit zero least significant.

Edge	(0, 1)	(2, 3)	(4, 6)	(5, 13)
Direction	0	0	1	3
Edge	(7, 15)	(8, 12)	(9, 11)	(10, 14)
Direction	3	2	1	2

Every vertex occurs exactly once, and all four coordinate directions occur. Label every edge zero. The resulting map is blind but cannot be executed: whatever the first queried coordinate is, it is the final target on some configuration. Making no query cannot implement a map whose target varies. This proves a sharp four-box separation, independently of the enumeration below.

8 Exact enumeration of executable maps

Let F_n count sliceable matchings of the labeled cube Q_n . Put $F_0 = 0$ as an auxiliary convention and $F_1 = 1$.

Theorem 8.1 (Inclusion–exclusion recurrence). *For every $n \geq 2$,*

$$F_n = \sum_{k=1}^{n-1} (-1)^{k+1} \binom{n}{k} F_{n-k}. \quad (8)$$

The number of distinct legal binary output maps is

$$L_n = 2^{2^{n-1}} F_n. \quad (9)$$

Proof. For each coordinate i , let A_i be the class of matchings that can be sliced first in direction i . Then $F_n = |\bigcup_i A_i|$. For a set K of $k < n$ possible first coordinates, the intersection $\bigcap_{i \in K} A_i$ consists exactly of independent choices of sliceable matchings on the 2^k faces obtained by fixing K .

One direction follows from Lemma 7.4. In the other direction, given the matchings on those faces, one may first query any chosen coordinate of K , query the remaining coordinates of K , and then execute the supplied face tree. Thus

$$\left| \bigcap_{i \in K} A_i \right| = F_{n-k}^{2^k}.$$

The intersection for all n coordinates is empty because no matching edge direction could remain. Inclusion–exclusion proves (8). Arbitrary edge labels are permitted by Theorem 7.3, proving (9). $\square$

n	F_n	M_n	Legal output maps L_n	Blind output maps B_n
1	1	1	2	2
2	2	2	8	8
3	9	9	144	144
4	232	272	59 392	69 632
5	206 065	589 185	$2^{16} F_5$	$2^{16} M_5$
6	212 181 312 096	—	$2^{32} F_6$	—

The M_n entries agree with A005271 [5] and were recomputed here by exact enumeration or a permanent dynamic program through $n = 5$. The row $n = 6$ intentionally does not import an unverified larger enumeration into the certificate calculation.

[write] Added 3 October 2026, batch 87. A005271 records $M_6 = 16\,332\,454\,526\,976$ (read at intake). It is not used anywhere in this report; it is consistent with $D_6 \leq M_6$ for the count D_6 of (25) (their ratio is about 0.1226). The column F_n and the ternary values below are not in the OEIS (Section 1.4).

For $n = 4$, a matching is nonsliceable precisely when it uses all four directions, since every three-dimensional matching is sliceable. Inclusion–exclusion gives the number of such matchings as

$$272 - 4 \cdot 9^2 + 6 \cdot 2^4 - 4 \cdot 1^8 = 40.$$

Hence exactly $40 \cdot 2^8 = 10\,240$ blind four-box output maps are not executable.

8.1 The q -ary version

For a general alphabet Q , replace cube edges by full coordinate lines of length q . Blind output maps correspond to partitions of Q^n into such lines with one guess label per line. The decision-tree proof is unchanged, with q children at each query. If $F_n^{(q)}$ counts recursively sliceable line partitions, then

$$F_1^{(q)} = 1, \quad F_n^{(q)} = \sum_{k=1}^{n-1} (-1)^{k+1} \binom{n}{k} \left(F_{n-k}^{(q)} \right)^{q^k}. \quad (10)$$

There are $q^{q^{n-1}} F_n^{(q)}$ legal output maps. In particular, the ternary values begin

$$1, \quad 2, \quad 21, \quad 33976, \quad 188162675402345.$$

This counts maps, not trees, so alternative query orders are not counted multiple times.

9 All-orders asymptotics for sliceable matchings

We now concentrate on the binary sequence F_n . Its doubly exponential scale is natural because an initial query creates two independent subproblems. The inclusion–exclusion terms encode overlaps between alternative initial queries.

Theorem 9.1 (Full inverse-power expansion). *There is a positive constant*

$$\gamma = \lim_{n \rightarrow \infty} 2^{-n} \log F_n = \sum_{k=2}^{\infty} 2^{-k} \log r_k, \quad r_k = \frac{F_k}{F_{k-1}^2}, \quad (11)$$

such that, for every integer $N \geq 1$, there are rational coefficients $c_1, \dots, c_N$ satisfying

$$\log F_n = \gamma 2^n - \log n + \sum_{j=1}^N \frac{c_j}{n^j} + O(n^{-N-1}). \quad (12)$$

The coefficients are recursively computable. The first terms are

$$\log F_n = \gamma 2^n - \log n - \frac{3}{2n} + \frac{25}{8n^2} - \frac{27}{4n^3} + O(n^{-4}), \quad (13)$$

$$F_n = \frac{e^{\gamma 2^n}}{n} \left(1 - \frac{3}{2n} + \frac{17}{4n^2} - \frac{12}{n^3} + O(n^{-4}) \right). \quad (14)$$

9.1 Ratio bounds and existence of the growth constant

By the union bound and the first two Bonferroni bounds for the sets A_i in the proof of Theorem 8.1,

$$nF_{n-1}^2 - \binom{n}{2} F_{n-2}^4 \leq F_n \leq nF_{n-1}^2. \quad (15)$$

Also $F_n \geq F_{n-1}^2$ by choosing one specified first coordinate. Thus $1 \leq r_n \leq n$. In fact,

$$\frac{n}{2} \leq r_n \leq n \quad (n \geq 2), \quad \frac{r_n}{n} \geq 1 - \frac{2}{n-1} \quad (n \geq 5). \quad (16)$$

The first inequality is checked directly at $n = 2, 3, 4$. If it holds at $n - 1$, then (15) gives

$$r_n \geq n \left(1 - \frac{n-1}{2r_{n-1}^2} \right) \geq n \left(1 - \frac{2}{n-1} \right) \geq \frac{n}{2}$$

for $n \geq 5$, proving the induction and the second assertion.

Since

$$2^{-n} \log F_n - 2^{-(n-1)} \log F_{n-1} = 2^{-n} \log r_n,$$

the scaled logarithms increase and are bounded by the convergent series $\sum_{n \geq 2} 2^{-n} \log n$. This proves (11); positivity follows already from $F_2 = 2$. It also gives the exact tail identity

$$\gamma 2^n - \log F_n = \sum_{j=1}^{\infty} 2^{-j} \log r_{n+j}. \quad (17)$$

9.2 The first correction terms

The next Bonferroni term bounds the error in (15) by $\binom{n}{3} F_{n-3}^8$. Dividing by $n F_{n-1}^2$ and using (16) yields

$$\frac{r_n}{n} = 1 - \frac{n-1}{2r_{n-1}^2} + O(n^{-4}). \quad (18)$$

For clarity, the normalized triple-intersection term is

$$\frac{\binom{n}{3}}{n r_{n-1}^2 r_{n-2}^4} = O(n^{-4}).$$

Starting with $r_n = n + O(1)$ from (16), repeated substitution into (18) improves the approximation one power at a time. It gives

$$r_n = n - \frac{1}{2} - \frac{1}{n} - \frac{23}{8n^2} + O(n^{-3}), \quad (19)$$

$$\log(r_n/n) = -\frac{1}{2n} - \frac{9}{8n^2} - \frac{41}{12n^3} + O(n^{-4}). \quad (20)$$

For example, writing $r_{n-1} = n - \frac{3}{2} - n^{-1} + O(n^{-2})$ gives

$$\frac{n-1}{2r_{n-1}^2} = \frac{1}{2n} + \frac{1}{n^2} + \frac{23}{8n^3} + O(n^{-4}),$$

which displays the coefficient that is most easily missed.

Substitute (20) into (17). The geometric moments

$$\sum_{j \geq 1} \frac{1}{2^j} = 1, \quad \sum_{j \geq 1} \frac{j}{2^j} = 2, \quad \sum_{j \geq 1} \frac{j^2}{2^j} = 6, \quad \sum_{j \geq 1} \frac{j^3}{2^j} = 26$$

give

$$\gamma 2^n - \log F_n = \log n + \frac{3}{2n} - \frac{25}{8n^2} + \frac{27}{4n^3} + O(n^{-4}).$$

This proves (13). Exponentiating gives (14).

9.3 A coefficient algorithm and proof to every order

Put $z = 1/n$. There is a unique formal power series $R(z) = 1 + \sum_{j \geq 1} a_j z^j$ satisfying

$$R(z) = \sum_{k \geq 1} \frac{(-1)^{k+1}}{k!} z^{2^k - k - 1} \frac{\prod_{h=0}^{k-1} (1 - hz)}{\prod_{j=1}^{k-1} \left[(1 - jz) R\left(\frac{z}{1 - jz}\right) \right]^{2^j}}. \quad (21)$$

The $k = 1$ term is 1. Every other term is divisible by z , and its coefficient of degree d involves coefficients of R of degree at most $d - 1$. Also $2^k - k - 1 \rightarrow \infty$, so every coefficient receives contributions from only finitely many k . These observations prove existence, uniqueness, and rational computability coefficient by coefficient.

To see that this is an asymptotic rather than merely a formal calculation, divide the exact recurrence by nF_{n-1}^2 . For a fixed k , its normalized term is

$$\frac{\binom{n}{k}}{n \prod_{j=1}^{k-1} r_{n-j}^{2j}} = O\left(n^{k+1-2^k}\right). \quad (22)$$

The Bonferroni inequalities sandwich the exact answer between any two consecutive inclusion–exclusion truncations. Their difference is precisely the next intersection term, bounded by (22). Given any desired power of $1/n$, choose a fixed truncation long enough that this error is smaller. In the remaining finite expression, the dependence on an error in each shifted ratio is multiplied by at least one extra power of $1/n$. Starting with $r_n/n = 1 + O(n^{-1})$, induction therefore improves the error by one power at each step. This proves

$$r_n/n = R(1/n) + O(n^{-N-1})$$

when R is truncated at degree N .

Finally define the formal series

$$C(z) = - \sum_{j \geq 1} 2^{-j} \left[\log(1 + jz) + \log R\left(\frac{z}{1 + jz}\right) \right]. \quad (23)$$

Each coefficient is a rational linear combination of geometric moments and is therefore well-defined and rational. Its coefficients are the c_j in (12). To justify termwise asymptotic expansion in (17), split the sum at $j = n/2$. In the initial part, Taylor remainders are bounded by a constant times $(1 + j)^{N+1} n^{-N-1}$ and are summable against 2^{-j} . The remaining tail is exponentially small, using $r_{n+j} \leq n + j$. This completes the all-orders proof.

The exponents at which successive intersection sizes first enter are

$$2^k - k - 1 = 0, 1, 4, 11, 26, \dots$$

Thus a high asymptotic order needs surprisingly few intersection types. This is an all-orders Poincaré expansion on a doubly exponential scale; no claim about beyond-all-orders terms is being made.

9.4 Effective bounds for the constant

For $N \geq 4$, let $g_N = 2^{-N} \log F_N$. From (16), (17), and concavity of the logarithm,

$$\begin{aligned} g_N + 2^{-N} \left[\log(N+1) + \log\left(1 - \frac{2}{N}\right) \right] &\leq \gamma \\ &\leq g_N + 2^{-N} \left[\log(N+1) + \frac{1}{N+1} \right]. \end{aligned} \quad (24)$$

For the upper bound, use $\log(N+j) \leq \log(N+1) + (j-1)/(N+1)$ and sum. For the lower bound, use $r_{N+j} \geq (N+j)(1 - 2/(N+j-1)) \geq (N+1)(1 - 2/N)$.

The exact recurrence through $N = 20$ gives rounded evaluations

$$0.4381576906086384 \lesssim \gamma \lesssim 0.4381578365013189.$$

The proved statement is the symbolic enclosure (24); these displayed decimals are rounded evaluations, not an interval-arithmetic certificate. A rough corresponding base is $e^\gamma \approx 1.54985$. The supplied checker records the exact recurrence data and the formulas used for the enclosure.

[write] Added 3 October 2026, batch 87. The checker's recorded run, `data/verification_report.json`, gives the two sides of (24) at $N = 20$ as 65-digit rounded evaluations,

$$0.43815769060863840388\dots \quad \text{and} \quad 0.43815783650131883608\dots$$

The two 16-digit decimals displayed above are outward roundings of these (the lower one rounded down, the upper one rounded up in its last digit). They remain rounded evaluations, not an interval-arithmetic certificate.

10 Executable blind maps are doubly exponentially rare

A small finite matching count already combines with the preceding analysis to give a uniform rarity theorem. Let D_6 count matchings of Q_6 that omit at least one coordinate direction. A prescribed set of k omitted directions separates the graph into 2^k independent copies of Q_{6-k} , so

$$\begin{aligned} D_6 &= \sum_{k=1}^5 (-1)^{k+1} \binom{6}{k} M_{6-k}^{2^k} \\ &= 2\,001\,589\,252\,896. \end{aligned} \tag{25}$$

This calculation uses only $M_1, \dots, M_5$, all recomputed by the accompanying exact program.

Theorem 10.1 (Explicit rarity bound). *For $n \geq 6$, the fraction of blind binary output maps that are executable satisfies*

$$\frac{L_n}{B_n} = \frac{F_n}{M_n} < \frac{2}{n+1} \left(\frac{10\,396\,884\,292\,704}{12\,009\,535\,517\,376} \right)^{2^{n-6}}. \tag{26}$$

In particular, this fraction tends to zero doubly exponentially.

Proof. Partition Q_n into 2^{n-6} six-dimensional faces and choose independently a matching counted by D_6 in each face. This gives $M_n \geq D_6^{2^{n-6}}$.

From $r_{n+j} \geq (n+j)/2$ and (17),

$$F_n \leq \frac{2}{n+1} e^{\gamma 2^n}.$$

Use the upper bound in (24) at $N = 6$ and the elementary strict inequality $e^{1/7} < 7/6$ to obtain

$$e^{64\gamma} \leq 7F_6 e^{1/7} < \frac{49F_6}{6}.$$

Thus

$$\frac{F_n}{M_n} < \frac{2}{n+1} \left(\frac{49F_6}{6D_6} \right)^{2^{n-6}}.$$

The numerator and denominator in (26) are exactly $49F_6$ and $6D_6$. Their strict order is an integer comparison, verified by the program. No numerical approximation to γ is needed. $\square$

This theorem concerns the uniform counting measure on extensional blind maps. It is not a claim about the frequency with which a particular programming language generates legal programs. It quantifies how weak the semantic blindness condition becomes as a substitute for actual executability.

11 How much does a coordinate budget reveal?

Return to arbitrary finite q , but now assume that I is countable. This removes the variable-evaluation measurability obstruction. Let $\tau = (b, a)$ be blind and measurable, and regard Q as the cyclic group $\mathbb{Z}/q\mathbb{Z}$. Define the residual

$$Z = x_{b(x)} - a(x) \pmod{q}.$$

For $K \subseteq I$, let X_K denote all values on K and put $\rho_K = \mu_I(b \in K)$. Entropies use natural logarithms.

Theorem 11.1 (Sharp information-budget inequalities). *The random variable Z is uniform on Q . For every measurable estimator $g(X_K)$,*

$$\mathbb{P}(g(X_K) = Z) \leq \frac{1}{q} + \left(1 - \frac{1}{q}\right) \rho_K, \quad (27)$$

$$H(Z | X_K) \geq (1 - \rho_K) \log q, \quad (28)$$

$$I(Z; X_K) \leq \rho_K \log q. \quad (29)$$

For $q = 2$ and every event A determined by X_K ,

$$\mu_I(E_\tau \triangle A) \geq \frac{1}{2} \mu_I(b \notin K). \quad (30)$$

Proof. For a fixed $i \notin K$ and $c \in Q$, the fiber $\{b = i, a = c\}$ is measurable and independent of the value at i in the pointwise invariance sense. Intersecting it with an event determined by X_K preserves that invariance. Integration in coordinate i therefore gives, for every $z \in Q$ and every such event D ,

$$\mathbb{P}(D, b = i, a = c, Z = z) = \frac{1}{q} \mathbb{P}(D, b = i, a = c).$$

Sum over $i \notin K$ and c . Conditional on X_K , the contribution of the external-target case is exactly a uniform subprobability of mass $1 - p(X_K)$, where

$$p(X_K) = \mathbb{P}(b \in K | X_K).$$

Consequently the conditional law of Z has the form

$$(1 - p(X_K)) U_q + p(X_K) \eta_{X_K},$$

where η_{X_K} is some probability on Q whenever $p > 0$. Each point mass is at most $(1 - p)/q + p$, proving (27). Concavity and nonnegativity of finite entropy give conditional entropy at least $(1 - p) \log q$, proving (28) after integration.

Taking $K = \emptyset$ proves that Z is uniform, so $H(Z) = \log q$ and (29) follows. When $q = 2$, on the external-target case the conditional probability of either success or failure is $1/2$. Hence

$$\mu_I((E_\tau \triangle A) \cap \{b \notin K\}) = \frac{1}{2} \mu_I(b \notin K),$$

which implies (30). $\square$

Example 11.2 (Sharpness). Take two data coordinates $i_{\text{in}} \in K$ and $i_{\text{out}} \notin K$, plus a control block inside K disjoint from both. The player reads only the control block. On a control event of probability ρ , it chooses i_{in} ; otherwise it chooses i_{out} . It always guesses zero. Conditional on X_K , the residual is known exactly on the first case and is uniform on the second. Thus all three bounds (27)–(29) are attained. A countable fair control block permits any $\rho \in [0, 1]$, by comparison of its base- q real with a threshold; finite control blocks give the corresponding q -adic probabilities.

These inequalities quantify an observer's information about the outcome, not the player's information while running the strategy. In the example, the observer's set K contains a coordinate that the player deliberately leaves unopened. For finite K , (30) gives a direct lower bound on the error of any finite-coordinate approximation to the success event.

12 Effective finite certificates and the Busy Beaver barrier

12.1 Finite averaging without measure theory

For a blind map on Q^n , its fibers are unions of full lines in the selected coordinate, with one successful point per q -point line. Therefore

$$\sum_{x \in Q^n} \mathbf{1}_{E_\tau}(x) = q^{n-1}. \quad (31)$$

For m players, the total of all scores is mq^{n-1} . Some configuration has score at most $\lfloor m/q \rfloor$. This is an integer identity amenable to a small exact checker.

A finite output-table certificate consists of q, n, m , the output pair for each player on each configuration in lexicographic order, and optionally a proposed counterexample configuration. The checker verifies dimensions and ranges, condition (1) for every one-coordinate replacement, and the exact score. A theorem connecting accepted tables to actual programs is a separate obligation: blind tables alone need not be executable, as the four-box example demonstrates.

12.2 Total oracle strategies have finite certificates

Theorem 12.1 (Uniform extraction under a totality promise). *Given programs for finitely many deterministic oracle strategies on $Q^{\mathbb{N}}$, promised to halt on every oracle and never to query their eventual targets, there is a uniform partial algorithm that halts on every such promised input and returns a finite prefix σ whose every extension gives team score at most $\lfloor m/q \rfloor$.*

Proof. First prove existence of a finite common horizon. Every individual halting run reads finitely many coordinates, has a finite running time, and outputs a finite target index. All oracles agreeing on its queried coordinates produce the same run and output. These finite-coordinate neighborhoods cover $Q^{\mathbb{N}}$. Compactness supplies a finite subcover, and hence finite common bounds on queried indices, target indices, and running time. Repeat for the finitely many players and take maxima.

The bounds can be found effectively under the promise. Search over pairs (L, T) of positive integers. For each of the q^L prefixes, simulate every player for at most T steps. Reject the candidate bounds if any simulation asks for a coordinate outside $\{0, \dots, L-1\}$, fails to halt within T , or outputs a target at least L . A sufficiently large pair passes by the preceding argument.

The accepted simulations produce finite output tables on Q^L . They are blind because the original programs are legal. By (31), exhaustive search finds a prefix with score at most $\lfloor m/q \rfloor$. Every extension produces the same runs and the same target values, since all targets lie within the prefix. Return that prefix. $\square$

The promise is substantive. This theorem is not a decision procedure for whether arbitrary oracle programs are total or legal. It also makes no useful polynomial or elementary running-time claim.

Theorem 12.2 (No computable length bound for the horizon). *For any effective programming formalism supporting ordinary Turing simulation, no total computable function of program length bounds the necessary input horizon of every total legal oracle strategy. This remains true for strategies that make no oracle queries.*

Proof. Given a program P , compile a strategy S_P that simulates P on blank input, counts its steps, and, if P halts after t steps, outputs target $t + 1$ and guess zero. The strategy makes no queries. Whenever P halts, S_P is total and legal, and a horizon including its target must be at least $t + 2$.

Suppose a total computable bound f existed. The length of the compiled program S_P is computable from P . If P halts, it must do so before the computable bound $f(|S_P|)$. Simulate P for that many steps: a failure to halt would establish that it never halts. This would decide the halting problem, a contradiction. $\square$

For an encoding equipped with an additive-overhead simulation compiler, this argument can be expressed as

$$H(n + c) \geq \text{BB}_{\text{time}}(n) + 2,$$

where H is the maximum horizon over total strategies of length at most its argument and BB_{time} is a *program-length running-time* Busy Beaver function. The compiler and encoding are hypotheses of this sharpened comparison. This is not the same numerical function as an n -state Rado printing-score Busy Beaver function. The distinction mirrors the explicit separation between machine semantics, maxima, and compiler assumptions in the inspected ProveIt core [7].

[write] Added 3 October 2026, batch 87. The Busy Beaver development in Computability/BusyBeaver formalizes blank-tape two-symbol machines, their halting scores and the domination theorem, with the compiler as a separately stated hypothesis; it says nothing about H or BB_{time} , and neither Theorem 12.1 nor Theorem 12.2 is formalized there or elsewhere in ProveIt. The Hilbert’s-tenth review cited in Section 1.4 read this section and agrees with its hypotheses: the promise in Theorem 12.1 is essential, and the displayed inequality is conditional on the encoding and the compiler.

13 Executed checks and reproducibility

The accompanying program `verify_certificates.py` uses only the Python standard library. It validates the four-box nonrealizable matching, recomputes every cube matching through dimension four, checks sliceability independently against the recurrence, and recomputes M_5 by an exact permanent dynamic program.

For the latter, bipartition the cube into even- and odd-parity vertices. Process even vertices in a fixed order, and use the bit mask of already matched odd vertices as a state. The recurrence sums over unused adjacent odd vertices. At dimension five there are only 2^{16} possible masks, rather than a search through all permutations. The result is $M_5 = 589185$.

The program also enumerates all $4^4 = 256$ two-box output tables, finding exactly eight blind ones. For all ordered teams of one through four such tables it checks (31) and the sharp minimax bound. The resulting counts by minimum score are:

Players	Ordered teams	Minimum 0	Minimum 1	Minimum 2
1	8	8	0	0
2	64	52	12	0
3	512	236	276	0
4	4096	988	2784	324

An additional explicit three-box team has player p target coordinate p and guess $1 - x_{p+1 \bmod 3}$. Each player reads one other coordinate. Its score is zero on 000 and 111, and two on the other six configurations. Thus its probability of at least two successes is $3/4$, attaining the first-moment upper bound $(3/2)/2$. This average-case fact does not contradict its failure to guarantee two successes.

The checker further validates 1,320 balanced-list constructions: $2 \leq q \leq 12$, $1 \leq r < q$, and $1 \leq m \leq 20$. It computes F_n through $n = 20$, verifies the exact integer comparison used in Theorem 10.1, and records rounded evaluations of (24). The separate script `formal_series.py` verifies the formal fixed-point identity and computes the coefficients of (21) and (23) through degree ten, using exact rational arithmetic. It independently checks the displayed coefficients; the next multiplicative coefficient after those in (14) is $2677/48$.

Verification boundary. All these finite tests were executed successfully. They are exact arithmetic checks except for explicitly labeled decimal evaluations of logarithms. They are not a proof-assistant verification of the infinite theorems, nor do they replace the proofs of the recurrence and asymptotic remainder estimates.

[write] **Added 3 October 2026, batch 87.** In the collection the programs are shipped as `code/verify_certificates.py` and `code/formal_series.py` (with the delivered `code/Makefile`), and their recorded outputs as `data/certificates.json`, `data/verification_report.json` and `data/formal_series_report.json`, all byte-identical to the delivery. Both programs write into the current directory, and the first one rewrites `certificates.json` too, so they are rerun on a scratch copy only; the report's `README.md` gives the commands. The intake's replay on such a copy (3 October 2026, Python 3.14.4 on Windows) passed all three commands of the `Makefile` target `verify` in a few seconds, and the three regenerated files equal the shipped ones after the CRLF line ends that Windows text mode writes are stripped.

14 Foundations and a concrete ProveIt integration plan

14.1 Set-sized objects and explicit ambient axioms

All configuration spaces in this note are sets Q^I for set-sized I . The index space $\mathbb{N} \sqcup 2^{\mathbb{N}}$, the set of strategies, and the σ -algebras used in the proofs are sets. No class of all ordinals, all surreal numbers, or all sets is being treated as a measurable sample space.

The measure-theoretic development is given in ordinary classical mathematics, with ZFC as a sufficient ambient theory. The uses of countable support, product probability, null-set completion, Radon–Nikodym, and compactness are stated explicitly. No equivalence with ZF, ZF plus dependent choice, or a subsystem of arithmetic has been proved here. In particular, the proof should not be transported into a choiceless model merely by reusing the same notation.

The finite counting and certificate results do not use an uncountable probability space. Their proofs employ finite sets, induction, and integer arithmetic. The asymptotic theorem adds real analysis of an explicit integer recurrence. These layers can therefore be formalized separately, instead of requiring the entire continuum theory before checking a finite certificate.

14.2 Relevant inspected repository interfaces

The pinned Busy Beaver core explicitly separates a Rado-style machine representation and score maxima from the compiler interface used in domination arguments [7]. This is the right model for Theorem 12.2: first state which programming model and size measure are used, then prove or assume the appropriate compilation theorem.

The finite-list-coding README documents validity guards, unique decoding, round-trip properties, and genuine first-order arithmetic predicates [8]. These are suitable interfaces for encoding output tables and

finite computation histories. A malformed table must not become a meaningful object just because a total decoder supplies a default value.

The closure-axiomatization README distinguishes semantic directions from a deductive-equivalence layer built on a first-order calculus [9]. That distinction is relevant to the eventual weak-foundation audit: a theorem proved in an ambient Lean type theory is not automatically a theorem that a specified first-order set theory proves internally. These are descriptions of inspected source interfaces, not a claim that the entire repository was rebuilt during this task.

14.3 Suggested modules and proof obligations

A practical implementation would start with four small mathematical modules rather than a monolithic formalization.

Finite blind maps. Define a table on functions $\text{Fin}(n) \rightarrow \text{Fin}(q)$, a coordinate-update operation, and predicate (1). Prove the finite identity (31), then its team bound. A Boolean checker needs a reflection theorem equating acceptance with the mathematical predicate.

Decision trees and partitions. Give queries and leaves an operational semantics with an explicit set of previously queried coordinates. Prove that legal executions induce blind maps. Formalize the matching or line-partition correspondence, the leaf-refinement argument, and the sliceability equivalence. The four-box example is a regression test against accidentally declaring the converse of blindness to be automatic.

Enumeration and asymptotics. Prove the intersection formula for first-query classes before deriving inclusion–exclusion. Treat F_n as the cardinality of an explicit finite family, then prove the recurrence agrees with the numerical evaluator. The exact rarity bound can be checked before the all-orders asymptotic theorem, since it needs only ratio inequalities and finite constants.

Countable support and probability. State the measurable codomain assumptions explicitly, establish the countable-support factorization, and prove finite-coordinate integration lemmas. The fresh replacement construction then supplies a measurable-space extension with a pushforward measure. The diagonal counterexample should be formalized before any automatic rule for evaluating a process at a random index is trusted.

No Lean source claiming these new theorems as already established is included. The supplied code is a working computational companion, while this section specifies the remaining proof-assistant work.

15 Further research questions and topics

The first question below is a question from Glazer’s paper. The others are precisely formulated directions suggested by this note; they are not all claimed to be previously recognized open problems.

- Q1. The unrestricted choiceless continuum problem.** Determine the ZF status of the three-player binary continuum game posed in [1]. A relevant next step is to compare any proposed strategy with the exact output-regularity obstruction here, without imposing measurability by assumption.

- Q2. Larger measurable structures.** Which versions of Theorem 3.1 remain true when cylinder measurability is replaced by full product-topology Borel measurability, the Baire property, or universal measurability? Each proposed replacement must specify the ambient measurable or topological structure on both configurations and target labels.
- Q3. Weak foundational strength.** Determine the weakest explicit choice and comprehension principles sufficient for the countable-support minimax proof. Separate a theorem about coded measurable maps from a theorem quantifying over all members of an abstract generated σ -algebra.
- Q4. Universal fair extensions.** Theorem 4.1 handles countably many output maps at once. Is there a useful maximal class of uncountably many maps whose success events admit a simultaneous countably additive fair extension? Which extra conditional-independence requirements obstruct such an extension?
- Q5. Joint extension classification.** For a finite team, characterize all extensions of the product measure to its success events, including their possible joint distributions. The single-event density classification in Theorem 6.2 suggests a finite-family theory with pointwise compatibility constraints.
- Q6. q -ary decision-tree asymptotics.** Starting from (10), derive a uniform-in- q expansion, identify the appropriate polynomial prefactor, and compare different alphabets. The fixed- q ratio method is a starting point, but uniformity in a growing alphabet is a separate problem.
- Q7. Beyond inverse powers.** Investigate the remainder after optimal truncation of (12), and whether the sparse intersection orders $2^k - k - 1$ govern identifiable smaller asymptotic scales. The all-orders result here does not settle such questions.
- Q8. Structural enumeration and database identification.** Identify the sliceable-matching sequence in the wider literature on decision-tree partitions, and determine whether an existing OEIS entry records it under another interpretation. Investigate symmetries, unlabeled versions, and counts refined by the set of permissible first queries.
- Q9. Optimal average-case teamwork.** For fixed (m, q, n) , determine the largest probability of at least t successes over legal decision trees. Compare it with the larger class of blind maps and locate the first parameter values where the optima differ. The exact 3/4 binary example gives a small benchmark.
- Q10. Complexity of realizability.** Given a succinct blind output map, what is the complexity of deciding whether it is executable? How does the answer change for an explicit truth table, a Boolean circuit, a matching list, or an oracle program with a supplied totality certificate?
- Q11. Sharper information costs.** Refine Theorem 11.1 when observation sets are adaptive, observation costs are nonuniform, or the box law has correlations. The product-law proof identifies precisely which missing-coordinate integration identity needs replacement.
- Q12. Kernel-checked certificate compilation.** Implement the finite reflection theorem and the decision-tree-to-table compiler, then connect the oracle horizon theorem to a formally specified machine model. Track compiler size and time overhead rather than asserting an encoding-independent Busy Beaver comparison.

[write] **Status of the questions, 3 October 2026, batch 87.** All twelve questions stay open as printed. **Q1** is Glazer’s question “Does ZF prove that $G([3], \mathbb{R}, \{0, 1\})$ is winnable?” from Section 5 of [1] (one arXiv version, v1 of 15 November 2022); the intake did not search the later literature for its status, and, as Appendix B says, this report does not

assert that it is unresolved everywhere. It is unrelated to the “Glazer’s Question 1” answered (as claimed, unrefereed) in Part VIII of the surreal report *polish-models-of-omnific-arithmetic*, which belongs to a different paper. For **Q8**, the database half was checked at intake: on 3 October 2026 no OEIS entry matched 1, 2, 9, 232, 206065 or 2, 21, 33976 (Section 1.4); identification in the wider literature on decision-tree partitions, and any OEIS submission, remain open.

[write] Status of Q9, 4 October 2026, batch 90. Q9 is answered for one family and otherwise stays open. Corollary 20.3 of Part II takes $m = n = N = 2a(2^r - 1)$ binary boxes with player i ’s target fixed to box i and the threshold $t = a2^r$: no legal compulsory strategy reaches t correct guesses with probability above $1 - 2^{-r}$, and the syndrome block of Section 20, in which each player reads the other hats of its block in a fixed order and never its own, attains it. The upper bound is the first-moment bound of Proposition 18.3, and it holds for every team of m blind binary output maps of Section 2, targets free: by Proposition 7.1 each such map succeeds on exactly half of the configurations, so the expected score is $m/2$ and the probability of at least t successes is at most $m/(2t)$. Hence, for $(m, q, n) = (N, 2, N)$ and this t , the optimum over legal decision trees and the optimum over blind maps coincide and both equal $1 - 2^{-r}$; the same holds for every $n \geq N$, the further boxes being left unread (this extension is the write’s observation, not the manuscript’s). The member $r = 2, a = 1$ is $(6, 2, 6)$ with $t = 4$ and optimum $3/4$; the three-box benchmark $(3, 2, 3)$, $t = 2$, of Section 13 is not in the family (3 is not of the form $2a(2^r - 1)$), although it too attains the first-moment bound $(3/2)/2$. Every other (m, q, n, t) remains open, and so does the question’s last request, the first parameter values at which the legal and blind optima differ: in this family they never do. Part II’s Research question 29.4 continues Q9 for fixed targets and binary hats.

[write] Status of Q2, Q5 and Q9, 4 October 2026, batch 92. The batch-87 note’s “All twelve questions stay open as printed” and the batch-90 note on Q9 are superseded as follows; the other questions are unchanged.

Q2 is answered for two of its three clauses by Part III: for blind outputs measurable for the full product-topology Borel σ -algebra, or universally measurable relative to it, the optimal guarantee is still $\lfloor m/q \rfloor$ (Theorem 39.2, through the completion regularity of Corollaries 32.3 and 32.6). The Baire-property clause stays open: Proposition 42.1 reduces Baire-property outputs to countably many coordinates on an invariant comeagre set, and the example after it shows that such a set can be null, so the averaging proof does not transfer. Part III’s direction R1 continues it.

Q5 is answered by Theorem 36.1: the extensions to a finite team’s success events correspond exactly to measurable densities on the success patterns supported on the feasible-pattern events, unique almost everywhere. Theorem 38.1 describes the joint laws as a weighted Minkowski sum of simplices cut out by Hall inequalities, and Theorem 41.1 extends the classification to countable teams by kernels.

Q9 is settled for a larger family by Part IV’s source 04 (the note at the end of Section 65): for $(m, q, n) = (N, 2, N)$ with $N = m(2^r - 1)$ and $t = m2^{r-1}$, for every $m \geq 1$, the optima over legal decision trees and over blind maps coincide and equal $1 - 2^{-r}$, also for every $n \geq N$ (the further boxes left unread; the write’s observation, as in the batch-90 note). This contains the family of the batch-90 note (even m) and Part I’s three-box benchmark ($m = 1, r = 2$). The batch-90 note’s sentence “Every other (m, q, n, t) remains open” was not correct when it was written; Remark 15.1 gives the counterexamples. There is also the elementary divisor family of Remark 15.2: for every $q \geq 2$ and $n \geq 1$, if $t \mid m$ and $\lfloor m/q \rfloor < t \leq m$, both optima equal $m/(qt)$. The general optimization problem outside the cases established here, and the first parameters separating the legal and blind optima, remain unclassified by this report. In every case established here the two optima coincide. Part II’s Research question 29.4 and Part IV’s Research question 68.5 continue it.

Q4 is refined, not answered, by Part III’s direction R2 (its Theorem 41.1 treats countable teams, as Theorem 4.1 did for one fair extension), and Q3 is related to R7. Q11 is not addressed: Part IV measures the cost of adaptive inspection in a different game and does not refine Theorem 11.1.

Remark 15.1 (The batch-90 sentence “Every other (m, q, n, t) remains open” was false when written). This remark is the write’s (batch 92), recorded so that the superseded sentence stays on record with its refutation.

- (a) For every (m, q, n) with $n \geq 1$, the optimum is 1 when $t \leq \lfloor m/q \rfloor$ and 0 when $t > m$, over legal decision trees and over blind maps alike. For the first case, the constant team of Theorem 3.1 (every player targets one fixed box, the guesses spread as evenly as possible over the q colours) reads no box and has at least $\lfloor m/q \rfloor$ correct guesses on every configuration; the second case is trivial.
- (b) For $(m, q, n) = (3, 2, 3)$ and $t = 2$ the optimum is $3/4$ over both classes. Each blind binary output map

succeeds on exactly half of the configurations (Proposition 7.1), so every team of three such maps has expected score $3/2$ and, by Markov's inequality, probability at most $(3/2)/2 = 3/4$ of at least two successes. The three-box team of Section 13 (player p targets box p and guesses $1 - x_{p+1 \bmod 3}$, reading one other box) is a legal decision tree and attains $3/4$.

Both facts were in Part I when the batch-90 note was written; that note itself observes that the three-box team attains the first-moment bound. Since batch 92 the family of the preceding note is settled as well, and it contains (b).

Remark 15.2 (A second overstatement of the open range). The batch-92 update initially said that the remaining open parameters included “every $q \geq 3$ with $\lfloor m/q \rfloor < t \leq m$ ”; its README made the same assertion. That sentence was false, and is retained here with a counterexample and the more general calculation. For finite $n \geq 1$, each blind output map succeeds with probability $1/q$: for each target i and predicted colour a , the fibre where the output is (i, a) is invariant under changing coordinate i , and exactly one of its q equally likely coordinate values is correct. Summing over the finitely many fibres gives this probability. Thus every m -player blind team has $\mathbb{E}S = m/q$, and $\mathbb{P}(S \geq t) \leq m/(qt)$ for $t > 0$. If $t \mid m$ and $\lfloor m/q \rfloor < t \leq m$, put $k = m/t < q$. Choose one fixed box and k distinct colours; assign exactly t players to each colour, all predicting that colour at that same box without inspecting anything. This legal team has score t on those k colours and zero otherwise, so $\mathbb{P}(S \geq t) = k/q = m/(qt)$. Hence the legal and blind optima coincide at that value for every $n \geq 1$. In particular $(m, q, n, t) = (1, 3, 1, 1)$ has optimum $1/3$, and more generally $t = m$ has optimum $1/q$. This argument concerns Q9's free-target box game. It does not change the own-hat, fixed-target minimum-depth classification of Section 65.

16 Conclusion

Two distinct mistakes are exposed by this analysis. On uncountable products, a measurable choice of target does not ensure a measurable success event. On finite products, invariance under changing the target does not ensure that the output rule can be executed without querying it.

The first distinction leads to an exact measurable-output minimax theorem, a simultaneous fair extension, and a sharp countable-target obstruction. The second leads to a combinatorial classification, a closed recurrence for extensional legal rules, an all-orders asymptotic expansion, and a quantitative rarity theorem. These are complementary rather than competing explanations: regularity limits which infinite outcomes can be assigned probabilities, while executability limits which finite semantic maps are operationally meaningful.

The next verification milestone is not another numerical experiment. It is a kernel-checked bridge from a legal decision tree to an accepted finite table, followed by the sliceability and inclusion–exclusion theorems. That would turn the strongest finite results of this note into reusable components in ProveIt, while leaving the genuinely set-theoretic questions visible and separate.

A Assumption and claim ledger

Result	Main hypotheses	Verification and scope
Sharp minimax	Standard Borel targets; cylinder-measurable blind output maps; finite alphabet and team	Full proof; success-event measurability not assumed.

Result	Main hypotheses	Verification and scope
Fair extension	Same output regularity; countable team	Full construction; marginal fairness, not independence or uniqueness.
Target collapse	Blindness; cylinder-measurable success, or completed measurability	Full proof; output-map measurability not assumed.
Diagonal example	Tagged continuum of targets; cylinder product probability	Full proof of inner/outer measures and extension classification.
Sliceability	Finite deterministic query model; binary outputs	Full equivalence proof; finite counterexample checked.
Enumeration	Extensional output maps, not program descriptions	Recurrence proved; matching counts independently checked through dimension four.
All-orders expansion	Explicit recurrence; ordinary real analysis	Remainder argument supplied; rational coefficients computed separately.
Rarity	Exact counts $M_1, \dots, M_5$ and ratio inequalities	Counts recomputed; final comparison is exact integer arithmetic.
Oracle extraction	Totality and legality on every oracle are promises	Existence and uniform partial algorithm proved; no practical complexity bound.
Busy Beaver comparison	Specified encoding and simulation interface	Uncomputable horizon theorem proved; additive-overhead inequality conditional on its stated compiler.

B Source and priority audit

The mathematical anchor is the full text of [1], including its final continuum-box question, inspected also as a rendered PDF page. The original Łoś–Marczewski article was inspected as scanned pages; its general extension theory is credited, while the special density classification here is proved directly. The OEIS A005271 entry was inspected for the identity of the all-matching sequence, not as a substitute for the executed small counts.

Repository files were read at the pinned commit, especially the initial machine and compiler description in the Busy Beaver core, the finite-list coding README, and the closure-axiomatization README. No repository-wide build was run, and no repository files were modified.

The LinkedIn page was not fully accessible; research interests were instead anchored in authored papers and the author profile in [3]. A complete dissertation-level priority comparison was not obtained, and exact-sequence searches did not establish whether the sliceable sequence already has a separate catalog entry. Consequently this note makes no claim of exhaustive literature coverage, no assertion that its formulations are historically first, and no assertion that the original unrestricted problem remains unresolved everywhere as of the preparation date.

[write] **Added 3 October 2026, batch 87.** The intake read the arXiv record and PDF of [1] (one version;

Section 1.4), the record of [2] (three versions, v1 of 19 December 2023 to v3 of 7 January 2024) and the entry A005271. The post cited as [3] was reachable at the cited samaritan-research.org address on 3 October 2026, with the title, authors, dates and Glazer’s author profile as cited; the same post is published on Epoch AI’s blog, <https://epoch.ai/blog/frontiermath-competition-setting-benchmarks-for-ai-evaluation>, which the batch-87 manuscript printed as Part VIII of polish-models-of-omnific-arithmetic cites; the intake did not establish which copy is the original. The remark above about a LinkedIn page names no address, and none is recorded here. The package’s checksum manifest SHA256SUMS (nine entries) was verified 9/9 at placement and is not shipped, by repository policy; the delivered PDF and README are not shipped either, and survive in the arrival commit’s archive. The priority search was not extended at intake.

Part II

No universal sublinear bound for measurable hat guessing

Finite coding, exact logarithmic excess, and arbitrarily prescribed growth

Research report prepared for Vladimir Reshetnikov, October 3, 2026

Abstract. Let S_n be the number of correct guesses among the first n players in the infinite binary hat game, with independent fair hats and compulsory guesses. We construct strategies in which each player inspects only finitely many other hats. For every prescribed positive function $g(n) = o(n)$, one such strategy satisfies

$$\frac{S_n - n/2}{g(n)} \longrightarrow +\infty, \quad \frac{S_n}{n} \longrightarrow \frac{1}{2} \quad \text{almost surely.}$$

A simpler schedule gives $S_n - n/2 = \log_2 n + O_\omega(1)$. These conclusions answer the two growth questions in Eldredge’s 2025 Remark 6.8, on a topic explicitly attributed there to Elliot Glazer and Charles Wang.

The construction adapts classical Hamming-code and XOR amplification methods from Ebert, Merkle, and Vollmer’s work on autoreducibility. We give an explicit transformation from partial predictions to compulsory guesses, prove a bound at every prefix, and obtain a deterministic piecewise-linear growth profile with bounded random error. Additional results include a sharp family of finite-team success probabilities, a conull but meager success set, effective versions, and a concrete formalization plan connecting the construction to ProveIt.

Status and attribution. The report supplies complete mathematical proofs and executable finite checks. It does not claim historical priority for the coding mechanism or the slow-density phenomenon: substantial prior art is identified explicitly. The infinite theorems have not been verified by a proof assistant. The finite checker is supporting evidence, not a substitute for their proofs.

Keywords: hat guessing; autoreducibility; Hamming codes; measurable prediction; asymptotic growth; finite certificates.

[write] Added 4 October 2026, batch 90. This Part is manuscript 02 of batch 90 of ProveIt’s incoming-reports intake, printed in full after Part I (provenance, the checks made at intake and a notation table: Sections 17.3 and 17.4). Like Part I it is AI-assisted, unrefereed and not formalized. To keep Part I’s numbers, its sections continue Part I’s: **the manuscript’s Section k is Section $k + 16$ here, its statement or equation $k.m$ is $(k + 16).m$, and its Appendices A–C are Appendices C–E.** Thus the manuscript’s Corollary 4.3 is Corollary 20.3, its Theorem 7.1 is Theorem 23.1 and its Theorem 8.1 is Theorem 24.1. Numbers such as “Remark 6.8” or “Theorem 6.6” that cite Eldredge [10] or Ebert–Merkle–Vollmer [11] are theirs and unchanged. Apart from this offset, the editorial changes listed in Section 17.3 and the dated **[write]** notes, no statement, proof, number or non-claim of the manuscript was changed.

17 The research intersection and the precise questions

The intersection considered here is prediction without access to the coordinate being predicted. It links Glazer’s work on box-game paradoxes and choice [1] with the finite prediction and certification problems in ProveIt’s report *Countable Information, Uncountably Many Boxes* [12]. It also meets computability theory through autoreducibility: computing selected bits of a sequence without asking for those same bits.

The direct published target is Eldredge’s discussion of the surplus

$$D_n := S_n - \frac{n}{2}.$$

That discussion credits Glazer and Wang with suggesting the topic. Remark 6.8 asks whether logarithmic or faster almost-sure growth is possible, and whether a deterministic sublinear scale can universally bound the lower growth of D_n [10]. We give explicit answers:

1. Exact logarithmic growth is possible, up to a bounded random additive error.
2. Every positive deterministic sublinear scale can be exceeded eventually by an arbitrarily large factor, almost surely.

Both conclusions hold for continuous strategies whose individual guesses depend on finitely many hats. The construction also maintains the limiting fraction of correct guesses at exactly $1/2$.

17.1 What is classical, and what this report establishes

The main coding ingredients have substantial prior art in Ebert–Merkle–Vollmer [11, Theorem 6.6]. Their theorem concerns partial autoreductions of computably random sequences and arbitrarily slowly vanishing densities of predicted coordinates. Their proof already combines Hamming codes with XOR aggregation of repeated rounds. We adapt that mechanism to compulsory guessing and give a direct construction with explicit bounds at every physical prefix. We do not claim that the coding mechanism or the slow-density phenomenon is new. Priority of this precise compulsory-guess formulation has not been established.

The role of the present report is therefore mathematical resolution and explicit synthesis: it proves the stated hat-game conclusions directly, identifies the older machinery supporting them, and adds concrete finite optimality, topological, quantitative, and verification statements. It does not infer novelty from a failure to find a matching title.

Item	Result or role	Status in this report
Hamming-code predictions	Rare common failure with many coordinated correct partial predictions	Classical ingredient, credited to [11]
Compulsory guessing	Pairing turns a partial correct/wrong guess into surplus $+1/-1$	Explicit transfer with proof
Finite-team problem	Exact optimal probability at a family of score thresholds	Derived theorem; direct ProveIt connection
Logarithmic question	$D_n = \log_2 n + O_\omega(1)$	Explicit answer to the stated question
Universal sublinear scale	For each $g = o(n)$, $D_n/g(n) \rightarrow \infty$ a.s.	Negative answer to the proposed bound
Probability and category	Conull meager success; residual density oscillation	Derived structural theorem
Formal verification	Finite exhaustive checks and a modular proof plan	No claim of a kernel-checked infinite proof

17.2 Source scope and the repository snapshot

We inspected ProveIt at commit

7c0f2d9f92c3d51ec85703bed1022924a4b7359b.

The most direct source is the report under

SetTheory/Cardinals/docs/reports/ordinals-and-order-types/
measurable-box-games/.

Its ninth further-research item asks for optimal average-case teamwork over legal finite decision trees. Our finite construction gives an exact answer for a family of parameters in the special case where each player’s target coordinate is fixed in advance.

The source review also identified the finite-list coding interfaces in `Logic/PeanoArithmetic/ListCoding/` [13]. Their usefulness is in encoding finite tables and verified computations; they do not by themselves prove an infinite probabilistic theorem.

The arXiv record inspected for [10] lists version 2, dated October 24, 2025. Statements about answering its questions refer to that specific text. No authorship, endorsement, or review by Glazer, Eldredge, or the authors of [11] is implied.

17.3 Provenance and place in this report

[write] Added 4 October 2026, batch 90. *Source.* This Part is a single manuscript, printed in full: manuscript 02 of batch 90 of ProveIt’s incoming-reports intake, delivered as the archive `hat_guessing_research.zip` (550,545 bytes; inner directory `hat_guessing_research/`, main file `hat_guessing_growth.tex` of 1,587 lines, a 22-page A4 PDF) in the arrival commit `a162e4386`, and placed in commit `12076b2e8` as Part II of this report, with its code, data, figure and audit under the prefix `02-hat-surplus-`. The author line reads “Research report prepared for Vladimir Reshetnikov” and carries no AI wording; like the other deliveries of the intake, it is AI-assisted, unrefereed and not formalized. It was placed here, not as a report of its own, because it names this report as its most direct source and answers Research question Q9 of Part I for a family of parameters (Corollary 20.3); its own subject, Eldredge’s growth questions, is new to ProveIt. No other manuscript was merged into this Part, so it needed no merge choices.

Pin. The pin `7c0f2d9f92c3d51ec85703bed1022924a4b7359b` (quoted in Section 17 and in [12]) is the batch-89 arrival commit “New research reports” of 3 October 2026, an ancestor of the placement. It holds this report as placed in `d151b39ca` but before its batch-87 write `5f02820a7`: the manuscript read the delivered text of Part I (unprefixed labels, no **[write]** notes), whose ninth further-research item, at line 914 there, is Research question Q9 of Section 15 here, unchanged in substance.

Repository claims, checked at the write. (i) The ninth further-research item asks for optimal average-case teamwork over legal finite decision trees: true. (ii) The finite construction “gives an exact answer for a family of parameters” when each target is fixed: true, and, as the manuscript itself says after Corollary 20.3, the optimum is the same over blind maps; the note after the questions of Section 15 records what this settles. (iii) The source audit (`02-hat-surplus-source_audit.txt`) found no infinite surplus-rate theorem in Part I: true, and at placement no tracked file of ProveIt mentioned Eldredge’s paper, arXiv:2508.02828, autoreducibility, Ebert–Merkle–Vollmer or hat guessing. (iv) The finite-list coding files of [13], `Logic/PeanoArithmetic/ListCoding/README.md`, `Lean/PAListCoding/Basic.lean` and `Lean/PAListCoding/Predicates.lean` below that project, exist and are byte-identical at the pin and at the write (blobs `74164ac6`, `b7e037f4`, `c9f8f650`); Section 28 describes them accurately as interfaces for encoding finite tables, not as proofs. The four formalization layers there are proposals: no Lean or Rocq file of ProveIt states any result of this Part. No repository claim is refuted.

Eldredge’s paper, checked at the write. On 4 October 2026 the arXiv record of [10] listed two versions, v1 of 4 August 2025 and v2 of 24 October 2025 (20 pages; math.PR, cross-listed to math.LO), and the HTML text of v2 was read. The manuscript’s account of it is accurate. Its Theorem 4.1 states that for every measurable strategy, almost surely, the lower density of the correctly guessing players is at most $1/2$ and the upper density at least $1/2$. The paragraph before its Proposition 6.6 credits the study of the excess $S_n - n/2$ to a suggestion of Elliot Glazer and Charles Wang (personal communication); Proposition 6.6 gives a measurable strategy with $S_n - n/2 \rightarrow +\infty$ almost surely,

and Remark 6.7 explains it, as Section 22 does, by an infinite expected number of players in bad blocks. Remark 6.8 states that the excess of that strategy is of order $\log \log n$ almost surely, of order $\log n / (\log \log n)^2$ for a variant, that such variations do not reach order $\log n$, and asks two questions it says it cannot answer: (1) is there a measurable strategy with $S_n - n/2$ of order $\log n$ almost surely, or faster, and (2) is there a function g such that every measurable strategy has $\liminf_n (S_n - n/2)/g(n) = 0$ almost surely, a property the remark attributes to Theorem 4.1 for $g(n) = n$ and conjectures possible for $g(n) = \log n$. Theorem 23.1 and Propositions 23.2–23.3 answer (1), and Theorem 24.1 with Corollary 24.2 answers (2) negatively for every positive $g = o(n)$; the notes after those statements give the scope, and the note after Remark 24.4 checks the audit’s remark on the wording of (2).

Prior art, partly checked. The ECCC record of [11] (report TR02-056 of 19 September 2002, by Ebert, Merkle and Vollmer, *On the Autoreducibility of Random Sequences*) exists, and its abstract concerns how dense the set of guessed bits can be in autoreductions of random sequences. The content of Theorem 6.6 of the SIAM paper, the manuscript’s principal priority reference, was not checked at the write; the manuscript’s own statement that the coding mechanism and the slow-density phenomenon are not new stands as its claim.

Review in the Hilbert’s-tenth research tree. After the placement, another session inventoried the six batch-90 archives at their arrival revision in `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-q` `ueue/review_polish_partx_c9bc70d8f.md` (commit bcc1a4438; member hashes in the companion `.json`). For this archive it read only the delivery `README.txt` and `code/README.txt`, in full, and routed it as finite-dependence hat strategies with asymptotic surplus whose finite XOR and parity diagnostics have size-dependent exhaustive scope, with no paid Diophantine source in those READMEs; it executed no program and did not read the manuscript. A second review there, `review_batch90_placement_12076b2e8.md` beside it (commit 3c25fbb55), authenticated the placement by byte comparison: every file placed for this Part equals its archive member, the CSV keeps its delivered CRLF bytes, and the host article and README were unchanged at the placement; it read none of these files as mathematics. No bound of the Hilbert’s-tenth programme changes, and no scope correction follows for this Part.

Editorial changes in printing this Part. All 55 delivered labels carry the prefix `mbg:hat:`, with every reference updated; the write added `mbg:hat:part`, `mbg:hat:sec:provenance`, `mbg:hat:sec:notation`, `mbg:hat:rem:wording` (on the manuscript’s Remark on the wording of Eldredge’s question) and `mbg:hat:q:finiteteam` (on Research question 29.4), and `mbg:part:one` for Part I. Sections, statements, equation tags and appendices are renumbered by the offset stated at the head of this Part. The manuscript’s title page became the Part heading, abstract, status box and keywords above. Its `\PP` and `\EE` are printed with Part I’s macros for the same symbols $\mathbb{P}$ and $\mathbb{E}$; `ProveIt` is set in Part I’s typewriter style rather than italic, and remarks in Part I’s upright style. Its bibliography entry for Glazer’s *A choiceless box game paradox* (arXiv:2211.10474, with the date 15 November 2022 and the DOI) is the same paper as Part I’s entry [1], cited by that number; its other four entries follow Part I’s as [10]–[13]. Figure 1 is the delivered `figures/block_prefix.pdf`, shipped as `figures/02-hat-surplus-block_prefix.pdf`; the manuscript included it only if present, and it is now included unconditionally, so a build without it fails instead of silently dropping it. The manuscript was set on A4 paper with Latin Modern; here it takes Part I’s US Letter layout and fonts, so line and page breaks differ from the delivered PDF.

17.4 Notation in this Part

[write] Added 4 October 2026, batch 90. Within Part II every symbol has the manuscript’s meaning, given in the middle column; the right column gives the meaning the same letter has in Part I, which is *not* meant here. No symbol was renamed.

Symbol	Meaning in Part II	<i>not</i> : meaning in Part I
legal, blind	a <i>legal compulsory strategy</i> has player i guess its own hat ξ_i with a guess unchanged when ξ_i alone changes: Part I’s blindness for a player whose target is fixed to its own box. For such a player, on finitely many hats, blindness and executability coincide (read every other hat), which is why Corollary 20.3 holds over both classes	“legal” = realizable by a decision tree that never queries its target (Theorem 7.3); blindness of a map with variable target, (1)

Symbol	Meaning in Part II	not: meaning in Part I
N, n, q	N the players of a finite team, each guessing its own binary hat; n a prefix length; $q(i)$ a visibility bound in Research question 29.2, and q a quotient in the proof of Lemma 21.1	m players, n boxes and the alphabet size q
measurable	on the countable product $\{0, 1\}^{\mathbb{N}_{>0}}$, for the product (or completed) σ -algebra	the cylinder σ -algebra Σ_I on an uncountable product
$S_n, D_n, D_u^{\text{loc}}$	correct guesses among the first n players; the surplus $S_n - n/2$; the surplus inside one block	$S(x)$ a team score; $D(u)$ in the proof of Theorem 3.1, the count D_6 of (25)
Z_i, Y_i	correctness indicator of player i ; $Y_i = Z_i - 1/2$	Z the residual of Section 11
r, t, a, M	code dimension; $t = 2^r - 1$ columns per round; a rounds; $M = at$ logical pairs (and $M(b)$ a convergence modulus in Section 27)	r list size, r_n , the label $r = x \upharpoonright \mathbb{N}$; t the success threshold of Q9 (here s or $a2^r$); $a(x)$ a guess map; M_n all perfect matchings
$h_j, h(n), g, H$	nonzero columns of $\mathbb{F}_2^r$; $h = \max\{g, \sqrt{n}\}$ in the proof of Theorem 24.1; the prescribed growth g ; a profile H in Research question 29.1	H the horizon of Section 12 and entropy in Section 11
$\sigma, U_{\ell j}, U_m$	the full syndrome (20.1); the visible syndrome; the sets of the Martin-Löf test in Proposition 27.1	σ in “ σ -algebra”; $U = (U_0, U_1, \dots)$ in Theorem 4.1
F, C	the piecewise-linear profile (22.1); the random debt (22.2)	F_n sliceable matchings; a group of players, Cantor space, the series $C(z)$
B_k, B_t, L_k, N_k, R_k	bad-block events; the sharp prefix constant of Proposition C.1; block lengths and endpoints; the thresholds (24.2)	B_n, L_n blind and legal map counts; L a prefix bound; $R(z)$ the series (21)
T_i, K	the flip of hat i (proof of Theorem 25.1); a block index	$T(Y, U)$ the replacement map of Theorem 4.1; K a coordinate set
$\mathbb{F}_2, \oplus, \text{FIN}, \mathcal{S}_g$	the two-element field; exclusive or; sequences with finitely many ones; the success set of Theorem 26.2	—
$\mathbb{P}, \mathbb{E}$	probability and expectation (the manuscript’s \PP, \EE)	the same
29.1–29.12	Part II’s Research questions	Part I’s questions Q1–Q12

18 The game and its elementary constraints

Let $\Omega = \{0, 1\}^{\mathbb{N}_{>0}}$, equipped with the fair product probability measure. Write $\xi_i(\omega) = \omega_i$ for player i ’s hat. The hats are independent, with each color having probability $1/2$.

Definition 18.1. A *legal compulsory strategy* is a family of functions $f_i : \Omega \rightarrow \{0, 1\}$ such that changing ξ_i alone never changes f_i . Player i ’s guess is $f_i(\omega)$. Put

$$Z_i = \mathbf{1}_{\{f_i = \xi_i\}}, \quad S_n = \sum_{i=1}^n Z_i, \quad D_n = S_n - \frac{n}{2}.$$

A strategy is *finite-information* if every f_i depends on a fixed finite set of coordinates not containing i .

A finite-information strategy is continuous in the product topology. Conversely, a continuous Boolean-valued function on Cantor space depends on finitely many coordinates: its two fibers are clopen, compactness supplies a finite cylinder cover, and a finite union of the coordinates in that cover determines the value. The finite set can grow with i ; a uniform bound is not assumed.

All strategies below are chosen before the hats are distributed. Players do not hear earlier guesses. The numerical ordering of the players determines the score S_n , not a restriction that a player may only inspect earlier hats.

Lemma 18.2 (Individual fairness). *For every measurable legal strategy and every i ,*

$$\mathbb{P}(Z_i = 1) = \frac{1}{2}, \quad \mathbb{E}S_n = \frac{n}{2}, \quad \mathbb{E}D_n = 0.$$

Proof. Condition on all hats except ξ_i . The guess is then fixed and ξ_i remains a fair independent bit, so the conditional probability of a correct guess is $1/2$. Integrating proves the first assertion. The others follow by finite additivity of expectation. $\square$

The correctness indicators Z_i need not be independent. Coordinating their dependence is the entire purpose of the construction.

Proposition 18.3 (Finite first-moment bound). *For N players and $s > 0$,*

$$\mathbb{P}(S_N \geq s) \leq \frac{N}{2s}.$$

Proof. The pointwise inequality $S_N \geq s \mathbf{1}_{\{S_N \geq s\}}$, followed by Lemma 18.2, gives the result. $\square$

For later use, it is convenient to represent the usual balanced two-player strategy in parity language. If a pair has hats u, v , the first player guesses v and the second guesses $u \oplus 1$. Exactly one is correct. Thus the pair contributes zero excess. If instead both players predict that their pair parity is b , their guesses are $v \oplus b$ and $u \oplus b$. Both are correct when $u \oplus v = b$, and both are wrong otherwise.

19 A transfer from partial predictions to compulsory guesses

The preceding observation provides a general bridge to autoreducibility. It also explains why a literature result allowing abstentions can be relevant to a game requiring every player to guess.

Definition 19.1. A legal partial predictor on logical bits $x_1, x_2, \dots$ consists of functions $p_j(x) \in \{0, 1, \perp\}$ whose value does not depend on x_j . The symbol $\perp$ means that the predictor passes. For the first m logical coordinates let C_m, W_m count, respectively, its correct and wrong nonpass predictions.

Theorem 19.2 (Pairing transformation). *Given any legal partial predictor, there is a legal compulsory strategy on physical pairs such that*

$$D_{2m} = C_m - W_m \quad \text{for every assignment and every } m.$$

Measurability, continuity, and effective finite-coordinate dependence are preserved. At an odd prefix, the excess differs from the preceding even-prefix excess by exactly $1/2$ in absolute value.

Proof. Use physical hats ξ_{2j-1}, ξ_{2j} to define the logical bit

$$x_j = \xi_{2j-1} \oplus \xi_{2j}.$$

Both players in pair j can compute all other logical bits without seeing their own hats. They evaluate p_j . If it passes, they use the balanced pair strategy. If it predicts b , both predict pair parity b .

A passed pair has one correct guess and contributes zero excess. A correct nonpass prediction gives two correct guesses and excess +1; a wrong one gives zero correct guesses and excess −1. Summing over the first m pairs proves the identity. The own-hat restriction is respected at every step. A finite logical query set uses the two hats of each queried pair and the player's mate; all these hats are visible. The claimed regularity properties follow by composition. Finally each individual correctness indicator, after subtracting $1/2$, is either $1/2$ or $-1/2$. $\square$

Corollary 19.3. *If a fixed partial predictor makes only finitely many wrong predictions almost surely and makes A_m nonpass predictions among the first m logical coordinates, then its compulsory lift satisfies*

$$D_{2m} = A_m - O_\omega(1).$$

Proof. Since $A_m = C_m + W_m$, Theorem 19.2 gives $D_{2m} = A_m - 2W_m$. The nondecreasing sequence W_m is eventually constant on the event in question. $\square$

Care is needed with quantifiers. A statement that each random sequence has some error-free autoreduction may choose that reduction using the sequence. It is not automatically a single strategy that works almost surely. Our direct block construction below chooses one fixed strategy in advance, permits finitely many erroneous blocks, and proves all its probability assertions without sequence-dependent hardwiring.

20 The finite syndrome block

The construction has two independent parameters:

$$r \geq 1, \quad a \geq 1, \quad t = 2^r - 1.$$

The block has a rounds, each containing t adjacent pairs. Thus there are

$$M = at \text{ logical pairs,} \quad N = 2at \text{ physical players.}$$

Label the nonzero elements of $\mathbb{F}_2^r$ by $h_1, \dots, h_t$, using increasing binary integer order if an explicit ordering is wanted. The columns $h_1, \dots, h_t$ repeat in every round.

Write $(u_{\ell j}, v_{\ell j})$ for the hats in pair j of round ℓ , and let

$$x_{\ell j} = u_{\ell j} \oplus v_{\ell j}.$$

The *full syndrome*, used for analysis, is

$$\sigma = \bigoplus_{\ell=1}^a \bigoplus_{j=1}^t x_{\ell j} h_j \in \mathbb{F}_2^r. \quad (20.1)$$

The strategy never asks a player to evaluate this expression using their own unknown hat.

20.1 The rule available to each player

For the pair (ℓ, j) , both players compute the visible syndrome

$$U_{\ell j} = \bigoplus_{(b,c) \neq (\ell,j)} x_{bc} h_c.$$

It excludes their entire pair, so both can compute it. They then use the following rule.

Visible value	Predicted pair parity	Action
$U_{\ell j} = 0$	1	Both guess mate's hat XOR 1
$U_{\ell j} = h_j$	0	Both guess mate's hat
$U_{\ell j} \notin \{0, h_j\}$	First 0, second 1	Balanced pair strategy

The first two rows are called *active*; the last is *neutral*. The rule is a finite Boolean computation and it is blind to each player's own hat.

Lemma 20.1 (Syndrome law). *Under independent fair physical hats, σ is uniform on $\mathbb{F}_2^r$. In particular, $\mathbb{P}(\sigma = 0) = 2^{-r} = 1/(t+1)$.*

Proof. The pair parities are independent fair bits. The linear map sending their vector to σ is surjective, because its columns include all basis vectors of $\mathbb{F}_2^r$. Each fiber of a surjective linear map from $\mathbb{F}_2^M$ to $\mathbb{F}_2^r$ has cardinality 2^{M-r} . Uniformity follows. $\square$

Theorem 20.2 (Exact finite score). *For every assignment of hats, the block has the following outcome:*

$$\begin{array}{c|c|c}
 & S_N & D_N \\
 \hline
 \sigma = 0 & 0 & -at \\
 \sigma = h_c \neq 0 & a(t+1) & a
 \end{array} \tag{20.2}$$

In the nonzero case, precisely the a pairs with column h_c are active. Each such pair has two correct guesses; every other pair has one.

Proof. For the actual pair parity $x = x_{\ell j}$,

$$U_{\ell j} = \sigma \oplus x h_j.$$

If $\sigma = 0$, then $U_{\ell j} = x h_j$. The active rule predicts parity $1 - x$, so both players are wrong. This holds for every pair.

If $\sigma = h_c \neq 0$ and $j = c$, then $U_{\ell j} = (1 - x) h_j$. The active rule predicts x , so both are correct. If $j \neq c$, then $U_{\ell j}$ is neither 0 nor h_j : either equality would force $\sigma \in \{0, h_j\}$. That pair is neutral. There are a active pairs, so the total score is $2a + (at - a) = a(t+1)$, proving all assertions. $\square$

Corollary 20.3 (Sharp finite teamwork). *For $N = 2a(2^r - 1)$, the largest possible probability that a legal compulsory strategy has at least $a2^r$ correct guesses is*

$$1 - 2^{-r}.$$

The finite syndrome block attains this maximum.

Proof. Proposition 18.3 bounds the probability by

$$\frac{N/2}{a2^r} = \frac{2^r - 1}{2^r}.$$

Theorem 20.2 and Lemma 20.1 give equality. $\square$

This optimum is attained by a legal finite decision procedure: each player reads the other hats in its block in a predetermined order, computes the rule, and guesses its fixed target. Therefore allowing more general blind output maps cannot improve this particular optimum. It supplies a family of exact instances of the finite-team research question in [12]; it does not classify all player counts or all thresholds.

[write] Added 4 October 2026, batch 90. The “finite-team research question in [12]” is Research question Q9 of Part I. In Part I’s model the corollary concerns $(m, q, n) = (N, 2, N)$ with the threshold $a2^r$, and its upper bound covers every team of blind output maps, targets free, since each blind binary map succeeds on exactly half of the configurations (Proposition 7.1). The note after the questions of Section 15 records what this settles and what stays open. For $r = 1$ the block is the two-hat-per-pair “all right or all wrong” team, with optimum $1/2$; for $r = 2$, $a = 1$ it is a six-player team reaching four correct guesses with probability $3/4$.

20.2 Why repeated rounds matter

The parameter r determines the failure probability. The parameter a determines how long the block lasts and how much positive surplus it contributes when it succeeds. Increasing a does not increase the failure probability.

An equivalent description aggregates the a pair parities carrying the same column:

$$X_j = \bigoplus_{\ell=1}^a x_{\ell j}.$$

Applying the ordinary Hamming syndrome to $X_1, \dots, X_t$ selects the same successful column in every round. This is the XOR amplification connection to [11]. The repeated-column description above makes the visibility rule and physical ordering explicit.

21 A bound at every prefix

An endpoint gain is insufficient for an assertion about all large n . A long interval between favorable endpoints could destroy the claimed growth. Repeating the column order in consecutive rounds gives an exact prefix count.

Lemma 21.1 (Prefix balance). *In a block with $\sigma = h_c \neq 0$, let D_u^{loc} be the excess among its first u physical players. For $0 \leq u \leq 2at$,*

$$\left| D_u^{\text{loc}} - \frac{u}{2t} \right| \leq \frac{3}{2}. \quad (21.1)$$

At every completed round the error is zero. If $\sigma = 0$, then $D_u^{\text{loc}} = -u/2$.

Proof. First take $u = 2p$, a whole number of pairs. Write $p = qt + j$ with $0 \leq j < t$, allowing the terminal case $p = at$ separately. Each full round contributes one unit of excess. In the unfinished round, the contribution is one exactly when its active column has already occurred. Consequently

$$D_{2p}^{\text{loc}} = q + \mathbf{1}_{\{c \leq j\}}.$$

Its difference from $p/t = q + j/t$ has absolute value less than one, and vanishes at full rounds. For odd $u = 2p + 1$, one more player adds $1/2$ or $-1/2$ to the excess, while the comparison line adds $1/(2t)$. The lower deviation is bounded below by $-3/2$, and the upper one by $3/2$. For completeness, if the active pair has not yet been completed, the even-prefix deviation is $-j/t$ with $j \leq t - 1$, so even a negative half-step is greater than $-3/2$; if it has been completed, the deviation is $1 - j/t$, and the positive half-step is at most $3/2$. The last assertion follows because every player is wrong in a zero-syndrome block. $\square$

Remark 21.2. The bound $3/2$ is deliberately convenient and uniform. Some parameter choices admit a smaller exact bound. No optimal constant is needed for any infinite conclusion.

Example 21.3 (Three columns, four rounds). Take $r = 2$, $t = 3$, $a = 4$, so there are 24 physical players. On a nonzero syndrome, one pair in every round is fully correct and the other two are balanced. The final score is 16, with excess 4. On syndrome zero all 24 players are wrong. The two possibilities have probabilities $3/4$ and $1/4$, and the expected score is 12, as required.

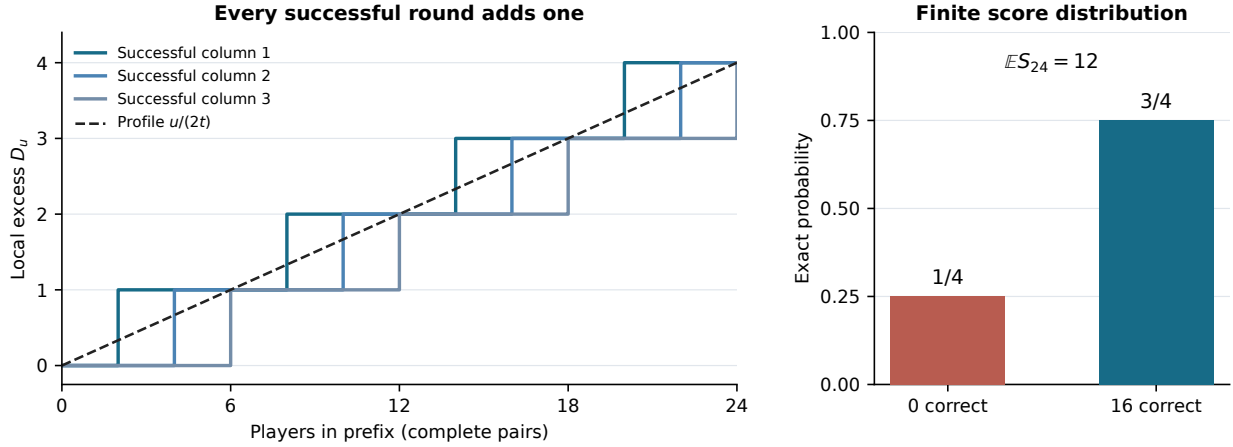


Figure 1: Exact finite-block quantities for $r = 2$, $a = 4$. The successful even-prefix paths follow the line $u/(2t)$ with uniformly bounded error. The score distribution includes a common failure outcome; the expected score remains $N/2$. These are exact formulas, not Monte Carlo estimates.

22 Concatenation and a deterministic growth profile

Choose positive integers r_k, a_k and put

$$t_k = 2^{r_k} - 1, \quad L_k = 2a_k t_k, \quad N_0 = 0, \quad N_k = \sum_{j=1}^k L_j.$$

Block k consists of players $N_{k-1} + 1, \dots, N_k$. Every player uses only hats in that block. Let B_k be its zero-syndrome event and put

$$p_k = \mathbb{P}(B_k) = \frac{1}{t_k + 1} = 2^{-r_k}.$$

The events B_k are independent, though their independence is not needed for the first Borel–Cantelli argument.

Define a deterministic continuous piecewise-linear function F on $[0, \infty)$ by

$$F(n) = \sum_{j < k} a_j + \frac{n - N_{k-1}}{2t_k} \quad \text{when } N_{k-1} \leq n \leq N_k. \quad (22.1)$$

The endpoint formulas agree, and $F(N_k) = \sum_{j \leq k} a_j$. When t_k is nondecreasing, F is increasing and concave.

Theorem 22.1 (Bounded random error). Suppose $\sum_k 2^{-r_k} < \infty$. Define the nonnegative extended random variable

$$C = \sum_{k=1}^{\infty} a_k (t_k + 1) \mathbf{1}_{B_k}. \quad (22.2)$$

Then $C < \infty$ almost surely, and after the final bad block, simultaneously for every physical prefix n ,

$$|D_n - F(n) + C| \leq \frac{3}{2}. \quad (22.3)$$

At every block endpoint, without any qualification on bad blocks,

$$D_{N_K} = \sum_{k \leq K} a_k - \sum_{k \leq K} a_k(t_k + 1)\mathbf{1}_{B_k}. \quad (22.4)$$

For every hat assignment and every n , one also has

$$D_n \leq F(n) + \frac{3}{2}. \quad (22.5)$$

Proof. The block excess is a_k in the good case and $-a_k t_k$ in the bad case. Rewriting it as $a_k - a_k(t_k + 1)\mathbf{1}_{B_k}$ proves the endpoint identity. Since $\sum_k \mathbb{P}(B_k) < \infty$, only finitely many B_k occur almost surely. Every corresponding summand is finite, so C is finite on that event.

Consider a prefix in a good block after the last bad one. The completed blocks contribute $\sum_{j < k} a_j - C$. Lemma 21.1 bounds the current contribution by its linear profile within $3/2$, proving (22.3). For the universal upper bound, the accumulated debt from completed bad blocks is nonnegative. A current good block satisfies the prefix bound, and a current bad block contributes $-u/2 \leq u/(2t_k)$. This proves (22.5) for all assignments. $\square$

Corollary 22.2 (Density and divergence). *Under the hypotheses of Theorem 22.1, assume also $t_k \rightarrow \infty$. Then*

$$D_n \rightarrow +\infty, \quad \frac{D_n}{n} \rightarrow 0, \quad \frac{S_n}{n} \rightarrow \frac{1}{2} \quad a.s..$$

Proof. Since $a_k \geq 1$, $F(N_k) \rightarrow \infty$; monotonicity gives $F(n) \rightarrow \infty$. To prove $F(n)/n \rightarrow 0$, fix $\varepsilon > 0$. All sufficiently late slopes $1/(2t_k)$ are at most ε , so $F(n) \leq A_\varepsilon + \varepsilon n$. Divide by n , let $n \rightarrow \infty$, and then let $\varepsilon \downarrow 0$. Theorem 22.1 now gives the asserted almost-sure limits. $\square$

Proposition 22.3 (A quantitative stabilization bound). *For any K , with probability at least*

$$1 - \sum_{k \geq K} 2^{-r_k},$$

there are no bad blocks from K onward. On that event, every $n \geq N_{K-1}$ obeys

$$|D_n - F(n)| \leq \sum_{j < K} a_j(t_j + 1) + \frac{3}{2}.$$

For $r_k = k$, the displayed failure bound is 2^{1-K} .

Proof. Use the union bound on $\bigcup_{k \geq K} B_k$. The finite initial debt is at most the displayed deterministic sum, and (22.3) applies afterward. $\square$

22.1 Why this does not violate expectation

For every finite n , $\mathbb{E}D_n = 0$. Nevertheless the preceding theorem gives $D_n \rightarrow \infty$ almost surely. The exact accounting is particularly transparent:

$$\mathbb{E}C = \sum_{k=1}^{\infty} a_k(t_k + 1)\mathbb{P}(B_k) = \sum_{k=1}^{\infty} a_k = \infty. \quad (22.6)$$

Tonelli's theorem applies because the summands are nonnegative. Thus the total debt is finite almost surely and has infinite expectation. A rare failure of a sufficiently large block produces a sufficiently large loss to maintain fairness at every finite endpoint. An interchange of an almost-sure limit with expectation would require an integrability hypothesis that is absent here.

[write] Added 4 October 2026, batch 90. The divergence $D_n \rightarrow +\infty$ almost surely in Corollary 22.2 is not new as a phenomenon: Eldredge's Proposition 6.6 [10] already gives a measurable strategy with this property, built from independent blocks of growing length that follow the pairs strategy, except that a player who sees only black hats in its block guesses white; his blocks have a bad case (all hats black, every guess wrong), a good case adding one to the excess, and a neutral case, and the first and second Borel–Cantelli lemmas make bad blocks finitely many and good blocks infinitely many. His Remark 6.7 gives the explanation of this subsection: almost surely finitely many players sit in bad blocks, yet their expected number is infinite. What the syndrome blocks add is a gain proportional to the block length (Theorem 20.2), the bound at every prefix (Lemma 21.1) and hence the deterministic profile with bounded random error of Theorem 22.1. The manuscript cites Eldredge's Proposition 6.6 only in its bibliography entry.

23 An exact logarithmic answer and explicit faster examples

Theorem 23.1 (Logarithmic excess). *There is a computable finite-information compulsory strategy such that*

$$D_n = \log_2 n + O_{\omega}(1) \quad \text{a.s.} \quad (23.1)$$

Here $O_{\omega}(1)$ means a finite bound depending on the hat assignment, valid for every sufficiently large n .

Proof. Take $r_k = k$ and $a_k = 1$. Then

$$N_K = 2 \sum_{k=1}^K (2^k - 1) = 2^{K+2} - 2K - 4.$$

For $N_{k-1} < n \leq N_k$, the profile $F(n)$ lies between $k - 1$ and k . On the same interval, $\log_2 n = k + O(1)$, uniformly as $k \rightarrow \infty$, since its endpoints are bounded positive multiples of 2^k . Therefore $F(n) = \log_2 n + O(1)$. Summability of 2^{-k} and Theorem 22.1 prove (23.1). All block boundaries and finite rules are computable. $\square$

This answers the logarithmic alternative itself, rather than only providing a rate larger than a logarithm.

[write] Added 4 October 2026, batch 90. Eldredge's first question in Remark 6.8 [10] (text checked at the write; Section 17.3) asks for a measurable strategy whose excess $S_n - n/2$ is of order $\log n$ almost surely, in his sense of random constants $0 < K_1 < K_2 < \infty$ and a random N_0 with $K_1 \log \log n \leq S_n - n/2 \leq K_2 \log \log n$ for $n \geq N_0$ (stated there for $\log \log n$, "or even some faster rate". Theorem 23.1 gives more than order $\log n$: $D_n/\log_2 n \rightarrow 1$ almost surely. Its strategy is continuous, hence measurable. Propositions 23.2 and 23.3 and Theorem 24.1 give the faster rates. Eldredge's own block strategy reaches orders $\log \log n$ and $\log n/(\log \log n)^2$ (his Remark 6.8).

Proposition 23.2 (Every fixed power below one). *For every $0 < \beta < 1$, a finite-information strategy satisfies*

$$D_n \asymp n^{\beta} \quad \text{a.s.},$$

where the comparison holds for all sufficiently large n . For computable β , one can use a computable schedule.

Proof. Take $r_k = k$ and

$$a_k = \left\lceil 2^{\beta k/(1-\beta)} \right\rceil.$$

Geometric-sum estimates give

$$\sum_{j \leq k} a_j \asymp 2^{\beta k/(1-\beta)}, \quad N_k \asymp 2^{k/(1-\beta)}.$$

The implicit constants depend only on β . For a prefix in block k , its index lies between endpoints whose ratio is bounded, and its profile lies between consecutive endpoint profiles whose ratio is also bounded. Consequently $F(n) \asymp n^\beta$ at every prefix. Equation (22.3) transfers this to D_n .

For the effective assertion one may avoid exact ceiling decisions at computable-real integer boundaries by choosing any effectively computable integer approximation within a fixed factor of the displayed a_k . Such approximations preserve all estimates. $\square$

Proposition 23.3 (A concrete nearly linear example). *There is a computable finite-information strategy with*

$$D_n \asymp \frac{n}{\log n} \quad a.s..$$

Proof. Set $r_k = k$, $a_k = 2^{2^k}$. Since the final term dominates each of the relevant sums,

$$N_k \asymp 2^k 2^{2^k}, \quad F(N_k) \asymp 2^{2^k}, \quad \log N_k \asymp 2^k.$$

For $N_{k-1} \leq n \leq N_k$, one has $\log n \asymp 2^k$. Nondecreasing t_j gives

$$F(n) \geq \frac{n}{2t_k} \asymp \frac{n}{\log n}.$$

For the reverse bound, write

$$F(n) = F(N_{k-1}) + \frac{n - N_{k-1}}{2t_k}.$$

The first term is $O(N_{k-1}/\log N_{k-1})$, which is $O(n/\log n)$ by eventual monotonicity of $x/\log x$. The second has that bound by $\log n \asymp 2^k$. Apply (22.3). $\square$

These examples are mathematical constructions, not low-resource algorithms. The number of visible hats used by a late player can be very large. The general theorem next shows that no fixed sublinear scale exhausts what is possible.

24 No deterministic sublinear universal bound

Theorem 24.1 (Arbitrary prescribed sublinear growth). *Let $g : \mathbb{N}_{>0} \rightarrow (0, \infty)$ satisfy $g(n)/n \rightarrow 0$. There is a finite-information continuous compulsory strategy such that*

$$\frac{D_n}{g(n)} \longrightarrow +\infty \quad \text{and} \quad \frac{D_n}{n} \longrightarrow 0 \quad a.s.. \quad (24.1)$$

The strategy can be chosen so that its excess tracks an increasing, concave, deterministic function $F = o(n)$ within a finite random additive constant.

Proof. Put $h(n) = \max\{g(n), \sqrt{n}\}$. Then $h = o(n)$, $h(n) \rightarrow \infty$, and $h \geq g$. Fix $r_k = k$ and $t_k = 2^k - 1$. For each $k \geq 2$, choose an integer R_k such that

$$n \geq R_k \implies \frac{h(n)}{n} \leq \frac{1}{k^2 t_k}. \quad (24.2)$$

Such an integer exists by sublinearity. A least suitable integer can be used, so no arbitrary selection among unrelated objects is necessary.

Choose a_k recursively, always finite and positive, so that

$$N_k = N_{k-1} + 2a_k t_k \geq R_{k+1}. \quad (24.3)$$

For example, once the thresholds are fixed, take the maximum of 1 and the ceiling of $(R_{k+1} - N_{k-1})/(2t_k)$. The resulting disjoint blocks define one strategy before any hats are sampled.

Consider an assignment with only finitely many bad blocks, and let C be its finite debt. For n in a later good block k , monotonicity $t_j \leq t_k$ gives

$$F(n) = \sum_{j < k} \frac{L_j}{2t_j} + \frac{n - N_{k-1}}{2t_k} \geq \frac{n}{2t_k}.$$

Using (22.3), (24.2), and $n \geq N_{k-1} \geq R_k$, we obtain

$$\frac{D_n}{h(n)} \geq \frac{n}{2t_k h(n)} - \frac{C + 3/2}{h(n)} \geq \frac{k^2}{2} - \frac{C + 3/2}{h(n)}. \quad (24.4)$$

Every block is finite, so $k \rightarrow \infty$ with n . Since $h(n) \rightarrow \infty$, the right side tends to infinity. Thus D_n is eventually positive and $D_n/g(n) \geq D_n/h(n) \rightarrow \infty$.

The probability of finitely many bad blocks is one, since $\sum_k 2^{-k} < \infty$. Finally F is increasing and concave with slopes tending to zero, so $F = o(n)$. Theorem 22.1 proves the second limit and the tracking assertion. $\square$

Corollary 24.2 (Failure of every proposed sublinear envelope). *There is no positive deterministic $g = o(n)$ such that every measurable legal strategy satisfies*

$$\liminf_{n \rightarrow \infty} \frac{D_n}{g(n)} \leq 0 \quad \text{a.s.}$$

There is also no such g for a universal finite upper bound on that lower limit. Both assertions remain false when the strategies are restricted to continuous finite-information strategies with $S_n/n \rightarrow 1/2$ almost surely.

Proof. For any proposed g , Theorem 24.1 supplies a strategy whose ratio tends to $+\infty$. $\square$

Remark 24.3 (Quantifier order). The theorem says

$$\forall g = o(n) \quad \exists \text{ a strategy depending on } g.$$

It does not say there is one strategy that dominates every sublinear function. Indeed our fixed strategy tracks its own deterministic $F = o(n)$, and $D_n/F(n) \rightarrow 1$, so taking $g = F$ prevents that stronger statement.

Remark 24.4 (The wording of the published second question). Eldredge's Remark 6.8 writes a universal equality for a normalized lower limit. Its cited density theorem gives a nonpositive lower limit, rather than equality for every strategy. Corollary 24.2 addresses the stronger meaningful obstruction: even among strategies with eventually positive excess and limiting success density $1/2$, every prescribed sublinear scale fails. Thus the equality wording does not affect the negative answer.

[write] Added 4 October 2026, batch 90. Checked against the text of v2 [10]. The source audit (02-hat-surplus-source_audit.txt, item 1) says that Eldredge’s Theorem 4.1 gives $\liminf D_n/n \leq 0 \leq \limsup D_n/n$, not equality for every measurable strategy. That is accurate, and Theorem 4.1 itself is not at fault: it is the density statement $L \leq \frac{1}{2} \leq U$, which is (25.1), because $D_n/n = S_n/n - \frac{1}{2}$. The wording issue is in the second question of Remark 6.8, which asks for g with $\liminf_n (S_n - n/2)/g(n) = 0$ almost surely for *every* measurable strategy and says that Theorem 4.1 shows this for $g(n) = n$. Theorem 4.1 gives only “ ≤ 0 ”, and equality does fail for some strategies. The write checked one, using this Part’s blocks: take every $r_k = 1$ (so each block of a_k pairs is all right or all wrong, each with probability $1/2$, independently) and the dominant-block condition $L_k \geq kN_{k-1}$ of (26.1). By the second Borel–Cantelli lemma infinitely many blocks are bad almost surely; at a bad endpoint $D_{N_k}/N_k \leq 1/(k+1) - 1/2$, as in the proof of Theorem 26.1, while $D_n/n \geq -1/2$ always; so $\liminf_n D_n/n = -1/2$ almost surely. This is a check made at the write, not a statement of the manuscript. Under the meaningful reading “ ≤ 0 ” Corollary 24.2 is the negative answer for every positive g with $g(n)/n \rightarrow 0$; under the literal reading the answer is negative a fortiori. Positive functions g with $\limsup_n g(n)/n > 0$ are outside the corollary’s statement.

25 The linear boundary remains unavoidable

For perspective, we give a short independent proof of the measurable density obstruction, crediting the theorem to [10, Theorem 4.1]. The argument needs no assumptions about the particular block construction.

Theorem 25.1 (Measurable density obstruction). *For every measurable legal compulsory strategy,*

$$\liminf_{n \rightarrow \infty} \frac{D_n}{n} \leq 0 \leq \limsup_{n \rightarrow \infty} \frac{D_n}{n} \quad a.s.. \quad (25.1)$$

In particular, no positive-probability event can support a strictly positive asymptotic linear lower surplus.

Proof. Let T_i flip hat i and leave all other hats fixed. The map preserves product probability. Put $Y_i = Z_i - 1/2$. Legality implies $Y_i \circ T_i = -Y_i$.

For any measurable event A , invariance under T_i yields

$$\mathbb{E}[\mathbf{1}_A Y_i] = \frac{1}{2} \mathbb{E}[(\mathbf{1}_A - \mathbf{1}_A \circ T_i) Y_i],$$

and hence

$$|\mathbb{E}[\mathbf{1}_A Y_i]| \leq \frac{1}{4} \mathbb{P}(A \triangle T_i A).$$

The right side tends to zero. Indeed, approximate A in measure by a finite-coordinate event A_0 . For all sufficiently large i , $T_i A_0 = A_0$, so $\mathbb{P}(A \triangle T_i A) \leq 2\mathbb{P}(A \triangle A_0)$. Finite-coordinate events approximate all events in the completed product sigma-field in measure.

Cesàro averaging now gives $\mathbb{E}[\mathbf{1}_A D_n/n] \rightarrow 0$. Apply this to $A = \{\liminf_n D_n/n > 0\}$. Since D_n/n lies in $[-1/2, 1/2]$, Fatou’s lemma applied after adding $1/2$ gives

$$\mathbb{E} \left[\mathbf{1}_A \liminf_n \frac{D_n}{n} \right] \leq 0.$$

Its integrand is strictly positive on A , forcing $\mathbb{P}(A) = 0$. Apply the same argument to $-D_n$ for the second inequality. $\square$

Theorems 24.1 and 25.1 give the exact qualitative distinction relevant to the proposed envelope: linear positive lower growth is impossible for measurable strategies, while every fixed sublinear candidate can be defeated by a continuous strategy.

[write] **Added 4 October 2026, batch 90.** Theorem 25.1 is Eldredge's Theorem 4.1 [10] restated for the surplus, as the manuscript says; it is inherited, not new. Eldredge requires each guess to be μ -measurable (almost everywhere equal to a Borel function) and bases his proof on the martingale convergence of $\mathbb{E}[X \mid \mathcal{F}_m]$ (his Theorem 4.2); the proof above, by flip invariance and finite-coordinate approximation, is a second route to the same statement.

26 Probability, Baire category, and the exact success set

The preceding almost-sure behavior does not imply that the same behavior is typical in the sense of Baire category. We can arrange an explicit separation while retaining arbitrary prescribed growth.

In the construction of Theorem 24.1, impose the additional requirement

$$L_k = 2a_k t_k \geq kN_{k-1}. \quad (26.1)$$

Increasing a_k can enforce this simultaneously with (24.3). Call this a *dominant-block schedule*.

Theorem 26.1 (Category behavior). *For the dominant-block strategy associated to any positive $g = o(n)$, the following hold:*

1. *Almost surely $D_n/g(n) \rightarrow +\infty$ and $S_n/n \rightarrow 1/2$.*
2. *For every hat assignment, $\limsup_n S_n/n \leq 1/2$.*
3. *On a comeager subset of Cantor space,*

$$\liminf_n \frac{S_n}{n} = 0, \quad \limsup_n \frac{S_n}{n} = \frac{1}{2}.$$

Proof. The first conclusion is Theorem 24.1. The second follows from (22.5) and $F(n)/n \rightarrow 0$. For the third, note that every B_k is a nonempty clopen finite-coordinate event, as is its complement. Because the blocks are disjoint, each of

$$\bigcap_K \bigcup_{k \geq K} B_k, \quad \bigcap_K \bigcup_{k \geq K} B_k^c$$

is a dense G_δ : any finite prescription can be extended to either type of pattern in a sufficiently late block.

At a bad block endpoint, all new guesses are wrong, so

$$\frac{S_{N_k}}{N_k} \leq \frac{N_{k-1}}{N_{k-1} + L_k} \leq \frac{1}{k+1}.$$

At a good block endpoint, at least $L_k/2$ new guesses are correct, so

$$\frac{S_{N_k}}{N_k} \geq \frac{L_k/2}{N_{k-1} + L_k} \geq \frac{k}{2(k+1)}.$$

On the comeager intersection of the two displayed sets, both kinds of endpoints occur infinitely often. Nonnegativity of S_n and the universal upper limit bound give the claimed limits. $\square$

Theorem 26.2 (Exact topological complexity). *For the same dominant-block strategy, let*

$$\mathcal{S}_g = \{\omega : D_n(\omega)/g(n) \rightarrow +\infty\}.$$

Then

$$\mathcal{S}_g = \{\text{only finitely many } B_k \text{ occur}\} = \{\omega : S_n(\omega)/n \rightarrow 1/2\}. \quad (26.2)$$

This set is dense, conull, meager, and Σ_2^0 -complete under continuous reductions.

Proof. If only finitely many B_k occur, the deterministic proof of Theorem 24.1 applies. If infinitely many occur, the bad endpoints satisfy

$$\frac{D_{N_k}}{N_k} \leq \frac{1}{k+1} - \frac{1}{2}.$$

Since $g(N_k)/N_k \rightarrow 0$, the ratio $D_{N_k}/g(N_k)$ tends to $-\infty$ along those endpoints. Their success densities also tend to zero. This proves both equalities in (26.2).

The event can be written as

$$\mathcal{S}_g = \bigcup_K \bigcap_{k \geq K} B_k^c.$$

Each inner intersection is closed with empty interior, and therefore nowhere dense. The union is F_σ and meager. It is dense because any finite prescription can be extended to good patterns in all sufficiently late blocks. It is conull by the first Borel–Cantelli lemma.

For completeness, choose in every block a fixed bad pattern (all hats zero) and a fixed good pattern (exactly one white hat in its first pair). Given $z \in \{0, 1\}^{\mathbb{N}_{>0}}$, use the bad pattern in block k if $z_k = 1$, and the good one if $z_k = 0$. This defines a continuous map Ψ , and

$$\Psi^{-1}(\mathcal{S}_g) = \text{FIN} := \{z : z_k = 1 \text{ for only finitely many } k\}.$$

The set FIN is Σ_2^0 -complete; a self-contained verification is given in Appendix D. Thus $\mathcal{S}_g$ has the claimed complexity. $\square$

The completeness statement is topological, or boldface. With an arbitrary g , the block schedule need not be computable, so no unqualified lightface complexity statement is being made. The exact logarithmic schedule of Theorem 23.1 is a different schedule and is not asserted to satisfy (26.1).

27 Effective construction and finite information costs

27.1 A computable schedule from a sublinearity modulus

Suppose a function g is supplied with a computable modulus M satisfying

$$n \geq M(b) \implies g(n)/n \leq 2^{-b}.$$

One can compute thresholds in (24.2). Choose b_k with $2^{-b_k} \leq 1/(k^2 t_k)$, and take

$$R_k = \max\{M(b_k), (k^2 t_k)^2\}.$$

Then both $g(n)/n$ and $1/\sqrt{n}$ satisfy the required bound. Taking integer ceilings in (24.3), and increasing them if (26.1) is desired, gives a computable sequence of finite blocks.

Given a player index, enumerate these increasing endpoints until its block is located. The block has finitely many hats, and the own-hat-excluding query set is computable. Thus the strategy is a total truth-table procedure with a computable, but generally unbounded, query bound.

An arbitrary computable function known abstractly to be $o(n)$ need not be accompanied by a computable convergence modulus. The theorem without that extra information remains an existence theorem for continuous strategies; it does not silently provide uniform effective threshold selection.

27.2 Algorithmically random assignments

Proposition 27.1 (An effective almost-sure refinement). *For a computable block schedule with $r_k = k$, every Martin-Löf random hat sequence has only finitely many bad blocks. Consequently all deterministic conclusions proved on that event hold on every such sequence.*

Proof. The events B_k are uniformly computable clopen events with $\mathbb{P}(B_k) = 2^{-k}$. The effectively open sets

$$U_m = \bigcup_{k \geq m+1} B_k$$

satisfy $\mathbb{P}(U_m) \leq 2^{-m}$. They form a Martin-Löf test. A sequence belonging to infinitely many B_k belongs to every U_m , so it fails this test. Martin-Löf random sequences avoid that intersection. $\square$

This is an effective form of the particular summable-event argument, not a new general result in algorithmic randomness. Its relationship to the stronger computable-randomness treatment in [11] is part of the prior-art connection.

27.3 What the finite algorithms cost

For a block of $N = 2at$ physical players, the simplest individual implementation reads all $N - 1$ other hats. It XORs visible pair parities weighted by r -bit columns and applies the local rule. This costs $O(Nr)$ elementary bit operations under a direct implementation, with $O(r)$ working storage beyond query indexing and counters. There is no communication of a shared syndrome.

For a centralized verifier that already has the full hat assignment, one can calculate group parities and the syndrome once and derive all guesses efficiently. That is a testing optimization, not the information available to a player. The independent verifier in this package explicitly checks own-hat blindness as well as agreement between a group-based and a directly visible-pair implementation.

The ability to make a block arbitrarily long at fixed failure probability is responsible for the arbitrary-growth theorem. It is also why that theorem does not supply an economical visibility bound as a function of player index. Quantitative restrictions on visibility constitute a separate research problem.

28 Verification and a concrete route into ProveIt

28.1 Executed finite checks

The archive contains a Python standard-library verifier and its generated data. The following table records the actual exhaustive cases. Prefix errors are computed as exact rational numbers.

r	a	t	Players N	Assignments	Maximum prefix error
1	1	1	2	4	0
1	2	1	4	16	0
1	3	1	6	64	0
1	4	1	8	256	0
1	5	1	10	1,024	0
2	1	3	6	64	1
2	2	3	12	4,096	1
2	3	3	18	262,144	1
3	1	7	14	16,384	9/7
Total assignments				284,052	

The completed run checked:

- 5,010,248 individual predictions against an independently implemented rule;
- 10,020,496 own-hat toggles across the two implementations;
- 3,997,918 prefixes of nonzero-syndrome assignments, including empty prefixes;
- the full syndrome distribution, exact endpoint score law, individual correctness probabilities, mean surplus, and second moment.

The measured run time was approximately 17.2 seconds in the execution environment. This timing is descriptive, not a performance guarantee.

For every parameter pair, the exact second moment is

$$\mathbb{E}[D_N^2] = a^2 \frac{t}{t+1} + a^2 t^2 \frac{1}{t+1} = a^2 t.$$

This additional check detects incorrect gain/loss normalization. The data also include explicit good and bad examples and a complete prefix path.

Finite enumeration can detect an erroneous strategy, a hidden own-coordinate dependency, or a wrong prefix constant. It does not prove the theorem for all r, a , and it cannot establish an almost-sure statement over infinitely many blocks. Those claims are proved in Sections 20–24.

[write] Added 4 October 2026, batch 90. In the collection the verifier and the figure script are shipped as `code/02-hat-surplus-verify_hat_strategy.py` and `code/02-hat-surplus-make_figures.py`, with the delivered `code/02-hat-surplus-README.txt`, `code/02-hat-surplus-Makefile` and the code's permissive licence `code/02-hat-surplus-LICENSE.txt`; the recorded outputs are `data/02-hat-surplus-verification.json`, `data/02-hat-surplus-examples.json` and `data/02-hat-surplus-example_prefix.csv`, and the figure `figures/02-hat-surplus-block_prefix.pdf` (with a PNG copy), all byte-identical to the delivery. Run in place, the verifier would write unprefix `verification.json`, `examples.json` and `example_prefix.csv` into this report's `data/`, and the figure script unprefix `block_prefix.pdf` and `.png` into `figures/`, so both are rerun on a scratch copy; the report's `README.md` gives the commands. The timing of about 17.2 seconds above is the delivered run's (Python 3.12.14). At the write (4 October 2026, Python 3.14.4 on Windows, on a copy) the verifier passed all nine cases, with the totals of the table and list above, in about 112 seconds on a loaded machine (37 seconds at placement); the regenerated `examples.json` and `example_prefix.csv` equal the shipped ones (the first after stripping the CR bytes Windows text mode writes), and `verification.json` does too apart from its completion time, Python version and elapsed-time fields. At placement the figure script, run on a copy with Matplotlib 3.10.8, reproduced the PDF figure up to its creation date, and the PNG at the same size with different bytes.

28.2 Formalization layers

A proof-assistant development can be divided into four small layers. The following are proposed interfaces, not claims that the files already exist.

1. **Finite semantics.** Represent a hat assignment as $\text{Fin}(N) \rightarrow \text{Bool}$. Define a strategy as a family of Boolean functions and own-hat blindness by invariance under a coordinate flip. Define score and twice the surplus as integers, avoiding fractions initially.
2. **Syndrome algebra and counting.** Represent columns in $\mathbb{F}_2^r$; prove the two visibility cases and the neutral case. Prove that the syndrome map is surjective and that all its fibers have the same cardinality. The exact block law then reduces to finite sums and a count of active columns.
3. **Physical order and prefixes.** Use the lexicographic finite order on $\text{Fin}(a) \times \text{Fin}(t) \times \text{Fin}(2)$. Prove the division-with-remainder formula for completed pairs. This isolates the part where a proof about endpoints could otherwise be mistaken for a proof about all prefixes.
4. **Infinite product probability and asymptotics.** Construct the disjoint finite blocks in countable product space. Apply the first Borel–Cantelli lemma, establish the exact finite debt identity, and then prove deterministic tracking and the threshold schedule. This layer can use a standard library of measure and asymptotic results.

The existing `ProveIt` finite-list interfaces can encode the parameter tables, predictions, and prefix certificates. A bridge to an executable checker would require a theorem showing that accepted certificates imply the intended finite statements. Merely evaluating a Boolean checker or translating a program to arithmetic would not establish that bridge.

28.3 A sensible order of formal work

The shortest useful first target is the finite theorem with arbitrary r, a , including blindness and the score law. The second is the prefix lemma. These two targets already validate the complete local mechanism. The third is the general concatenation theorem; once available, logarithmic and arbitrary-growth schedules become deterministic corollaries.

This order preserves a clear trust boundary. No claim about an unformalized probability argument is promoted to kernel-checked status merely because a finite arithmetic component has been verified.

29 Further research questions

The following are proposed follow-up problems. They are not presented as a list of universally recognized open problems, and their precise literature status should be checked before claiming a new solution. They are chosen to continue the explicit connections among prediction, asymptotics, computability, and formal verification.

Research question 29.1 (Which deterministic profiles admit bounded random error?). Characterize increasing concave $H = o(n)$ for which a compulsory finite-information strategy satisfies

$$D_n = H(n) + O_\omega(1) \quad \text{a.s..}$$

Our construction realizes profiles with slopes $1/(2(2^{r_k} - 1))$, integral gains per block, and summable bad-block probabilities. Determine whether these restrictions are artifacts of the construction or reflect a genuine obstruction to bounded additive error.

Research question 29.2 (Information cost versus guaranteed growth). Suppose player i may inspect at most $q(i)$ other hats. What excess profiles remain attainable? In particular, determine whether a uniform bound on $q(i)$ permits $D_n \rightarrow +\infty$ almost surely. Distinguish a bound on the number of queries from a bound on their distance in the player ordering.

Research question 29.3 (Polynomially bounded visibility). For an explicit target such as $D_n \asymp n/\log n$, seek the smallest possible maximum queried index as a function of n . The current construction is computable but its large blocks give weak quantitative use bounds. An efficient construction would strengthen the computational content of the result.

Research question 29.4 (The complete finite-team optimization problem). For arbitrary N and threshold $s > N/2$, determine

$$\sup_{\text{legal strategies}} \mathbb{P}(S_N \geq s).$$

Corollary 20.3 settles $(N, s) = (2a(2^r - 1), a2^r)$. For the remaining parameters, identify exactly when the first-moment bound is attainable. This is the most direct continuation of ProveIt's finite-team question.

Research question 29.5 (Classify equality cases). For the parameters in Corollary 20.3, equality in the first-moment bound forces the score to be supported on $\{0, s\}$. Classify legal strategies with that distribution, up to relabeling players and hat colors. Determine how much of the syndrome geometry follows from extremality alone.

Research question 29.6 (More than two colors). Develop an analogue of the compulsory pairing transformation for q equiprobable colors. A neutral group can have exactly one correct guess, but an active group has a different gain/loss balance from the binary case. Determine whether every positive $o(n)$ scale can be dominated above the baseline n/q by finite-information strategies.

Research question 29.7 (Nonuniform and dependent hats). Which parts of the theorem survive unequal known marginal probabilities, or specified dependencies? Individual fairness and uniform syndrome counting would both need replacement. A useful first target is an independent binary product with biases uniformly separated from zero and one.

Research question 29.8 (Precise distribution of the random debt). For the logarithmic schedule,

$$C = \sum_{k \geq 1} 2^k \mathbf{1}_{B_k}, \quad \mathbb{P}(B_k) = 2^{-k}.$$

Determine sharp tail asymptotics and possible oscillations caused by the dyadic weights. Extend this to regularly varying choices of a_k . The distinction between almost-sure finiteness and infinite mean is already settled, but finer distributional questions are not addressed here.

Research question 29.9 (Best finite-horizon confidence guarantees). Optimize the visibility and block lengths needed to ensure a specified lower surplus after a specified index with probability at least $1 - \delta$. Proposition 22.3 supplies a direct bound, but it may be far from optimal when the cost of the finite initial debt is included.

Research question 29.10 (Which pairs of measure and category behavior occur?). The dominant-block construction is successful on a conull meager set and has residual lower density zero. Characterize the possible Borel complexities and residual density spectra of continuous strategies with prescribed almost-sure excess growth. Can similarly strong growth be arranged on a set that is both conull and comeager?

Research question 29.11 (Effective randomness beyond the present test). Determine the weakest standard randomness notion sufficient for every computable schedule in this family. Compare that answer with the computable-randomness results of [11], taking care to distinguish a fixed strategy with finitely many errors from a sequence-dependent error-free autoreduction.

Research question 29.12 (A complete formal certificate chain). Implement the four layers in Section 28, including a proved correspondence between an executable finite checker and the mathematical semantics. Then formalize both the rate theorem and the measure/category separation without introducing an unproved probability or asymptotic axiom.

29.1 Suggested priorities

The best immediate formal target is the arbitrary-parameter finite block theorem and prefix count. The best finite experimental target is the optimization problem outside the dyadic family. The most substantial structural targets are profile realization and information-cost lower bounds. These directions ask for results not supplied merely by repeating the arbitrary-growth scheduling argument.

[write] **Status of the questions, 4 October 2026, batch 90.** All twelve Research questions of this section stay open as printed; the intake searched no literature for them. Research question 29.4 is the continuation, for fixed targets and binary hats, of Part I's Q9, which Corollary 20.3 settles for one family (see the note after the questions of Section 15). Research question 29.6, on more than two colours, meets Part I's q -ary setting; Research question 29.12 and the formalization layers above are proposals, and no Lean or Rocq file of ProveIt implements any of them.

[write] **Status of Research questions 29.2, 29.4, 29.5 and 29.10, 4 October 2026, batch 92.** The preceding note is superseded for these four; the others stay open as printed. **29.2 is answered**, qualitatively, by Part IV, by both of its sources independently: a uniform bound on the number of queries excludes $D_n \rightarrow +\infty$ almost surely, indeed with positive probability (Corollary 48.4, where the divergence events are also meagre; Theorem 61.3); for a count budget that is eventually at least one, almost-sure divergence is possible exactly when the budget is unbounded (Theorem 54.2); and count and distance budgets differ (Theorems 54.3–54.4). In expectation the exact uniform threshold is two queries (Theorems 53.3 and 60.1). Which excess profiles a given budget permits stays open (Research questions 57.3, 57.4, 57.6 and 68.1). **29.4 is answered at minimum depth:** by Theorem 65.4 a strategy of per-player depth $s - 1$ attains the first-moment bound exactly when $(N, s) = (m(2^r - 1), m2^{r-1})$, and Theorem 65.5 attains it for all these pairs, odd m included (Corollary 20.3 is the case $m = 2a$); by Theorem 65.2 no saturating strategy has smaller depth. For other pairs, with greater depth, it stays open (Research question 68.5). **29.5 is answered for extremizers of minimum depth:** their failure sets are affine, each nonzero parity-check column repeated m times (Theorems 65.3–65.4). Extremizers of greater depth, among them the syndrome block of Section 20 for $r \geq 2$, are not classified. **29.10** stays open: Proposition 53.4 shows the conull meagre behaviour of Theorems 26.1–26.2 for the critical-cost construction, not a success set that is conull and comeagre.

C The exact prefix constant

The convenient bound $3/2$ in Lemma 21.1 can be replaced by an optimal parameter-dependent constant.

Proposition C.1. *For $t = 2^r - 1$, the largest possible value of*

$$\left| D_u^{\text{loc}} - \frac{u}{2t} \right|$$

over all nonzero-syndrome assignments and all prefixes in a block with $a \geq 1$ is

$$B_t = \frac{3(t-1)}{2t}.$$

Proof. For $t = 1$, every pair in a good block is fully correct, so the error is zero. Assume $t \geq 3$. At an even prefix with $p = qt + j$ completed pairs, the error is

$$e_{2p} = \mathbf{1}_{\{c \leq j\}} - \frac{j}{t}.$$

Its absolute value is at most $(t - 1)/t$.

At the next odd prefix, add $z - 1/2 - 1/(2t)$, where z is the next player's correctness indicator. There are three cases. If $j + 1 = c$, the next pair is active and $z = 1$; the resulting absolute error is at most $(t - 1)/(2t)$. If $j + 1 < c$, then $j \leq t - 2$, the active pair is still ahead, and using $z \in \{0, 1\}$ gives the lower bound

$$-\frac{t-2}{t} - \frac{1}{2} - \frac{1}{2t} = -\frac{3(t-1)}{2t}.$$

The corresponding upper bound is smaller. If $c \leq j$, then $j \geq 1$, and the largest possible error is

$$1 - \frac{1}{t} + \frac{1}{2} - \frac{1}{2t} = \frac{3(t-1)}{2t}.$$

The lower bound in that case is smaller in absolute value. This proves the bound.

For attainment, set exactly one of the hats in the very first column-1 pair to one, and all other hats to zero. Then $\sigma = h_1$, and the first pair is fully correct. The first player in the next, neutral pair also guesses correctly, so $D_3^{\text{loc}} = 3/2$. Its deviation from $3/(2t)$ equals B_t . $\square$

The exhaustive data attain $B_1 = 0$, $B_3 = 1$, and $B_7 = 9/7$, matching this general proof.

D A self-contained completeness fact for FIN

We include the elementary descriptive-set-theoretic fact used in Theorem 26.2.

Proposition D.1. *The subset FIN of Cantor space consisting of sequences with finitely many ones is Σ_2^0 -complete under continuous reductions from Cantor space.*

Proof. It is F_σ , since

$$\text{FIN} = \bigcup_m \bigcap_{n \geq m} \{z : z_n = 0\}.$$

Let $A = \bigcup_j C_j$ be any F_σ subset of Cantor space. By taking finite unions, assume the closed sets C_j are increasing. For a point x , and each stage n , let $m_n(x)$ be the least $j \leq n$ such that the cylinder determined by the first n bits of x meets C_j ; if there is no such j , put $m_n(x) = n + 1$.

Each m_n depends only on a finite prefix. Moreover $m_{n+1}(x) \geq m_n(x)$: shrinking a cylinder cannot make it meet a previously excluded closed set, and a newly introduced index is no smaller than the previous default. Define a bit sequence $z(x)$ by setting

$$z_n(x) = \begin{cases} 1, & m_{n+1}(x) > m_n(x), \\ 0, & m_{n+1}(x) = m_n(x). \end{cases}$$

This map is continuous.

If $x \in C_J$, every cylinder around x meets C_J . Thus $m_n(x) \leq J$ for $n \geq J$. Being nondecreasing, the sequence $m_n(x)$ is eventually constant, so $z(x)$ has finitely many ones. If $x \notin A$, then for each fixed J , closedness of C_J implies some sufficiently short neighborhood of x misses it. Since the C_j are increasing, $m_n(x) > J$ for all sufficiently large n . Therefore $m_n(x) \rightarrow \infty$ and changes infinitely often, so $z(x) \notin \text{FIN}$. Hence $z^{-1}(\text{FIN}) = A$. $\square$

E Reproducibility and source notes

E.1 Files and commands

The package contains the complete LaTeX source, compiled PDF, the finite verifier, exact JSON results, a CSV prefix example, and the figure-generation script. Build commands are recorded in `README.txt` and `Makefile`.

From the package directory:

```
python3 code/verify_hat_strategy.py
python3 code/make_figures.py
pdflatex -interaction=nonstopmode -halt-on-error hat_guessing_growth.tex
pdflatex -interaction=nonstopmode -halt-on-error hat_guessing_growth.tex
pdflatex -interaction=nonstopmode -halt-on-error hat_guessing_growth.tex
```

The verifier needs only the Python standard library. Regenerating the plots additionally needs Matplotlib. The complete mathematical proofs are in the LaTeX source; they do not depend on the image files.

[write] Added 4 October 2026, batch 90. These commands and file names are the delivery's. In the collection the manuscript's source and PDF, `hat_guessing_growth.tex` and `.pdf`, and the delivered top-level `README.txt` are not shipped (this Part is the source, built as part of the report's `article.tex`); they survive in the archive of the arrival commit `a162e4386`. The shipped names are listed in the note of Section 28, and the delivered `code/02-hat-surplus-Makefile` and `code/02-hat-surplus-README.txt` still use the delivery names; the report's `README.md` gives reruns that work with the shipped names, on a scratch copy.

E.2 Source distinctions

The main literature checks used the primary arXiv text of [10], the official journal record and author-uploaded final text of [11], and the exact repository snapshot. The older ECCC preprint is also listed for access. Some hosting-page metadata describes an earlier conference version of the autoreducibility paper; the final text itself identifies the three authors, the 2003 SIAM publication, and the DOI given below.

The key conclusion is a direct theorem with a complete proof. The historical question of whether this exact compulsory-guess formulation appeared elsewhere is separate, and remains unverified. The report should be cited or circulated with that distinction intact.

Part III

Adaptive box games beyond cylinder measurability

Exact extension laws and the atomic–diffuse boundary

Prepared for Vladimir Reshetnikov, October 4, 2026

Abstract. A player observes a configuration of finitely colored boxes, chooses an unread box, and predicts its color. Even a continuous choice of box can have a nonmeasurable success event when the set of boxes is uncountable. We give an exact classification of all countably additive extensions of the product measure to the success events of a finite team with blind, completed-product-measurable outputs. The parameters are measurable probability densities on the finite set of success patterns, subject only to explicit constraints imposed by fixed targets and shared diffuse targets. The corresponding joint laws form a finite polytope with sharp marriage-type inequalities.

For one player, let α be the mass of the atomic part of the distribution of its chosen box. With q equally likely colors, its success event has inner measure α/q and outer measure $1 - (q - 1)\alpha/q$, and every intervening extension probability occurs. Thus success is measurable exactly when the target law is purely atomic. Classical completion regularity extends the sharp guaranteed-score bound $\lfloor m/q \rfloor$ to full product-topology Borel and universally measurable outputs. We also establish a countable-team kernel version, explain the remaining Baire-property gap, and propose further research directions.

Contribution and status. The article resolves the full-Borel and universal-measurability clauses of Question 2 and the finite-team extension problem of Question 5 in the inspected ProveIt box-game report. It includes complete mathematical proofs, independent proof review, and exact finite checks. The results are not proof-assistant certified or externally refereed. Classical ingredients are attributed; historical priority of the particular extensions is not established. Glazer’s unrestricted choiceless problem is not resolved.

[write] **Added 4 October 2026, batch 92.** This Part is manuscript 03 of batch 92 of ProveIt’s incoming-reports intake, printed in full after Part II (provenance, the checks made at the write and a notation table: Sections 30.3 and 30.4). Like Parts I and II it is AI-assisted, unrefereed and not formalized. Its sections continue Part II’s: **the manuscript’s Section k is Section $k + 29$ here, its statement $k.m$ is $(k + 29).m$, its equation (k) is (III. k), its Appendices A–B are Appendices F–G and its Figure 1 is Figure 2.** Thus the manuscript’s Theorem 7.1 is Theorem 36.1, its Theorem 8.1 is Theorem 37.1 and its Theorem 10.2 is Theorem 39.2. Where the manuscript names “Section 15”, “Question 2” or “Question 5” of the inspected report, it means Section 15 and Research questions Q2 and Q5 of Part I, whose numbers are unchanged. Apart from these renumberings, the editorial changes listed in Section 30.3 and the dated [write] notes, no statement, proof, number or non-claim of the manuscript was changed.

30 Research setting and the questions answered

Elliot Glazer’s work on box games studies a tension between unrestricted choice and the intuition that an unseen random value should not be predictable. His paper *A choiceless box game paradox* and his dissertation give precise game models and analyze their set-theoretic strength [1, 14]. In particular, the paper asks whether ZF proves a winning strategy for three players, continuum many binary boxes, and a requirement that at most one player fail. That question concerns unrestricted strategies in a choiceless foundation. The present article studies regular output maps in ordinary ZFC, a related but different problem.

The direct repository anchor is *Countable Information, Uncountably Many Boxes*, filed in ProveIt as

SetTheory/Cardinals/docs/reports/ordinals-and-order-types/
measurable-box-games/article.tex.

We inspected this file at commit 1afa38bf984377eb85a5569cf6376054994a17a5 [15, 16]. Its Section 15 lists two questions particularly suited to a precise extension:

- Q2.** Determine whether its cylinder-measurable minimax theorem extends to full product-topology Borel measurability, the Baire property, or universal measurability.
- Q5.** Characterize all product-measure extensions to the success events of a finite team, including the possible joint distributions.

These are questions proposed by the repository report. They must not be attributed to Glazer as if they were the numbered questions in his own papers.

30.1 The main mathematical outcome

The essential reduction is to an ordinary measurable random finite set

$$\Gamma(x) \subseteq \{0, 1\}^m.$$

Its elements are the success patterns compatible with the information already fixed by countably many coordinates. Some chosen boxes belong to a countable set containing every target atom; their success bits are already measurable. The other chosen boxes have diffuse labels. They can be changed to realize any compatible color assignment without affecting a prescribed countably supported event. Players choosing the same diffuse box must receive consistent answers, which produces the constraints defining Γ .

The extension theorem then says that every extension, and only an extension, has densities

$$h_s \geq 0, \quad \sum_s h_s = 1, \quad h_s(x) = 0 \quad \text{when } s \notin \Gamma(x),$$

up to null sets. These densities can depend on the entire completed base measurable space, not just the original coordinate support of the strategies. This distinction is needed to classify the full extensions rather than only their unconditional success laws.

30.2 What is classical, and what is added here

Ingredient or conclusion	Provenance and scope
Countable-coordinate factorization	Classical product-measure theory; see Fremlin, 254P and 4A3N [19, 20]. We give the version needed for blind outputs.
Borel completion regularity	Classical work of Kakutani, with broader results and closely related proofs in Gryllakis–Koumoullis [17, 18]. It is not asserted to be a new theorem.
Measure-extension machinery	The general subject goes back at least to Łoś–Marczewski [4]. Our finite-partition proof is given directly.

Ingredient or conclusion	Provenance and scope
Selection-law inequalities	The finite polytope is a finite random-set selection polytope; Artstein provides the general selection-distribution perspective [21]. We prove the finite characterization used here.
Cylinder minimax and target collapse	Already in the inspected ProveIt report [15]. We do not claim those statements again as new.
Full extension classification	Application-specific result proved here: an exact support criterion for every density, with explicit treatment of diffuse target collisions. This answers the report's Q5.
Atomic-mass formula	Quantitative refinement proved here: the complete interval of success probabilities and the exact measurability criterion.
Broader regularity and countable teams	The full-Borel and universal clauses of Q2 follow; the kernel theorem extends the finite classification to countable teams. The Baire-property minimax clause remains open in this article.

The article supplies proofs rather than inferring correctness from a repository claim. Neither authorship nor endorsement by Glazer is implied. We make no assertion that a comprehensive literature search has established historical priority, and we do not label the results a breakthrough.

30.3 Provenance and place in this report

[write] Added 4 October 2026, batch 92. *Source.* This Part is a single manuscript, printed in full: manuscript 03 of batch 92 of ProveIt's incoming-reports intake, delivered as the archive `adaptive_box_games_research.zip` (485,758 bytes; inner directory `adaptive_box_games/`, main file `adaptive_box_games.tex` of 1,207 lines, a 29-page US Letter PDF) in the arrival commit 9dc8db274, and placed in commit e38f368c2 as Part III of this report, with its checker, the checker's methods text and its recorded output under the prefix `03-adaptive-extensions-`. The author line reads "Prepared for Vladimir Reshetnikov" (the PDF metadata "Research article prepared for Vladimir Reshetnikov") and carries no AI wording; like the other deliveries of the intake, it is AI-assisted, unrefereed and not formalized. It was placed here, not as a report of its own, because it answers two named questions of Part I: Q2 for full product-topology Borel and for universally measurable outputs (Theorem 39.2; the Baire-property clause stays open, Section 42), and Q5 completely (Theorem 36.1, with Theorem 38.1 for the joint laws). No other manuscript was merged into this Part, so it needed no merge choices.

Pin. The pin 1afa38bf984377eb85a5569cf6376054994a17a5 (the macro of Section 30 and of [15, 16]) is the commit "Jointly recode U21 control words to reach 467 and 466 operations" of 4 October 2026, 09:47 (UTC-7), an ancestor of the placement. It holds this report after its batch-87 write and before the batch-90 placement: the manuscript read Part I only, with its **[write]** notes and the label prefix `mbg:` (article blob 0749d641), and not Part II.

Repository claims, checked at the write. (i) Section 15 of the inspected report lists Q2 (Borel measurability, the Baire property, universal measurability) and Q5 (classification of the extensions to a finite team's success events, including the joint laws) as quoted in Section 30: true, and both are Part I's Section 15 unchanged; they are the report's own questions, not Glazer's, as the manuscript says. (ii) "The report's intake note still listed all twelve directions as open" (Appendix G): true at the pin; since then the batch-90 note on Q9 and the batch-92 notes after Section 15 have qualified it. (iii) The cylinder minimax theorem, the target collapse and the diagonal example are already in the report and are not claimed again (Sections 30, 39 and 40): true; they are Theorem 3.1, Corollary 3.3, Theorem 5.1 and Section 6. (iv) The report gives finite blind maps that no legal probing tree executes (Section 31): true (Theorem 7.3 and the four-box example of Section 7). No repository claim is refuted, and nothing is retracted.

Relation to Parts I and II. Three statements re-prove Part I's, as the manuscript says or as the write checked: Lemma 32.1 is Lemma 2.4; the fairness identity of Lemma 33.2 is the fibre identity (3) in the proof of Theorem 3.1; Theorem 39.1 is Theorem 4.1 for finite teams with outputs measurable for the completion rather than for Σ_I . Theorem 39.2

and the list version of Section 39 extend Theorem 3.1 and Corollary 3.3 from cylinder-measurable outputs to completion-measurable, full Borel and universally measurable ones. Theorems 35.2, 36.1 and 37.1 generalize Theorem 6.1 (the diagonal case $\alpha = 0$, as Section 40 says), Theorem 6.2 (one success event) and Theorem 5.1 (whose measurable-success direction is the “ $\Rightarrow$ ” half of (III.11), in the form “the target law is purely atomic”). Theorems 38.1, 41.1 and 41.2 and Proposition 42.1 have no counterpart in Parts I–II. Two contacts with Part II: the comeagre null invariant set of Section 42 and the conull meagre success sets of Theorems 26.1–26.2 show the same divergence of measure and category, on different objects; and Theorem 41.2 is a worst-case statement over configurations for adaptive targets, whereas Theorem 25.1 (Eldredge’s Theorem 4.1) is an almost-sure statement for fixed own-hat targets. Neither implies the other. Part IV shares no theorem with this Part: its players’ targets are fixed (their own hats), and its subject is the cost of inspection.

Review in the Hilbert’s-tenth research tree. Before the placement, another session inventoried this archive with four others at their arrival revision in `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_latest_five_9dc8db274.md` (commit 3c25fbb55; member hashes in the companion `.json`). For this archive it read only the delivery `README.txt`, lines 1–97, authenticated the members by hash and routed the package as measurable envelopes and extension laws for adaptive blind targets, with the boundary that success events need not be measurable in the starting completion and that no algorithmic compiler follows from an extension of probability measures. It ran no program and did not review the manuscript; it records that the arithmetic bounds of the Hilbert’s-tenth programme are unchanged. No scope correction follows for this Part.

Checks at the write. The checker, run on a scratch copy (Python 3.14.4, Windows), passed in about a second, and its output equals the shipped `data/03-adaptive-extensions-finite_results.json` after the CR bytes of Windows text mode are stripped. The sources named in Appendix G were not re-read at the write; the attributions are the manuscript’s.

Editorial changes in printing this Part. All 62 delivered labels carry the prefix `mbg:ext:`, with every reference updated; the write added `mbg:ext:part`, `mbg:ext:sec:provenance` and `mbg:ext:sec:notation`. Sections, statements and appendices are renumbered by the offset stated at the head of this Part, and the consecutively numbered equations are tagged (III.k). The manuscript’s title page became the Part heading, abstract and status box above. Its macro `\F` for the enlarged σ -algebra $\mathcal{F}$ (script F) is printed with a new macro, because Part II’s `\F` is the blackboard $\mathbb{F}$ of $\mathbb{F}_2$; its `\Prob` is printed with Part I’s $\mathbb{P}$; its pin macro `\snap` is renamed. Its colours `ink` and `accent` and its result box are Part I’s, whose shades differ slightly (the manuscript’s are 183449, 247B83 and a box fill EFF6F7). The figure is drawn by the manuscript’s own `pgfplots` code, unchanged. Its bibliography entries for Glazer’s *A choiceless box game paradox* and for Łoś–Marczewski’s *Extensions of measure* are the same papers as Part I’s entries [1] and [4], cited by those numbers (the manuscript’s entries add the DOI 10.4064/fm-36-1-267-276 and the scan <https://matwbn.icm.edu.pl/ksiazki/fm/fm36/fm36127.pdf> for the latter); its entry for Glazer’s dissertation is shared with Part IV; its other nine entries follow Part II’s with keys `ext:`. The manuscript was set in Latin Modern; here it takes Part I’s fonts, so line and page breaks differ from the delivered PDF.

30.4 Notation in this Part

[write] Added 4 October 2026, batch 92. Within Part III every symbol has the manuscript’s meaning, given in the middle column; the right column gives a different meaning the same letter has elsewhere in the report, which is *not* meant here. The only renamed symbols are the macros listed in Section 30.3; the printed symbols are the manuscript’s.

Symbol	Meaning in Part III	<i>not</i> : meaning elsewhere
X, Σ, μ	Q^I ; the cylinder σ -algebra; its product probability, also on the completion	the same objects as Part I’s X_I, Σ_I, μ_I ; <i>not</i> Part IV’s hat X_i
$\mathcal{A}, \mathcal{B}(X)$	the completion of Σ ; the full product-topology Borel σ -algebra	Part IV’s strategy $\mathcal{A}, \mathcal{A}_*$; Part IV’s bound B
$\mathcal{F}$	$\sigma(\mathcal{A}, E_1, \dots, E_m)$, the enlarged σ -algebra	the field $\mathbb{F}_2$ of Parts II and IV; Part I’s F_n

Symbol	Meaning in Part III	<i>not</i> : meaning elsewhere
regular, blind	$\mathcal{A}$ -measurable output; Part I's blindness (1)	Part I's outputs are Σ_I -measurable; Parts II and IV's <i>legal</i> (own hat unseen)
E_p, E_s, E_H, Ω_m	success sets; success-pattern cells; their unions; patterns $\{0, 1\}^m$	Part II's hat space Ω ; Part IV's $\Omega = \{-1, 1\}^{\mathbb{N}}$
Γ, D_s, D_H, L_H	feasible patterns; the events “ s feasible”, “some pattern of H feasible”, “all feasible patterns in H ”	Part I's $D(u)$, D_6 ; the surplus D_n of Parts II and IV; Part I's L_n , Part II's L_k , Part IV's L_m
J, N_J, G	countable support, a null set, the invariant good set	Part IV's geometric variable G and visibility graph G
$\lambda_p, A_p, \alpha, \beta, \theta$	target law; its atoms; atomic mass; internal success mass (biased colours; nonblind outputs)	Part IV's α (activation probability) and α_d
$C, n_{C,v}, \ell, u, t$	external target group (as Part I's C); guesses of colour v in it; least and largest feasible scores; a threshold	Part II's $t = 2^r - 1$ and debt C ; Part IV's all-wrong set C
$T, w_T, \Delta(T), K$	a feasible-pattern set with its weight and simplex (Section 38); the realization map $T(Y, Z)$ and a kernel (Section 41)	Part I's $T(Y, U)$ (same role as $T(Y, Z)$); Part II's T_i, K ; Part IV's T_n, T_j, T_a
h_s, S, q	extension densities (as Part I's h); the score; the alphabet size	Part IV's $h(i)$ and its count budget $q(i)$
R1–R12	this Part's research directions	Part I's Q1–Q12; Research questions 29.1–29.12 and 57.1–57.12, 68.1–68.12

31 Configurations, output maps, and the base probability space

Fix an integer $q \geq 2$ and put $Q = \{0, \dots, q-1\}$. Let I be a nonempty standard Borel space. Its elements are labels for boxes. The configuration space is

$$X = Q^I,$$

with the product topology of the discrete spaces Q . There are two different structures in this definition: the standard Borel structure on the set of labels I , and the product topology on Q^I . No topology on I is used to construct that product topology.

Let

$$\Sigma = \sigma\{x : x_i = v \mid i \in I, v \in Q\}$$

be the cylinder σ -algebra, and let μ be its product probability, with all coordinates independent and uniform. Write $\mathcal{A}$ for its completion, and use μ also for the completed measure. Let $\mathcal{B}(X)$ denote the full product-topology Borel σ -algebra. We will prove

$$\Sigma \subseteq \mathcal{B}(X) \subseteq \mathcal{A}. \quad (\text{III.1})$$

For uncountable I , the first inclusion can be strict. The second inclusion is a completion-regularity theorem, not a definition.

For a set $E \subseteq X$, its inner and outer measures throughout this article are

$$\mu_*(E) = \sup\{\mu(A) : A \in \mathcal{A}, A \subseteq E\}, \quad \mu^*(E) = \inf\{\mu(A) : A \in \mathcal{A}, E \subseteq A\}.$$

Using Σ instead of its completion gives the same numbers.

Definition 31.1 (Blind output). A player is represented by a total map

$$\tau = (b, a) : X \longrightarrow I \times Q.$$

Here $b(x)$ is the chosen box and $a(x)$ the predicted color. For $i \in I$ and $v \in Q$, let $x^{i \leftarrow v}$ replace only coordinate i . The output is *blind* if

$$\tau(x) = (i, c) \implies \tau(x^{i \leftarrow v}) = (i, c) \quad (v \in Q). \quad (\text{III.2})$$

It is *regular* in the main theorems if it is measurable from $(X, \mathcal{A})$ to the standard Borel space $I \times Q$.

Every total deterministic probing strategy that ultimately predicts an unread box induces a blind map: changing that box changes none of the observations on the original run, so the run and its output remain the same. The converse is not assumed. The repository report gives finite blind maps that are not executable by legal probing trees [15].

For player p , set

$$E_p = \{x : x_{b_p(x)} = a_p(x)\}.$$

These success sets are subsets of X ; they are not initially assumed measurable. For a finite team, let $\Omega_m = \{0, 1\}^m$ and define the actual success-pattern cells

$$E_s = \{x : (\mathbf{1}_{E_1}(x), \dots, \mathbf{1}_{E_m}(x)) = s\} \quad (s \in \Omega_m).$$

They form a finite partition of X . The enlarged σ -algebra is

$$\mathcal{F} = \sigma(\mathcal{A}, E_1, \dots, E_m).$$

An *extension* always means a countably additive probability ν on $\mathcal{F}$ satisfying $\nu \upharpoonright \mathcal{A} = \mu$.

Remark 31.2 (Why a random target is delicate). For each fixed i , the function $x \mapsto x_i$ is measurable. This does not imply that $x \mapsto x_{b(x)}$ is measurable when b has an uncountable range. An uncountable union of measurable target fibers need not belong to the cylinder σ -algebra, and even full Borel regularity of the output does not fix the issue. Section 40 gives a continuous example with inner success measure zero and outer success measure one.

32 Classical support reduction, with the required exceptional-set control

32.1 Countable support and the Borel sandwich

Lemma 32.1 (Literal factorization). *Every $A \in \Sigma$ has the form $\pi_J^{-1}(A_J)$ for a countable $J \subseteq I$ and a Borel set $A_J \subseteq Q^J$. Every cylinder-measurable map to a standard Borel space factors measurably through a countable subproduct.*

Proof. Sets of the displayed form constitute a σ -algebra: take the union of the countably many supports when forming a countable union. They contain every finite-coordinate cylinder. For maps, take preimages of a countable family of Borel sets separating points in the target. A common countable support of those preimages determines the value of the map. To see that the factor is measurable, compose with the section that places color zero outside J . $\square$

The following is a finite-alphabet instance of classical completion regularity [17, 18]. We include its proof because it identifies exactly which null sets later may be changed.

Lemma 32.2 (Open-set sandwich). *For every open $U \subseteq Q^I$ there are a countable $J \subseteq I$ and open sets $V_J, W_J \subseteq Q^J$ such that*

$$\pi_J^{-1}(V_J) \subseteq U \subseteq \pi_J^{-1}(W_J), \quad \mu_J(W_J \setminus V_J) = 0.$$

The gap $W_J \setminus V_J$ is also nowhere dense.

Proof. Let t be the supremum of the measures of countable unions of basic finite-coordinate cylinders contained in U . Choose such unions approaching t and take their union V . Then $V \subseteq U$, $\mu(V) = t$, and $V = \pi_J^{-1}(V_J)$ for a countable J and open V_J .

If $O \subseteq U$ is a basic cylinder, maximality gives $\mu(O \setminus V) = 0$. Write O as the conjunction of its restrictions in J and in a finite set outside J . The latter finite pattern has strictly positive probability. Independence therefore implies

$$\mu_J(\pi_J(O) \setminus V_J) = 0.$$

The projection $W_J = \pi_J(U)$ is open. Since Q^J is second countable, it is a countable union of sets of the form $\pi_J(O)$ with $O \subseteq U$ a basic cylinder. Thus $\mu_J(W_J \setminus V_J) = 0$, and the claimed sandwich follows.

Full support of μ_J also implies $W_J \subseteq \overline{V_J}$: an open portion of W_J disjoint from $\overline{V_J}$ would have positive measure. Hence the gap lies in the boundary of the open set V_J , a closed nowhere dense set. $\square$

Corollary 32.3 (Completion regularity in the needed form). *Every $B \in \mathcal{B}(X)$ differs from a cylinder-measurable set by a subset of a cylinder-measurable null set. The null set can also be chosen meager. In particular, (III.1) holds.*

Proof. Sets with this approximation property form a σ -algebra. Complements retain the same exceptional set; countable unions use the union of the exceptional sets and their supports. Lemma 32.2 supplies the property for open sets. The lift of a nowhere dense subset of Q^J is nowhere dense in Q^I , since projection is an open continuous surjection. $\square$

Consequently there is exactly one probability on $\mathcal{B}(X)$ with the prescribed cylinder law: every Borel sandwich forces its value. Its completion is precisely $\mathcal{A}$. The freedom in the subsequent classification concerns adjoining success events which can be non-Borel, not choosing different Borel product measures.

32.2 A common invariant good set

Lemma 32.4 (Normal form for a countable family). *Let (f_p) be a finite or countable family of $\mathcal{A}$ -measurable maps into standard Borel spaces. There exist a countable $J \subseteq I$, a Borel null set $N_J \subseteq Q^J$, and Borel maps f_p^J such that, with*

$$G = X \setminus \pi_J^{-1}(N_J),$$

the following hold:

- (i) $f_p(x) = f_p^J(x \upharpoonright J)$ for all p and all $x \in G$;
- (ii) G is invariant under changing any finite set of coordinates in I ;
- (iii) if all f_p are full product-topology Borel, N_J can also be meager, so G is both conull and comeager.

Proof. Embed each target Borel-isomorphically into a Borel subset of $2^{\mathbb{N}}$. Approximate the preimages of the bit-coordinate sets by cylinder-measurable sets, with cylinder-null error envelopes. A countable union produces one support J and one null envelope on which all coded approximations might fail. Away from

it the approximating bits are the code of the actual value, so the Borel inverse coding map gives a Borel factor. Give it an arbitrary fixed value on the exceptional set. For full Borel maps use the null-and-meager approximations of Corollary 32.3.

Finally saturate the exceptional set in Q^J under the group of coordinate permutations with finite support. This group is countable: J is countable and each coordinate has only finitely many permutations. Its transformations preserve the uniform product measure and are homeomorphisms, so the saturation is still Borel null, and still meager in the Borel case. Every change of finitely many coordinates of Q^J can be effected by such a permutation at that configuration. Changes outside J do not affect membership in G at all. This proves (ii) without taking an uncountable union of null sets. $\square$

Remark 32.5 (The factors need not be blind everywhere). The original outputs are blind everywhere. Their arbitrary Borel definitions on N_J need not be blind. All uses of blindness below take place on G , which is invariant under the relevant changes. Ignoring this qualification would leave a gap in the averaging argument.

Corollary 32.6 (Universally measurable outputs). *If the preimage of every target Borel set under an output map is universally measurable relative to $\mathcal{B}(X)$, then that output is $\mathcal{A}$ -measurable and satisfies Lemma 32.4.*

Proof. The canonical Borel product probability is one of the Borel probabilities used in the universal completion. Its completion is $\mathcal{A}$. $\square$

Here universal measurability refers explicitly to the full Borel structure of the product topology on X , and to the standard Borel structure on the target. The Baire *property*, which concerns category rather than measure, will be treated separately in Section 42.

33 Target atoms and measurable internal success

For a regular target map b_p , its probability law is the ordinary pushforward $\lambda_p = (b_p)_*\mu$ on I . Define

$$A_p = \{i \in I : \lambda_p(\{i\}) > 0\}, \quad \alpha_p = \lambda_p(A_p).$$

Every A_p is countable: for each positive integer n only finitely many points have mass at least $1/n$. On a standard Borel space these point atoms account for the atomic part of the measure; the remaining part is diffuse.

Choose J from Lemma 32.4 for the outputs and enlarge it to contain all A_p . Extend the old exceptional set cylindrically to the enlarged subproduct. The good set remains conull and invariant, and every output on it depends only on $x \upharpoonright J$.

We call $b_p(x) \in J$ an *internal target*, and $b_p(x) \notin J$ an *external target*. These terms refer only to this proof. They do not say which boxes a legal strategy has physically inspected. Because J is countable and includes all point atoms,

$$\mu(b_p \in J) = \alpha_p. \tag{III.3}$$

In particular, adding further countably many labels changes the internal-target event only on a null set.

Lemma 33.1 (Diffuse targets escape a prescribed countable set). *For every countable $L \subseteq I$ and a finite or countable team,*

$$\mu\{x : \exists p (b_p(x) \notin J \text{ and } b_p(x) \in L)\} = 0.$$

Proof. For each p and each $i \in L \setminus J$, $\mu(b_p = i) = 0$. There are only countably many pairs (p, i) . $\square$

Lemma 33.2 (Internal fairness). *For every regular blind player,*

$$\mu(G \cap \{b_p \in J\} \cap E_p) = \frac{\alpha_p}{q}.$$

The set on the left belongs to $\mathcal{A}$.

Proof. For $i \in J$ and $c \in Q$, consider the measurable set

$$H_{p,i,c} = G \cap \{b_p = i, a_p = c\}.$$

Blindness and invariance of G make this set invariant under changing coordinate i . Therefore its intersections with the q possible values of x_i have equal measure. Equivalently, integrate the indicator over that finite coordinate in the countable subproduct. Thus

$$\mu(H_{p,i,c} \cap \{x_i = c\}) = \frac{1}{q} \mu(H_{p,i,c}).$$

Summing over the countably many pairs (i, c) and using (III.3) gives the result. All the displayed sets are countable unions of measurable sets. No integration of an external random evaluation has been used. $\square$

The countable set J is not canonical. The eventual extension laws are canonical: enlarging J changes the feasible-pattern description only on a null set, by Lemma 33.1.

34 The finite feasible-pattern object

Fix a team of m regular blind players, and keep the normal form above. At a configuration $x \in G$, let

$$P_{\text{in}}(x) = \{p : b_p(x) \in J\}, \quad P_{\text{out}}(x) = \{1, \dots, m\} \setminus P_{\text{in}}(x).$$

Partition $P_{\text{out}}(x)$ into groups C whose members choose the same box. For each such group and each color v , set

$$n_{C,v}(x) = |\{p \in C : a_p(x) = v\}|.$$

Definition 34.1 (Feasible patterns). For $x \in G$, a vector $s \in \Omega_m$ belongs to $\Gamma(x)$ if:

(i) for every $p \in P_{\text{in}}(x)$,

$$s_p = \mathbf{1}_{\{x_{b_p(x)} = a_p(x)\}};$$

(ii) for every external target group C , there is a color $v_C \in Q$ such that

$$s_p = \mathbf{1}_{\{a_p(x) = v_C\}} \quad (p \in C).$$

On $X \setminus G$, define $\Gamma(x) = \Omega_m$. For $s \in \Omega_m$, write

$$D_s = \{x : s \in \Gamma(x)\}.$$

The exceptional-set convention ensures $E_s \subseteq D_s$ pointwise, and only alters null sets. The set $\Gamma(x)$ is always nonempty. It is determined by $x \upharpoonright J$ on G . Each D_s is measurable, indeed cylinder-measurable after using the Borel normal form: only finitely many target-equality tests and color tests are needed, together with countably many possible internal coordinates. Equality of two targets is Borel because the diagonal of a standard Borel space is Borel.

The shared-target constraint is indispensable. For example, three players choosing the same external binary box and guessing $(0, 0, 1)$ can produce only

$$\Gamma(x) = \{(1, 1, 0), (0, 0, 1)\}. \quad (\text{III.4})$$

Assigning each player a separately sampled fair answer would incorrectly admit impossible patterns such as $(1, 0, 1)$.

For $H \subseteq \Omega_m$, use the abbreviations

$$E_H = \bigcup_{s \in H} E_s, \quad D_H = \{x : \Gamma(x) \cap H \neq \emptyset\}.$$

Thus D_H is the measurable event that some pattern in H is feasible. The measurable event that every feasible pattern lies in H is

$$L_H = \{x : \Gamma(x) \subseteq H\} = X \setminus D_{\Omega_m \setminus H}.$$

35 The envelope lemma

The next statement is the main bridge between the pointwise box game and ordinary measure-extension theory. It identifies all measurable obstructions to realizing a pattern.

Lemma 35.1 (Exact measurable obstruction). *For $A \in \mathcal{A}$ and $s \in \Omega_m$,*

$$A \cap E_s = \emptyset \implies \mu(A \cap D_s) = 0. \quad (\text{III.5})$$

More generally, $A \cap E_H = \emptyset$ implies $\mu(A \cap D_H) = 0$ for every $H \subseteq \Omega_m$.

Proof. Choose $B, N \in \Sigma$ with $\mu(N) = 0$ and $A \triangle B \subseteq N$. Take a common countable support L for B and N . Suppose, towards a contradiction, that $\mu(A \cap D_s) > 0$.

By removing N , $X \setminus G$, and the null set from Lemma 33.1, choose

$$x \in A \cap D_s \cap G \setminus N$$

such that every selected external target lies outside L . For each external target group, choose a color witnessing feasibility of s and change that box to this color. Only finitely many coordinates are changed. None belongs to $J \cup L$. Call the result y .

The point y remains in G , and all output maps are unchanged, since they depend only on J there. All internal success bits remain unchanged. The chosen external colors give exactly pattern s , so $y \in E_s$. On the other hand, x and y agree on L , so $y \notin N$ and $y \in B$. Hence $y \in A$, a contradiction.

For a set H of patterns, apply the preceding conclusion to each $s \in H$ and take the finite union. $\square$

Theorem 35.2 (Inner and outer probabilities of every pattern event). *For every $H \subseteq \Omega_m$,*

$$\mu_*(E_H) = \mu\{\Gamma \subseteq H\}, \quad \mu^*(E_H) = \mu\{\Gamma \cap H \neq \emptyset\}. \quad (\text{III.6})$$

Proof. Since the actual success pattern always belongs to $\Gamma(x)$, we have

$$L_H \subseteq E_H \subseteq D_H.$$

If a measurable set B contains E_H , its complement is disjoint from E_H . Lemma 35.1 implies $\mu(D_H \setminus B) = 0$, whence $\mu(B) \geq \mu(D_H)$. This proves the outer-measure formula. Apply it to the complementary event to obtain the inner-measure formula. $\square$

In particular, the formula describes the size of all threshold-success events before any probability extension has been chosen. It also shows precisely when such an event is already measurable: its lower and upper expressions must be equal.

36 Classification of all finite-team extensions

Theorem 36.1 (Full density classification). *Every probability extension ν of μ to $\mathcal{F}$ corresponds to a family of $\mathcal{A}$ -measurable functions $(h_s)_{s \in \Omega_m}$ satisfying, almost everywhere,*

$$h_s \geq 0, \quad \sum_{s \in \Omega_m} h_s = 1, \quad h_s = 0 \text{ on } X \setminus D_s. \quad (\text{III.7})$$

Conversely, every such family defines an extension, by

$$\boxed{\nu \left(\bigcup_{s \in \Omega_m} (A_s \cap E_s) \right) = \sum_{s \in \Omega_m} \int_{A_s} h_s d\mu, \quad A_s \in \mathcal{A}.} \quad (\text{III.8})$$

The family is unique up to equality almost everywhere. No countable-generation hypothesis on $\mathcal{A}$ is needed.

Proof. Necessity. Given an extension, define finite measures on the base by

$$\lambda_s(A) = \nu(A \cap E_s) \quad (A \in \mathcal{A}).$$

They satisfy $0 \leq \lambda_s \leq \mu$. The Radon–Nikodym theorem gives $\lambda_s = h_s \mu$. Summing the measures gives $\sum_s \lambda_s = \mu$, so the densities sum to one almost everywhere. Since $E_s \subseteq D_s$, $\lambda_s(X \setminus D_s) = 0$, yielding the support condition. Radon–Nikodym uniqueness gives uniqueness of the densities. This theorem applies to any finite measure space, including the completed uncountable product.

Description of the enlarged sets. The sets of the form

$$\bigcup_s (A_s \cap E_s), \quad A_s \in \mathcal{A},$$

form a σ -algebra. Complements and countable unions can be taken separately within each cell of the finite partition (E_s) . They contain $\mathcal{A}$ and the success events, so they are exactly $\mathcal{F}$.

Well-definedness. Suppose (A_s) and (B_s) describe the same set. Then

$$(A_s \triangle B_s) \cap E_s = \emptyset$$

for every s . Lemma 35.1 gives

$$\mu((A_s \triangle B_s) \cap D_s) = 0.$$

As h_s vanishes outside D_s , the two integrals in (III.8) agree.

Countable additivity. Let $F_n \in \mathcal{F}$ be pairwise disjoint, with $F_n = \bigcup_s (A_{n,s} \cap E_s)$. For fixed s and $n \neq k$, the base set $A_{n,s} \cap A_{k,s}$ is disjoint from E_s . Its $h_s \mu$ measure is therefore zero by the same envelope argument. Removing these countably many null overlaps makes the sets $A_{n,s}$ pairwise disjoint for the measure $h_s \mu$. Ordinary countable additivity on the base then gives

$$\nu \left(\bigcup_n F_n \right) = \sum_s \int_{\bigcup_n A_{n,s}} h_s d\mu = \sum_n \sum_s \int_{A_{n,s}} h_s d\mu = \sum_n \nu(F_n).$$

Finally, the density sum gives total mass one and $\nu(A) = \mu(A)$ for $A \in \mathcal{A}$. This proves sufficiency. $\square$

The theorem directly answers Question 5 of the inspected report. It provides the measure on every event of the enlarged σ -algebra, not only the vector $(v(E_p))$ or the distribution of the number of correct guesses.

Remark 36.2 (Why densities live on the full base). It would be incorrect to require every h_s to depend only on the original strategy support J . In a one-player diffuse example, both outcomes are feasible almost everywhere. For a fixed coordinate k outside that support, the choice

$$h_1(x) = \mathbf{1}_{\{x_k=0\}}, \quad h_0 = 1 - h_1$$

defines a valid extension, correlating success with that additional base event. Each particular finite family of densities has a countable support modulo null sets, but that support can be larger than the one used by the outputs.

Corollary 36.3 (All bounded payoff ranges). *For any function $\psi : \Omega_m \rightarrow \mathbb{R}$, the set of extension expectations of ψ of the actual success pattern is exactly*

$$\left[\int \min_{s \in \Gamma(x)} \psi(s) d\mu(x), \quad \int \max_{s \in \Gamma(x)} \psi(s) d\mu(x) \right]. \quad (\text{III.9})$$

Both endpoints occur, and every intermediate value occurs.

Proof. At almost every x , the densities in Theorem 36.1 define a probability supported on $\Gamma(x)$, so its conditional payoff is between the indicated extrema. A measurable choice of the first minimizing, or first maximizing, pattern attains the endpoint: there are finitely many patterns and their feasibility sets are measurable. Constant convex mixtures of the two density families attain the whole interval. $\square$

For $\psi = \mathbf{1}_H$, this also proves directly that every probability between the inner and outer values in (III.6) is obtained by an extension.

Corollary 36.4 (Uniqueness). *The extension is unique if and only if $\Gamma(x)$ is a singleton almost everywhere. For blind regular players with $q \geq 2$, this is equivalent to every target law being purely atomic.*

Proof. A singleton feasible set forces the density vector; conversely, two feasible patterns on a positive-measure set allow distinct measurable density choices. If every target is internal, the success vector is fixed. If some target is external, its nonempty group contains a player guessing a color c ; choosing c or a different color changes that player's success bit. Thus an external group creates at least two patterns. Use (III.3) and the finite union over players. $\square$

37 The exact atomic-mass law for one player

Theorem 37.1 (Atomic–diffuse formula). *Let $\tau = (b, a)$ be one regular blind player and let*

$$\alpha = \sum_{i: \mu(b=i) > 0} \mu(b=i)$$

be the mass of the atomic part of its target distribution. Then

$$\mu_*(E) = \frac{\alpha}{q}, \quad \mu^*(E) = \frac{\alpha}{q} + 1 - \alpha = 1 - \frac{q-1}{q}\alpha. \quad (\text{III.10})$$

Every value in this closed interval is the success probability in some countably additive extension. Moreover,

$$E \in \mathcal{A} \iff \alpha = 1. \quad (\text{III.11})$$

When these equivalent conditions hold, $\mu(E) = 1/q$.

Proof. On the internal-target event the success bit is fixed and its success portion has measure α/q , by Lemma 33.2. On the external-target event, of measure $1 - \alpha$, both success and failure are feasible: set the selected color equal to or different from the guess. Consequently the lower feasibility event for success is exactly the internal success event, up to a null set; the upper feasibility event also includes every external-target point. Theorem 35.2 gives (III.10), and Corollary 36.3 gives attainability.

The interval has width $1 - \alpha$. In a complete finite measure space a subset has equal inner and outer measures exactly when it is measurable: choose measurable inner and outer approximations with equal limiting measure and use completeness for the intervening null difference. This proves (III.11). $\square$

The formula is stronger than a countable-target obstruction: it measures exactly how much probability remains undetermined. A purely atomic target law can have an uncountable pointwise range on a null set, so the criterion concerns its law rather than literal global countability of the range.

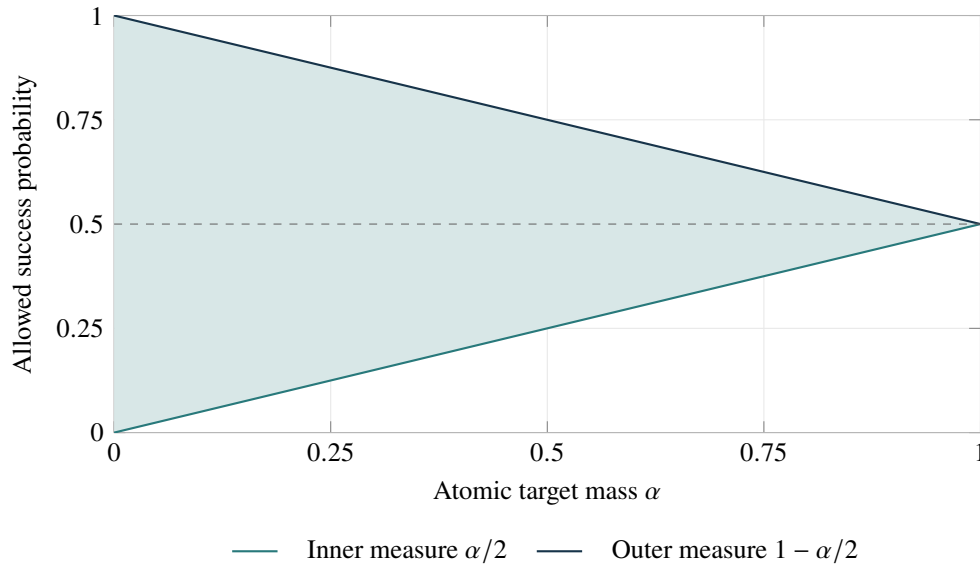


Figure 2: Binary case. The shaded interval at each α is the exact set of extension probabilities. The fair value $1/2$ always occurs; uniqueness holds only at $\alpha = 1$. This figure is the theorem's exact geometry, not a numerical estimate.

Atomic mass α	Binary success interval	Width
0	$[0, 1]$	1
$1/4$	$[1/8, 7/8]$	$3/4$
$1/2$	$[1/4, 3/4]$	$1/2$
$3/4$	$[3/8, 5/8]$	$1/4$
1	$\{1/2\}$	0

37.1 Nonuniform independent colors

The geometric extension theorem does not require uniform probabilities. Suppose coordinate i has probabilities $p_i(v) > 0$ for every $v \in Q$, independently across coordinates. The same open-set proof works because every finite pattern has positive probability. Finite coordinate permutations need not preserve the measure, but they

preserve null sets: on a finite set of coordinates all likelihood ratios are bounded above and below by positive finite constants. The invariant normal form and the envelope argument therefore survive unchanged.

For one blind player define $A_b = \{i : \mu(b = i) > 0\}$ and

$$\beta = \sum_{i \in A_b} \sum_{v \in Q} p_i(v) \mu(b = i, a = v).$$

The fiber $\{b = i, a = v\}$ is insensitive to coordinate i , so this is exactly the measure of internal success. The analogue of (III.10) is

$$\mu_*(E) = \beta, \quad \mu^*(E) = \beta + 1 - \alpha. \quad (\text{III.12})$$

The same uniqueness criterion holds. The uniform minimax value in Section 39, however, is a worst-case combinatorial statement obtained using an auxiliary uniform product law; it is not a claim that every nonuniform marginal success probability is $1/q$.

38 The joint-law polytope and exact inequalities

The full extension space contains arbitrary measurable density functions. Its projection to unconditional success-pattern laws is finite-dimensional and has a particularly explicit description.

For every nonempty $T \subseteq \Omega_m$, put

$$w_T = \mu\{x : \Gamma(x) = T\}, \quad \Delta(T) = \{p \in \mathbb{R}^{\Omega_m} : p_s \geq 0, \sum_s p_s = 1, p_s = 0 \text{ for } s \notin T\}.$$

Only finitely many T occur, and $\sum_T w_T = 1$. The set $\Delta(T)$ is the probability simplex on T , embedded in the common pattern-law space $\mathbb{R}^{\Omega_m}$.

Theorem 38.1 (Complete finite law description). *The set $\mathcal{P}$ of all joint pattern laws $(v(E_s))_{s \in \Omega_m}$ is*

$$\mathcal{P} = \sum_{\emptyset \neq T \subseteq \Omega_m} w_T \Delta(T). \quad (\text{III.13})$$

A probability vector p belongs to $\mathcal{P}$ if and only if, for every $H \subseteq \Omega_m$,

$$\sum_{s \in H} p_s \leq \sum_{T: T \cap H \neq \emptyset} w_T = \mu\{\Gamma \cap H \neq \emptyset\}. \quad (\text{III.14})$$

Equivalently, every such H satisfies

$$\sum_{s \in H} p_s \geq \mu\{\Gamma \subseteq H\}.$$

Proof. If h is an extension density, average it over each measurable set $\{\Gamma = T\}$ of positive measure. Its average is a probability vector supported on T . Thus its unconditional law has the form in (III.13). Conversely, assign any chosen vector from $\Delta(T)$ as a constant density on $\{\Gamma = T\}$ and use Theorem 36.1. This proves the Minkowski-sum formula.

Necessity of (III.14) is immediate: mass assigned to H can only come from feasible sets intersecting H . For sufficiency, use the finite transportation network whose left nodes are the nonempty sets T with supply w_T , whose right nodes are patterns s with demand p_s , and whose permitted edges are $T \rightarrow s$ for $s \in T$. A flow of total mass one is exactly a decomposition in (III.13).

For completeness, the cut condition can be checked directly. Give source-to- T edges capacity w_T , allowed T -to- s edges capacity 2, and s -to-sink edges capacity p_s . A cut of capacity below one cannot cross an edge of capacity 2. If H is the collection of pattern nodes on the sink side, then all T meeting H must also be on that side. The cut capacity is therefore at least

$$\sum_{T: T \cap H \neq \emptyset} w_T + \sum_{s \notin H} p_s \geq 1$$

by (III.14). The finite max-flow/min-cut theorem gives a unit flow. The lower inequalities are the upper inequalities for complementary pattern sets. $\square$

This is a finite instance of the marriage-type characterization of random-set selection distributions [21]. Our application permits randomized conditional selections, as the extension densities show. Restricting to deterministic selections on a fixed atomic probability space would in general produce a smaller set, so that restriction is not being made here.

Remark 38.2 (Pattern laws and player marginals are different spaces). The vector of individual success probabilities is the image of p under

$$p \mapsto \left(\sum_s s_1 p_s, \dots, \sum_s s_m p_s \right).$$

Its attainable region is consequently

$$\sum_T w_T \text{conv}(T) \subseteq [0, 1]^m.$$

The summands $\text{conv}(T)$ need not be simplices. The simplex terminology in (III.13) applies to distributions over full patterns, not to their projection onto individual marginals.

38.1 Explicit score formulas

Let $S(x) = \sum_{p=1}^m \mathbf{1}_{E_p}(x)$, and for $x \in G$ let $S_{\text{in}}(x)$ be the measurable internal score. Define

$$\ell(x) = S_{\text{in}}(x) + \sum_C \min_{v \in Q} n_{C,v}(x), \quad (\text{III.15})$$

$$u(x) = S_{\text{in}}(x) + \sum_C \max_{v \in Q} n_{C,v}(x), \quad (\text{III.16})$$

where the sums run over distinct external target groups. These are respectively the smallest and largest feasible scores. Values on the null exceptional set can be defined arbitrarily. Corollary 36.3 yields the exact expected-score interval

$$[\mathbb{E}\ell, \mathbb{E}u]. \quad (\text{III.17})$$

Internal fairness makes its endpoints

$$\frac{1}{q} \sum_{p=1}^m \alpha_p + \mathbb{E} \sum_C \min_v n_{C,v} \quad \text{and} \quad \frac{1}{q} \sum_{p=1}^m \alpha_p + \mathbb{E} \sum_C \max_v n_{C,v}.$$

For a threshold t , the exact possible range of $\nu(S \geq t)$ is

$$[\mu\{\ell \geq t\}, \mu\{u \geq t\}]. \quad (\text{III.18})$$

These formulas account for target collisions, so they can be much sharper than bounds using only separate player marginals.

39 Full Borel and universal minimax

Theorem 39.1 (A fair extension). *Every finite team of regular blind outputs admits an extension with*

$$\nu(E_p) = 1/q \quad (1 \leq p \leq m).$$

Proof. At each $x \in G$, retain the internal bits and assign an independent uniform color to each distinct external target group. This induces a measurable probability vector $h_s(x)$ supported on $\Gamma(x)$. It is a finite sum over finite color assignments; all group-equality tests are measurable. Any assignment on the exceptional null set is harmless. Theorem 36.1 gives an extension.

For player p , its internal success contribution is α_p/q . Conditional on the base configuration, it succeeds with probability $1/q$ whenever its target is external. Thus

$$\nu(E_p) = \frac{\alpha_p}{q} + \frac{1 - \alpha_p}{q} = \frac{1}{q}.$$

Independence is imposed between distinct external *boxes*, not between players who may choose the same box. $\square$

Theorem 39.2 (Sharp guaranteed score under broader regularity). *For every nonempty standard Borel label space I , every $q \geq 2$, and every $m \geq 1$,*

$$\boxed{\max_{\substack{\tau_1, \dots, \tau_m \\ \text{regular and blind}}} \min_{x \in Q^I} S(x) = \lfloor m/q \rfloor.} \quad (\text{III.19})$$

The same value holds if the class is restricted to full product-topology Borel outputs or to universally measurable outputs. The lower bound is attained by legal strategies that inspect no boxes.

Proof. Under the fair extension, S is a genuine measurable random variable with $\mathbb{E}_\nu S = m/q$. If every configuration had score at least $\lfloor m/q \rfloor + 1$, this expectation would be at least that larger integer. Hence some configuration has score at most $\lfloor m/q \rfloor$.

For attainment, fix one box i_0 and distribute the m constant guesses as evenly as possible among the q colors. Every color is named at least $\lfloor m/q \rfloor$ times, so every configuration receives at least that many correct guesses. The outputs are constant, hence Borel, universally measurable, regular, and blind. They are implemented without any observation. Corollaries 32.3 and 32.6 give the upper bounds in the two narrower classes. $\square$

For the three-player binary continuum game, regular blind players can guarantee one correct guess and cannot guarantee two. This is the precise negative conclusion for the specified regularity classes. It is not a resolution of Glazer's unrestricted ZF question.

Corollary 39.3 (Witnesses outside arbitrary prescribed null exceptions). *Put $k = \lfloor m/q \rfloor$ and $L = \{x : S(x) \leq k\}$. For the fair extension,*

$$\nu(L) \geq 1 - \frac{m}{q(k+1)} > 0. \quad (\text{III.20})$$

Consequently $\mu^(L)$ is at least the same number. For every $N \in \mathcal{A}$ with $\mu(N) = 0$, there is a losing witness $x \in L \setminus N$.*

Proof. The inequality $S \geq (k+1)\mathbf{1}_{X \setminus L}$ gives (III.20). An extension probability never exceeds the base outer measure. Since $\nu(N) = 0$, positive extension mass of L also gives a point outside N . $\square$

The null set in this corollary may be chosen after the strategies are specified. The corollary gives an existence and outer-measure statement, not an algorithm for finding a configuration or a claim that the losing event is Borel.

39.1 Fixed-size lists

If each player gives a set of r colors, with $1 \leq r < q$, replace the external compatibility condition by $s_p = \mathbf{1}_{\{v_C \in A_p(x)\}}$. The envelope and density proofs are identical, and the fair marginal is r/q . The sharp guaranteed score is therefore

$$\lfloor mr/q \rfloor.$$

For the lower bound, let player p , numbered from zero, use the r consecutive residues $pr, pr+1, \dots, pr+r-1$ modulo q , all at one fixed box. The mr consecutive residues distribute as evenly as possible over the colors. The cylinder version of this statement was already in the prior report; the conclusion here is its completed, full-Borel, and universal extension.

40 Examples: what the extension freedom means

40.1 Continuous output, maximally nonmeasurable success

Take a countable infinite set J_0 , let $C = Q^{J_0}$ with its compact metric product topology, and let

$$I = J_0 \sqcup C$$

be a tagged disjoint union. Give it the standard Borel structure of that topological disjoint union. For $x \in Q^I$, set

$$b(x) = \text{the } C\text{-tagged label } x \upharpoonright J_0, \quad a(x) = 0.$$

The output is continuous. It is blind because its chosen target is in the C component, whereas it depends only on the J_0 coordinates. The target law is diffuse: every single sequence in Q^{J_0} has product probability zero. Theorem 37.1 gives

$$\mu_*(E) = 0, \quad \mu^*(E) = 1.$$

Its success is the diagonal evaluation assertion that the box labeled by the observed sequence itself contains zero. The existing repository report already contains a binary diagonal example of this type [15]; the purpose here is to identify it as the endpoint $\alpha = 0$ of the general formula. Nonmeasurability of evaluation on an uncountable Bernoulli cube has related classical examples, such as Janson's Appendix G [22].

The player can be understood operationally as first revealing the entire countable set J_0 , then choosing the box carrying the resulting label. This uses the broad probing convention in which an entire set of boxes may be observed. A model requiring termination after finitely many individual queries is a stricter notion of executability.

40.2 Every atomic mass is realized

Add a separate label $\star$ to the preceding label space. Choose a Borel set $A \subseteq Q^{J_0}$ of product probability $\alpha \in [0, 1]$. Such sets exist because this countable product is an atomless standard probability space; explicitly, use the base- q expansion of a uniform point of $[0, 1]$ and a threshold. Define

$$b(x) = \begin{cases} \star, & x \upharpoonright J_0 \in A, \\ \text{the } C\text{-tagged label } x \upharpoonright J_0, & x \upharpoonright J_0 \notin A, \end{cases} \quad a(x) = 0.$$

It is Borel and blind. Its only possible positive-mass target atom is $\star$, of mass α . Thus every interval in (III.10), including every binary interval in Figure 2, occurs in a concrete example.

40.3 Shared targets force exact correlations

In the binary case $q = 2$, let three players use the same diffuse diagonal target and the guesses $(0, 0, 1)$. The feasible patterns are exactly those in (III.4). Every extension has a parameter $t \in [0, 1]$ such that

$$\nu(110) = t, \quad \nu(001) = 1 - t.$$

Its individual success probabilities are

$$(t, t, 1 - t),$$

and its expected score is $1 + t$. The fair extension has $t = 1/2$, while other extensions give different marginal probabilities. Regardless of the extension, every actual configuration has either one or two successes. This shared-target constraint persists even though each separate success event has inner measure zero and outer measure one.

If instead all three guesses are zero, the only patterns are 000 and 111. In the fair extension, successes are perfectly correlated. A statement that the players' successes are independent would therefore be false even though every marginal is $1/2$.

40.4 Distinct diffuse targets allow every joint law

Use the label space

$$I = J_0 \sqcup (\{1, \dots, m\} \times Q^{J_0})$$

with tags, and let player p choose $(p, x \upharpoonright J_0)$ and guess zero. The target labels are diffuse and always distinct. Every success pattern can be obtained by setting these m boxes independently to a matching or nonmatching color, so $\Gamma(x) = \Omega_m$ almost everywhere. Hence every probability distribution on Ω_m is the joint success law of an extension. The full theorem allows, more generally, every measurable conditional distribution given the base space.

In particular, one extension makes every player succeed almost surely and another makes every player fail almost surely, while both preserve the original measure of every base-measurable event. This does not change the pointwise game or create a strategy that succeeds everywhere. Both all-success and all-failure configurations still exist. The distinction concerns what the original measurable structure determines about a nonmeasurable event. Classical saturated-partition extensions provide related constructions [23]; here the feasible-pattern theorem determines when that freedom is available to adaptive targets.

40.5 What remains fixed in every extension

Every extension preserves the joint law of every fixed finite or countable collection of box coordinates, as well as every full product-topology Borel event. It also preserves every completed base event. A varying target can evade every fixed countable coordinate set almost surely without there being one common null set on which it evades *all* coordinates simultaneously. Indeed, each actual target is some coordinate. This order of quantifiers explains how diffuse random evaluation can be unconstrained while all fixed-coordinate laws remain fixed.

41 A complete classification for countably many players

The finite theorem answers the repository's Q5. The following extension treats countably many players and shows that the same geometry remains complete when probability vectors are replaced by kernels.

Let $(b_p, a_p)_{p \geq 1}$ be a countable family of regular outputs. For the classification in this section, blindness is not required. Choose one countable support J and good set G as before, including every atom of every target law. Let $\Omega_\infty = \{0, 1\}^{\mathbb{N}_{>0}}$. Define $\Gamma(x)$ by fixing all internal success bits and requiring one consistent color for every external target group. Put $\Gamma(x) = \Omega_\infty$ on the exceptional set.

For a binary word w of length n , let $[w] \subseteq \Omega_\infty$ be its cylinder, and let D_w be the event that w is a feasible prefix. These are base-measurable. The finite alphabet gives

$$\Gamma(x) = \bigcap_{n \geq 1} \{s : s \upharpoonright \{1, \dots, n\} \text{ is feasible at } x\}. \quad (\text{III.21})$$

Each set in the intersection is clopen in Ω_∞ , and $\Gamma(x)$ is nonempty and compact. To check the equality, note that for any fixed external group the colors compatible with its first n members form a decreasing sequence of nonempty subsets of finite Q . Some color is therefore compatible with the entire group. The graph of Γ is measurable for $\mathcal{A} \otimes \mathcal{B}(\Omega_\infty)$ by (III.21).

Write $S_\infty(x) = (\mathbf{1}_{E_p}(x))_{p \geq 1}$ for the actual success sequence. A base-measurable probability kernel K means that $K(x, \cdot)$ is a probability on Ω_∞ for every x , and $x \mapsto K(x, B)$ is $\mathcal{A}$ -measurable for every Borel $B \subseteq \Omega_\infty$.

Theorem 41.1 (Countable kernel classification). *All extensions of μ to $\sigma(\mathcal{A}, E_p : p \geq 1)$ correspond, up to almost-everywhere equality of kernels, to base-measurable probability kernels satisfying*

$$K(x, \Gamma(x)) = 1 \quad \text{for almost every } x. \quad (\text{III.22})$$

The associated extension satisfies

$$\nu(A \cap \{S_\infty \in B\}) = \int_A K(x, B) d\mu(x) \quad (A \in \mathcal{A}, B \in \mathcal{B}(\Omega_\infty)). \quad (\text{III.23})$$

The base probability space need not be standard or countably generated.

Proof. Necessity without a disintegration assumption on the base. Given an extension, let k_w be the Radon–Nikodym density of

$$A \mapsto \nu(A \cap \{S_\infty \in [w]\}).$$

There are countably many finite words. After removing one common base-null set, all identities

$$k_\emptyset = 1, \quad k_w \geq 0, \quad k_w = k_{w0} + k_{w1}$$

hold pointwise. They determine a unique probability on the compact binary sequence space for each remaining x , by the usual countable product extension theorem. Set $K(x, [w]) = k_w(x)$ and give the kernel any fixed value on the common null set. The monotone-class theorem extends measurability from cylinders to all Borel sets. It also proves (III.23) for all B .

Apply the finite feasible-pattern necessity argument to each prefix. It gives $k_w = 0$ almost everywhere outside D_w . Removing a further countable union of null sets makes all these conditions simultaneous. For every n , the kernel then gives probability one to the permitted length- n prefixes. Continuity of probability for the decreasing intersection in (III.21) proves (III.22). Agreement on the countably many cylinder values gives almost-everywhere uniqueness of the entire kernel.

Sampling a proposed kernel. Conversely, suppose K satisfies (III.22). On an auxiliary completed product probability space, take Y with law μ and an independent sequence of uniform random numbers. Recursively choose the next bit, conditional on a prefix w , with probabilities

$$\frac{K(Y, [w0])}{K(Y, [w])}, \quad \frac{K(Y, [w1])}{K(Y, [w])}$$

when the denominator is positive, and choose a default bit otherwise. This is an explicitly measurable sequential construction. It produces a random sequence Z with conditional law $K(Y, \cdot)$. In particular, $Z \in \Gamma(Y)$ almost surely. The construction uses only the measurable functions $K(\cdot, [w])$, not a regular conditional probability theorem for the base space.

Realizing the sampled sequence. For $Y \in G$ and $Z \in \Gamma(Y)$, use the least player index choosing each external target as its representative. For each representative, choose the least color compatible with all members of its group and their bits in Z . Such a color exists by feasibility. The condition for a color to work is a countable conjunction of measurable equality and bit tests, so these color choices are measurable. Replace each selected external coordinate by its chosen color, leaving every other coordinate unchanged, and call the result $T(Y, Z)$. On the exceptional set where these conditions fail, define $T(Y, Z) = Y$.

For every fixed label i , the coordinate T_i is measurable: partition into the countably many events on which a specified player is the least representative targeting i , and use the measurable selected colors. This gives coordinatewise, hence cylinder, measurability; full Borel measurability of T is not being assumed.

Only countably many coordinates are replaced, and every one lies outside J . Thus T fixes J identically. On the good event all outputs are unchanged, because they factor through J , and the actual success sequence of T is exactly Z .

Preservation of the completed base. For any fixed countable $L \subseteq I$, Lemma 33.1 shows that almost surely no selected external target lies in L . Consequently

$$T(Y, Z) \upharpoonright L = Y \upharpoonright L \quad \text{almost surely.}$$

Every cylinder-measurable set has such a support. If $A \in \mathcal{A}$, choose cylinder sets B, N with $A \triangle B \subseteq N$ and $\mu(N) = 0$, and take a common support for both. The displayed equality proves that $\mathbf{1}_A(T) = \mathbf{1}_A(Y)$ almost surely. It also proves measurability of the pullback of A : the possible difference lies in a measurable null set of the completed auxiliary space.

The pullback of every success-coordinate event agrees almost surely with the corresponding bit event of Z , and is therefore measurable. The collection of all subsets of X with measurable pullback under T is a σ -algebra containing the base and all success events. Its pushforward probability gives an extension. Finally, for $A \in \mathcal{A}$ and Borel $B \subseteq \Omega_\infty$,

$$\mathbb{P}(T \in A, S_\infty(T) \in B) = \mathbb{P}(Y \in A, Z \in B) = \int_A K(x, B) d\mu(x).$$

This proves (III.23). The sets $A \cap \{S_\infty \in [w]\}$ form a generating π -system, so these values uniquely determine the extension. $\square$

The countable proof requires finite Q in the compact-feasibility step. For an infinite alphabet, an infinite group may have every finite prefix realizable without the whole success sequence being realizable. For example, if one group lists every color once, the all-zero sequence has every finite prefix realizable but is not itself realizable. That case is a separate problem.

41.1 Asymptotic minimax

For blind players, independently assigning uniform colors to distinct external groups produces a kernel with all marginals $1/q$. The internal contribution is again given by Lemma 33.2. Hence one common extension satisfies $\nu(E_p) = 1/q$ for every player of a countable team.

Theorem 41.2 (Sharp lower asymptotic density). *For every countable regular blind team there is a configuration x with*

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \sum_{p=1}^n \mathbf{1}_{E_p}(x) \leq \frac{1}{q}.$$

The optimum guaranteed lower asymptotic success density is exactly $1/q$. In particular, no such team is eventually correct on every configuration.

Proof. Under the common fair extension, every success average has expectation $1/q$. Fatou's lemma gives

$$\int \liminf_n \frac{1}{n} \sum_{p=1}^n \mathbf{1}_{E_p} d\nu \leq \frac{1}{q}.$$

If the lower limit were strictly larger than $1/q$ at every point, its integral would be strictly larger: the sets where it exceeds $1/q$ by at least $1/k$ form a countable cover, one of positive measure. This contradiction proves existence of x .

For sharpness, cycle through the q constant guesses at one fixed box. The success frequency tends to $1/q$ on every configuration. Eventual correctness everywhere would make every lower limiting average equal to one, so it is impossible. $\square$

This conclusion is a worst-case statement. It does not preclude a strategy that performs much better on a set assigned full measure by a particular extension, or on a selected family of configurations. Such qualifications matter when comparing it with almost-sure prediction results.

42 The Baire-property clause: a reduction, not a solution

The word “Baire” can denote two different objects. The *Baire σ -algebra* is generated by continuous real-valued functions; for the compact finite-alphabet cube it agrees with the cylinder σ -algebra [20, 4A3N]. The *Baire property* means differing from an open set by a meager set. It is the latter, larger regularity condition that remains in the repository's Q2. A Baire-property output map here means that the inverse image of every target Borel set has the Baire property; this definition does not require a chosen topology on a standard Borel target.

Proposition 42.1 (Category normal form). *Every set with the Baire property in Q^I differs from a countably supported Borel set by a subset of a countably supported Borel meager set. A countable family of Baire-property output maps to standard Borel spaces therefore factors through one countable subproduct on a common comeager set, which may be made invariant under finite coordinate changes.*

Proof. For an open U , Lemma 32.2 already gives an appropriate countably supported approximation and a meager envelope for its difference. If F is closed nowhere dense, apply that lemma to the dense open set $X \setminus F$. Its lower open approximation V is dense: otherwise a nonempty open part disjoint from $\bar{V}$ would meet $X \setminus F$ in a nonempty open subset of the nowhere dense sandwich gap. Hence F is contained in the countably supported closed nowhere dense set $X \setminus V$. A general meager set is contained in a countable union of such closed nowhere dense envelopes.

Combine an open approximation with the envelope for the meager error. Encode standard Borel output spaces into $2^{\mathbb{N}}$ and treat the countably many coordinate preimages, as in Lemma 32.4. Saturation under the countable finite-coordinate permutation group preserves meagerness. These are standard category-support facts; see also Fremlin, 4A2E and 4A3S [20]. $\square$

This proposition is insufficient for the measure-based minimax proof. The following elementary example isolates the obstruction. In $2^{\mathbb{N}}$ let

$$B_n = \{x : x_n = x_{n+1} = \cdots = x_{2n-1} = 0\} \quad (n \geq 1), \quad H = \bigcap_{k \geq 1} \bigcup_{n \geq k} B_n.$$

Each union is open dense: extend any finite cylinder by a sufficiently late block of zeros. The Baire category theorem makes H comeager. But

$$\mu\left(\bigcup_{n \geq k} B_n\right) \leq \sum_{n \geq k} 2^{-n} \longrightarrow 0,$$

so H is null. It is invariant under finite coordinate changes, since sufficiently late blocks avoid any fixed finite set of coordinates.

Thus even an invariant comeager good set can have product measure zero. A category factorization cannot be inserted into the earlier expectation argument as though it held almost everywhere. The Baire-property minimax clause remains unanswered here. We do not infer either a positive strategy or an impossibility theorem for that class.

43 Foundations, verification, and a route into ProveIt

43.1 The exact logical scope

All arguments are in ordinary ZFC. The spaces are sets; no proper-class probability space or class-sized family of coordinates is used. The label set is standard Borel, so its cardinality is at most the continuum. The support lemmas themselves hold for arbitrary index sets, but the output factorization relies on a countable family separating target values and on Borel coding of that target space. One cannot replace the specified target measurable structure by an arbitrary uncountable discrete one without checking these hypotheses again.

Several steps use familiar classical principles: countably many choices of supports and null envelopes; Radon–Nikodym; product extension; and the Baire category theorem in the category comparison. We have not established their exact strength over a choiceless or reverse-mathematical base. Consequently the conclusions do not answer the original question of what ZF alone proves about unrestricted strategies.

The central geometric classification has a slightly broader scope than the game corollaries. Blindness is used in Lemma 33.2, hence in the value α/q and the fair/minimax consequences. It is *not* needed for Lemma 35.1, Theorem 36.1, the joint-law polytope, or Theorem 41.1. Those results also classify arbitrary regular adaptive evaluations. For one such nonblind output, replace α/q by its actual measurable atomic success mass

$$\theta = \mu\{b \in J, x_b = a\}.$$

The interval is then $[\theta, \theta + 1 - \alpha]$. For example, a map choosing a fixed coordinate and simply reporting its observed value is not blind and has success probability one; the blind fair-value conclusion would be inapplicable.

Full support of finite coordinate laws matters for the Borel-completion bridge and for null-preservation under finite changes. The biased finite-color extension in (III.12) already handles unequal strictly positive

probabilities. Zero probabilities and nonproduct laws need a separate analysis; they should not be included by an informal limiting argument.

43.2 Executed finite checks

The supplied program `verify_finite.py` uses Python’s standard library, integer arithmetic, and exact rational numbers. It performs exhaustive small checks and independent finite-polytope checks. The results are recorded in `finite_results.json`; the companion text explains the enumeration and its scope.

A blind binary output on the finite n -cube pairs every vertex with the neighbor obtained by changing its chosen coordinate. Blindness makes these pairs a perfect matching, and the guessed bit is an edge label. Conversely, every edge-labeled perfect matching gives a blind output. This correspondence, already discussed in the prior repository report, justifies exhaustive enumeration rather than sampling a convenient subclass.

Coordinates n	Configurations	Perfect matchings	Blind labeled maps
1	2	1	2
2	4	2	8
3	8	9	144

Every enumerated map succeeds on exactly half the configurations. All $8^3 = 512$ ordered triples on two coordinates were checked. Their worst-case scores have the distribution

$$\#\{\text{teams with worst score } 0\} = 236, \quad \#\{\text{teams with worst score } 1\} = 276.$$

No triple guarantees two successes, and the constant guesses $(0, 0, 1)$ at one fixed coordinate attain the optimal guarantee one.

The collision tests verify the shared-target patterns in (III.4), all eight patterns for three distinct binary targets, and a ternary example in which the unused color makes all three guesses wrong. Two mixed law polytopes are also checked exactly, with patterns ordered as 00, 01, 10, 11:

$$\begin{aligned} \mathcal{P}_A &= \frac{1}{2}\Delta(\{00, 11\}) + \frac{1}{3}\Delta(\{01, 10\}) + \frac{1}{6}\Delta(\{00, 01, 10, 11\}), \\ \mathcal{P}_B &= \frac{1}{4}\Delta(\{00\}) + \frac{1}{4}\Delta(\{01, 11\}) + \frac{1}{2}\Delta(\{10, 11\}). \end{aligned}$$

Their Hall descriptions have respectively eight and four vertices. Exact Gaussian elimination enumerates those vertices and verifies equality with the convex hulls of weighted simplex-vertex sums. A separate integer maximum-flow implementation checks Hall membership on every probability-vector lattice point with denominators 12 and 8, respectively. The numbers of feasible points are 142 of 455 and 15 of 165. Finally, 15 rational specializations of the atomic-mass interval are checked for $q \in \{2, 3, 5\}$ and five values of α .

These checks are finite consistency evidence and reproducible examples. They do not prove completion regularity, the envelope lemma, Radon–Nikodym classification, or the countable kernel theorem. On a finite target space every target law is atomic; the formal diffuse weights in the polytope tests are not being misrepresented as diffuse finite-space distributions.

[write] Added 4 October 2026, batch 92. In the collection the program is shipped as `code/03-adaptive-extensions-verify_finite.py`, the companion text as `code/03-adaptive-extensions-finite_checks_README.txt` and the results as `data/03-adaptive-extensions-finite_results.json`, all byte-identical to the delivery; the companion text and the delivered command (Appendix G) use the delivery names. The program writes

its JSON to standard output, so it is rerun with an explicit redirection to a scratch file; the report’s `README.md` gives the commands. At the write (Python 3.14.4, Windows, on a copy) it passed and reproduced the shipped results, which on Windows differ only by CR line ends. The counts above (2, 8 and 144 blind maps; 236 and 276 triples; 142 of 455 and 15 of 165 lattice points; 15 interval cases) are those of the shipped JSON. The 236/276 split agrees with the table of Section 13, computed independently by Part I’s checker.

43.3 A focused formalization plan

The inspected report already connects finite probing semantics to ProveIt’s certificate-oriented work [15]. This article suggests a separate sequence of proof obligations rather than a monolithic formalization:

1. **Finite output semantics.** Define a finite output table, the precise blindness condition, the success vector, and the grouped-target feasible-pattern predicate. Prove the edge-labeled matching characterization in the binary case and verify small tables with finite reflection.
2. **Abstract finite-partition extension lemma.** For a finite partition (E_s) and measurable envelopes (D_s) satisfying (III.5), formalize Theorem 36.1 independently of any box game. This isolates the Radon–Nikodym and well-definedness arguments from product geometry.
3. **Completed product normal forms.** Prove literal countable support, completed support, and the finite-change-invariant good-set lemma. Record explicitly whether a function is pointwise factored or only factored off a named null set.
4. **Game-specific envelope computation.** Formalize diffuse target escape and the finite coordinate replacement that proves Lemma 35.1. This is the central application-specific step.
5. **Finite law polytope.** Connect the feasible-pattern predicate to a finite rational transportation problem when the weights are supplied by exact certificates. This can be verified separately from computing those weights in an infinite game.
6. **Countable kernel construction.** Build kernels from compatible Radon–Nikodym cylinder densities, then verify the countable diffuse-coordinate replacement. The common-null-set construction and the individual almost-sure preservation of every base event must be explicit.

These are proposed development stages, not names of existing implemented modules. No Lean or Rocq files are supplied, no repository files were changed, and no proof-assistant build was performed for this article. The mathematical proof status and the finite computational status are recorded separately in Appendix F.

44 Further research questions and topics

The following are proposed directions after the results above. Except where expressly tied to a cited question, they are not claimed to be established open problems in the literature. They are ordered roughly from direct extensions of the present argument toward broader foundational questions.

- R1. The remaining Baire-property minimax problem.** Does the guarantee $\lfloor m/q \rfloor$ remain optimal for blind outputs having the Baire property relative to the full product topology? Proposition 42.1 reduces the outputs to countably many coordinates on an invariant comeager set. A solution still needs a category argument that replaces averaging, or a counterexample exploiting a comeager null set. This is the unresolved clause of the inspected report’s Q2.
- R2. Simultaneous extensions for uncountable teams.** Which uncountable families of outputs admit one countably additive extension making every player fair? A union of uncountably many supports or

target atoms need not be countable, so the current escape argument may fail. Seek necessary and sufficient conditions on the family of target maps or on the collection of their supports. This refines the prior report's Q4.

- R3. Countably infinite and general standard Borel color spaces.** For countably infinite colors, characterize the realizable infinite success fibers without relying on compactness of a finite color set. For general standard Borel colors, separate exact guesses, finite lists, and measurable acceptance sets. The finite-prefix example following Theorem 41.1 shows that mere prefix compatibility is insufficient in the countably infinite case.
- R4. Correlated configuration laws.** Find a condition on a nonproduct probability law under which diffuse selected coordinates can be changed while preserving all prescribed base-measurable events almost surely. The support-and-escape portion of the envelope lemma is largely measure-theoretic, whereas completion regularity and fixed-target fairness depend on the law. The correct replacement may involve conditional support rather than unconditional independence.
- R5. Additional invariance requirements on the enlarged measure.** The classification preserves every original base event. Determine which density families also permit a specified group of configuration symmetries to act measurably and preserve the extended measure. Enlarging the σ -algebra by all symmetry translates of success events may add compatibility constraints not visible in a single finite team. Distinguish finite coordinate changes from arbitrary coordinate permutations.
- R6. Measurable evaluation and target topology.** Characterize pairs of configuration and label structures for which joint evaluation is measurable. Under such hypotheses every regular output automatically has measurable success, eliminating the diffuse ambiguity. Compare restricting the allowed targets, strengthening the observable σ -algebra, and requiring a jointly measurable family of coordinate maps.
- R7. Weak-choice strength and coded versions.** Determine which choice and comprehension principles suffice for the normal form, the diffuse escape lemma, and the finite-partition extension separately. A theorem about explicitly coded Borel outputs may have a different foundational strength from one quantifying over an abstract generated σ -algebra. This is related to the report's Q3 and to Glazer's foundational program, but the present ZFC proof supplies no exact equivalence.
- R8. Executable strategies and effective witnesses.** If outputs are given by terminating finite-query algorithms, can the losing witness and feasible-pattern certificate be extracted with useful complexity bounds? For transfinite or arbitrary Borel probing rules, what extra representation of the strategy supports such extraction? Blindness alone is insufficient to answer the operational question.
- R9. The complexity of the law polytope.** The generic Hall description contains exponentially many pattern subsets, and there are already 2^m patterns. Exploit the partition of players into common-target groups to produce smaller extended formulations, efficient separation oracles, or exact descriptions for bounded collision-group size. Distinguish computation with known weights w_T from the potentially harder task of deriving them from strategies.
- R10. Observables with extension-independent laws.** Theorem 35.2 decides when a finite success-pattern event is already measurable, and Corollary 36.3 decides when a scalar payoff has a forced expectation. For countable teams, characterize tail events and asymptotic statistics whose entire distributions are the same under every supported kernel. Examples should distinguish a fixed expectation from a fixed law.

R11. General adaptive evaluations beyond guessing games. Apply the envelope theorem without blindness to randomly indexed coordinate statistics, dynamically selected data records, or independently sampled fields. Determine when the atomic/diffuse decomposition controls more general loss functions, especially where several outputs share indices. The regularity assumptions must remain explicit; the theorem does not automatically apply to arbitrary statistical data models.

R12. Formal proof and literature comparison. Carry out the staged formalization above, and compare the resulting general finite-partition lemma and countable kernel theorem with the precise hypotheses of existing extension and random-set selection results. Historical priority of the application-specific formulas should be assessed independently of successful formal verification.

The nearest concrete verification milestone is the abstract finite-partition lemma together with the game envelope lemma. Those two pieces would certify the exact atomic-mass formula and the whole finite joint law classification before committing to the more extensive countable-kernel development.

[write] **Status of the directions, 4 October 2026, batch 92.** All twelve directions stay open as printed; the write searched no literature for them. R1 is the clause of Part I's Q2 that this Part leaves open (see the note after the questions of Section 15); R2 refines Q4, whose countable case Theorem 4.1 and, for kernels, Theorem 41.1 treat; R7 is related to Q3. Part IV does not bear on them: its targets are fixed and its subject is the cost of inspection.

F Assumption and result ledger

Result	Main hypotheses	Status and boundary
Open Borel sandwich	Finite discrete colors; independent full-support coordinate laws; set-sized product	Classical ingredient, proved directly; attributed to completion-regularity literature.
Normal form	Countably many completed-base measurable standard Borel outputs	Full proof, including one invariant exceptional null set.
Finite envelope	Finite team; countable common support modulo null; target atoms included	Full proof; blindness unnecessary.
Density classification	Finite actual partition and exact measurable envelopes	Full Radon–Nikodym and countable-additivity proof; general machinery specialized to the computed envelopes.
Law polytope	Finite team and feasible-pattern weights	Full finite proof; classical random-set/transport geometry with game-specific feasible sets.
Atomic-mass formula	Regular blind output; uniform finite colors	Full proof; exact interval, attainability, and measurability equivalence.
Biased-color formula	Independent strictly positive finite coordinate laws	Full argument; replaces α/q by atomic success mass β .
Borel/universal minimax	Finite blind regular team; auxiliary uniform law	Full proof; resolves two clauses of repository Q2.

Result	Main hypotheses	Status and boundary
Countable kernels	Countably many regular outputs; finite colors	Full construction from cylinder densities and diffuse replacements; blindness unnecessary.
Asymptotic guarantee	Countably many regular blind outputs	Exact worst-case lower density $1/q$; no independence claim.
Baire-property version	Category-measurable outputs	Support reduction proved; numerical minimax remains open here.
Finite computations	Enumerated cubes and exact rational feasibility polytopes	Executed reproducible checks; not a verification of infinite proofs.
Original Glazer question	Unrestricted legal strategies in ZF	Not settled by this article.

G Sources, attribution, and reproducibility

The user-specified LinkedIn profile was used to identify the intended researcher; his authored papers and dissertation ground the mathematical connection. The box-game arXiv record lists version 1 submitted November 15, 2022. We inspected its stated question through the paper and the repository’s explicit discussion, and consulted the dissertation’s corresponding box-game chapter [1, 14]. The present paper does not rely on any unverified recent research announcement.

The repository audit used the pinned commit stated in Section 30. In its box-game report, the two questions answered here are explicitly research directions proposed by that report. The report’s intake note still listed all twelve directions as open. We read its definitions, the cylinder minimax proof, its fair-extension and target-collapse arguments, its diagonal example, and its question list. The new article therefore builds on a precise existing result rather than presenting a renamed copy of the earlier minimax theorem.

Completion regularity is credited to Kakutani and its subsequent development. Gryllakis–Koumoullis’s Lemma 2.6 contains a closely related maximal countable union argument, and its Theorem 2.9 gives a broader product result [18]. Fremlin supplies standard factorization and category-support references [19, 20]. The general one-set interval of extension probabilities and its countable additivity belong to Łoś–Marczewski’s theory [4, Theorems 1, 2, 4]; finite saturated-partition extensions are also illustrated by Vidmar [23, Lemma 7]. The finite selection polytope and its inequalities are presented with Artstein’s prior work in view [21]. The specific mathematical work here is the exact envelope computation for adaptive targets, its atomic-mass consequence, and the resulting resolution of the repository’s questions under the stated regularity conventions.

The source is self-contained LaTeX and uses standard TeX Live packages. Compile it with

```
latexmk -pdf -interaction=nonstopmode -halt-on-error
adaptive_box_games.tex
```

The finite checks run with Python 3.9 or later:

```
python3 verify_finite.py > finite_results.json
```

No external numerical libraries, network access, or random seed are needed to reproduce those checks. The figure is drawn from its exact formulas in the LaTeX source. The delivered PDF was compiled and visually

inspected. Neither the infinite theorems nor the whole article have a proof-assistant certificate, and the literature review is not a proof of historical novelty.

[write] Added 4 October 2026, batch 92. These commands and file names are the delivery's. In the collection the manuscript's source and PDF, `adaptive_box_games.tex` and `.pdf`, and the delivered `README.txt` are not shipped (this Part is the source, built as part of the report's `article.tex`); they survive in the archive of the arrival commit 9dc8db274. The shipped names are in the note of Section 43, and the report's `README.md` gives a rerun with those names into a scratch directory. The remark about a LinkedIn profile names no address, and none is recorded here. The pin is discussed in Section 30.3, where the sentence above about the report's intake note is qualified.

Part IV

Two queries in expectation: sharp information thresholds and the exact inspection cost of infinite hat guessing

Source 06 (the base): *Two Queries in Expectation: Sharp Information Thresholds for Infinite Hat Guessing*. Research article and reproducibility package, October 4, 2026.

Source 04: *The Exact Inspection Cost of Infinite Hat Guessing: Geometric query laws, bounded-depth obstructions, and affine rigidity of optimal finite teams*. Research report prepared for Vladimir Reshetnikov, 4 October 2026.

Abstract (source 06). In the compulsory binary hat game, let D_n be the number of correct guesses among the first n players minus $n/2$. We resolve the bounded-inspection question posed in a recent ProveIt research report. For every strategy with a common finite bound on adaptive queries, both events $D_n \rightarrow +\infty$ and $D_n \rightarrow -\infty$ are null and meager. The probability conclusion needs no restriction on query distance or on the number of players who inspect one hat.

We obtain an exact threshold for deterministic expected query cost: the least uniform expected budget permitting divergent positive excess is 2, whether success is required with positive probability or almost surely. Below 2, a decision-tree path estimate forces uniformly bounded depth. At 2, an early-stopping implementation of Eldredge's existing block strategy has $\mathbb{E}Q_i < 2$ for every player, $\sup_i \mathbb{E}Q_i = 2$, and $\mathbb{P}(Q_i > t) \leq 2^{-t}$, while $D_n \rightarrow +\infty$ almost surely. Explicit schedules give exact asymptotic equivalents, including $D_n \sim \log \log n$.

Further results include an exact classification of eventually positive query-count budgets, allowing arbitrarily sparse large values; a separation between count and distance budgets; a finite-chromatic visibility obstruction allowing infinite neighborhoods; finite sign bounds from sparse Fourier supports; and the sharp one-query success probability $3/4$. Private free randomization changes the expected-cost threshold: every budget strictly greater than 1 becomes sufficient.

Contribution and verification status (source 06). The article contains full proofs and reproducible finite checks. The threshold, classification, and obstruction results are developed here as answers to the repository's follow-up questions. The block mechanism is explicitly credited to Eldredge, and the analytic sign inequality is classical. Historical priority is not established. These are unrefereed mathematical results, not a claimed proof-assistant formalization or an endorsed contribution by Elliot Glazer.

Abstract (source 04). Let $D_n = S_n - n/2$ be the excess number of correct guesses in the countably infinite binary hat game with independent fair hats. Players use deterministic adaptive decision trees, never inspect their own hats, and make compulsory guesses. We prove that a uniform bound on decision-tree depth forces $\mathbb{P}(D_n \rightarrow +\infty) = 0$, resolving the uniform-query question in the preceding ProveIt research report.

More precisely, let G have the geometric distribution $\mathbb{P}(G = r) = 2^{-r}$, $r \geq 1$. Every strategy with positive

probability of divergent excess satisfies

$$\sup_i \mathbb{E}\phi(Q_i) \geq \mathbb{E}\phi(G)$$

for every nonnegative nondecreasing cost function ϕ , where Q_i is player i 's number of inspected hats. One explicitly evaluated version of Eldredge's existing block strategy attains equality simultaneously for all these costs, while $D_n \rightarrow +\infty$ almost surely. Thus the exact uniform expected-query threshold is 2, the optimal tail envelope is 2^{-r} , and the critical exponential-moment parameter is $\log 2$.

We also prove a sharp variance bound in terms of total expected queries, derive precise growth and fluctuation laws for an explicit schedule, and classify all finite parameter pairs attaining the first-moment bound at the minimum possible inspection depth. Their failure sets are affine codes with replicated simplex parity-check columns; for an odd number of players at strict majority, these are affine Hamming codes. Complete proofs, finite verification code, a source audit, and a concrete formalization roadmap are provided.

Status (source 04). This is an AI-assisted research manuscript with independently reviewed mathematical arguments and executable finite checks. The positive block mechanism is due to Eldredge, and the analytic tools are classical. The cost classification and finite refinements are proved here; global priority and proof-assistant certification are not asserted.

Keywords (source 04): infinite hat games; adaptive queries; Boolean Fourier analysis; hypercontractivity; geometric costs; perfect codes; formal verification.

[write] Added 4 October 2026, batch 92. This Part merges two manuscripts of batch 92 of ProveIt's incoming-reports intake that answer the same question independently: manuscript 06 (*the base*, labels `mbg:qry:`) and manuscript 04 (labels `mbg:insp:`). Neither is an edition of the other, and neither cites the other. Like Parts I–III they are AI-assisted, unrefereed and not formalized; provenance, the merge decisions and a notation table are in Sections 45.5, 45.6 and 45.7. **Source 06's Section k is Section $k + 44$ here (Sections 45–58) and source 04's Section k is Section $k + 58$ (Sections 59–69); in both, statement or equation $k.m$ is $(k + \text{offset}).m$,** and source 04's tagged displays $(k.m a)$ are $(k + 58).m a$. Thus source 06's Theorem 9.3 (the threshold) is Theorem 53.3, and source 04's Theorems 2.1, 4.2 and 7.4 are Theorems 60.1, 62.2 and 65.4. Every statement of both sources is printed, in its own place, so that both numberings survive. Where source 04 states a result that source 06 has already proved, a **[write]** note there says so; a proof that repeats source 06's argument is replaced by a pointer, and a genuinely different proof is printed as source 04's second route. Numbers such as "Proposition 6.6" or "Remark 6.8" that cite Eldredge [10] are his and unchanged; "Research question 13.2" of the predecessor is Research question 29.2 of Part II. Apart from this, the editorial changes listed in Section 45.5 and the dated **[write]** notes, no statement, proof, number or non-claim of either manuscript was changed.

45 The research intersection and the results

45.1 Why this is an intersection with Glazer's work

The question is how coordinated predictors can succeed without observing the coordinate they must predict. Glazer's work on choiceless box games and his dissertation place this question at the meeting point of choice principles, regularity, and probability [1, 14]. There is also a direct connection to the particular statistic studied here: Eldredge introduces excess correct guesses in response to a suggestion by Elliot Glazer and Charles Wang [10, p. 14]. Thus the choice of topic rests on mathematical sources, independently of the accessibility of the LinkedIn profile supplied in the request.

The relevant ProveIt source is *No Universal Sublinear Bound for Measurable Hat Guessing* [26]. Its further-research section asks whether bounding the number of inspected hats uniformly still permits $D_n \rightarrow +\infty$ almost surely, and asks that query count be distinguished from query distance. We settle that qualitative

question, extend it to arbitrary count budgets, and find an exact expected-cost boundary. We also connect the finite certificates to the repository’s decision-tree and arithmetic-coding work [27, 28].

45.2 What is already known

Eldredge proves that measurable strategies can have divergent positive excess, using finite blocks of increasing size. His Remark 6.8 discusses growth rates. The existing *ProveIt* report develops much faster prescribed sublinear growth using Hamming-code constructions [10, 26]. Coding methods for related partial prediction problems have older antecedents in autoreducibility [11]. None of these facts should be confused with the bounded-query conclusion proved below.

Our positive construction uses Eldredge’s rule, with a different emphasis: we specify the order of queries, prove its exact cost distribution, and combine it with a lower bound for every deterministic query algorithm. The resulting threshold theorem is an additional conclusion, not a claim to have invented its underlying block rule.

45.3 A map of the conclusions

Throughout, hats are independent fair bits and every player must guess. *Deterministic* allows arbitrary fixed algorithms and fixed parameters, but no free random seed. A query reveals the hat of one other player.

Restriction	Proven consequence
At most $d < \infty$ adaptive queries per player	$D_n \rightarrow +\infty$ and $D_n \rightarrow -\infty$ each occur on a null, meager set.
Deterministic algorithms with $\sup_i \mathbb{E}Q_i < 2$	The same impossibility, because expected cost forces a common depth bound.
Deterministic algorithms with $\sup_i \mathbb{E}Q_i = 2$	Almost-sure positive divergence is possible, with geometric cost tails.
Count budget $q(i) \geq 1$ eventually	Almost-sure divergence is possible exactly when q is unbounded.
Count and distance budget $h(i) \rightarrow \infty$	Both constraints can be met simultaneously with almost-sure divergence.
Merely unbounded distance budget	Insufficient in general; an explicit counterexample has a three-colorable visibility graph.
Finite-chromatic visibility graph	No positive-probability divergence for measurable strategies, even with infinite neighborhoods; no nonmeager divergence for Baire-measurable rules.
One query per player, finite team	The greatest possible probability of strictly positive excess, allowing team size to vary, is exactly $3/4$.
Free private random bits	Every uniform expected budget $C > 1$ permits almost-sure divergence; the deterministic threshold 2 does not apply.

45.4 Scope of the source comparison

The repository snapshot inspected for this article is

9fe62865dc30d71256183466b744b98ebf7e64d6.

The earlier hat manuscript is inside docs/incoming/hat_guessing_research.zip. The companion box-games manuscript is at SetTheory/Cardinals/docs/reports/ordinals-and-order-types/measurable-box-games/article.tex. We compared the directly relevant manuscripts and the cited literature. We do not infer worldwide novelty from the absence of a matching search result. In particular, the unrestricted choiceless question in Glazer’s paper is a different problem and is not resolved here.

45.5 Provenance and place in this report

[write] **Added 4 October 2026, batch 92.** *Sources.* This Part is built from two manuscripts of batch 92 of Provelt’s incoming-reports intake, placed in commit e38f368c2. **Source 06** (the base) was delivered as the archive hat_query_thresholds_package.zip (478,565 bytes; inner directory hat_query_thresholds/, main file hat_query_thresholds.tex of 1,919 lines, a 26-page US Letter PDF) in the arrival commit afd7ffabb; its programs, outputs and source notes carry the prefix 05-hat-queries-. Its title page has no author line, only “Research article and reproducibility package”, and its **PDF metadata name the author as “ChatGPT research report”**, the only AI wording in this report’s sources. **Source 04** was delivered as hat_inspection_frontier.zip (491,169 bytes; inner directory hat_inspection_frontier/, main file hat_inspection_frontier.tex of 2,319 lines, a 31-page A4 PDF) in the arrival commit 9dc8db274; its program, output, build script, source audit and proof-status record carry the prefix 04-hat-inspection-. Its author line reads “Research report prepared for Vladimir Reshetnikov”, and its status box calls it “an AI-assisted research manuscript”. Both are AI-assisted, unrefereed and not formalized. The file prefixes continue this report’s local sequence (Parts I–III are 01 to 03) in arrival order, so source 06 has the prefix 05. Both were placed here, not as a report of their own, because both answer Research question 29.2 of Part II, “Information cost versus guaranteed growth” (source 06 all three of its clauses, source 04 its uniform-bound clause); source 04 also settles Part I’s Q9 for a further family and partly answers Research questions 29.4 and 29.5 (the notes after the questions of Sections 15 and 29).

Pin. Both manuscripts were written against 9fe62865dc30d71256183466b744b98ebf7e64d6 (source 06’s macro of Section 45, source 04’s revision of Section 59), the commit “Merge branch ‘main’” of 4 October 2026, 09:56 (UTC–7), an ancestor of the placement. It contains the batch-90 archive docs/incoming/hat_guessing_research.zip (arrival a162e4386) but not its placement as Part II (12076b2e8), so both manuscripts read Part II as the delivered hat_guessing_growth.tex, whose Research question 13.2 is Research question 29.2 here, and Part I at the same text as Part III’s pin (article blob 0749d641). That archive path no longer exists; the batch-90 placement retired the archive, which survives in its arrival commit.

Repository claims, checked at the write. Source 06 (Section 45 and its delivered SOURCE_NOTES.txt): the question “Information cost versus guaranteed growth” is in hat_guessing_growth.tex at the pin, asking about count budgets, a uniform bound and count versus distance: true. “The earlier repository report also contains stronger unrestricted growth constructions”: true (Theorems 23.1 and 24.1). The box-games report supplies finite-query context and the list-coding project interfaces, neither proving anything here: accurate. Source 04 (Section 59 and the delivered SOURCE_AUDIT.txt): the predecessor archive’s blob ea9919e06, its 550,545 bytes and arrival time, the member’s 60,488 bytes, its Theorem 3.2 (now Theorem 19.2), its Section 11.3 on the cost of the direct implementation (now Section 27.3) and Research question 13.2 at its lines 1278–1284: all true, and the predecessor contains no hypercontractive obstruction, threshold or cost frontier. Part I’s article blob, its questions Q9–Q12 and the functions of its checker verify_certificates.py (blob ed035fe98, unchanged now) are as described. The list-coding files Logic/PeanoArithmetic/ListCoding/README.md, Lean/PAListCoding/Basic.lean and Lean/PAListCoding/Predicates.lean below that project (blobs 74164ac6, b7e037f4, c9f8f650, unchanged now) declare the noncomputable encode and decode, the seven named theorems, validFormula with its specification and the guarded predicates listed in Section 66: true. Source 04’s sentence that the box-game report “separately asks about finite teamwork and the costs of adaptive observations” and that its finite results “address a precise family within

that program” holds for Q9; neither source refines Theorem 11.1 as Q11 asks. No repository claim is refuted, and nothing is retracted. No Lean or Rocq file of ProveIt states any result of this Part, and the formalization routes of Sections 56 and 66 are proposals.

Reviews in the Hilbert’s-tenth research tree. Before the placement, another session inventoried source 04 with four other archives in Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_latest_five_9dc8db274.md (commit 3c25fbb55): it read only the delivery README.txt, lines 1–96, authenticated the members by hash and routed the package as sharp deterministic adaptive query costs with the expected-cost threshold 2 and a finite affine-code classification, with the boundary that the charged resource is observed hats, not arithmetic gates, and that free public randomness changes the model. The same session then reviewed source 06 in review_hat_threshold_afd7ffabb.md beside it (commit b560cee86): it read the delivery README.txt and SOURCE_NOTES.txt in full and the manuscript’s lines 216–578, 795–1380 and 1466–1663 (here Sections 46 to 49 through Corollary 49.3 and its paragraph, Sections 51 to 54 through the proof of Theorem 54.3, and Sections 55–56) and found “no finding in the inspected proof”; it verified the checksum file 8/8 and ran no program, did not check the sharper constant attributed to O’Donnell, the bibliography or novelty, and states that it does not supersede the README-only intake of source 04. Not reviewed there: Section 45, Proposition 49.4, Section 50, Theorem 54.4, Proposition 54.5, Sections 57–58, and all of source 04. Both reviews leave the arithmetic bounds of that programme unchanged; no scope correction follows for this Part.

Checks at the write. The three programs, run on scratch copies (Python 3.14.4, Windows), passed: source 04’s verifier (“Passed 19 exhaustive cases, 59546 hat assignments”, about 6 s) and source 06’s block verifier (about 18 s) reproduced their shipped JSON files, and source 06’s orbit verifier (about 1 s) reproduced its file except the field elapsed_seconds (0.393909 delivered), once the CR bytes of Windows text mode are stripped. At placement both of source 06’s programs also passed under python -0, its checksum file SHA256SUMS.txt matched 8/8 (dropped by repository policy), and an independent program (not shipped) confirmed source 04’s trace teams for $(r, m) = (1, 1), (1, 3), (2, 1), (2, 2), (3, 1), (3, 2), (4, 1)$ (exactly $s - 1$ queries, $\mathbb{P}(S = s) = n/(2s)$, $\mathbb{P}(S = 0) = 2^{-r}$), the block law with $\mathbb{E}Q_S = 2 - 4^{1-k}$ for $k \leq 6$, and the value $3/4$ of the three-player cycle. Eldredge’s credit to Glazer and Wang, which source 06 places on “p. 14” and source 04 in the paragraph before his Proposition 6.6, was found at placement in that paragraph of the v2 HTML text (page numbers differ between renderings); Part II’s write had read the same text (Section 17.3). The other works cited (Austrin–Håstad, Bonami, Bonisoli, O’Donnell, Lietz–Winkel, the Samaritan Research page and Glazer’s dissertation) were not checked at the write.

Editorial changes in printing this Part. All delivered labels carry the prefixes mbg:qry: (source 06, 66 labels) and mbg:insp: (source 04, 71 labels), with every reference updated; the write added mbg:qry:part, mbg:qry:sec:provenance, mbg:qry:sec:merge and mbg:qry:sec:notation. Sections, statements and equations are renumbered by the offsets stated at the head of this Part. The two title pages became the Part heading, the two abstracts, the two status boxes and source 04’s keywords above; source 04’s front table of contents is the report’s. Both manuscripts’ \PP, and source 04’s \EE, are printed with Part I’s $\mathbb{P}$ and $\mathbb{E}$; source 04’s \ind with an argument is printed with the new macro \indic, since Part I’s \ind takes none (the printed $\mathbf{1}_{\{\cdot\}}$ is unchanged); its \as, “almost surely” in words, is printed as those words, Part II’s \as being “a.s.”; its unused \supp, \Z, \Tr, \Cov, \abs and \depth are not defined. ProveIt is set in Part I’s typewriter style rather than italic (in source 04 also where it spelled \textit{ProveIt}), remarks in Part I’s upright style. Two renamings unify source 04 with source 06: its correctness sign ε_{ig_i} , which source 04 called Y_i in Section 60 but X_i in Sections 61 and 64, is Y_i throughout, as in source 06, where X_i is the hat; and the second member of a block pair, source 04’s “D player” (a letter that clashes with the surplus D_n), is the “T player”, source 06’s name (whose verifier, in turn, says D). One typographical error is corrected: in (63.18) source 04 printed $\frac{3}{2} - \frac{1}{2}, 4^{1-k_b}$, with a comma for a thin space; the product $\frac{3}{2} - \frac{1}{2} 4^{1-k_b}$ is meant, as $k_b(3 - 4^{1-k_b})/(2k_b)$ shows. Bibliography: source 06’s entries for Glazer’s *A choiceless box game paradox*, Eldredge’s paper and Ebert–Merkle–Vollmer’s are the same works as entries [1], [10] and [11], and are cited by those numbers, as is source 04’s Eldredge entry (both cite v2; source 06 adds that its PDF is dated 28 October 2025 and points also to Eldredge’s Section 7.3); Glazer’s dissertation is Part III’s entry [14]; the predecessor manuscript (source 06’s PriorHat, source 04’s PriorGrowth), the box-game report (RepoBoxes, BoxGames) and the list-coding project (ListCoding in both) are one entry each, [26], [27] and [28], pinned at the same commit. Source 04’s bibliography lists one work its text never cites, Butler, Hajiaghayi, Kleinberg and Leighton’s *Hat guessing games* [32], a study of finite hat-guessing games; it is kept, cited here, and was not checked. Source 06 was set in Latin Modern on US Letter, source 04 in Latin Modern on A4 with roman page numbers before its text; here both take Part I’s layout and fonts, so line and page breaks differ from the delivered PDFs.

45.6 How the two sources are merged

[write] **Added 4 October 2026, batch 92.** The two manuscripts prove the same central answer by the same construction: a common bound on query depth excludes divergence; a reachable path of length L costs at least $2 - 2^{1-L}$ expected queries; Eldredge’s block, evaluated partner first with early stopping, has a truncated geometric query law; the same schedules give the same rates; the exact uniform expected-query threshold is 2. Beyond that core they are complementary. Source 06 is the base because on the shared spine it has the weaker hypotheses and the stronger conclusion (any visibility graph with finitely many colours, Baire-measurable rules, divergence events null *and* meagre) and because it answers all three clauses of Research question 29.2. Source 04’s proofs use deterministic decision trees, the hypothesis of source 06’s threshold theorem as well, so nothing printed from source 04 depends on a stronger hypothesis than the statement it is placed beside. To keep both numberings, source 06 is printed first and whole (Sections 45–58), then source 04 (Sections 59–69). The shared results:

Result	Source 06	Source 04	Printed
bounded depth excludes divergence	Corollary 48.4 (null and meagre; colourings)	Theorem 61.3 (null; sign bound and martingale convergence)	both statements; source 04’s proof as a second route
fourth-moment and sign bounds	proof of Lemma 49.2	Lemmas 61.1–61.2	source 04’s statements (cited by its later proofs); its derivation of the sharper constant; the rest a pointer
path cost; depth from the mean	Lemma 51.1, Corollary 51.2	Lemma 62.1, Corollary 62.3	source 04’s statements (its lemma states the stochastic domination); proofs as pointers
every nondecreasing cost	—	Theorem 62.2, Corollary 62.5	source 04 only
block law	Proposition 52.2	Lemma 63.1	both statements; source 04’s proof a pointer
query law of the block	Proposition 52.3	Proposition 63.2 (adds the envelope for every cost and the exact exponential moment)	both, with proofs
schedule rates	Theorem 53.1, ε , $\lceil \cdot \rceil$	Theorem 63.3, $p = 1 + \varepsilon$, $\lfloor \cdot \rfloor$	both; source 04’s proof, needed by its Section 63, as a second route
counting lemma	Lemma 53.2	Lemma 63.4	both, with their different proofs
threshold 2	Theorem 53.3	Theorem 60.1, mean case	both
low-degree Fourier mass	Proposition 49.4	Proposition 64.7	both (different constants)
three-player one-query team, 3/4	Theorem 50.2 (optimal)	Theorem 65.5, $r = 2$, $m = 1$	both
free randomness	Theorem 55.2 (private coins)	Section 60 (a public coin)	both; complementary

Only source 04 has the lower bound for every nondecreasing cost, the central limit theorem, realized and maximal costs and the exact variance of the schedule (Section 63), the variance bound and the strong law (Section 64) and the finite-team classification (Section 65). Only source 06 has the finite-chromatic obstruction, category, the sparse-support

bounds and the one-query optimum, the shape of trees below the threshold, count and distance budgets, acyclic visibility and private randomness. The research questions of Sections 57 and 68 overlap; the note after the latter pairs them.

45.7 Notation in this Part

[write] Added 4 October 2026, batch 92. Both sources' letters are kept, except the two renamings of Section 45.5. The middle column gives the meaning in this Part (with the source, when only one uses the letter), the right column a different meaning the letter has elsewhere in the report, which is *not* meant.

Symbol	Meaning in Part IV	<i>not</i> : meaning elsewhere
legal	a guess unchanged when the player's own hat changes; deterministic decision trees never query the own hat	Part I's "legal" (a tree never querying a variable target)
X_i, ε_i, x_i	player i 's hat: as ± 1 (06; 04 writes ε_i), as a bit $x_i \in \{0, 1\}$ (04, and 06 in Section 52)	Part I's X_I , Part III's X (configuration spaces)
Y_i, f_i, g_i	correctness sign $X_i f_i$ (06) = $\varepsilon_i g_i$ (04); prediction sign (06); guess sign (04)	—
S_n, D_n, F_n, T_n	correct guesses; surplus $S_n - n/2$; $F_n = 2D_n$ (06) and $T_n = 2D_n$ (04), the same quantity	Part I's F_n (sliceable matchings); Part III's D_s
$Q_i, q(i), h(i), C$	queries of player i ; count budget; distance budget; an expected cap (06)	Part I's alphabet Q and its size q ; Part II's $h(n)$
$C_\phi, \mathcal{A}, \mathcal{A}_*$	uniform cost $\sup_i \mathbb{E}\phi(Q_i)$ of a strategy $\mathcal{A}$; the optimal strategy (04)	Part III's completion $\mathcal{A}$
G	a geometric variable, $\mathbb{P}(G = j) = 2^{-j}$ (both); a visibility graph and G_M a positive-block count (06)	Part III's good set G
S, T players, S_j, T_j	the two roles in a block pair (04's "D player" renamed T)	S_n, T_n ; 06's sign flips T_a ; 04's endpoints T_B
m, k, d	06: pairs per block, depth; 04: k half block size or depth, $m = F $ (Section 61) or the replication (Section 65), $d = 2s - n$	Part I's m players
$L_m, R_k, N_K, N_M, \mu_K, \mu_M$	blocks per stage (06, 04); players and expected gains through a stage	Part II's L_k, R_k, N_k (block lengths, endpoints, thresholds); Part III's μ
$B, B_n, H_K, L_K, L_\infty$	06's debt; 04's $\sum_{i \leq n} \mathbb{E}Q_i$, good-block count and losses	Part II's B_k, H
C, δ, s, n (Section 65)	all-wrong set, its density, threshold, team size	Part I's group C , Part II's debt C
K, Tr	the field with 2^r elements and its absolute trace (Section 65); K also a stage index	Part III's kernel K
$A_\pm, \alpha, \alpha_d$	divergence events; activation probability (06) or a variance proportion (04); the constant of Research question 57.1	Part III's atomic mass α
Inf, Var, $\mathbf{1}_{\{\cdot\}}$	influence, variance, indicator	—
57.1–57.12, 68.1–68.12	the two sources' Research questions	Part I's Q1–Q12, Part II's 29.1–29.12, Part III's R1–R12

46 The game, its algorithms, and three notions of cost

Let

$$\Omega = \{-1, 1\}^{\mathbb{N}}$$

with the fair product probability measure and product topology. Write $X_i(x) = x_i$. A legal measurable prediction is $f_i : \Omega \rightarrow \{-1, 1\}$ independent of coordinate i : changing only x_i does not change $f_i(x)$. Put

$$Y_i = X_i f_i, \quad S_n = \sum_{i=1}^n \mathbf{1}_{\{Y_i=1\}}, \quad F_n = \sum_{i=1}^n Y_i = 2S_n - n = 2D_n. \quad (46.1)$$

The independence of X_i from the information available to player i gives

$$\mathbb{E}Y_i = 0, \quad \mathbb{E}F_n = \mathbb{E}D_n = 0. \quad (46.2)$$

The variables Y_i need not be independent. Indeed, all the interesting constructions exploit their dependence.

Definition 46.1 (A deterministic query algorithm). For each player i , a binary decision tree specifies which other coordinate to query next, as a function of the answers already received. Leaves specify the prediction. No node queries i . The number of queries on input x is $Q_i(x)$.

A tree of depth d can query different coordinates on different branches, so at most d queries is more general than dependence on a predetermined set of d hats. Nevertheless it involves at most $2^d - 1$ possible queried coordinates. Its prediction is continuous. For cost lower bounds we may also allow algorithms that halt almost surely; the lower bound itself will force everywhere bounded query depth when the expected budget is below 2.

There are three separate resources:

1. A *count budget* $q(i)$ requires $Q_i(x) \leq q(i)$ on every input.
2. A *distance budget* $h(i)$ permits player i to query coordinate j only when $|j - i| \leq h(i)$.
3. An *expected budget* controls $\mathbb{E}Q_i$, with expectation over the hat assignment, and also over independent coins if such coins are explicitly allowed.

Distance $h(i)$ allows up to $2h(i)$ possible hats; it is not itself a query-count bound of $h(i)$. Our simultaneous positive constructions will satisfy both stronger inequalities separately.

We write

$$A_+ = \{x : F_n(x) \rightarrow +\infty\}, \quad A_- = \{x : F_n(x) \rightarrow -\infty\}.$$

These are measurable when the predictions are measurable. They have the Baire property when the predictions are Baire measurable, and they are Borel when all predictions are continuous.

Remark 46.2 (Two tempting invalid shortcuts). Equation (46.2) does not prevent $F_n \rightarrow +\infty$ almost surely. An unbounded negative contribution on rarer and rarer events can maintain zero expectation. Also, changing one hat may affect infinitely many predictions. We cannot simply declare A_+ a tail event and apply a zero-one law. The conditioning arguments below avoid that assumption.

47 From conditional obstructions to probability zero

We first isolate the step that upgrades a finite probability bound to an almost-sure obstruction.

Lemma 47.1 (Cylinder density). *Let $A \subseteq \Omega$ be measurable. Suppose some $c > 0$ satisfies*

$$\mathbb{P}(A \mid C) \leq 1 - c$$

for every nonempty cylinder fixing finitely many initial coordinates. Then $\mathbb{P}(A) = 0$.

Proof. Finite unions of initial-coordinate cylinders approximate every measurable set in probability. One proof is to observe that the class of sets approximable this way is a sigma-algebra: complements preserve approximation; for a countable union, first truncate the union in measure, then approximate its finitely many members. This class contains all cylinders and hence the product sigma-algebra. Completion adds only null modifications, which do not affect the argument.

The assumed inequality holds also for a finite union B of cylinders of a common length, by adding over its disjoint atoms. Thus

$$\mathbb{P}(B \setminus A) \geq c\mathbb{P}(B).$$

If $\mathbb{P}(A) > 0$, choose such a B with $\mathbb{P}(A \triangle B) < \delta$, where $\delta < c\mathbb{P}(A)/(1+c)$. Then

$$\delta > \mathbb{P}(B \setminus A) \geq c\mathbb{P}(B) \geq c(\mathbb{P}(A) - \delta),$$

a contradiction. $\square$

Lemma 47.2 (Conditional sign criterion). *Suppose Y_i are bounded by 1 in absolute value. For every cylinder C fixing coordinates $1, \dots, m$, suppose a constant $c > 0$, independent of m, C, n , satisfies*

$$\mathbb{P}\left(\sum_{i=m+1}^n Y_i \leq 0 \mid C\right) \geq c \quad (n > m).$$

Then $\mathbb{P}(A_+) = 0$. The corresponding nonnegative inequality implies $\mathbb{P}(A_-) = 0$.

Proof. On each of the displayed events $F_n \leq m$. Under the conditional probability on C , the probability that these events occur infinitely often is at least c : the probability of their union over $n \geq N$ is at least c , and then one lets $N \rightarrow \infty$. Consequently $\mathbb{P}(A_+ \mid C) \leq 1 - c$. Lemma 47.1 finishes the proof. Apply the same argument to $-Y_i$ for the other direction. $\square$

Conditioning works particularly well for blind prediction. After fixing the first m hats, each Y_i with $i > m$ still has conditional mean zero: its owner hat is still an independent fair bit. The first m scores may lose this property, but their total is bounded by m . This is why we delete owners, rather than trying to delete every prediction affected by a fixed hat.

48 Finite colors obstruct both measure and category success

Definition 48.1 (Visibility graph). An undirected graph G on $\mathbb{N}$ is a visibility graph for a strategy if f_i depends only on hats at neighbors of i . Neighborhoods may be infinite. An r -coloring is a map $c : \mathbb{N} \rightarrow \{1, \dots, r\}$ with different colors at the two ends of every edge.

Theorem 48.2 (Finite-chromatic obstruction). *Suppose the strategy has a visibility graph with a finite r -coloring. Then:*

(a) *For every finite set I of scored players,*

$$\mathbb{P}\left(\sum_{i \in I} Y_i \leq 0\right) \geq 2^{-r}, \quad \mathbb{P}\left(\sum_{i \in I} Y_i \geq 0\right) \geq 2^{-r},$$

provided the predictions are measurable.

(b) *For measurable predictions, $\mathbb{P}(A_+) = \mathbb{P}(A_-) = 0$.*

(c) For predictions with the Baire property, both A_+ and A_- are meager.

No degree bound, finite-information assumption, or bound on the number of observers of a hat is needed.

Proof. For $a = (a_1, \dots, a_r) \in \mathbb{F}_2^r$, define a homeomorphism preserving product measure by

$$(T_a x)_j = (-1)^{a_{c(j)}} x_j.$$

Flipping the bit $a_{c(i)}$ changes X_i to its negative and leaves every hat inspected by f_i unchanged. Therefore the average over the group annihilates every score:

$$\sum_{a \in \mathbb{F}_2^r} Y_i(T_a x) = 0, \quad \sum_{a \in \mathbb{F}_2^r} \sum_{i \in I} Y_i(T_a x) = 0. \quad (48.1)$$

At least one value in the latter sum is nonpositive and at least one is nonnegative. Averaging the corresponding indicator inequality and using measure preservation proves (a).

For (b), fix a cylinder C on the first m hats and modify T_a so that it flips only coordinates greater than m . These maps preserve C and its conditional probability. Equation (48.1) still holds for every owner $i > m$. It follows that

$$\left| 2^{-r} \sum_a F_n(T_a x) \right| \leq m \quad (n > m, x \in C).$$

Thus the finitely many points $T_a x$ cannot all belong to A_+ . Their indicator average is at most $1 - 2^{-r}$. Integration gives $\mathbb{P}(A_+ | C) \leq 1 - 2^{-r}$, and Lemma 47.1 proves nullity. The negative direction is identical.

For (c), if A_+ were nonmeager and had the Baire property, it would be comeager in some nonempty basic cylinder C . Use the same cylinder-preserving transformations. Each $T_a^{-1} A_+$ would be comeager in C . Their finite intersection would be nonempty by the Baire category theorem. A point in that intersection would make all the sequences $F_n(T_a x)$ tend to $+\infty$, contradicting their bounded average. Apply the same reasoning to A_- . $\square$

Lemma 48.3 (Bounded outgoing visibility admits finitely many colors). *Suppose each player can inspect a predetermined set of at most b hats. The undirected graph obtained by joining two players whenever either may inspect the other has a $(2b + 1)$ -coloring.*

Proof. In each finite induced subgraph on U , the number of edges is at most $b|U|$, since every edge has an outgoing witness in U . Its average degree is at most $2b$. The same holds in every induced subgraph, so successive removal of a vertex of degree at most $2b$, followed by reverse greedy coloring, gives $2b + 1$ colors. For the countable graph, consider the finitely branching tree of proper colorings of the first n vertices. Every level is nonempty by the finite argument. An infinite branch gives the claimed coloring. $\square$

Corollary 48.4 (The bounded-query question). *If each player makes at most d adaptive queries, for one common finite d , then A_+ and A_- are null and meager.*

Proof. The union of all coordinates queried by one depth- d decision tree has size at most $2^d - 1$. Apply Lemma 48.3 with that bound, and then Theorem 48.2. The rules are continuous. For $d = 0$, the graph has no edges and one color suffices. $\square$

This already settles the repository question without an analytic inequality. The potentially large number of colors is irrelevant for nullity and category. The next sections give substantially better finite probability bounds.

[write] **Added 4 October 2026, batch 92.** This corollary answers the uniform-bound clause of Part II’s Research question 29.2 negatively, and more strongly than asked: probability zero, not merely “not almost surely”. Source 04 proves its probability part independently as Theorem 61.3, by the analytic route that the paragraph after Corollary 49.3 sketches (a sign bound for the conditioned score of bounded degree, then martingale convergence); its proof is printed there as a second route. Category is source 06’s alone.

49 Quantitative bounds from Boolean degree

For a finite coordinate set V , the Walsh characters are $\chi_A(x) = \prod_{j \in A} x_j$, $A \subseteq V$. Every real function on the finite cube has the unique expansion

$$P = \sum_{A \subseteq V} \widehat{P}(A) \chi_A, \quad \mathbb{E}P^2 = \sum_A \widehat{P}(A)^2.$$

Its degree is the largest $|A|$ with nonzero coefficient.

Lemma 49.1 (Decision-tree degree). *A Boolean rule computed by at most d adaptive queries has Walsh degree at most d . Consequently every finite sum of its correctness scores has degree at most $d + 1$.*

Proof. At the first query j , with two remaining subtree functions g_+ , g_- , the rule is

$$f = \frac{1+x_j}{2} g_+ + \frac{1-x_j}{2} g_-.$$

Induction on depth proves the bound. Replacing x_j^2 by 1 in a multilinear representation cannot increase degree. Multiplication by the owner coordinate increases degree by at most one, and summing does not increase it. $\square$

49.1 The classical sign estimate, with a short elementary proof

The following weaker form of the classical hypercontractive sign bound has a convenient self-contained proof. The sharper constant is recorded immediately afterward. These analytic inequalities are established background, not new results of this article [24, 25].

Lemma 49.2 (A sign bound). *If P is nonzero, centered, and has degree at most k , then*

$$\mathbb{P}(P > 0), \mathbb{P}(P < 0) \geq \frac{1}{4 \cdot 9^k}.$$

Proof. First prove

$$\|P\|_4^2 \leq \sum_A 3^{|A|} \widehat{P}(A)^2. \quad (49.1)$$

Induct on the number of coordinates, writing $P = A + x_j B$. With $a = \|A\|_4^2$ and $b = \|B\|_4^2$, Cauchy–Schwarz gives

$$\mathbb{E}P^4 = \mathbb{E}A^4 + 6\mathbb{E}A^2B^2 + \mathbb{E}B^4 \leq a^2 + 6ab + b^2 \leq (a + 3b)^2.$$

The inductive estimates for A, B give (49.1). Hence $\|P\|_4 \leq 3^{k/2} \|P\|_2$. Interpolation by Hölder’s inequality yields

$$\|P\|_2 \leq \|P\|_1^{1/3} \|P\|_4^{2/3}, \quad \|P\|_1 \geq 3^{-k} \|P\|_2.$$

Since P is centered, $\mathbb{E}P_+ = \mathbb{E}P_- = \|P\|_1/2$. Another use of Cauchy–Schwarz gives

$$\frac{1}{2} \|P\|_1 = \mathbb{E}(P \mathbf{1}_{\{P>0\}}) \leq \|P\|_2 \mathbb{P}(P > 0)^{1/2}.$$

Squaring proves the positive estimate; replace P by $-P$ for the negative estimate. $\square$

O'Donnell's Theorems 9.22 and 9.24 state the sharper classical bounds

$$\|P\|_1 \geq e^{-k} \|P\|_2, \quad \mathbb{P}(P > 0), \mathbb{P}(P < 0) \geq \frac{1}{4} e^{-2k} \quad (49.2)$$

for centered nonzero P [25]. The same elementary last step derives the second inequality from the first.

[write] Added 4 October 2026, batch 92. Source 04 states the same estimates as Lemmas 61.1 and 61.2, credits the sharper constant to Austrin and Håstad [31] rather than to O'Donnell, and derives $\|P\|_1 \geq e^{-d} \|P\|_2$ from the degree- d hypercontractive inequality by letting $p \downarrow 2$; that derivation is printed there.

Corollary 49.3 (Uniform finite-horizon obstruction). *For any finite set of players whose rules use at most d adaptive queries,*

$$\mathbb{P}(F \leq 0), \mathbb{P}(F \geq 0) \geq \frac{1}{4} e^{-2(d+1)}, \quad F = \sum_i Y_i.$$

The weaker constant $1/(4 \cdot 9^{d+1})$ follows solely from the proof above.

Proof. Use Lemma 49.1, centering, and (49.2). If F vanishes identically, both nonstrict inequalities have probability 1. Hats observed outside the scored team are simply additional independent variables. $\square$

After fixing finitely many hats and deleting their owners, the same degree bound and conditional centering remain valid. Thus Lemma 47.2 gives a second proof of the probability part of Corollary 48.4.

49.2 A necessary aggregate Fourier tail

The degree argument has a stable version. This explains why a small probability of a long inspection cannot simply be ignored in a large team.

Proposition 49.4 (Fourier-tail obstruction). *Let F be a nonzero centered function on a finite cube. Write $F = P + R$, where P is its Walsh projection to degrees at most k , and put*

$$\delta = \frac{\|R\|_2^2}{\|F\|_2^2}.$$

Then

$$\mathbb{P}(F \leq 0), \mathbb{P}(F \geq 0) \geq \frac{1}{4} \left(e^{-k} \sqrt{1 - \delta} - \sqrt{\delta} \right)_+^2. \quad (49.3)$$

In particular, if centered finite scores F_n satisfy $\mathbb{P}(F_n > 0) \rightarrow 1$, then for each fixed k ,

$$\liminf_{n \rightarrow \infty} \frac{\|(F_n)_{>k}\|_2^2}{\|F_n\|_2^2} \geq \frac{1}{1 + e^{2k}},$$

with the ratios taken at the eventually nonzero scores.

Proof. Orthogonality gives $\|P\|_2 = \sqrt{1 - \delta} \|F\|_2$. The triangle inequality, (49.2), and $\|R\|_1 \leq \|R\|_2$ imply

$$\|F\|_1 \geq \left(e^{-k} \sqrt{1 - \delta} - \sqrt{\delta} \right)_+ \|F\|_2.$$

Apply the centered positive-part argument from Lemma 49.2 directly to F . For the final assertion, a subsequence with $\delta < 1/(1 + e^{2k})$ bounded away from that value would keep the right side of (49.3) bounded away from zero. $\square$

50 Stronger finite bounds from sparse supports

The generic degree estimate disregards the special form $Y_i = X_i f_i$. The owner coordinate in every monomial lets us produce finite collections of sign flips that give better constants.

Theorem 50.1 (Sparse-support cancellation). *Let I be a finite scored team with finite-coordinate prediction rules. Write the nonzero Walsh support of each legal rule as $\mathcal{A}_i$, and suppose*

$$b_i := \sum_{A \in \mathcal{A}_i} (|A| + 1) \leq B, \quad B \geq 1.$$

Let Q be the smallest power of 2 strictly greater than B . For arbitrary real weights w_i , the score $F = \sum_{i \in I} w_i X_i f_i$ satisfies

$$\mathbb{P}(F \leq 0), \mathbb{P}(F \geq 0) \geq Q^{-1}. \quad (50.1)$$

No nondegeneracy assumption is needed.

Proof. Choose a finite V containing the owners and every coordinate on which one of the prediction rules depends. Form a hypergraph $\mathcal{H}$ consisting of the distinct nonempty sets $\{i\} \cup A$, $i \in I$, $A \in \mathcal{A}_i$. For every nonempty $U \subseteq V$,

$$\sum_{\substack{E \in \mathcal{H} \\ E \subseteq U}} |E| \leq \sum_{i \in I \cap U} b_i \leq B|U|. \quad (50.2)$$

Indeed, every edge contained in U has an owner in U ; counting all its possible owners only overcounts. The left side is the sum of vertex degrees in the induced hypergraph. It therefore has a vertex incident to at most $\lfloor B \rfloor$ edges. Repeatedly remove such a vertex, retaining its removal order.

Write $Q = 2^r$. Assign labels $v_j \in \mathbb{F}_2^r$ in reverse removal order. Each edge whose other vertices have already been assigned forbids exactly one label for the current vertex: the sum of those other labels. There are at most $\lfloor B \rfloor < Q$ such forbidden labels. Choose any permitted one. Every edge then has

$$\sum_{j \in E} v_j \neq 0.$$

For $a \in \mathbb{F}_2^r$, define

$$(T_a x)_j = (-1)^{a \cdot v_j} x_j.$$

Character orthogonality gives

$$\sum_a \chi_E(T_a x) = \chi_E(x) \sum_a (-1)^{a \cdot \sum_{j \in E} v_j} = 0 \quad (E \in \mathcal{H}).$$

Hence $\sum_a F(T_a x) = 0$ pointwise. At least one group value is nonpositive and one is nonnegative. Averaging their indicators and using the invariance of the fair cube proves (50.1). $\square$

50.1 Predetermined sets of visible hats

If a rule depends on at most d predetermined hats, then, including even the subsets with zero coefficient,

$$b_i \leq \sum_{A \subseteq [d]} (|A| + 1) = (d + 2)2^{d-1}.$$

For $d = 0$ the value is 1. Theorem 50.1 yields

$$\mathbb{P}(F \leq 0), \mathbb{P}(F \geq 0) \geq c_d := 2^{-d - \lfloor \log_2(d+2) \rfloor} \geq \frac{1}{(d+2)2^d}. \quad (50.3)$$

50.2 Adaptive trees

For a depth- d tree, $d \geq 1$, expand the tree down to level $d - 1$. There are at most 2^{d-1} branches at that level. The remaining function on each branch is a constant or a signed single coordinate. Its product with the branch's path indicator has weighted support at most

$$\sum_{t=0}^{d-1} \binom{d-1}{t} (t+2) = (d+3)2^{d-2}.$$

Here the weight of a monomial on t path variables and at most one bottom variable is at most $t+2$, after including the owner. Short branches can be extended in this calculation by constant subtrees; repeated variables and cancellation only reduce the bound. Consequently

$$b_i \leq B_d^{\text{ad}} := (d+3)2^{2d-3},$$

and therefore

$$\mathbb{P}(F \leq 0), \mathbb{P}(F \geq 0) \geq c_d^{\text{ad}} := 2^{-2d+2-\lfloor \log_2(d+3) \rfloor} \geq \frac{4}{(d+3)4^d}. \quad (50.4)$$

When $d = 0$, use $1/2$. Fixing some hats and reexpanding the remaining rules does not increase the weighted support, so these constants also apply conditionally after deletion of fixed owners.

d	Fixed visible set: c_d	Adaptive tree: c_d^{ad}	Status
0	1/2	1/2	Sharp
1	1/4	1/4	Sharp
2	1/16	1/16	Proven lower bounds
3	1/32	1/64	Proven lower bounds
4	1/64	1/256	Proven lower bounds
5	1/128	1/2048	Proven lower bounds
6	1/512	1/8192	Proven lower bounds

50.3 The exact one-query value

Theorem 50.2 (One query: sharp success probability). *Over all nonempty finite teams and legal deterministic strategies with at most one query per player,*

$$\sup \mathbb{P}(D > 0) = \frac{3}{4}.$$

With zero queries the supremum is $1/2$.

Proof. The upper bound $3/4$ is (50.4) at $d = 1$. For attainment, take three players in a directed cycle, each predicting the opposite of the next player's hat. In sign notation,

$$F = -X_1 X_2 - X_2 X_3 - X_3 X_1.$$

If all hats agree, $F = -3$; otherwise $F = 1$. The first event has probability $2/8 = 1/4$, and the second has probability $3/4$. The zero-query sum is a signed sum of independent hats, so global complementation negates it; hence its strict-positive probability is at most $1/2$. One player attains equality. $\square$

An alternative four-transformation proof for the one-query bound labels the three colors of its visibility graph by the three nonzero vectors of $\mathbb{F}_2^2$. Singletons and edge endpoint sums then all have nonzero labels. This also makes the small finite certificate particularly easy to verify.

For a quantitative consequence of (50.3), if strict positive excess has probability at least $1 - \eta$, then

$$d + \log_2(d + 2) \geq \log_2(1/\eta).$$

Writing $L = \log_2(1/\eta)$ gives the explicit necessary condition

$$d \geq L - \log_2(L + 2).$$

If $d > L$ this is automatic; otherwise use $\log_2(d + 2) \leq \log_2(L + 2)$. We do not claim these finite bounds are optimal beyond one query.

[write] Added 4 October 2026, batch 92. The three-player cycle is Part I's three-box team of Section 13 (player p guesses $1 - x_{p+1 \bmod 3}$) and source 04's trace team for $r = 2, m = 1$ (Theorem 65.5). Theorem 50.2 shows that no team with one query per player does better, so source 04's lower bound $p_1 \geq 3/4$ in Research question 68.2 is an equality.

51 Expected query cost: the lower half of the exact threshold

The number 2 comes from the geometry of a binary decision tree under fair independent answers. This observation turns an average resource restriction into a worst-case restriction.

Lemma 51.1 (A deep path costs almost two queries in expectation). *Let a deterministic query algorithm have a feasible execution path with at least $L \geq 1$ queries. Under independent fair hats,*

$$\mathbb{E}Q \geq \sum_{j=0}^{L-1} 2^{-j} = 2 - 2^{1-L}. \quad (51.1)$$

Consequently, if $0 \leq C < 2$ and $\mathbb{E}Q \leq C$, every execution has at most

$$d(C) = \lfloor 1 - \log_2(2 - C) \rfloor \quad (51.2)$$

queries.

Proof. For each $j < L$, agreeing with the first j answers on that execution forces the algorithm to make query $j + 1$. These answers specify at most j independent hats, so their event has probability at least 2^{-j} . Thus $\mathbb{P}(Q \geq j + 1) \geq 2^{-j}$. Summing these tail probabilities gives (51.1). Repeated queries do not invalidate the estimate: they specify fewer fresh hats and can only increase the prefix probability. Equivalently, they can first be eliminated.

Solving $2 - 2^{1-L} \leq C$ for the integer L gives (51.2). An infinite execution would have arbitrarily long feasible finite prefixes and would therefore force $\mathbb{E}Q \geq 2$. The same argument handles algorithms initially assumed only to halt almost surely; a nonterminating computation after a finite query transcript would occur on a positive-probability cylinder and is already excluded by that assumption. $\square$

Corollary 51.2 (Uniform expectation strictly below two is impossible). *For deterministic legal query algorithms,*

$$\sup_i \mathbb{E}Q_i < 2 \implies \mathbb{P}(A_+) = \mathbb{P}(A_-) = 0,$$

and both divergence events are meager.

Proof. Choose a common $C < 2$ bounding the expectations. Lemma 51.1 gives a common finite depth bound $d(C)$, so apply Corollary 48.4. $\square$

Uniform expected cap C	Forced depth cap	Positive divergence
1	1	Null and meager
3/2	2	Null and meager
7/4	3	Null and meager
15/8	4	Null and meager
31/16	5	Null and meager
2	No finite cap follows	Almost sure success is attainable

Proposition 51.3 (The shape of trees below the threshold). *In a reduced finite deterministic binary decision tree, where each queried hat is fresh along its path, $\mathbb{E}Q < 2$ if and only if its internal vertices form a single path. In this case, if its depth is L , then*

$$\mathbb{E}Q = 2 - 2^{1-L},$$

with the value 0 when $L = 0$.

Proof. If the internal vertices form a path, exactly one answer continues at each step before the final query, giving tail probabilities 2^{-j} and the stated sum. Otherwise, consider the first vertex with two internal children. If it is at depth k , the path reaching it has probability 2^{-k} . The expected cost before reaching or avoiding that vertex is $2 - 2^{1-k}$; the conditional expected remaining cost at that vertex is at least 2, since both answers force another query. The total is therefore at least

$$2 - 2^{1-k} + 2^{-k} \cdot 2 = 2.$$

$\square$

The critical construction below uses precisely these path-shaped decision trees, with depths tending to infinity. Every individual mean is below 2, but the means have no common bound strictly below 2.

[write] Added 4 October 2026, batch 92. Lemma 51.1 and Corollary 51.2 are source 04's Lemma 62.1 and Corollary 62.3, proved there by the same path argument; source 04 states the lemma as the stochastic domination $Q \geq \min(G, L)$ and extends the corollary to every nonnegative nondecreasing cost (Theorem 62.2). Proposition 51.3 has no counterpart there.

52 The efficient implementation of a known block strategy

For this section use bits $x_i \in \{0, 1\}$, with 0 black and 1 white. Form a block of $2m$ players in the logical order

$$S_1, T_1, S_2, T_2, \dots, S_m, T_m.$$

The letter T here names a partner role, not a sign transformation. The baseline pairs strategy has S_j copy T_j 's hat and T_j guess the opposite of S_j 's hat. It gives exactly one correct guess per pair.

Eldredge's block rule modifies this baseline when a player sees only black hats [10, Proposition 6.6]. The following operational description computes that same prediction rule while often stopping after its first query.

Definition 52.1 (Early-stopping block algorithm). Player T_j queries S_j 's hat and predicts its opposite. Player S_j follows this decision tree:

1. Query the partner T_j . If its hat is 1, predict 1 and stop.
2. If its hat is 0, inspect the other $2m - 2$ visible hats in a fixed order, stopping at the first 1. If one is found, predict 0.
3. If none is found, predict 1.

No query refers to the player's own hat.

Proposition 52.2 (Exact block law and prefixes). *Let Δ_m be the surplus contributed by a complete block. Then*

$$\Delta_m = \begin{cases} -m, & \text{all hats are black,} \\ 1, & \text{there is exactly one white hat, on an } S\text{-player,} \\ 0, & \text{otherwise.} \end{cases} \quad (52.1)$$

Thus

$$\mathbb{P}(\Delta_m = -m) = 4^{-m}, \quad \mathbb{P}(\Delta_m = 1) = m4^{-m}, \quad \mathbb{P}(\Delta_m = 0) = 1 - (m+1)4^{-m}. \quad (52.2)$$

On every block except the all-black block, the surplus at completed-pair boundaries is nondecreasing. On such a nonbad block, relative to the complete-block endpoint, the surplus at every logical prefix differs by at most 1.

Proof. An S -player departs from copying its partner only when every visible hat is black. If the entire block is black, all players predict white; every pair then contributes -1 . If the only white hat is on S_j , that player's modified prediction is correct and its partner's opposite prediction is also correct. Their pair contributes $+1$; all other pairs remain neutral.

If the only white hat belongs to a T -player, its opposite-partner prediction was already white. Its S -partner copies white and is wrong, so the pair is neutral. If there are at least two white hats, each S -player sees a white hat and uses the baseline copying rule. All pairs are neutral. This proves the pointwise law; counting the all-black configuration and the m favorable singleton configurations gives (52.2).

On a nonbad block there is at most one positive pair and no negative pair. A partially completed pair contributes either $1/2$ or $-1/2$. In a positive block every S -player is correct. Before the exceptional pair the surplus is 0 at pair boundaries, after it the surplus is 1, and its first member is correct. The difference from the final value is therefore in $[-1, 1/2]$. In a neutral block it is in $[-1/2, 1/2]$. $\square$

In particular $\mathbb{E}\Delta_m = -m4^{-m} + m4^{-m} = 0$. For example, when $m = 2$, the three surplus values are $-2, 1, 0$ with probabilities $1/16, 2/16, 13/16$. The mechanism gains one on some rare configurations and loses much more on even rarer configurations.

Proposition 52.3 (Exact query distribution). *For an S -player in a block of m pairs,*

$$Q_S \stackrel{d}{=} \min(G, 2m - 1),$$

where G has geometric law $\mathbb{P}(G = t) = 2^{-t}$, $t = 1, 2, \dots$. Consequently

$$\mathbb{E}Q_S = 2 - 2^{-(2m-2)}, \quad Q_T = 1, \quad (52.3)$$

$$\mathbb{P}(Q_i > t) \leq 2^{-t} \quad (t = 0, 1, 2, \dots). \quad (52.4)$$

For S -players the tail inequality is equality when $t < 2m - 1$ and is zero afterward. Every player has expected cost strictly below 2, and the expected cost averaged over a pair is

$$\frac{3}{2} - 2^{-(2m-1)} < \frac{3}{2}.$$

For $0 < \lambda < \log 2$, uniformly in m and the player,

$$\mathbb{E}e^{\lambda Q_i} \leq \frac{e^\lambda}{2 - e^\lambda}. \quad (52.5)$$

Proof. To continue after its first query, an S -player must see a black partner. To continue after t queries, all t inspected hats must have been black. They are distinct independent fair bits. This has probability 2^{-t} , until the $2m - 1$ visible hats are exhausted. The tail-sum formula gives (52.3). Stochastic domination by G gives (52.5) from

$$\mathbb{E}e^{\lambda G} = \sum_{t \geq 1} (e^\lambda/2)^t = \frac{e^\lambda}{2 - e^\lambda}.$$

The partner role uses exactly one query. □

The cost statement counts individual hats, including the partner. It assumes neither free parity queries nor a public signal. This distinction matters: an implementation that first inspects every visible hat computes the same guesses but misses the expected-cost conclusion completely.

[write] Added 4 October 2026, batch 92. Definition 52.1 and Propositions 52.2–52.3 are source 04’s implementation of Section 63 with its Lemma 63.1 and Proposition 63.2, found independently (source 04 writes k for the number of pairs, here m). Source 04’s proposition adds that, for blocks of unbounded size, $\sup_i \mathbb{E}\phi(Q_i) = \mathbb{E}\phi(G)$ for every nonnegative nondecreasing ϕ , so the bound (52.5) is the exact supremum. The block rule is Eldredge’s [10, Proposition 6.6], as both sources say. Source 06’s verifier calls the T role D , as source 04 did.

53 Infinite schedules, exact rates, and the sharp threshold

Choose a real parameter $0 < \varepsilon \leq 1$. For every integer $m \geq 2$, stage m consists of

$$L_m = \left\lceil \frac{4^m}{m^{1+\varepsilon}} \right\rceil \quad (53.1)$$

consecutive disjoint blocks of m pairs, each using Definition 52.1. Stages occur in increasing order. For rational ε , these schedules can be made computable using integer power comparisons; the especially simple $\varepsilon = 1$ schedule uses only exact integer arithmetic.

Theorem 53.1 (Almost-sure growth with uniformly geometric query tails). *The stage strategy satisfies*

$$D_n \rightarrow +\infty \quad a.s., \quad \sup_i \mathbb{E}Q_i = 2, \quad \mathbb{E}Q_i < 2 \text{ for each } i,$$

and the uniform bounds (52.4) and (52.5). More precisely, at every physical prefix,

$$D_n \sim \frac{(\log_4 n)^{1-\varepsilon}}{1-\varepsilon} \quad (0 < \varepsilon < 1), \quad (53.2)$$

$$\frac{D_n}{\log \log n} \rightarrow 1 \quad (\varepsilon = 1), \quad (53.3)$$

almost surely, with natural logarithms in (53.3). The worst-case number of queried hats and their maximum distance from their owner are both $O(\log n)$ for player n .

53.1 Only finitely many losses, infinitely many gains

Proof of divergence in Theorem 53.1. Call an all-black block bad and a block with the favorable singleton white hat positive. The sum of bad-block probabilities is

$$\sum_{m \geq 2} L_m 4^{-m} \leq \sum_{m \geq 2} m^{-1-\varepsilon} + \sum_{m \geq 2} 4^{-m} < \infty.$$

The first Borel–Cantelli lemma therefore gives only finitely many bad blocks almost surely. On the other hand, the sum of positive-block probabilities is

$$\sum_{m \geq 2} L_m m 4^{-m} = \sum_{m \geq 2} m^{-\varepsilon} + O(1) = \infty.$$

Distinct blocks use disjoint hats, so their positive events are independent. There are infinitely many positive blocks almost surely. For completeness, independence gives probability at most $\exp(-\sum p_b)$ of no positive block on a finite interval of blocks; letting the endpoint tend to infinity and then intersecting over starting points proves the assertion.

After the last bad block, completed-pair surpluses never decrease and increase by 1 at each positive block. Odd physical prefixes differ from completed-pair values by $1/2$. This proves divergence at all prefixes. Proposition 52.3 gives the cost bounds, and its means approach 2 as $m \rightarrow \infty$. $\square$

53.2 A self-contained strong law for the gains

Lemma 53.2 (Independent indicator counting). *Let $B_1, B_2, \dots$ be independent indicators, let $M_j = \sum_{\ell \leq j} \mathbb{E} B_\ell$, and suppose $M_j \rightarrow \infty$. Then*

$$\frac{\sum_{\ell \leq j} B_\ell}{M_j} \longrightarrow 1 \quad \text{a.s..}$$

Proof. Let j_k be the first index for which $M_{j_k} \geq k^2$. Because each increment of M_j is at most 1, $k^2 \leq M_{j_k} < k^2 + 1$. Independence gives variance at most M_{j_k} , so for any fixed $\delta > 0$,

$$\mathbb{P} \left(\left| \sum_{\ell \leq j_k} B_\ell - M_{j_k} \right| > \delta M_{j_k} \right) \leq \frac{1}{\delta^2 M_{j_k}} \leq \frac{1}{\delta^2 k^2}.$$

Borel–Cantelli, followed by a countable sequence of positive δ 's tending to zero, proves the limit along j_k . For $j_k \leq j \leq j_{k+1}$, both the partial indicator sum and its expectation are nondecreasing, and $M_{j_{k+1}}/M_{j_k} \rightarrow 1$. Squeezing gives the full limit. $\square$

Proof of the rates in Theorem 53.1. Let G_M be the number of positive blocks through stage M . Its mean is

$$\mu_M = \sum_{m=2}^M L_m m 4^{-m} = \sum_{m=2}^M m^{-\varepsilon} + O(1). \quad (53.4)$$

Lemma 53.2 implies $G_M/\mu_M \rightarrow 1$ almost surely. At completed block endpoints after the last bad block, the total surplus is the number of positive blocks minus

$$B = \sum_{\text{bad blocks } b} m(b),$$

where $m(b)$ is its number of pairs. We have $B < \infty$ almost surely. After the final bad block, the same identity holds at completed-pair endpoints with the gain count recording the exceptional pair of each positive block.

The total number of physical players through stage M is

$$N_M = \sum_{m=2}^M 2mL_m \sim \frac{8}{3} \frac{4^M}{M^\varepsilon}. \quad (53.5)$$

To check the constant, divide $\sum_{m=2}^M 4^m/m^\varepsilon$ by $4^M/M^\varepsilon$. The term at $m = M - j$ tends to 4^{-j} , whose sum is $4/3$. The part $m < M/2$ is exponentially negligible; in the remaining part it is dominated by $2^\varepsilon 4^{-j}$. The rounding contribution to N_M is $O(M^2)$ and is negligible. Multiplication by 2 gives $8/3$.

Thus, uniformly for a physical index in stage M ,

$$\log_4 n = M + O(\log M).$$

The means in adjacent stages have ratio tending to one, because $\mu_M - \mu_{M-1} = M^{-\varepsilon} + O(M4^{-M})$. After the last bad block, all completed-pair scores within a stage lie between its two endpoint scores; the unfinished-pair error is at most $1/2$. The endpoint asymptotics therefore hold at every prefix. Finally,

$$\mu_M \sim \frac{M^{1-\varepsilon}}{1-\varepsilon} \quad (0 < \varepsilon < 1), \quad \mu_M = \log M + O(1) \quad (\varepsilon = 1).$$

These formulas and (53.5) prove (53.2)–(53.3). Each query remains in a block of $2M$ players and there are at most $2M - 1$ of them. This proves both $O(\log n)$ resource statements. $\square$

53.3 The exact expected-cost threshold

Theorem 53.3 (Critical deterministic expected budget). *Let $\mathcal{S}_+$ be the class of deterministic legal query strategies with $\mathbb{P}(D_n \rightarrow +\infty) > 0$, and let $\mathcal{S}_1$ require that probability to equal 1. Then*

$$\inf_{\mathcal{S} \in \mathcal{S}_+} \sup_i \mathbb{E}Q_i = \inf_{\mathcal{S} \in \mathcal{S}_1} \sup_i \mathbb{E}Q_i = 2. \quad (53.6)$$

The value 2 is attained by an everywhere finite, computable strategy whose query tails satisfy (52.4).

Proof. Corollary 51.2 excludes every value below 2, even for positive-probability success. The $\varepsilon = 1$ strategy of Theorem 53.1 attains 2 with almost-sure success. $\square$

[write] Added 4 October 2026, batch 92. Source 04 proves the same threshold as the mean case of Theorem 60.1, by the same two halves (Lemma 62.1 with Theorem 61.3 below, the early-stopping block with its schedule above), and strengthens the lower half to every nonnegative nondecreasing cost: tails 2^{-r} , second moment 6, exponential moments $z/(2-z)$ for $1 \leq z < 2$ and none for $z \geq 2$ (Corollary 62.5). Theorem 53.1 and source 04's Theorem 63.3 are the same theorem, with $p = 1 + \varepsilon$ and $\lfloor 4^k/k^p \rfloor$ blocks of k pairs in stage k for $\lceil 4^m/m^{1+\varepsilon} \rceil$; source 04's proof, a second route through Kronecker's lemma, is printed there because its later results use its notation.

53.4 Why fairness and almost-sure growth coexist

Although B is almost surely finite,

$$\mathbb{E}B = \sum_{m \geq 2} L_m m 4^{-m} = \infty. \quad (53.7)$$

Every finite endpoint has equal expected positive gain and negative debt. Along almost every sample, only finitely many losses actually occur and infinitely many gains remain. Passing the limit through expectation would require an integrability condition that fails.

The same distinction explains why the geometric query tail does not contradict the degree obstruction. The queries are usually short, but their deterministic depths are unbounded. Rare long branches support an essential part of the aggregate score's variance. Proposition 49.4 makes the latter assertion quantitative: success with probability tending to one requires a definite fraction of Fourier mass above every fixed degree.

Proposition 53.4 (Category behavior at the critical cost). *For the contiguous stage strategy of Theorem 53.1, the set A_+ is conull and meager. There is a comeager set on which*

$$\limsup_n D_n = +\infty, \quad \liminf_n D_n = -\infty.$$

Proof. Conullity is already proved. For positive integers K, N , the sets

$$U_{K,N}^+ = \bigcup_{n \geq N} \{D_n > K\}, \quad U_{K,N}^- = \bigcup_{n \geq N} \{D_n < -K\}$$

are open, since every score uses finitely many hats. They are dense. Given any cylinder, first complete the finitely many blocks meeting its fixed coordinates. Then assign sufficiently many subsequent blocks positive configurations to reach a score above K , or all-black configurations to reach a score below $-K$. The endpoint can be chosen beyond N , and each instruction fixes only finitely many additional hats. The intersection of all these open dense sets is comeager and has the two asserted limit behaviors. It is disjoint from A_+ . $\square$

[write] **Added 4 October 2026, batch 92.** This is the phenomenon of Part II's Theorems 26.1–26.2 (a conull meagre success set) for the critical-cost construction. It does not answer Part II's Research question 29.10, which asks whether strong growth can hold on a set that is conull *and* comeagre: the construction here gives a meagre success set, and by Corollary 48.4 every strategy of bounded depth has meagre divergence sets.

54 Prescribed budgets: count and distance are different

54.1 Arbitrarily sparse high-budget players

Lemma 54.1 (Two ordered matchings). *Take two disjoint infinite subsequences of $\mathbb{N}$, and pair consecutive members within each subsequence. At any physical prefix, at most one pair from each subsequence has begun but not finished. The total surplus from all such unfinished pairs has absolute value at most 1.*

Proof. In the increasing enumeration of one subsequence, a prefix contains an initial segment, which has at most one unpaired final member. That member contributes $1/2$ or $-1/2$. There are two subsequences. $\square$

Theorem 54.2 (Exact classification of count budgets). *Let $q : \mathbb{N} \rightarrow \mathbb{N} \cup \{0\}$ be finite valued and satisfy $q(i) \geq 1$ for all sufficiently large i . The following are equivalent:*

- (i) *q is unbounded.*
- (ii) *There is a deterministic legal strategy with $Q_i(x) \leq q(i)$ for every player and every input, such that $D_n \rightarrow +\infty$ almost surely.*
- (iii) *There is such a strategy with positive-probability divergence.*

The strategy in (ii) can additionally satisfy $\mathbb{P}(Q_i > t) \leq 2^{-t}$ for all i, t . If q is computable and unbounded, an everywhere terminating computable strategy can be constructed from it.

Proof. If q is bounded, Corollary 48.4 excludes (iii). The implication (ii) to (iii) is immediate.

Assume q is unbounded. Let every player with $q(i) = 0$ use a constant guess. There are finitely many such players, so their total surplus is a bounded random offset; their locations need not be known in advance. Use the $\varepsilon = 1$ list of blocks from (53.1). For each block of m pairs, select $2m$ previously unused indices, all beyond the indices of the preceding block, each satisfying

$$q(i) \geq 2m - 1.$$

This is possible: a finite-valued unbounded sequence has infinitely many indices above every fixed threshold. Leave at least two unselected guard indices between successive blocks, ensuring that the unselected positive-budget subsequence is infinite.

Pair consecutive selected indices within each block and run Definition 52.1 on them. They may be far apart physically; only the number of hats queried is constrained. Pair all unselected indices with positive budget consecutively in their own increasing enumeration and give them the baseline neutral pairs strategy. Their cost is 1. Selected players use at most $2m - 1 \leq q(i)$ queries.

Selected blocks depend on disjoint hats. The same two summability arguments as in Section 53 give finitely many bad blocks and infinitely many positive blocks almost surely. Completed neutral pairs contribute zero. Completed selected pairs outside bad blocks contribute 0 or 1, with infinitely many 1's. The total negative contribution of bad blocks is finite almost surely. The argument of Lemma 54.1 bounds the unfinished-pair error by 1 at every physical prefix, besides the bounded contribution of the zero-budget players. Therefore $D_n \rightarrow +\infty$. The query-tail distribution of each selected rule is unchanged by relocating its coordinates; neutral and zero-budget players also satisfy the stated tail bound.

For computable q , choose eligible indices by increasing search, using the first possible choices and designated guard indices. Each search terminates by unboundedness. To determine the role of a given input index i , construct blocks until their endpoint passes i . The positive-budget complement is decidable by this procedure; its next member can also be found by search because the guards and eventual positivity ensure infinitude. Thus no bound on the locations of the finitely many zero budgets is needed. The resulting finite prediction algorithm terminates on every input. $\square$

The eventual lower bound $q(i) \geq 1$ is material. It lets the unused players be paired without creating an uncontrolled random score. The theorem does not classify arbitrary budgets having infinitely many zero entries.

54.2 A budget tending to infinity permits local construction

Theorem 54.3 (Simultaneous count and distance constraints). *Let $h : \mathbb{N} \rightarrow \mathbb{N} \cup \{0\}$ satisfy $h(i) \rightarrow \infty$. There is a deterministic strategy such that every query by player i satisfies*

$$Q_i(x) \leq h(i), \quad |j - i| \leq h(i),$$

and $D_n \rightarrow +\infty$ almost surely. The uniform geometric query tail and the expectation bound $\mathbb{E}Q_i < 2$ are retained.

Proof. Again take the $\varepsilon = 1$ list of blocks. Before placing a block of length $2m$, move to a sufficiently late odd starting position at which every position in the next $2m$ places has $h(i) \geq 2m - 1$. Such a position exists

because $h(i) \rightarrow \infty$. Fill each intervening even-length gap with adjacent neutral pairs. If either member of a filler pair has budget zero, let both members guess constantly instead. Only finitely many filler pairs are exceptional, and their total surplus is bounded.

Every active query is within its contiguous block, so both inequalities hold. Neutral queries have count and distance 1, and exceptional players make no queries. Inserting the filler pairs changes completed-pair surplus by only a bounded total. The same block argument proves almost-sure divergence and preserves the query-cost bounds. If h is computable, a finite search for the next admissible interval suffices; a computable modulus of convergence is not needed. $\square$

54.3 The same unbounded numerical budget can permit one task and forbid the other

Theorem 54.4 (Count–distance separation). *Define*

$$h(i) = \begin{cases} k + 1, & i = 4^k, \quad k \geq 1, \\ 1, & \text{otherwise.} \end{cases} \quad (54.1)$$

As a query-count budget, h permits $D_n \rightarrow +\infty$ almost surely. As a distance budget, every measurable legal strategy has $\mathbb{P}(A_+) = \mathbb{P}(A_-) = 0$, and every Baire-measurable such strategy has meager divergence events.

Proof. The sequence is unbounded and everywhere at least 1, so the count conclusion follows from Theorem 54.2. For the distance conclusion, form the undirected allowed graph: join i, j if $|i - j| \leq h(i)$ or $|i - j| \leq h(j)$. Give every special index 4^k color 3, and every other index color 1 or 2 according to its parity.

An edge between two ordinary indices has length 1, so it joins different parity colors. No two special indices can be adjacent: for consecutive special indices the gap is $3 \cdot 4^k$, larger than both $k + 1$ and $k + 2$; all other gaps are larger. An edge incident to exactly one special index also joins different colors. This is a proper three-coloring. Theorem 48.2 applies even if each player reads its entire allowed neighborhood. $\square$

The distinction is structural. In the count model, selected high-budget players can inspect one another across arbitrary gaps. In the distance model, the high-radius centers remain isolated from one another. Merely observing that h is unbounded does not identify which model is intended.

54.4 One-way visibility produces independent correctness

Proposition 54.5 (Acyclic directed visibility). *Orient a visibility edge $i \rightarrow j$ when player i may inspect hat j . If the directed graph has no directed cycle, then for measurable legal rules the signs Y_i are independent fair signs. This remains true with infinite sets of visible hats.*

Proof. For any nonempty finite set I of owners, its induced directed graph has a vertex i with no incoming edge. No Y_j , $j \in I \setminus \{i\}$, depends on X_i , and f_i does not depend on X_i . Conditioning on all other hats therefore gives

$$\mathbb{E} \prod_{j \in I} Y_j = 0.$$

All nonempty joint moments vanish. For any finite assignment of signs y_i , expand

$$\mathbb{P}(Y_i = y_i \text{ for all } i \in I) = \mathbb{E} \prod_{i \in I} \frac{1 + y_i Y_i}{2} = 2^{-|I|}.$$

This is precisely independence and fairness. $\square$

In particular, past-only and future-only inspection give ordinary symmetric random-walk scores, even if the number of available hats is unbounded. Such a walk reaches every positive and negative integer almost surely. An elementary proof uses the finite gambler's-ruin calculation: starting at 0, the probability of hitting $-a$ before b is $b/(a+b)$, obtained by stopping the bounded fair walk at its first exit from $(-a, b)$. The exit is almost surely finite, since a sufficiently long run of equal steps forces it and such runs have a fixed positive probability on disjoint trials. Let $b \rightarrow \infty$, and then intersect over integer a ; reverse signs for positive levels. Hence neither divergence event has positive probability. This elementary observation emphasizes the role of directed cycles in coordinated surplus.

[write] Added 4 October 2026, batch 92. With Corollary 48.4, this section answers the three clauses of Part II's Research question 29.2 qualitatively: a uniform bound on the number of queries excludes $D_n \rightarrow +\infty$ almost surely, indeed with positive probability; for a count budget $q(i)$ that is eventually at least one, almost-sure divergence is possible exactly when q is unbounded (Theorem 54.2); and a count budget differs from a distance budget (Theorems 54.3–54.4). The question's first clause, which excess *profiles* remain attainable under a given $q(i)$, is answered only as “divergent or not”; the attainable growth rates under a budget, budgets with infinitely many zeros and general distance profiles stay open (Research questions 57.3, 57.4 and 57.6).

55 Free private randomness changes the expected-cost problem

The word deterministic in Theorem 53.3 cannot be omitted. Randomization is harmless for a fixed worst-case depth bound, but it changes the expected query budget.

Proposition 55.1 (Bounded depth with independent random seeds). *If random seeds are independent of all hats and every realized prediction algorithm uses at most the same finite d queries, then $\mathbb{P}(A_+) = \mathbb{P}(A_-) = 0$, where probability includes both hats and seeds.*

Proof. Condition on all seeds. The resulting deterministic strategy satisfies Corollary 48.4. Integrate the resulting zero probabilities. For this statement we assume the randomized strategy is jointly measurable, as ordinary query algorithms are. $\square$

Theorem 55.2 (Expected budgets arbitrarily close to one). *For every $C > 1$, there is a legal strategy using independent private random bits at the players, with*

$$\sup_i \mathbb{E}Q_i < C, \quad D_n \rightarrow +\infty \quad \text{a.s..}$$

No communication or shared random seed is required.

Proof. Choose a dyadic $0 < \alpha < \min\{1, C-1\}$, so activation can be sampled using finitely many fair private bits, and use the $\varepsilon = 1$ stage schedule. Each S -player independently activates its early-stopping rule with probability α ; otherwise it simply copies its partner. Every T -player always predicts the opposite of its partner.

On a block that is not all black, no pair has negative surplus. A favorable singleton white hat on an S -player yields +1 exactly when that player activates, an event of probability $\alpha m 4^{-m}$. All-black blocks are still the only possible negative blocks; their probabilities remain 4^{-m} , regardless of the activation decisions. Thus there are only finitely many negative blocks and infinitely many positive blocks almost surely. Block events are independent because both their hats and their private coins are disjoint. The same prefix proof gives positive divergence.

A T -player uses one query. For an S -player,

$$\mathbb{E}Q_S = (1 - \alpha) \cdot 1 + \alpha(2 - 2^{-(2m-2)}) < 1 + \alpha < C.$$

This proves the claim. $\square$

The lower bound from a fixed deterministic branch no longer applies: the algorithm may decide to enter a long branch with a small probability supplied by a free private coin. If a coin were instead generated by observing an additional hat, that observation would have to be charged as a query. The theorem explicitly concerns the free-randomness model. It proves sufficiency for every $C > 1$; it does not determine the optimal randomized threshold.

[write] Added 4 October 2026, batch 92. Source 04 treats the other convention, a free *public* coin with costs averaged over it, in Section 60: mixing the optimal strategy with probability ϵ and a zero-query strategy gives uniform expected cost at most 2ϵ with divergence of probability at least ϵ , so its positive-probability lower bound fails in that model. The two observations are complementary; the sharp randomized thresholds are Research questions 57.2 and 68.6.

56 Finite certificates, reproducibility, and formalization

56.1 Executed checks

The package contains two independent Python programs using only the standard library. Their outputs use exact integers and rational numbers for the mathematical checks. Assertions in these programs are explicit runtime checks, rather than checks disabled by optimized Python execution.

The block and query verifier. `verify_hat_queries.py` executes the actual query algorithms through a guarded oracle. It exhaustively checks every physical hat assignment for $m = 1, \dots, 8$.

Quantity	Executed count
Physical assignments	87,380
Primary player executions	1,339,848
Reexecutions after flipping the owner's hat	1,339,848
Queries in primary executions	2,009,628

The verifier checks that no player queries its own hat or repeats a query, and that both the guess and the complete query transcript are unchanged when the owner's hat is flipped. It compares the executable rule with an independently expressed declarative rule, checks individual fairness and the complete three-case block law, verifies every claimed prefix bound, and computes the exact query histogram and geometric tail at each player position. It also checks the integer stage accounting for $\varepsilon = 1$ through stage 80. The result is `hat_query_verification.json`.

The orbit verifier. `verify_orbit_bounds.py` exhausts all one-query profiles on teams of size $1, \dots, 5$. There are $2n$ distinct rules for each player: the two constants, and a signed copy of any of the other $n - 1$ hats. Hence there are $(2n)^n$ profiles.

n	Profiles	Profile/assignment pairs	Minimum $\mathbb{P}(F \leq 0)$
1	2	4	1/2
2	16	64	1/2
3	216	1,728	1/4
4	4,096	65,536	1/2
5	100,000	3,200,000	1/4
Total	104,330	3,267,332	

It also constructs the labels of Theorem 50.1 for nonlinear fixed-visibility examples of bounds 2 and 3, and for a genuinely adaptive depth-2 multiplexer example. It checks exact Walsh reconstruction, the support-incidence bound on every induced vertex set, each nonzero support-label sum, and every actual score orbit. These examples account for another 5, 120 assignment/group pairs. The result is `orbit_verification.json`.

56.2 What the computations establish

Both programs passed. The finite enumeration is useful because an own-coordinate dependency, a factor-of-two error in surplus, or a prefix error can invalidate an otherwise plausible infinite proof. The tests exercise exactly those risks. They do not prove the all-parameter theorem, the probabilistic limit, the category statement, or historical priority. Those mathematical claims stand on their proofs in the article.

To reproduce the certificates and compile the article, run:

```
python3 verify_hat_queries.py
python3 verify_orbit_bounds.py
pdflatex -interaction=nonstopmode -halt-on-error hat_query_thresholds.tex
pdflatex -interaction=nonstopmode -halt-on-error hat_query_thresholds.tex
```

The main TeX source includes its bibliography and requires no downloaded image assets or external bibliography database.

[write] Added 4 October 2026, batch 92. These commands and file names are the delivery's. In the collection the programs are shipped as `code/05-hat-queries-verify_hat_queries.py` and `code/05-hat-queries-verify_orbit_bounds.py`, their outputs as `data/05-hat-queries-hat_query_verification.json` and `data/05-hat-queries-orbit_verification.json`, and the source notes as `05-hat-queries-SOURCE_NOTES.txt`, all byte-identical to the delivery; the manuscript's source and PDF, the delivered `README.txt` and the checksum file are not shipped and survive in the archive of the arrival commit `afd7ffabb`. By default each program writes its report beside itself under the delivery name, so run in place it would add an unprefixed file to the report's `code/`; the report's `README.md` reruns them on a scratch copy, or with `-output` into a scratch directory. The counts above are those of the shipped JSON files, reproduced at the write (Section 45.5).

56.3 A practical route to proof-assistant verification

The existing ProveIt list-coding project supplies finite-list validity, decoding, and round-trip interfaces [28]. Those interfaces can encode finite trees and certificates, but do not by themselves validate a probabilistic or Fourier assertion. A suitable development would separate the following obligations:

1. **Query semantics and blindness.** Represent a finite binary decision tree with a forbidden owner coordinate; prove that evaluation and its transcript are invariant under changing the owner hat. Prove the depth and path-probability lemmas by induction.
2. **Finite score and block law.** Use the integer score $F = 2D$ to avoid fractions in the executable semantics. Formalize the three configurations of Proposition 52.2, then prove the exact query-tail formula and the pair-prefix bound for arbitrary m .
3. **Finite group certificates.** A certificate gives a binary vector for each coordinate. A checker verifies that every displayed score-support sum is nonzero. Its soundness theorem should derive orbit cancellation and the finite sign probability bound. The certificate is meaningful only after the displayed Walsh representation is also verified.
4. **Infinite measure and category.** Prove the finite-color theorem using the product-space group action,

then prove the cylinder-density step and the Baire intersection step. Alternatively, formalize the conditional sign criterion.

5. **Scheduling and limits.** Formalize disjoint blocks, independence, the two summability calculations, the indicator-counting lemma, and the two-matchings prefix transfer. Finally connect the deterministic schedules to their asymptotic equivalents.

The shortest complete mathematical milestone is the bounded-query nullity theorem via finite colorings and cylinder density. The shortest useful finite milestone is the exact one-query $3/4$ theorem. The expected-cost threshold then requires only the path-cost lemma and the already isolated positive block construction. No claim is made that these additional formalization steps have been completed.

57 Further research questions

The following questions are proposed continuations. Except for the explicitly cited starting point, they are questions formulated by this article, not a claim that each is a recognized longstanding open problem.

Research question 57.1 (Optimal finite constants after one query). Determine

$$\alpha_d = \inf_{\text{finite legal teams with depth at most } d} \mathbb{P}(F \leq 0)$$

for $d \geq 2$, and its asymptotic behavior as $d \rightarrow \infty$. Is the factor d in the sparse-support bound necessary? Does adaptive choice of queries improve the optimum over a predetermined visible set at the same count?

Research question 57.2 (The exact randomized expected-cost threshold). Theorem 55.2 gives almost-sure divergence under every uniform expected cap $C > 1$. Is 1 optimal when random bits are free and private? Is the boundary value attained? Compare positive probability with almost-sure divergence, and compare private coins, shared coins, and randomness generated by charged hat queries.

Research question 57.3 (Growth at the deterministic critical budget). Under the sharp uniform cap $\sup_i \mathbb{E}Q_i \leq 2$, how quickly can D_n tend to infinity almost surely? Our explicit schedules give sublogarithmic rates with sharp equivalents. Can one obtain $D_n \asymp \log n$, or faster, while retaining the geometric tail $\mathbb{P}(Q_i > t) \leq 2^{-t}$? The earlier unrestricted-growth theorem does not impose this expected-cost constraint.

Research question 57.4 (Count budgets with infinitely many zeros). Classify arbitrary finite-valued $q(i) \geq 0$, dropping the eventual lower bound 1 in Theorem 54.2. How do the frequency and placement of zero-query players interact with the capacity of the high-budget players to compensate for their fluctuations?

Research question 57.5 (A structural visibility criterion). Characterize directed visibility graphs that admit almost-sure divergent positive surplus. Finite colorability of the undirected visibility graph is an obstruction, and acyclicity of the directed graph forces independent correctness. Which genuinely cyclic structures suffice? Can the problem be reduced to disjoint finite subgames with suitable gain and loss probabilities?

Research question 57.6 (Distance budgets without convergence). Theorems 54.3 and 54.4 leave a gap between $h(i) \rightarrow \infty$ and mere unboundedness. Find a useful necessary and sufficient condition on an arbitrary distance profile. In particular, can one express it through the arrangement of intervals $[i - h(i), i + h(i)]$, rather than through a difficult optimization over strategies?

Research question 57.7 (Sharp support-label certificates). Theorem 50.1 uses a greedy upper bound on the dimension of binary labels. Determine the smallest label space for the score hypergraph of a given strategy. Which improvements survive conditioning on arbitrary finite cylinders? An efficient algorithm with a short certificate could improve finite bounds without formalizing general hypercontractivity.

Research question 57.8 (Spectral structure of critical-cost success). How sharp is Proposition 49.4 for legal hat scores? Compute the degree distribution of the efficient block construction and identify exactly where its negative-debt variance resides. Can aggregate Fourier information distinguish rates attainable with the same query-tail bound?

Research question 57.9 (More colors and biased hats). For q equally likely colors, a deterministic branch of length L has probability q^{-L} , so the corresponding path-cost scale is $q/(q-1)$. Does that scale govern any sharp expected-query threshold for excess over n/q , or do compulsory neutral constructions require additional information? For biased hats, first specify the correct benchmark; blindly subtracting $n/2$ no longer expresses individual fairness.

Research question 57.10 (Average cost across the team). The threshold 2 concerns the supremum over individual expected costs. It is different from

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n \mathbb{E}Q_i.$$

Our active pairs already have mean cost below $3/2$, and sparse neutral padding changes the team average further. Determine the sharp average-cost constraints compatible with almost-sure growth and prescribed rates.

Research question 57.11 (Fluctuations and inverse waiting times). For the critical schedule, positive blocks are independent rare events, whereas negative debt is finite almost surely but has infinite expectation. Obtain fluctuation theorems around (53.2)–(53.3), and asymptotic laws for the first index after which the excess permanently exceeds a given level. Such laws should distinguish random gain fluctuations from the heavy distribution of the last bad block.

Research question 57.12 (Foundational and effective strength). Locate the precise logical strength of the countable coloring, cylinder-density, and category arguments. Does the bounded-depth nullity theorem admit a particularly weak formalization when strategies are given by computable trees? Can one obtain effective null or meager covers uniformly from a bound on the trees? A proof should distinguish a classical null set from an effectively null set, and a finite certificate from its soundness theorem.

[write] **Status of the questions, 4 October 2026, batch 92.** All twelve questions stay open as printed; the write searched no literature for them. Several coincide with source 04's questions of Section 68, as the note there lists. Two of source 04's results bear on them without answering them: its Theorem 65.5 gives finite teams of depth $k = 2^{r-1} - 1$ with $\mathbb{P}(F \leq 0) = 2^{-r}$, so $\alpha_k \leq 1/(2(k+1))$ for those k in question 57.1, against the lower bounds of (50.4); and its Section 63 computes the endpoint central limit theorem, the variance and the realized and maximal costs of the critical schedule, the first steps of questions 57.10–57.11.

58 Conclusions and contribution ledger

The repository's bounded-inspection question has a negative answer in a stronger form: under any fixed adaptive query bound, divergent excess has probability zero and occurs on a meager set. The exact qualitative classification for nonzero count budgets is unboundedness, even when the high-budget players are arbitrarily sparse. Distance budgets obey a different criterion, as witnessed by the explicit three-colorable counterexample.

The central quantitative boundary is (53.6). A uniform deterministic expected cost strictly below two forces bounded depth. At two, an efficient implementation of a known construction gives almost-sure divergence with exponentially decaying query tails. Free private randomization changes that optimization problem.

Ingredient or conclusion	Attribution and status
Pairs strategy and rare-event block rule	Established mechanism from Eldredge [10]; restated and implemented with explicit early stopping.
Boolean hypercontractivity and one-sided sign inequality	Classical; Bonami and the exact textbook statements in O'Donnell [24, 25].
Finite-color null and category obstruction	Proved here, including arbitrary measurable infinite-neighborhood rules and the cylinder-preserving argument.
Sparse-support bounds and the one-query optimum	Proved here; finite examples and exhaustive small-team enumeration executed. Historical priority not established.
Expected-cost threshold 2	Proved here by combining the universal path lower bound with the optimized known block construction.
Unbounded count-budget classification and distance separation	Proved here, with all-prefix and computability details.
Exact schedule equivalents and geometric query tails	Derived and proved here for the displayed implementations and schedules.
Private-randomness sufficiency for every $C > 1$	Proved here; the optimal randomized boundary is left open.
Proof-assistant certification	Not performed; the executable finite checks and the proposed formalization obligations are stated separately.

59 The question and the research intersection

The central conclusion is that *two expected inspections per player are exactly sufficient*, with the expectation interpreted uniformly across players and the strategies required to be deterministic functions of the hats. This is an exact threshold, not an asymptotic order estimate. A stronger theorem identifies the entire optimal family of increasing query costs with a geometric law.

The game is simple to state. Infinitely many players receive independent fair black or white hats. Each player may inspect other hats, possibly choosing the next inspection based on earlier observations. No player may inspect their own hat or communicate after the hats are assigned. Every player must guess. If S_n counts the correct guesses among the first n players, individual fairness gives $\mathbb{E}S_n = n/2$. Nevertheless, a suitable strategy can make $S_n - n/2$ tend to $+\infty$ on almost every hat assignment. Eldredge proved this last assertion in Proposition 6.6 of [10].

This topic has a direct, documented connection to Elliot Glazer: the paragraph preceding that proposition credits Glazer and Charles Wang with suggesting the study of excess correct guesses. Glazer's institutional biography also identifies set theory, formal systems, choice paradoxes, and finite and infinite puzzles among his interests [35]. Lietz and Winkel's work gives another explicit connection between a Glazer puzzle and finite-inspection infinite hat games [34]. We rely on these accessible sources; the LinkedIn page supplied with the request did not return readable content during the source review.

The ProveIt connection is equally specific. Its earlier manuscript *No Universal Sublinear Bound for Measurable Hat Guessing* asks in Research question 13.2 whether a uniform bound on the number of inspected hats permits $D_n \rightarrow +\infty$ almost surely [26]. The present article answers that question negatively,

and strengthens the conclusion to probability zero. The repository’s older measurable-box-games report separately asks about finite teamwork and the costs of adaptive observations [27]; our finite optimality results address a precise family within that program.

59.1 What the preceding work already established

The predecessor already supplies strategies with logarithmic excess and strategies exceeding any prescribed positive sublinear scale, using increasingly large syndrome blocks. Those growth conclusions are not claimed as new here. Eldredge’s rare all-black blocks already establish the underlying possibility of almost-sure divergent excess. The improvement here is to identify the optimal *cost of reading the hats*, including the distinction between a maximum depth and an expectation. The lower bound requires a uniform sign estimate for Boolean polynomials, followed by a conditional argument that excludes positive-probability divergence.

Classical hypercontractivity supplies the analytic estimate [30, 31]. We include a direct proof of a sufficient version, so the main argument can be read without a separate reference on Boolean analysis. The finite-field construction uses standard binary simplex geometry. Once the adaptive-tree argument has forced affinity, the classification of its constant-weight dual is a classical coding theorem of Bonisoli [33], whose binary case is also proved directly below.

Component	Mathematical role	Attribution/status
Individual fairness	$\mathbb{E}S_n = n/2$, despite dependence between correctness indicators	Classical
Rare exceptional blocks	Almost-sure divergent excess from summable failures and infinitely many gains	Eldredge [10]
Bounded-degree sign bounds	A centered polynomial of fixed degree cannot be positive with probability arbitrarily close to one	Classical analytic tool
Conditional obstruction	Bounded query depth implies $\mathbb{P}(D_n \rightarrow +\infty) = 0$, without a locality assumption	Proved application
Geometric cost frontier	Simultaneous sharp lower and upper bounds for every increasing query cost	Proved classification
Variance and finite depth	Sharp expected-query variance bound; optimal finite-team depth and failure-set structure	Proved refinements
Formal verification	Explicit finite checks and audited interfaces for later kernel verification	Infinite proof not formalized

59.2 Snapshot and the scope of the novelty claim

The repository was inspected at the fixed revision

9fe62865dc30d71256183466b744b98ebf7e64d6.

The immediate predecessor occurs in `docs/incoming/hat_guessing_research.zip`. The accompanying `SOURCE_AUDIT.txt` records the relevant paths, source identifiers, original questions, and trust boundaries. The scoped repository and literature review did not locate the geometric cost classification stated here. This is a limited bibliographic finding, not a proof of historical priority.

The concrete accomplishment is a proof of the named repository question together with an exact quantitative refinement. We do not claim to settle an unrelated famous open problem, or to have obtained an endorsement from the researchers whose work motivates the article.

[write] **Added 4 October 2026, batch 92.** Sections 59–69 are source 04 (Section 45.5 gives its provenance). Here “the preceding ProveIt research report” and *No Universal Sublinear Bound for Measurable Hat Guessing* [26] are Part II, its Research question 13.2 is Research question 29.2, and “the older measurable-box-games report” [27] is Part I, whose finite-teamwork question is Q9. The archive docs/incoming/hat_guessing_research.zip was retired by Part II’s placement and survives in its arrival commit a162e4386; the accompanying audit is shipped as 04-hat-inspection-SOURCE_AUDIT.txt, with the proof-status record 04-hat-inspection-PROOF_STATUS.txt. Source 06 (Sections 45–58), written independently against the same revision, answers the same repository question by the same construction; the answer is not this source’s alone, and neither source establishes priority over the other. The answer to Research question 29.2 is recorded after Section 54, the further finite results after the questions of Sections 15 and 29. The LinkedIn page mentioned above is named by no address, and none is recorded here.

60 The model and the main statements

60.1 Hats, guesses, and decision trees

Work on the product probability space $\Omega = \{0, 1\}^{\mathbb{N}_{>0}}$ with independent fair coordinates x_i . A deterministic adaptive strategy for player i is a binary query tree. Each internal node names a coordinate $j \neq i$, the observed value selects the next branch, and a leaf returns the guessed bit. The tree may refer to arbitrarily distant coordinates. All trees are fixed before the hats are assigned.

Let $Q_i(x)$ be the number of queries on input x . We permit almost-surely terminating trees when stating lower bounds. The construction proving sharpness uses finite trees and terminates on every input. Repeated inspections of a previously read coordinate can always be removed by retaining its answer; a *reduced query implementation* below makes no such repeated inspections. Unnecessary queries can only increase the costs being minimized.

Write $\hat{x}_i$ for player i ’s output and set

$$S_n = \sum_{i=1}^n \mathbf{1}_{\{\hat{x}_i = x_i\}}, \quad D_n = S_n - \frac{n}{2}, \quad T_n = 2D_n.$$

For Boolean Fourier arguments it is convenient to use the signs $\varepsilon_i = (-1)^{x_i}$ and $g_i = (-1)^{\hat{x}_i}$. Then

$$Y_i = \varepsilon_i g_i = 2\mathbf{1}_{\{\hat{x}_i = x_i\}} - 1, \quad T_n = \sum_{i=1}^n Y_i.$$

The guess g_i is independent of the coordinate ε_i . Thus $\mathbb{E}Y_i = 0$, $\mathbb{E}T_n = 0$, and $\mathbb{E}S_n = n/2$. No independence between different Y_i ’s is asserted.

The *reachable depth* d_i is the supremum of the lengths of consistent query paths. A uniform depth bound means that one finite integer k satisfies $d_i \leq k$ for every player. A uniform expected cost bound is a bound on $\sup_i \mathbb{E}Q_i$. These are different requirements.

60.2 A complete frontier, stated before its proof

Let G be geometric with success probability $1/2$, on the positive integers:

$$\mathbb{P}(G = r) = 2^{-r} \quad (r \geq 1), \quad \mathbb{P}(G > r) = 2^{-r} \quad (r \geq 0).$$

For a nonnegative nondecreasing function $\phi : \mathbb{N} \rightarrow [0, \infty]$, define the uniform cost

$$C_\phi(\mathcal{A}) = \sup_{i \geq 1} \mathbb{E}\phi(Q_i).$$

Here $\mathbb{N}$ includes 0 when it is the domain of a cost function.

Theorem 60.1 (Exact geometric frontier). *In the deterministic adaptive binary hat model:*

1. *If the reachable query depths are uniformly bounded, then $\mathbb{P}(D_n \rightarrow +\infty) = 0$.*
2. *Every strategy $\mathcal{A}$ with $\mathbb{P}(D_n \rightarrow +\infty) > 0$ satisfies*

$$C_\phi(\mathcal{A}) \geq \mathbb{E}\phi(G) \quad \text{for every nonnegative nondecreasing } \phi.$$

3. *There exists one strategy $\mathcal{A}_*$, using only finite trees, for which $D_n \rightarrow +\infty$ almost surely and*

$$C_\phi(\mathcal{A}_*) = \mathbb{E}\phi(G) \quad \text{for all these } \phi \text{ simultaneously.}$$

The same strategy satisfies $S_n/n \rightarrow 1/2$ almost surely.

The lower bound is proved in the sections on the conditional obstruction and decision-tree costs. The upper bound is the explicit early-stopping implementation in the construction section. Allowing infinite values of $\mathbb{E}\phi(G)$ makes the theorem include genuinely impossible uniform costs.

Uniform cost	Exact optimum for divergent excess	Attainment
$\sup_i \mathbb{E}Q_i$	2	$\mathcal{A}_*$
$\sup_i \mathbb{E}Q_i^2$	6	$\mathcal{A}_*$
$\sup_i \mathbb{E}Q_i^3$	26	$\mathcal{A}_*$
$\sup_i \mathbb{P}(Q_i > r), r \geq 0$	2^{-r}	$\mathcal{A}_*$
$\sup_i \mathbb{E}z^{Q_i}, 1 \leq z < 2$	$z/(2-z)$	$\mathcal{A}_*$
$\sup_i \mathbb{E}2^{Q_i}$	$+\infty$	No finite bound possible

The critical mean assertion is $\sup_i \mathbb{E}Q_i = 2$. In the realizing strategy, every individual expectation can be strictly below 2. What is impossible is a *common gap*: $\sup_i \mathbb{E}Q_i \leq 2 - \epsilon$ for some $\epsilon > 0$.

Theorem 60.2 (Two further quantitative conclusions). *For any almost-surely terminating strategy with finite expected query costs,*

$$\text{Var}(T_n) \leq n + \sum_{i=1}^n \mathbb{E}Q_i.$$

The constant is sharp. For a finite team of N players attaining

$$\mathbb{P}(S_N \geq s) = \frac{N}{2s}, \quad s > N/2,$$

every uniform worst-case query bound is at least $s - 1$. Such a finite optimum can be attained at depth $s - 1$ if and only if

$$N = m(2^r - 1), \quad s = m2^{r-1} \quad \text{for some integers } m, r \geq 1.$$

Every minimum-depth extremizer has an affine failure set whose parity-check matrix contains each nonzero binary column exactly m times. An explicit nonadaptive trace strategy realizes every allowed parameter pair.

60.3 Two distinctions that the hypotheses must preserve

First, divergence is stronger than eventual positivity. A bounded-depth strategy can have positive probability that $D_n > 0$ eventually while the excess stays bounded; Remark 61.4 gives an explicit example. The theorem concerns unbounded persistent growth.

Second, the source of randomness matters for query cost. The theorem counts randomness obtained by inspecting hats. If a free public coin is allowed and its cost is ignored, run $\mathcal{A}_*$ with public probability ϵ , and otherwise use a zero-query strategy. The uniform expected cost is at most 2ϵ , while divergent excess has probability at least ϵ . Thus the positive-probability lower bound is false for this different cost convention. A theorem with external randomization requires a budget conditional on the random seed or a separate analysis.

The order in which inspections are performed is local to each player; it does not give players access to other players' guesses or transcripts. None of the proofs uses post-assignment communication.

[write] Added 4 October 2026, batch 92. Theorem 60.1(1) is the probability part of source 06's Corollary 48.4 (proved here as Theorem 61.3); its part (2), for every nondecreasing cost, is this source's alone (Theorem 62.2); its part (3) is Propositions 63.2 and Theorems 63.3–64.5; its mean case is source 06's Theorem 53.3. Theorem 60.2 is this source's alone (Theorem 64.2 and Section 65). The public-coin remark above and source 06's Theorem 55.2 on private coins are complementary (the note after it).

61 Bounded inspection cannot produce a divergent surplus

Throughout this section the hats $(\varepsilon_i)_{i \in \mathbb{N}}$ are independent and uniform on $\{-1, 1\}$. A legal guess $g_i \in \{-1, 1\}$ is a measurable function of the hats other than ε_i . Write

$$Y_i = \varepsilon_i g_i, \quad T_n = \sum_{i=1}^n Y_i = 2S_n - n,$$

where S_n counts the correct guesses. Thus $\mathbb{E}Y_i = 0$, although the variables Y_i need not be independent or form martingale differences. An inspection algorithm is a deterministic binary decision tree: the next inspected coordinate and the eventual guess depend on the answers already received. It never inspects its own hat. We count individual hat inspections, and denote this count by Q_i . Unreachable branches can be deleted and repeated inspections can be removed without changing the guess or increasing its cost. These conventions also apply to trees which terminate almost surely rather than on every input.

61.1 A dimension-free polynomial lemma

For a finite set of coordinates, write $\chi_A(\varepsilon) = \prod_{j \in A} \varepsilon_j$ for the Walsh characters. The Walsh degree of $f = \sum_A \widehat{f}(A) \chi_A$ is the largest $|A|$ for which $\widehat{f}(A) \neq 0$. A depth- k decision tree computes a function of Walsh degree at most k : each leaf is specified by at most k answers, and its indicator is a product of at most k factors $(1 \pm \varepsilon_j)/2$. Consequently, if every player uses depth at most k , then T_n has degree at most $k + 1$. It depends on finitely many coordinates even when inspected indices are chosen adaptively: each of the first n trees has finitely many nodes.

We give the elementary hypercontractive estimate needed below. This also isolates precisely which standard analytic ingredient enters the argument; see the hypercontractivity discussion in [30, 31].

[write] Added 4 October 2026, batch 92. The degree bound above is source 06's Lemma 49.1, and Lemmas 61.1–61.2 state the estimates proved inside source 06's Lemma 49.2 (inequality (49.1) and its last step). They are printed here because this source's later proofs cite them. The proofs are source 04's own: the induction runs for the noise operator

$\mathcal{T}_{1/\sqrt{3}}$ rather than for the weighted coefficients of (49.1), an equivalent form of the same Bonami induction; the second paragraph of the proof of Lemma 61.2 derives the sharper constant, which source 06 cites.

Lemma 61.1 (A fourth-moment bound). *If f is a real Walsh polynomial of degree at most d , then*

$$\|f\|_4 \leq 3^{d/2} \|f\|_2, \quad \|f\|_1 \geq 3^{-d} \|f\|_2.$$

Both estimates are independent of the number of coordinates.

Proof. Let $\mathcal{T}_\rho$ multiply the Walsh coefficient of χ_A by $\rho^{|A|}$. We first prove

$$\left\| \mathcal{T}_{1/\sqrt{3}} h \right\|_4 \leq \|h\|_2 \quad (61.1a)$$

by induction on the number of coordinates. Decompose $h(x', x_N) = a(x') + x_N b(x')$ and put $A = \mathcal{T}_{1/\sqrt{3}} a$, $B = \mathcal{T}_{1/\sqrt{3}} b$. Averaging first over x_N gives

$$\mathbb{E}(\mathcal{T}_{1/\sqrt{3}} h)^4 = \mathbb{E} \left(A^4 + 2A^2 B^2 + \frac{1}{9} B^4 \right).$$

Cauchy–Schwarz and the induction hypothesis bound the right side by

$$\|a\|_2^4 + 2 \|a\|_2^2 \|b\|_2^2 + \frac{1}{9} \|b\|_2^4 \leq (\|a\|_2^2 + \|b\|_2^2)^2 = \|h\|_2^4.$$

The zero-coordinate case is immediate, proving the induction. Now apply this result to $h = \sum_A 3^{|A|/2} \widehat{f}(A) \chi_A$. Then $\mathcal{T}_{1/\sqrt{3}} h = f$, and Parseval gives $\|h\|_2 \leq 3^{d/2} \|f\|_2$. Finally, Hölder interpolation gives $\|f\|_2 \leq \|f\|_1^{1/3} \|f\|_4^{2/3}$, which yields the asserted first-moment lower bound. $\square$

Lemma 61.2 (Both signs have positive probability). *Let $P \neq 0$ be a real Walsh polynomial of degree at most d with $\mathbb{E}P = 0$. Then*

$$\min\{\mathbb{P}(P < 0), \mathbb{P}(P > 0)\} \geq \frac{1}{4 \cdot 9^d}. \quad (61.2a)$$

The standard general hypercontractive inequality improves the right side to $e^{-2d}/4$.

Proof. Set $P_+ = \max(P, 0)$ and $P_- = \max(-P, 0)$. The zero mean gives $\mathbb{E}P_+ = \mathbb{E}P_- = \|P\|_1/2$. Cauchy–Schwarz therefore gives, for either sign,

$$\frac{1}{4} \|P\|_1^2 \leq \|P\|_2^2 \mathbb{P}(P > 0), \quad \frac{1}{4} \|P\|_1^2 \leq \|P\|_2^2 \mathbb{P}(P < 0).$$

Lemma 61.1 proves the displayed bound.

For completeness, the sharper constant follows from the usual degree- d hypercontractive estimate $\|P\|_p \leq (p-1)^{d/2} \|P\|_2$ for $p > 2$. Interpolating between L^1 and L^p gives

$$\|P\|_1 \geq (p-1)^{-dp/(2(p-2))} \|P\|_2.$$

Letting $p \downarrow 2$ yields $\|P\|_1 \geq e^{-d} \|P\|_2$, and the preceding Cauchy–Schwarz argument applies. This sign-balance argument is standard; see Austrin–Håstad [31, Theorem 2.6 and Corollary 4.2] for this Boolean estimate and its applications. $\square$

61.2 The conditional obstruction

The preceding lemma immediately gives a uniform upper bound below one on $\mathbb{P}(T_n > 0)$. That observation alone rules out almost-sure divergence, but does not rule out divergence on an event of intermediate probability. Conditioning on finitely many hats closes this gap.

Theorem 61.3 (Uniform bounded depth). *Suppose that every player uses a deterministic legal decision tree of depth at most the same finite integer k . Then*

$$\mathbb{P}(T_n \longrightarrow +\infty) = \mathbb{P}(T_n \longrightarrow -\infty) = 0.$$

No bound on the number of players who inspect a given hat is required.

[write] **Added 4 October 2026, batch 92.** This is the probability part of source 06's Corollary 48.4, there proved by finite colourings (with category as well). The proof below is source 04's, a second route: the analytic one that source 06 sketches after Corollary 49.3, with source 06's Lemma 47.1 replaced by the martingale convergence theorem.

Second proof (source 04). Let $c_k = 1/(4 \cdot 9^{k+1})$ and fix a cylinder event C specifying the values of a finite set F of hats. Put $m = |F|$. On this conditional probability space, split

$$T_n = U_n + V_n, \quad U_n = \sum_{\substack{i \leq n \\ i \notin F}} \varepsilon_i g_i, \quad V_n = \sum_{\substack{i \leq n \\ i \in F}} \varepsilon_i g_i.$$

For $i \notin F$, its own hat is still an independent fair bit and the guess still does not depend on it. Hence $\mathbb{E}(U_n \mid C) = 0$. Fixing coordinates does not increase Walsh degree, so U_n has conditional degree at most $k + 1$. Also $|V_n| \leq m$. If U_n vanishes identically, its probability of being nonpositive is one; otherwise Lemma 61.2 applies. In either case,

$$\mathbb{P}(T_n \leq m \mid C) \geq \mathbb{P}(U_n \leq 0 \mid C) \geq c_k. \quad (61.3a)$$

It follows that $\mathbb{P}(T_n \leq m \text{ infinitely often} \mid C) \geq c_k$: the probability of a limsup of events is at least the limsup of their probabilities. For $A = \{T_n \rightarrow +\infty\}$ we obtain

$$\mathbb{P}(A \mid C) \leq 1 - c_k \quad (61.3b)$$

for every finite cylinder C .

Let $\mathcal{F}_m = \sigma(\varepsilon_1, \dots, \varepsilon_m)$. The last bound holds on every atom of $\mathcal{F}_m$, so $\mathbb{E}(\mathbf{1}_{\{A\}} \mid \mathcal{F}_m) \leq 1 - c_k$ almost surely for every m . The upward martingale convergence theorem gives $\mathbb{E}(\mathbf{1}_{\{A\}} \mid \mathcal{F}_m) \rightarrow \mathbf{1}_{\{A\}}$ almost surely. Thus $\mathbf{1}_{\{A\}} \leq 1 - c_k$ almost surely and $\mathbb{P}(A) = 0$. Applying the same argument to the legal guesses $-g_i$ proves the claim about negative divergence. The sharper constant in Lemma 61.2 can replace c_k throughout. $\square$

Remark 61.4 (Divergence is stronger than an eventual strict majority). It would be false to replace divergence by eventual positivity in Theorem 61.3. Let the first two players guess $+1$. Partition the remaining players into consecutive pairs (a, b) , with $g_a = \varepsilon_b$ and $g_b = -\varepsilon_a$. Each pair has total score zero. On the probability-1/4 event $\varepsilon_1 = \varepsilon_2 = +1$, every subsequent prefix has score 1, 2, or 3. This depth-one strategy has an eventual strict majority with positive probability, but its surplus stays bounded. The conditional proof above avoids any unsupported assertion that eventual positivity is a tail event.

62 The geometric lower frontier for inspection costs

The obstruction has an exact consequence for inspection complexity. Let G be a geometric random variable with success probability $1/2$, supported on the positive integers:

$$\mathbb{P}(G = s) = 2^{-s} \quad (s \geq 1), \quad \mathbb{P}(G > r) = 2^{-r} \quad (r \geq 0).$$

We first record the cost of merely having a long reachable path.

Lemma 62.1 (A path forces geometric cost). *If a deterministic decision tree has a reachable path of length d , then its inspection count Q stochastically dominates $\min(G, d)$. In particular,*

$$\mathbb{P}(Q > r) \geq 2^{-r} \quad (0 \leq r < d), \quad \mathbb{E}Q \geq 2 - 2^{1-d}.$$

For $d = 0$ the second inequality reads $\mathbb{E}Q \geq 0$.

[write] **Added 4 October 2026, batch 92.** This is source 06's Lemma 51.1 (with d for L), stated as a stochastic domination. Source 04's proof is source 06's argument: following the first $r < d$ answers of the path has probability at least 2^{-r} and forces $Q > r$, which is the domination; summing the tails gives the mean bound; repeated queries only raise the probability of a consistent prefix; and every finite prefix of a reachable infinite branch has positive probability, so a tree of unbounded depth has $\mathbb{E}Q \geq 2$. It is not printed a second time; the proof of Lemma 51.1 serves.

Theorem 62.2 (Universal monotone-cost lower bound). *Consider a deterministic legal strategy for which $\mathbb{P}(T_n \rightarrow +\infty) > 0$. For every nonnegative nondecreasing function $\varphi : \{0, 1, 2, \dots\} \rightarrow [0, \infty]$,*

$$\sup_i \mathbb{E}\varphi(Q_i) \geq \mathbb{E}\varphi(G). \quad (62.2a)$$

In particular, for each integer $r \geq 0$,

$$\sup_i \mathbb{P}(Q_i > r) \geq 2^{-r}. \quad (62.2b)$$

Proof. Theorem 61.3 implies that the reachable depths of the players' trees cannot have a common finite bound. Consequently, for every d there is a player with a reachable path of length at least d . Lemma 62.1 gives, for that player, $\mathbb{E}\varphi(Q_i) \geq \mathbb{E}\varphi(\min(G, d))$. Taking the supremum over players and then letting $d \rightarrow \infty$ proves the first assertion by monotone convergence. Alternatively, applying the same lemma to a path longer than r proves the tail assertion directly. It matters that one sufficiently deep player satisfies all the first d tail bounds simultaneously; no interchange of a supremum and an infinite sum is being assumed. $\square$

Corollary 62.3 (A uniform gap below two is impossible). *If $\sup_i \mathbb{E}Q_i \leq C < 2$, then $\mathbb{P}(T_n \rightarrow +\infty) = 0$. More precisely, every player's reachable depth is at most*

$$\lfloor 1 - \log_2(2 - C) \rfloor.$$

Proof. Lemma 62.1 bounds any reachable depth d by $2 - 2^{1-d} \leq C$, which rearranges to the displayed inequality. Theorem 61.3 then applies. $\square$

Remark 62.4 (Pointwise strictness does not give a uniform gap). The hypothesis is a common bound $C < 2$. The collection of individual inequalities $\mathbb{E}Q_i < 2$ does not imply it: the expectations may approach two as the players' finite worst-case depths grow. The positive construction in this article exploits exactly this distinction.

Corollary 62.5 (Moments, generating functions, and tail envelopes). *Every strategy satisfying the hypothesis of Theorem 62.2 obeys*

$$\sup_i \mathbb{E}Q_i \geq 2, \quad \sup_i \mathbb{E}Q_i^2 \geq 6, \quad \sup_i \mathbb{E}Q_i^s \geq \sum_{j \geq 1} j^s 2^{-j} \quad (s > 0).$$

For $1 \leq z < 2$ it obeys

$$\sup_i \mathbb{E}z^{Q_i} \geq \frac{z}{2-z},$$

whereas $\sup_i \mathbb{E}z^{Q_i} = \infty$ for every $z \geq 2$. Moreover, a common tail bound $\mathbb{P}(Q_i > r) \leq b_r$ rules out positive-probability divergence if $b_r < 2^{-r}$ at even one integer $r \geq 0$.

Proof. Apply Theorem 62.2 with $\varphi(q) = q^s$ and $\varphi(q) = z^q$. The geometric series gives $\mathbb{E}z^G = \sum_{j \geq 1} (z/2)^j = z/(2-z)$ for $z < 2$; its divergence for $z \geq 2$ proves the corresponding assertion. Differentiation at $z = 1$, or direct summation, gives $\mathbb{E}G = 2$ and $\mathbb{E}G^2 = 6$. Finally, if $b_r < 2^{-r}$, no player can have a reachable path longer than r , by Lemma 62.1. Uniform bounded depth applies. $\square$

Remark 62.6 (How the positive construction completes the frontier). A successful construction for which $\mathbb{P}(Q_i > r) \leq 2^{-r}$ for every i, r has Q_i stochastically dominated by G . Combined with Theorem 62.2, this proves equality in every worst-player cost bound above, simultaneously for every nonnegative nondecreasing φ . Thus the constructive upper bound establishes an exact stochastic frontier, including the mean and all increasing moment costs. The lower bound concerns deterministic inspection algorithms. Free external random bits can select a large decision tree with a small probability without paying for the binary answers in its selection; the path-probability proof does not apply to that different cost model.

63 An optimal inspection implementation of Eldredge's blocks

The upper construction uses the finite block strategy in [10, Proposition 6.6]. Its probabilistic mechanism, including the existence of almost surely divergent excess, is due to Eldredge. The point here is to implement that strategy by short adaptive decision trees, compute their full cost distribution, and give a schedule with exact asymptotics. All decisions below are deterministic functions of observed hats. No external random bits, communication, or shared inspection transcript are available. Different players who inspect the same hat each pay for their own inspection.

63.1 The finite block and its early stopping algorithm

Use 0 for black and 1 for white. A block of $2k$ consecutive players is divided into adjacent ordered pairs. The first member of each pair is an S player and the second a T player. Ordinarily S copies the partner's color and T guesses its opposite. This gives exactly one correct guess in every pair, for every hat assignment.

Eldredge's exception instructs any player who sees only black hats elsewhere in the block to guess white. The following implementation computes exactly the resulting guesses.

1. A T player inspects the partner and guesses the opposite color. No further inspection is needed: if the exception applies, the partner is black and the prescribed guess is already white.
2. An S player first inspects the partner. If the partner is white, the player guesses white immediately. If the partner is black, the player inspects the remaining $2k - 2$ hats in a fixed order. At the first white hat, the player stops and guesses black. If every inspected hat is black, the player guesses white.

In particular, every player halts on every assignment, and no player inspects their own hat. The ordering of the remaining hats can be increasing player index; it has no probabilistic significance.

Lemma 63.1 (Exact block score). *Let Z be the number of correct guesses in this block minus k . Then*

configuration	Z	probability
all hats black	$-k$	4^{-k}
exactly one white hat, belonging to S	1	$k4^{-k}$
all other configurations	0	$1 - (k+1)4^{-k}$.

In a block of the second or third kind, the centered score is nondecreasing at pair boundaries. In a block of the first kind, its values at successive pair boundaries are $-1, -2, \dots, -k$.

[write] Added 4 October 2026, batch 92. This is source 06's Proposition 52.2 for $m = k$ pairs, and source 04's proof is that proposition's case analysis: all black, every pair wrong twice; a single white hat on an S player, its pair right twice; a single white hat on a T player, or two or more white hats, every pair right once; one all-black and k favourable configurations among the 4^k . The case analysis also gives the assertions about pair boundaries, including the values $-1, \dots, -k$ in an all-black block, where every pair contributes -1 . It is not printed a second time.

The case distinction is useful because it controls every prefix of the block, not only its total. Passing from an even prefix to the following odd prefix changes the centered score by exactly $1/2$ or $-1/2$.

Proposition 63.2 (The full inspection distribution). *For a T player, $Q = 1$. For an S player in a block of size $2k$,*

$$Q \stackrel{\text{law}}{=} \min\{G, 2k-1\}, \quad \mathbb{P}(G = j) = 2^{-j} \quad (j = 1, 2, \dots).$$

Consequently, for integer $r \geq 0$,

$$\mathbb{P}(Q > r) = \begin{cases} 2^{-r}, & 0 \leq r < 2k-1, \\ 0, & r \geq 2k-1, \end{cases} \quad \mathbb{E}Q = 2 - 4^{1-k}.$$

For a family containing arbitrarily large blocks and every nonnegative nondecreasing function $\phi : \{0, 1, \dots\} \rightarrow [0, \infty)$,

$$\sup_i \mathbb{E}\phi(Q_i) = \mathbb{E}\phi(G), \quad (63.1)$$

where either side may be infinite. In particular,

$$\sup_i \mathbb{E}Q_i = 2, \quad \sup_i \mathbb{P}(Q_i > r) = 2^{-r}, \quad (63.2)$$

$$\sup_i \mathbb{E}Q_i^a = \sum_{j=1}^{\infty} j^a 2^{-j} \quad (a > 0), \quad (63.3)$$

$$\sup_i \mathbb{E}e^{\theta Q_i} = \begin{cases} \frac{e^{\theta}}{2 - e^{\theta}}, & 0 \leq \theta < \log 2, \\ +\infty, & \theta \geq \log 2. \end{cases} \quad (63.4)$$

Every individual mean is strictly less than 2.

Proof. For S , the inspected hats are distinct independent fair bits, listed with the partner first. The procedure stops at the first white hat, or after all $2k-1$ available hats have been inspected. This gives the stated law. The tail-sum formula gives

$$\mathbb{E}Q = \sum_{r=0}^{2k-2} 2^{-r} = 2 - 2^{-(2k-2)}.$$

The variables $\min\{G, 2k - 1\}$ increase to G . Apply monotone convergence to ϕ ; T's constant cost 1 cannot increase the supremum. The remaining identities follow by choosing the indicated cost functions. For the exponential moment, the series is geometric with ratio $e^\theta/2$. $\square$

Thus the same implementation realizes the entire geometric cost envelope, including second moment 6 and every finite higher moment. Combined with the lower cost bound proved earlier, equation (63.1) attains all of those lower bounds simultaneously. This is stronger than merely giving an implementation of mean cost at most two.

63.2 A deterministic schedule and exact excess rates

Fix $1 < p \leq 2$. For each $k = 2, 3, \dots$, stage k consists of

$$R_k = \left\lfloor \frac{4^k}{k^p} \right\rfloor \quad (63.5)$$

independent consecutive blocks, each of size $2k$. Let

$$N_K = 2 \sum_{k=2}^K k R_k, \quad \mu_K = \sum_{k=2}^K k R_k 4^{-k}. \quad (63.6)$$

Here independence refers to the independent hat coordinates of disjoint blocks, not to any extra randomization of the strategy.

Theorem 63.3 (Exact rates at every prefix). *For the schedule (63.5), almost surely*

$$D_{N_K} \sim \mu_K. \quad (63.7)$$

More precisely, the following limits hold over all integer prefixes n :

$$\frac{D_n}{(\log_4 n)^{2-p}/(2-p)} \rightarrow 1, \quad 1 < p < 2, \quad (63.8)$$

$$\frac{D_n}{\log \log n} \rightarrow 1, \quad p = 2. \quad (63.9)$$

Unsubscripted logarithms are natural. In both cases, $D_n \rightarrow +\infty$ and $S_n/n \rightarrow 1/2$ almost surely.

[write] Added 4 October 2026, batch 92. This is source 06's Theorem 53.1 with $p = 1 + \varepsilon$, for the schedule with $\lfloor 4^k/k^p \rfloor$ instead of $\lceil 4^m/m^{1+\varepsilon} \rceil$ blocks in stage k ; rounding does not affect the rates. Lemma 63.4 is source 06's Lemma 53.2, proved there by Chebyshev's inequality along a subsequence, here by Kronecker's lemma. The proof below is source 04's second route; its notation (H_K, L_K, μ_K, N_K) and equations are used by the rest of this section.

We give the details, including the passage from endpoints to arbitrary prefixes. The distinction matters because eventual growth at a selected subsequence alone would not prove eventual excess at every time.

Lemma 63.4 (A strong law for success counts). *Let $I_1, I_2, \dots$ be independent Bernoulli variables with means a_b , and put $A_B = \sum_{b \leq B} a_b$. If $A_B \rightarrow \infty$, then*

$$\frac{\sum_{b \leq B} I_b}{A_B} \rightarrow 1 \quad \text{almost surely.}$$

Proof. Discard a finite initial segment so that $A_{b-1} > 0$. Since $\text{Var}(I_b) \leq a_b = A_b - A_{b-1}$,

$$\sum_b \frac{\text{Var}(I_b)}{A_b^2} \leq \sum_b \int_{A_{b-1}}^{A_b} \frac{du}{u^2} < \infty.$$

The convergence theorem for independent centered series shows that $\sum_b (I_b - a_b)/A_b$ converges almost surely. Kronecker's lemma then gives $\sum_{b \leq B} (I_b - a_b)/A_B \rightarrow 0$. $\square$

Proof of Theorem 63.3. Call a block bad, good, or neutral according to Lemma 63.1. The expected total number of bad blocks is

$$\sum_{k=2}^{\infty} R_k 4^{-k} \leq \sum_{k=2}^{\infty} k^{-p} < \infty.$$

By the first Borel–Cantelli lemma, only finitely many bad blocks occur almost surely. If $B_{k,r}$ is the indicator that block r of stage k is bad, their total weighted loss therefore satisfies

$$L_{\infty} := \sum_{k=2}^{\infty} \sum_{r=1}^{R_k} k B_{k,r} < \infty \quad \text{almost surely.} \quad (63.10)$$

Let H_K count good blocks through stage K , and let L_K be the same partial sum as in (63.10). Good-block indicators are independent across blocks and have mean $k4^{-k}$ in stage k . Moreover,

$$\mu_K = \sum_{k=2}^K k^{1-p} + O(1) \sim \begin{cases} K^{2-p}/(2-p), & 1 < p < 2, \\ \log K, & p = 2. \end{cases} \quad (63.11)$$

The bounded error follows from the summability of $k4^{-k}$ and the error of at most one introduced by each floor. In particular $\mu_K \rightarrow \infty$. Lemma 63.4 yields $H_K/\mu_K \rightarrow 1$ almost surely. At stage endpoints the score identity is exact:

$$D_{N_K} = H_K - L_K. \quad (63.12)$$

Since $L_K \rightarrow L_{\infty} < \infty$ almost surely, this proves (63.7).

To relate stages to player indices, the geometrically weighted sum gives

$$N_K \sim \frac{8}{3} 4^K K^{1-p}. \quad (63.13)$$

For completeness, the floor errors in N_K are $O(K^2)$. In the main sum, write $k = K - j$ and divide by $4^K K^{1-p}$. The terms with fixed j converge to 4^{-j} , and their sum tends to $\sum_{j \geq 0} 4^{-j} = 4/3$; the geometrically decaying remainder is negligible. The factor two in N_K yields (63.13). Hence

$$K \sim \log_4 n \quad \text{uniformly for } N_{K-1} \leq n \leq N_K. \quad (63.14)$$

On the probability-one event already obtained, there is a last bad block. After it, D_n is nondecreasing at even indices by Lemma 63.1. Thus, for all sufficiently large K and $N_{K-1} \leq n \leq N_K$,

$$D_{N_{K-1}} - \frac{1}{2} \leq D_n \leq D_{N_K} + \frac{1}{2}.$$

Also $\mu_{K-1}/\mu_K \rightarrow 1$. Squeezing between these endpoints and using (63.11)–(63.14) proves (63.8). When $p = 2$, one uses $\log K = \log \log n - \log \log 4 + o(1)$, which gives (63.9). Finally these normalizing functions are $o(n)$, so $S_n/n = 1/2 + D_n/n \rightarrow 1/2$. $\square$

The rates in this theorem are exact equivalents, with specified leading constants. The earlier repository report discussed in the introduction already addresses the unrestricted growth questions raised in [10, Remark 6.8]. The refinements here concern inspection costs and exact asymptotics of this particular block family.

63.3 Normal fluctuations and rare losses

Proposition 63.5 (Central limit theorem at stage endpoints). *For $1 < p \leq 2$,*

$$\frac{D_{N_K} - \mu_K}{\sqrt{\mu_K}} \xrightarrow{\text{law}} \mathcal{N}(0, 1). \quad (63.15)$$

Nevertheless,

$$\mathbb{E}D_{N_K} = 0, \quad \text{Var}(D_{N_K}) = \sum_{k=2}^K k(k+1)R_k 4^{-k} \sim \frac{K^{3-p}}{3-p}. \quad (63.16)$$

Thus the normal limit describes typical fluctuations without asserting convergence of moments.

Proof. The variance of H_K is

$$v_K = \sum_{k=2}^K R_k (k 4^{-k})(1 - k 4^{-k}) = \mu_K - O(1),$$

because

$$\sum_{k=2}^{\infty} R_k k^2 16^{-k} \leq \sum_{k=2}^{\infty} k^{2-p} 4^{-k} < \infty.$$

The centered Bernoulli summands have absolute value at most one, while $v_K \rightarrow \infty$. The Lindeberg condition is therefore eventually empty for each fixed positive truncation parameter. The Lindeberg–Feller theorem gives $(H_K - \mu_K)/\sqrt{\mu_K} \xrightarrow{\text{law}} \mathcal{N}(0, 1)$. Equation (63.10) implies $L_K/\sqrt{\mu_K} \rightarrow 0$ almost surely, so (63.12) and Slutsky’s theorem prove (63.15). Independence between H_K and L_K is neither claimed nor needed.

For one block, Lemma 63.1 gives $\mathbb{E}Z = 0$ and $\mathbb{E}Z^2 = k(k+1)4^{-k}$. Summing variances of independent blocks proves the exact identity in (63.16). The floor errors are summable after weighting by $k(k+1)4^{-k}$, and $\sum_{k \leq K} k^{2-p} \sim K^{3-p}/(3-p)$, giving its equivalent. $\square$

This distinction explains why almost sure positive divergence is compatible with zero expected excess at every deterministic endpoint. The total weighted loss L_{∞} is finite almost surely but has infinite mean:

$$\mathbb{E}L_{\infty} = \sum_{k=2}^{\infty} k R_k 4^{-k} = \infty.$$

Every finite stage still balances its rare negative contribution against its positive one. One cannot pass an expectation through the almost sure limit or infer concentration of the whole score from concentration of the good-block count.

63.4 The inspection cost actually paid along the line

The uniform bound controls the most expensive individual distribution. The cost averaged over actual players is smaller, because half the players always make exactly one inspection.

Theorem 63.6 (Almost sure average inspection cost). *Let $A_n = \sum_{i=1}^n Q_i$ be the number of inspection operations made by the first n players under (63.5). Then*

$$\frac{A_n}{n} \longrightarrow \frac{3}{2} \quad \text{almost surely.} \quad (63.17)$$

This statement concerns all prefixes n , and the convergence holds on the same probability-one event as the score asymptotics after intersecting finitely many probability-one events.

Proof. Enumerate all blocks by $b = 1, 2, \dots$; write k_b for their half-sizes, C_b for their total inspection costs, and $T_B = 2 \sum_{b \leq B} k_b$ for their endpoint indices. The variables C_b are independent across blocks. Proposition 63.2 gives

$$\mathbb{E}C_b = k_b(3 - 4^{1-k_b}), \quad \frac{\mathbb{E}C_b}{2k_b} = \frac{3}{2} - \frac{1}{2}4^{1-k_b}. \quad (63.18)$$

[write] Corrected 4 October 2026, batch 92. The manuscript printed the second identity as $\frac{\mathbb{E}C_b}{2k_b} = \frac{3}{2} - \frac{1}{2}, 4^{1-k_b}$, a comma where a product is meant; dividing the first identity by $2k_b$ gives the product printed here.

All individual second moments are at most $\mathbb{E}G^2 = 6$; for T the second moment is one. Cauchy–Schwarz, which does not require independence within a block, gives

$$\mathbb{E}C_b^2 \leq (2k_b) \sum_{i \text{ in block } b} \mathbb{E}Q_i^2 \leq 14k_b^2.$$

The schedule implies $k_b = O(\log(b+1))$: before stage k , the preceding stage alone has order $4^{k-1}/(k-1)^p$ blocks. Since $T_B \geq 4B$,

$$\sum_{b=1}^{\infty} \frac{\text{Var}(C_b)}{T_b^2} \leq C \sum_{b=1}^{\infty} \frac{\log^2(b+1)}{b^2} < \infty$$

for a finite constant C . The independent-series convergence theorem followed by Kronecker’s lemma now yields

$$\frac{\sum_{b \leq B} (C_b - \mathbb{E}C_b)}{T_B} \longrightarrow 0 \quad \text{almost surely.}$$

Because $k_b \rightarrow \infty$, the weighted averages of (63.18) tend to $3/2$. This proves $A_{T_B}/T_B \rightarrow 3/2$.

Finally, the deterministic worst total cost of a block of half-size k is

$$k(2k-1) + k = 2k^2.$$

For $T_B \leq n < T_{B+1}$ the uncompleted block therefore contributes at most $2k_{B+1}^2 = O(\log^2(n+1)) = o(n)$ additional inspections, and contains only $2k_{B+1} = o(n)$ players. The endpoint limit consequently extends to every n , proving (63.17). $\square$

The number $3/2$ in (63.17) is an attained average for this implementation. It is not asserted to be the least possible average among all successful strategies. The sharp statement established by the cost envelope concerns the supremum of individual expected costs and, more generally, the supremum of each nondecreasing inspection cost functional.

Proposition 63.7 (The longest realized inspection). *For the same schedule, let $M_n = \max_{1 \leq i \leq n} Q_i$. Then*

$$\frac{M_n}{\log_2 n} \longrightarrow 1 \quad \text{almost surely.}$$

Proof. For $N_{K-1} \leq n \leq N_K$, every player among the first n belongs to a stage of index at most K , so $M_n \leq 2K-1$. Equation (63.13) gives $\log_2 n = 2K + O(\log K)$ uniformly in this interval. Thus $\limsup M_n/\log_2 n \leq 1$ deterministically.

Fix $0 < \epsilon < 1$ and put $d_K = \lfloor 2(1-\epsilon)K \rfloor$. For all sufficiently large K , $1 \leq d_K \leq 2K-1$. Choose one S player from each block of stage K . Their costs are independent across blocks, and each has probability $2^{-(d_K-1)} \geq 4^{-(1-\epsilon)K}$ of being at least d_K . The probability that none reaches d_K is consequently at most

$$(1 - 4^{-(1-\epsilon)K})^{R_K} \leq \exp\left(-c \frac{4^{\epsilon K}}{K^p}\right)$$

for some $c > 0$ and all large K . These probabilities are summable. Borel–Cantelli implies $M_{N_K} \geq d_K$ eventually, almost surely. For $N_{K-1} \leq n \leq N_K$, monotonicity gives $M_n \geq M_{N_{K-1}} \geq d_{K-1}$, so $\liminf M_n / \log_2 n \geq 1 - \epsilon$. Intersect the probability-one events for a countable sequence of positive ϵ tending to zero. $\square$

Thus the realized average tends to $3/2$ while the largest inspection grows like $\log_2 n$. The cost distribution has a uniformly light tail, but its support must continue expanding along the line.

64 Inspection, dependence, and the density of correct guesses

Divergence of the surplus does not imply a limiting success frequency above $1/2$. Inspection complexity also gives a quantitative bound on the possible dependence among the scores, and makes this distinction precise.

64.1 An exact covariance identity

For a Boolean function g , let

$$\partial_j g = \frac{g(\varepsilon^{j,+}) - g(\varepsilon^{j,-})}{2}, \quad \text{Inf}_j(g) = \mathbb{E}(\partial_j g)^2, \quad \text{Inf}(g) = \sum_j \text{Inf}_j(g).$$

Here $\varepsilon^{j,+}$ and $\varepsilon^{j,-}$ denote the two configurations obtained by setting coordinate j to $+1$ and -1 . Thus $\text{Inf}_j(g)$ is the probability that flipping that coordinate changes g . The definitions apply to measurable functions of countably many hats, with a possibly infinite total influence.

Lemma 64.1 (Mutual inspection controls covariance). *For distinct players i, j ,*

$$\mathbb{E}(Y_i Y_j) = \mathbb{E}[(\partial_j g_i)(\partial_i g_j)], \quad |\mathbb{E}(Y_i Y_j)| \leq \sqrt{\text{Inf}_j(g_i) \text{Inf}_i(g_j)}.$$

For any deterministic almost-surely terminating inspection algorithm computing g_i ,

$$\text{Inf}_j(g_i) \leq \mathbb{P}(\text{player } i \text{ inspects hat } j), \quad \text{Inf}(g_i) \leq \mathbb{E}Q_i.$$

Proof. Condition on all hats except $\varepsilon_i, \varepsilon_j$. Because each guess ignores its own hat, it has the form

$$g_i = a + \varepsilon_j b, \quad g_j = c + \varepsilon_i d,$$

where a, b, c, d depend on neither of these two hats, and $b = \partial_j g_i$, $d = \partial_i g_j$. Averaging $\varepsilon_i \varepsilon_j g_i g_j$ over the two independent fair bits leaves exactly bd . A further expectation and Cauchy–Schwarz prove the first two assertions.

If flipping hat j changes the output of a deterministic algorithm, the algorithm must inspect j . Otherwise the original input and the flipped input follow the same path and have the same output. For an almost-surely terminating tree, the exceptional null set and its image under a coordinate flip are still null, so the argument remains valid almost surely. Taking probabilities proves the individual influence bound. Summing over j bounds total influence by the expected number of distinct inspected hats, which is at most $\mathbb{E}Q_i$. $\square$

Theorem 64.2 (A sharp variance bound). *For any finite set of players I and real coefficients $(a_i)_{i \in I}$,*

$$\text{Var}\left(\sum_{i \in I} a_i Y_i\right) \leq \sum_{i \in I} a_i^2 \left(1 + \sum_{j \in I \setminus \{i\}} \text{Inf}_j(g_i)\right) \leq \sum_{i \in I} a_i^2 (1 + \mathbb{E}Q_i). \quad (64.2a)$$

In particular, if $B_n = \sum_{i=1}^n \mathbb{E}Q_i$, then

$$\text{Var}(T_n) \leq n + B_n. \quad (64.2b)$$

Proof. Since $\mathbb{E}Y_i = 0$ and $Y_i^2 = 1$, the variance is

$$\sum_{i \in I} a_i^2 + 2 \sum_{\substack{i, j \in I \\ i < j}} a_i a_j \mathbb{E}(Y_i Y_j).$$

By Lemma 64.1, the summand for a pair is at most $2|a_i a_j| \sqrt{\text{Inf}_j(g_i) \text{Inf}_i(g_j)}$. The inequality $2uv \leq u^2 + v^2$ bounds this by

$$a_i^2 \text{Inf}_j(g_i) + a_j^2 \text{Inf}_i(g_j).$$

Summing over unordered pairs gives the first inequality; the influence–inspection bound gives the second. $\square$

Remark 64.3 (Sharpness and cancellation). The factor $1 + k$ cannot be reduced when the allowed depth is k . Partition the players into blocks of size $m = k + 1$, and let each player guess the product of the other k hats in the same block. All m scores in a block equal its parity, and different block parities are independent fair signs. At complete block endpoints,

$$\text{Var}(T_n) = mn = (k + 1)n,$$

attaining Theorem 64.2. There is no comparable positive lower bound: opposite-copy pairs have total score identically zero. More generally, covariance is possible only through mutual dependence. If at least one of $\text{Inf}_j(g_i)$ and $\text{Inf}_i(g_j)$ vanishes for each distinct pair, the scores are pairwise uncorrelated and $\text{Var}(T_n) = n$, even though higher-order dependence may remain.

64.2 Finite-sample and almost-sure consequences

Corollary 64.4 (Concentration and convergence in mean square). *For $t > 0$,*

$$\mathbb{P}(|T_n| \geq t) \leq \frac{n + B_n}{t^2}, \quad \mathbb{P}(T_n \geq t) \leq \frac{n + B_n}{n + B_n + t^2}.$$

Equivalently, for $\epsilon > 0$,

$$\mathbb{P}\left(\left|\frac{S_n}{n} - \frac{1}{2}\right| \geq \epsilon\right) \leq \frac{n + B_n}{4\epsilon^2 n^2}.$$

If $B_n = o(n^2)$, then $S_n/n \rightarrow 1/2$ in mean square.

Proof. The two-sided estimate is Chebyshev’s inequality. For the one-sided estimate, let $\sigma^2 = \text{Var}(T_n)$ and apply Markov’s inequality to $(T_n + u)^2$, with $u \geq 0$:

$$\mathbb{P}(T_n \geq t) \leq \frac{\sigma^2 + u^2}{(t + u)^2}.$$

Taking $u = \sigma^2/t$ gives Cantelli’s bound $\sigma^2/(\sigma^2 + t^2)$; its monotonicity in σ^2 allows Theorem 64.2 to be substituted. Finally,

$$\mathbb{E}\left(\frac{S_n}{n} - \frac{1}{2}\right)^2 = \frac{\text{Var}(T_n)}{4n^2} \leq \frac{n + B_n}{4n^2},$$

which proves the remaining assertions. $\square$

Theorem 64.5 (Uniformly bounded mean inspection gives fair density). *If $\sup_i \mathbb{E}Q_i \leq K < \infty$, then*

$$\frac{T_n}{n} \longrightarrow 0, \quad \frac{S_n}{n} \longrightarrow \frac{1}{2} \quad \text{almost surely.}$$

The same conclusion holds under the more general sufficient condition

$$\sum_{i \geq 1} \frac{1 + \mathbb{E}Q_i}{i^2} < \infty. \quad (64.5a)$$

Proof. Under the uniform bound, Chebyshev's inequality gives

$$\sum_{r \geq 1} \mathbb{P}(|T_{r^2}| > \epsilon r^2) \leq \frac{K+1}{\epsilon^2} \sum_{r \geq 1} \frac{1}{r^2} < \infty.$$

Borel–Cantelli, applied to a countable sequence of positive ϵ tending to zero, implies $T_{r^2}/r^2 \rightarrow 0$ almost surely. For $r^2 \leq n < (r+1)^2$, bounded score increments give

$$|T_n - T_{r^2}| \leq 2r + 1.$$

Division by n completes the proof in the uniform case.

For the general condition, put $c_i = 1 + \mathbb{E}Q_i$, $C_n = \sum_{i \leq n} c_i$, and fix $a > 1$. For the eventually increasing sequence $n_r = \lfloor a^r \rfloor$, geometric summation gives a finite constant A_a such that

$$\sum_r \frac{C_{n_r}}{n_r^2} = \sum_i c_i \sum_{r: n_r \geq i} \frac{1}{n_r^2} \leq A_a \sum_i \frac{c_i}{i^2} < \infty.$$

Chebyshev and Borel–Cantelli therefore give $T_{n_r}/n_r \rightarrow 0$ almost surely. Between consecutive n_r , bounded increments imply $\limsup_n |T_n|/n \leq a - 1$ almost surely. Taking the intersection of these full-measure events for $a = 1 + 1/\ell$, $\ell \geq 1$, proves the result. $\square$

Remark 64.6 (What the critical construction achieves). For the construction with $\sup_i \mathbb{E}Q_i = 2$, Theorem 64.2 gives $\text{Var}(T_n) \leq 3n$, and Theorem 64.5 gives limiting success density exactly $1/2$. Its almost-sure divergent surplus is therefore a sublinear phenomenon. It is fully consistent with $\mathbb{E}T_n = 0$ for every finite n : the variables T_n are not uniformly integrable, and rare losses can balance the positive outcomes at each finite stage.

64.3 A robust form of the degree obstruction

The preceding obstruction is not restricted to exact degree bounds. It also forces the normalized variance of a highly successful score to leave every fixed collection of low Walsh degrees.

Proposition 64.7 (Spectral escape). *Let $X \not\equiv 0$ be a mean-zero square-integrable function of independent fair bits, and let L be its orthogonal projection onto Walsh degrees at most d . Set $\delta = \mathbb{P}(X < 0)$. Then*

$$\frac{\|L\|_2^2}{\|X\|_2^2} \leq 2 \cdot 3^d \sqrt{\delta}. \quad (64.7a)$$

Thus, if $\mathbb{P}(T_n > 0) \rightarrow 1$, then for every fixed d ,

$$\frac{\sum_{|A| \leq d} \widehat{T}_n(A)^2}{\text{Var}(T_n)} \longrightarrow 0$$

along the nonzero scores. In particular, this conclusion is necessary for almost-sure positive divergence.

Proof. The assertion is trivial if $L = 0$. Otherwise orthogonal projection, Hölder's inequality, and Lemma 61.1 give

$$\|L\|_2^2 = \mathbb{E}(XL) \leq \|X\|_{4/3} \|L\|_4 \leq 3^{d/2} \|X\|_1^{1/2} \|X\|_2^{1/2} \|L\|_2.$$

After division and squaring,

$$\frac{\|L\|_2^2}{\|X\|_2^2} \leq 3^d \frac{\|X\|_1}{\|X\|_2}.$$

Since $\mathbb{E}X = 0$, Cauchy–Schwarz yields

$$\|X\|_1 = 2\mathbb{E}X_- \leq 2\|X\|_2 \sqrt{\mathbb{P}(X < 0)},$$

which proves the estimate. In countably many coordinates the same fourth-moment bound for a fixed-degree projection follows by finite-coordinate approximation and Fatou's lemma. Finally, $\mathbb{P}(T_n > 0) \rightarrow 1$ implies $\mathbb{P}(T_n < 0) \rightarrow 0$, so the estimate applies for each fixed d . Almost-sure positive divergence implies the stated probability convergence by bounded convergence. $\square$

Remark 64.8 (An approximate-degree consequence). If a mean-zero score has a proportion at least $\alpha > 0$ of its variance in degrees at most d , then Proposition 64.7 gives the explicit lower bound $\mathbb{P}(X < 0) \geq \alpha^2/(4 \cdot 9^d)$. For example, an L^2 approximation by degree at most d with relative error at most $\eta < 1$ guarantees $\alpha \geq 1 - \eta^2$ and hence

$$\mathbb{P}(X < 0) \geq \frac{(1 - \eta^2)^2}{4 \cdot 9^d}.$$

This concerns the approximation of the aggregate score, not separate approximations to individual guesses. Uniformly small inspection tails for each player do not by themselves provide such an aggregate relative-error estimate.

65 Sharp inspection cost for finite first-moment extremizers

This section studies finite teams, with no restriction to linear prediction rules. The lower bounds apply to adaptive inspections. Besides a sharp cost bound, we obtain a rigidity statement: attaining that bound forces the all-wrong configurations to form an affine binary code of a specific type. The coding construction is classical in its ingredients; the statements below distinguish those ingredients from the inspection-cost conclusions.

65.1 Model and equality in the first-moment bound

There are $n \geq 1$ players, and the hat vector $x = (x_1, \dots, x_n)$ is uniform on $\{0, 1\}^n$. Player j predicts $g_j(x_{-j}) \in \{0, 1\}$ using a deterministic decision tree that may inspect coordinates other than j . Let $Q_j(x)$ be the number of inspected coordinates. Repeated inspections can be deleted, so we assume that a coordinate is queried at most once on any execution. Define

$$H_j(x) = \mathbf{1}_{\{g_j(x_{-j})=x_j\}}, \quad S(x) = \sum_{j=1}^n H_j(x), \quad Q(x) = \sum_{j=1}^n Q_j(x).$$

Fix an integer s with $n/2 < s \leq n$. Blindness to one's own hat gives $\mathbb{E}H_j = 1/2$, and therefore

$$\mathbb{P}(S \geq s) \leq \frac{\mathbb{E}S}{s} = \frac{n}{2s}. \quad (65.1)$$

We call equality in (65.1) *first-moment saturation*. This is an equality condition, not an assertion that the bound is attainable for arbitrary n, s .

Lemma 65.1 (The two possible scores). *First-moment saturation holds if and only if $S(x) \in \{0, s\}$ for every x . In this case, with*

$$C = \{x : S(x) = 0\}, \quad d = 2s - n, \quad \delta = \mathbb{P}(C) = \frac{d}{2s},$$

the set C is nonempty and contains no two adjacent vertices of the binary cube.

Proof. Subtracting $s\mathbb{P}(S \geq s)$ from $\mathbb{E}S$ yields

$$\mathbb{E}[S\mathbf{1}_{\{S < s\}}] + \mathbb{E}[(S - s)\mathbf{1}_{\{S \geq s\}}].$$

Both terms are nonnegative. They vanish exactly when the two-score condition holds; every cube point has positive probability. The formula for δ follows from $\mathbb{E}S = n/2$. If $c \in C$ and $c^{(i)}$ is obtained by flipping coordinate i , player i makes the same prediction at both inputs and hence is correct at $c^{(i)}$. Thus $c^{(i)} \notin C$. $\square$

65.2 A local lower bound that permits adaptive queries

Theorem 65.2 (Inspection cost at every all-wrong input). *Suppose first-moment saturation holds. At every $c \in C$ and for every coordinate i , at least $s - 1$ other players inspect hat i on their executions at c . Consequently,*

$$Q(c) \geq n(s - 1) \quad (c \in C). \quad (65.2)$$

In particular, a uniform per-player decision-tree depth bound k must satisfy $k \geq s - 1$. Without any uniform depth assumption,

$$\mathbb{E}[Q \mid C] \geq n(s - 1), \quad (65.3)$$

$$\mathbb{E}Q \geq 2\delta n(s - 1) = \frac{nd(s - 1)}{s}. \quad (65.4)$$

Proof. Fix $c \in C$ and i . By Lemma 65.1, $S(c^{(i)}) = s$. Player i is one of those s correct players. Each of the other $s - 1$ correct players, say j , has the same own hat at c and $c^{(i)}$, but changes from an incorrect prediction to a correct one. Therefore $g_j(c_{-j}) \neq g_j(c_{-j}^{(i)})$.

A deterministic decision tree that does not inspect coordinate i on its execution at c follows precisely the same path at $c^{(i)}$: every queried answer on that path is unchanged. It would produce the same output, a contradiction. Thus each of these $s - 1$ players inspects i . Summing over i counts every inspection at c once, proving (65.2), the depth bound, and (65.3).

For the unconditional bound, count changes of prediction along cube edges. Each of the $n|C|$ undirected edges with one endpoint in C has exactly $s - 1$ prediction changes by players other than the coordinate being flipped. There is no double counting of these edges because C is independent. A prediction change across an edge forces that coordinate to be inspected at *both* endpoints, by the same decision-tree argument. Equivalently, if

$$I_j(x) = \#\{i \neq j : g_j(x_{-j}) \neq g_j(x_{-j}^{(i)})\},$$

then $I_j(x) \leq Q_j(x)$. Summing the contributions of just the edges incident to C gives

$$\sum_x Q(x) \geq \sum_x \sum_j I_j(x) \geq 2n|C|(s - 1).$$

Division by 2^n proves (65.4). $\square$

At equality in the uniform depth bound $k = s - 1$, the query graph at any $c \in C$ has both its in-degrees and its out-degrees equal to $s - 1$. Moreover, every queried coordinate at that input changes the querying player's prediction when flipped. These conclusions follow because all inequalities in the local counting proof are then equalities.

Independent randomization does not weaken the theorem. Conditional on the random seed, (65.1) holds for the resulting deterministic strategy. Saturation after averaging forces saturation for almost every seed. Applying the theorem seed by seed gives the same bounds, with expectations over hats and randomness. A stated uniform depth bound must hold for those deterministic realizations.

65.3 Fourier rigidity and the failure-set geometry

For $A \subseteq [n]$, write $\chi_A(x) = (-1)^{\sum_{i \in A} x_i}$ and $\widehat{f}(A) = \mathbb{E}[f\chi_A]$. These characters are an orthonormal basis for functions on the uniform binary cube. The following argument includes the needed Fourier identities.

Theorem 65.3 (Geometry at the minimum depth). *Suppose first-moment saturation holds and every player has decision-tree depth at most $s - 1$. For $f = \mathbf{1}_{\{C\}}$, every nonconstant nonzero Fourier coefficient has degree exactly s . Consequently:*

1. *C has uniform projections onto every set of at most $s - 1$ coordinates;*
2. *every input outside C has exactly $d = 2s - n$ neighbors in C ;*
3. *every player inspects exactly $s - 1$ hats at every input.*

For odd n and $s = (n + 1)/2$, the closed radius-one balls centered at C partition the cube: C is a binary one-error-correcting perfect code.

Proof. Put $F = 2S - n$. First-moment saturation gives

$$F = d - 2sf. \quad (65.5)$$

Each signed correctness function $2H_j - 1 = (-1)^{x_j + g_j(x_{-j})}$ has degree at most $k + 1$ when the prediction tree has depth at most k . Indeed, the indicator of a leaf is a product of at most k factors $(1 \pm \chi_{\{i\}})/2$; the signed prediction is a signed sum of these indicators. Multiplication by $\chi_{\{j\}}$ increases the degree by at most one. Thus $\deg f \leq s$ under the present depth assumption.

Independence of C gives $\mathbb{E}[(f(x) - f(x^{(i)}))^2] = 2\delta$ for every i . Since flipping i negates exactly the characters containing i , Parseval's identity yields

$$\sum_A |\widehat{f}(A)|^2 = \frac{n\delta}{2}, \quad \sum_{A \neq \emptyset} \widehat{f}(A)^2 = \delta(1 - \delta) = \frac{n\delta}{2s}. \quad (65.6)$$

The average nonconstant Fourier degree is therefore exactly s . All those degrees are at most s , so every nonzero one equals s . This also supplies a second proof of the depth lower bound: without assuming $k = s - 1$, the same average degree implies $k + 1 \geq s$.

If $J \subseteq [n]$ has $|J| \leq s - 1$, expansion of the indicator of any specified J -pattern into characters gives

$$\mathbb{P}(C \text{ and } x_J = a) = \delta 2^{-|J|}.$$

This proves the uniform-projection assertion, often described as an orthogonal-array property.

Let $\mathcal{A}$ be cube adjacency: $(\mathcal{A}f)(x) = \sum_i f(x^{(i)})$. Its action on a character is $\mathcal{A}\chi_A = (n - 2|A|)\chi_A$. The constant term of f is δ , and all its other terms have degree s . Hence

$$\mathcal{A}f = n\delta + (n - 2s)(f - \delta) = d(1 - f). \quad (65.7)$$

This says that C has no internal edges and that each outside vertex has exactly d neighbors in C .

Finally, put $k = s - 1$. If a player's tree had a leaf of depth $\ell < k$, that leaf would fix ℓ other hats and the prediction. On C the player's own hat would then be fixed to the opposite bit. But the ℓ queried hats together with the player's own hat are at most k coordinates, whose projection from C is uniform. The purportedly excluded pattern must therefore occur, a contradiction. All leaves have depth k . When $d = 1$, equation (65.7) says that every vertex outside C has one neighbor in C , which proves the perfect-code assertion. $\square$

The depth hypothesis matters. Saturation by itself only makes C independent; it does not make C a perfect code. In particular, the perfect-code conclusion is a property of inspection-efficient extremizers, rather than a consequence of the first moment alone.

Theorem 65.4 (Affine rigidity and the exact parameter criterion). *A first-moment-saturating strategy with per-player depth at most $s - 1$ exists if and only if there are integers $r \geq 1$ and $m \geq 1$ such that*

$$n = m(2^r - 1), \quad s = m2^{r-1}. \quad (65.8)$$

For every such strategy, its all-wrong set C is affine. After permuting coordinates, it is a coset of the kernel of a binary $r \times n$ matrix containing each nonzero column of $\{0, 1\}^r$ exactly m times. In particular, for odd n at the strict-majority threshold, an extremizer of minimum depth has an affine Hamming failure code.

Necessity. Theorem 65.3 supplies the uniform projections through size $k = s - 1$, and every leaf has depth exactly k . Consider one leaf for player j , querying the set J of k other coordinates and reached by the pattern $x_J = a$. The conditional density of C on the s coordinates $A = J \cup \{j\}$ has the form

$$\mathbb{E}[f \mid x_A] = \delta + \widehat{f}(A)\chi_A(x). \quad (65.9)$$

All other nonconstant characters supported on A vanish by the Fourier conclusion. At this leaf, C excludes the value of x_j equal to the leaf's prediction. Substitution of that excluded pattern into (65.9) gives $\widehat{f}(A) = \varepsilon_A \delta$ for some $\varepsilon_A \in \{-1, 1\}$. Consequently $\mathbb{E}[\chi_A \mid C] = \varepsilon_A$, and therefore every point of C satisfies the affine parity equation $\chi_A = \varepsilon_A$.

Take these equations for every leaf of every player, and let D be their common solution set. We have $C \subseteq D$. Conversely, if $x \in D$, follow each player's tree to its actual leaf. The parity equation associated with that leaf, together with its queried pattern, forces that player's own bit to be the opposite of the prediction. Every player is wrong, so $x \in C$. Thus $C = D$, a nonempty affine solution space.

Write $C = a + V$, with V a linear subspace of codimension r . Its density is $\delta = 2^{-r}$, and $r \geq 1$. Let $L = V^\perp$. Direct summation over $a + V$ gives

$$\widehat{f}(A) = \begin{cases} 2^{-r} \chi_A(a), & \mathbf{1}_A \in L, \\ 0, & \mathbf{1}_A \notin L, \end{cases}$$

where $\mathbf{1}_A \in \{0, 1\}^n$ denotes the binary incidence vector of A . Every nonzero word of L therefore has weight s . The identity $\delta = d/(2s) = 2^{-r}$ gives $s = d2^{r-1}$ and $n = d(2^r - 1)$, proving (65.8) with $m = d$.

The remaining constant-weight classification is the binary case of Bonisoli's theorem [33]; we include its short Fourier proof here. Choose a full-rank generator matrix for L and let M_b count its columns equal to $b \in \{0, 1\}^r$. For every nonzero u , the word obtained by multiplying this matrix by u has weight s , so

$$\sum_b M_b (-1)^{u \cdot b} = n - 2s = -m, \quad \sum_b M_b = n.$$

Fourier inversion on $\{0, 1\}^r$ now gives $M_0 = 0$ and $M_b = m$ for every $b \neq 0$. This proves the claimed description of C . $\square$

65.4 An explicit trace construction attaining every allowed pair

The next construction completes the sufficiency proof and gives actual inspection sets and prediction formulas. When $m = 1$, its syndrome-zero set is the usual binary Hamming code; the proof below establishes every property used here without relying on a coding-theoretic black box.

Theorem 65.5 (Replicated trace strategy). *For every $r, m \geq 1$, the parameters in (65.8) admit a nonadaptive strategy using exactly $s - 1$ inspections per player and satisfying*

$$\mathbb{P}(S = s) = 1 - 2^{-r} = \frac{n}{2s}, \quad \mathbb{P}(S = 0) = 2^{-r}.$$

For $m = 1$, this gives $n = 2^r - 1$, $s = 2^{r-1}$, and the sharp depth $k = 2^{r-1} - 1$.

Proof. Let K be the field with $q = 2^r$ elements, and let

$$\text{Tr}(z) = z + z^2 + \cdots + z^{2^{r-1}} \in \{0, 1\}$$

be its absolute trace. The trace is linear over the two-element field. It is nonzero because its defining polynomial is nonzero and has degree $2^{r-1} < q$, so it cannot vanish at all q field elements. Consequently exactly $q/2$ elements have trace one. Fix $c \in K$ with $\text{Tr}(c) = 1$.

Index the players by $(v, a) \in K^* \times \{1, \dots, m\}$, and put

$$Z(x) = \sum_{w \in K^*} \sum_{b=1}^m x_{w,b} \frac{c}{w} \in K.$$

Player (v, a) predicts the unique value of its own bit that would make $\text{Tr}(vZ) = 1$. Explicitly, all arithmetic in the prediction formula is modulo two:

$$g_{v,a} = 1 + \sum_{(w,b) \neq (v,a)} \text{Tr}\left(\frac{vc}{w}\right) x_{w,b}. \quad (65.10)$$

The coefficient of its own hat in $\text{Tr}(vZ)$ is $\text{Tr}(c) = 1$, so this is a blind and well-defined prediction. It needs precisely the other hats whose displayed coefficients are one. As w runs over K^* , the element vc/w runs over K^* ; precisely $q/2$ of those values have trace one. Counting m copies and removing the player's own hat leaves $mq/2 - 1 = s - 1$ inspected hats.

Player (v, a) is correct exactly when $\text{Tr}(vZ) = 1$. If $Z = 0$, no player is correct. If $Z \neq 0$, multiplication by Z permutes K^* , so exactly $q/2$ choices of v , each with m copies, are correct. Thus $S = mq/2 = s$.

Finally, Z is a surjective linear map from the binary hat space to the r -dimensional binary vector space K : its coefficients include every nonzero element of K . All its fibers have cardinality 2^{n-r} , and a uniform hat vector makes Z uniform on K . Therefore $\mathbb{P}(Z = 0) = 1/q$. The displayed score distribution follows, and Theorem 65.2 proves that no strategy saturating the same first-moment bound can have smaller uniform depth. $\square$

For $m = 1$, the coefficients c/w are distinct and nonzero. A nonzero syndrome Z can be eliminated by flipping exactly the unique hat whose coefficient equals Z . This directly verifies the perfect-code property of the failure set. For general m , each nonzero syndrome has exactly $m = d$ such single-hat corrections, in agreement with (65.7).

The small odd strict-majority cases are

r	1	2	3	4
n	1	3	7	15
s	1	2	4	8
$k = s - 1$	0	1	3	7
$\mathbb{P}(C)$	1/2	1/4	1/8	1/16.

The minimum-depth classification is narrower than the unrestricted problem of optimizing the score distribution: it imposes a sharp inspection budget and derives both arithmetic restrictions and affine structure from equality in that budget.

[write] Added 4 October 2026, batch 92. *What this section settles elsewhere in the report.* (i) Part I's Q9, with m players, n boxes and $q = 2$: by Theorems 65.4 and 65.5, for $(m, q, n) = (N, 2, N)$ with $N = m(2^r - 1)$ and the threshold $t = s = m2^{r-1}$, the best probability of at least t successes over legal decision trees is $1 - 2^{-r}$, attained by the trace strategy (player i targets box i and reads $s - 1$ other boxes, a legal nonadaptive tree); the first-moment bound (65.1) holds for every team of blind binary output maps with free targets, since each succeeds on exactly half of the configurations (Proposition 7.1), so the blind optimum is the same. This holds for every $m \geq 1$. The even case $m = 2a$ is the family of Part II's Corollary 20.3, attained there by the syndrome block, which reads all $N - 1$ other hats of its block rather than $s - 1$; the case $m = 1$, $r = 2$ is Part I's three-box team of Section 13, and $r = 1$ is $(m, 2, m)$ with $t = m$ and optimum $1/2$. As in the batch-90 note, the same optimum holds for every $n \geq N$, the further boxes being left unread (the write's observation). (ii) Part II's Research question 29.4: the first-moment bound is attainable at depth $s - 1$ exactly for these pairs (N, s) ; with greater depth, for other pairs, it stays open (Research question 68.5). (iii) Research question 29.5: for the parameters of Corollary 20.3 the extremizers of minimum depth $s - 1$ exist ($m = 2a$) and all have an affine failure set with each nonzero parity-check column repeated m times; extremizers of greater depth, such as the syndrome block when $r \geq 2$, are not classified. The notes after the questions of Sections 15 and 29 record this.

66 A concrete route to formalization in ProveIt

The results in this article are mathematical proofs, not additions to a proof-assistant library. Their connection with ProveIt is nevertheless specific: the repository already distinguishes finite executable certificates, their mathematical meanings, and the arithmetic formulas that represent those meanings [29, 28]. The earlier hat-guessing manuscript asks for an information-cost theorem and a complete certificate chain [26]; the box-game report separately identifies the missing decision-tree compiler and reflection theorem [27]. The following development would address those interfaces without treating an executable experiment as a proof of an infinite probability statement.

66.1 Operational semantics before query bounds

The first module should use a finite binary tree whose leaves carry guesses and whose internal nodes carry natural-number labels of queried hats. The two children correspond to the two observed colors. A tree is valid for player i when every reachable query avoids i ; in a normalized tree, no coordinate is queried twice along one branch. Labels may repeat on different branches. There should be no restriction on the distance of a query, on the number of players that can inspect one hat, or on a common finite set containing every player's observations.

Evaluation on a configuration returns both a guess and its number of queries. The distinction is necessary: two trees can compute the same Boolean function with different costs. The basic structural induction should prove the following facts.

1. Evaluation terminates, and changing player i 's own hat changes neither its guess nor its query transcript.
2. Only the finite collection of labels appearing in the tree can affect evaluation.
3. If the tree has depth at most q , its sign-valued output has a multilinear representation of degree at most q .
4. A reachable branch of length h gives $\mathbb{P}(Q \geq r) \geq 2^{1-r}$ for $1 \leq r \leq h$. For a tree without repeated queries along a branch, the probability of following any prescribed $r - 1$ answers is exactly 2^{1-r} .

The degree proof is the recursion that combines the two child polynomials using the indicators of the queried sign. Multilinear reduction uses $x_j^2 = 1$, and cannot increase degree. Multiplying the guess sign by the player's own sign gives the correctness sign, with degree at most $q + 1$. This is an algebraic invariant of evaluation, not an assumption about independence of different players' successes.

The fourth assertion supplies the finite foundation of the cost frontier. For a nonnegative nondecreasing cost function Φ , prove first the exact finite identity

$$\mathbb{E}\Phi(Q) = \Phi(0) + \sum_{r=1}^h (\Phi(r) - \Phi(r-1))\mathbb{P}(Q \geq r).$$

It then gives a lower bound by the corresponding truncated geometric sum. The infinite geometric value is introduced only afterward, by monotone convergence. Keeping these two steps separate handles finite and infinite cost frontiers with the same finite theorem.

66.2 Finite cube algebra and the obstruction

For the first n players, the union of their tree labels and their own coordinates is finite. Thus their score can be analyzed on an ordinary finite Boolean cube, even when they query hats with arbitrarily large indices. The second module should establish the orthogonality of Walsh characters, the relevant fourth-moment inequality for bounded-degree polynomials, and the resulting two-sided sign bound for nonzero centered polynomials. All expectations at this stage are finite averages.

The proof must also cover a polynomial that is identically zero. A statement about a strictly negative value cannot include that case, whereas the one-sided event needed by the obstruction can. Recording this distinction in the lemma statements avoids silently dividing by a zero second moment.

The crucial interface is stability under fixing finitely many hat colors. Substituting constants cannot increase polynomial degree. Moreover, for a player whose own coordinate has not been fixed, its correctness sign still has mean zero: its own fair bit remains independent of its guess. If a finite set F has been fixed, only players in F can contribute to the conditional mean of the score. Their total absolute contribution to the excess is at most $|F|/2$, uniformly in the prefix length.

A suitable finite theorem therefore has three explicit parameters: the depth bound, the fixed-coordinate cylinder, and the prefix length. It supplies a positive lower bound, depending only on depth, for a score that does not exceed the cylinder's fixed additive allowance. It requires no bound on the number of times an observed hat is reused across the team. The conditional formulation is essential to the final probability-zero conclusion; an unconditional sign bound alone would only rule out almost-sure divergence.

66.3 The countable probability layer

The third module works on the fair product space of countably many hats. Finite-tree evaluations are cylinder-measurable, so the excess variables and the event $D_n \rightarrow +\infty$ are measurable. The finite-cube theorem transfers to every finite-coordinate cylinder. A cylinder approximation lemma, or equivalently convergence of conditional expectations with respect to the first m coordinates, then rules out a positive-measure divergence event.

This step should not invoke a zero-one law for tail events. In the permitted query model, changing one early hat can alter infinitely many later guesses, so the divergence event has not been assumed invariant under finite coordinate changes. Approximation of an arbitrary measurable event by finite-coordinate information is the appropriate statement.

The positive construction needs a different probability interface. Its bad blocks have summable probabilities, whereas independent good blocks have a divergent sum of probabilities. The two Borel–Cantelli conclusions must be proved or imported with those respective hypotheses. A separate deterministic prefix lemma extends block-end divergence to every physical prefix. Endpoint calculations alone do not establish that conclusion.

Finally, the geometric query law is verified on each finite block. Its uniform tail bound supplies the upper cost estimate, and increasing block sizes approach the geometric value. This yields the sharp mean and exponential-moment boundaries through the same cost identity; a separate probabilistic construction for each cost is unnecessary.

66.4 What the existing coding library supplies

The inspected `PAListCoding.Basic` module contains `encode`, `decode`, and proofs named `encode_valid`, `encode_length`, `encode_nth`, `encode_injective`, `valid_iff_existsUnique_encode`, `encode_decode`, and `decode_encode`. It also exposes an actual arithmetic validity formula and its standard-model specification. The inspected `PAListCoding.Predicates` module supplies guarded relations including `HasLength`, `Entry`, `Concat`, `NoDuplicates`, `Occurrences`, `SumElements`, and `ProductElements`, together with arithmetic formula interfaces [28].

These interfaces can represent a finite tagged serialization of a query tree and a finite transcript. A new validity predicate must check the node tags, child references, queried labels, and the exclusion of the player’s own coordinate. A compiler correctness theorem would then relate the decoded object to the operational semantics above. Existing list round-trip theorems justify recovering the serialized list; they do not already prove the hat-specific semantics.

Two limitations deserve explicit treatment. First, the inspected Lean encoding and decoder are declared noncomputable. Their mathematical contracts can support a proof, but an efficient external certificate format requires its own executable implementation and a proved bridge. Second, the exported formula specifications concern truth in the standard natural-number model. They should not be described as internal PA derivations of every existence or correctness assertion without an additional syntactic proof.

66.5 What finite verification can certify

A companion checker can exhaust small cubes, compare tree evaluation with the stated block rule, reject an own-hat query, and compute exact distributions of scores, prefixes, and costs. Its rational arithmetic can check the finite geometric sums and the boundary cases of the cost identity. These tests are useful for detecting an incorrect branch, a query-count convention, or an off-by-one error in a prefix.

They do not establish hypercontractivity for arbitrary degree, measurable event approximation, either

Borel–Cantelli theorem, or the infinite cost frontier. Nor does a successful run prove that the checker’s acceptance predicate entails the mathematical statement it is meant to certify. The existing box-game Python checker has the same distinction between execution and a proved reflection theorem [27].

The practical order is therefore: verify tree semantics and finite geometric costs; prove the finite cube obstruction and its cylinder version; connect the audited serialization interfaces; and only then assemble the countable probability theorems. Each stage has a theorem-level contract that can be reviewed independently, and none requires presenting the current manuscript as already kernel-checked.

67 Finite verification and reproducibility

The supplied program code/`verify_finite.py` uses only the Python standard library and exact integer arithmetic. Its final run completed all 19 cases listed below. It exhausts every hat assignment for each listed case; it does not sample assignments or search over all possible strategies.

Family	Parameter cases	Hat assignments checked
Exceptional blocks, $t = 2, 4, \dots, 12$	6	5,460
Trace strategies, $r = 1, 2, 3, 4, m = 1$	4	32,906
Replicated trace strategies	9	21,180
Total	19	59,546

The replicated cases are $r = 1, m = 2, \dots, 6$; $r = 2, m = 2, 3, 4$; and $r = 3, m = 2$. Across the cases, the program performed 838,498 comparisons under flipping a player’s own hat, verifying that both the computed output and, where adaptive, the transcript remain unchanged. For the exceptional blocks it checked 30,927 individual nonnegative pair increments and the same number of pair prefixes. For the trace strategies it checked 73,367 failure-boundary edge sensitivities.

The independent descriptions compared by the checker are the operational query rules, the direct classification of block configurations, and the linear syndrome characterization. This catches an implementation that produces a correct final count while illegally reading its own hat, as well as a wrong branch order or an incorrect query count. The trace checks also verify uniform syndrome fibers, the exact number of all-wrong neighbors, and the minimum-depth query counts.

The complete exact distributions and individual case counts are stored in `data/verification.json`. That file is written only after every assertion succeeds. To reproduce the checks from the extracted package directory, run

```
python3 code/verify_finite.py
```

To rebuild this standalone article, run

```
pdflatex hat_inspection_frontier.tex
pdflatex hat_inspection_frontier.tex
```

A third run may be needed if a local TeX installation changes the table-of-contents pagination; the supplied build script runs three. No bibliography database, generated figure, or network connection is required.

The finite enumeration verifies the supplied implementations and their finite identities. The probability-zero obstruction, the general affine-rigidity theorem, and the infinite cost frontier depend on the proofs in the article. No enumeration of these finite cases is offered as a substitute for those arguments.

[write] **Added 4 October 2026, batch 92.** These commands and file names are the delivery's. In the collection the program is shipped as `code/04-hat-inspection-verify_finite.py`, its output as `data/04-hat-inspection-verification.json` and the build script as `code/04-hat-inspection-build.sh`, byte-identical to the delivery; the script builds the unshipped `hat_inspection_frontier.tex`, which with the delivered PDF and `README.txt` survives in the archive of the arrival commit 9dc8db274 (this Part is the source, built as part of the report's `article.tex`). The program always writes `../data/verification.json` relative to its own directory, so run in place it would add an unprefixed file to the report's `data/`; the report's `README.md` reruns it on a scratch copy. At the write it passed there and reproduced the shipped output (Section 45.5). The table above and the counts of `04-hat-inspection-PROOF_STATUS.txt` agree with the shipped JSON.

68 Further research questions

The exact geometric frontier settles the existence question for uniform deterministic query costs. It leaves the following quantitative and structural problems. These are proposed research targets; except where an earlier source is explicitly cited, they are not advertised as a list of already recognized open problems.

Research question 68.1 (Growth at the optimal cost frontier). Characterize the increasing functions $h(n) \rightarrow \infty$ for which a finite-tree strategy can satisfy

$$D_n \asymp h(n) \quad \text{almost surely,} \quad \sup_i \mathbb{P}(Q_i > r) \leq 2^{-r} \quad (r \geq 0).$$

In particular, can the optimal geometric tail coexist with $D_n \asymp \log n$, or with D_n growing like a positive power of n ? The present schedule gives explicit powers of $\log n$ smaller than the first power. The predecessor's much faster construction [26] does not supply this optimal cost envelope.

Research question 68.2 (The general depth–confidence curve). Let p_k be the supremum of $\mathbb{P}(S_N > N/2)$ over all finite teams and deterministic adaptive strategies of depth at most k . The polynomial argument gives an explicit exponential gap from 1. The trace family gives

$$p_{2^{r-1}-1} \geq 1 - 2^{-r} = 1 - \frac{1}{2(k+1)}.$$

Determine the correct asymptotic behavior of $1 - p_k$. In particular, is the gap necessarily of order $1/k$, or can it decay much faster? The exponential example for arbitrary polynomials does not automatically respect the own-hat restriction.

Research question 68.3 (Stability of finite optimality). Suppose a finite strategy comes within ϵ of the first-moment bound. Which quantitative approximation to the failure-set and query-graph structure follows? An effective stability theorem should measure both the probability mass outside the ideal two-point score law and the distance from the relevant equitable partition. Exact equality is rigid enough for the arguments in this article; the useful scale of the error terms is unresolved here.

Research question 68.4 (The inspection penalty for nonlinear perfect codes). For odd team sizes at strict majority, the affine-rigidity theorem excludes every nonaffine perfect code from the minimum-depth class. For a prescribed nonaffine perfect code, determine the least depth of a probability-optimal strategy having that failure set. Is one extra query ever sufficient, and can the extra depth grow with the block length? The ordinary perfect-code geometry alone does not determine the answer.

Research question 68.5 (Beyond the classified minimum-depth pairs). The exact parameter criterion settles existence at depth $s - 1$. For parameters outside that family, determine whether the first-moment bound can still be attained with greater depth, and how much greater. When the probability bound is unattainable even with full visibility, determine the optimum and its inspection cost. This retains the unresolved part of the finite-team questions in [27, 26].

Research question 68.6 (Charged and uncharged external randomness). Determine the sharp cost frontier when protocols may use a public random seed. Distinguish expectations conditional on the seed from unconditional expectations, and charge random bits explicitly in a third variant. The simple mixture in the model section already separates the positive-probability criterion from almost-sure success under an unconditional cost convention.

Research question 68.7 (Visibility distance and computational time). The cost theorem counts inspected hats, with no bound on their indices. For a target growth law h , simultaneously minimize the number of queries, the farthest inspected index, and the time needed to compute the next queried coordinate. These are different resources. The explicit block schedule has logarithmic block lengths relative to its physical position, but faster excess profiles may require a different compromise.

Research question 68.8 (More colors and biased hats). Extend the theorem to q equiprobable colors, with baseline n/q , or to a nonuniform product measure with an explicitly chosen fairness criterion. A depth- d decision path then has probabilities governed by the individual branch masses. Determine whether a single extremal stopping-time distribution still governs all increasing costs. The binary paired neutral block has a special exact-balance property that cannot simply be assumed for another alphabet.

Research question 68.9 (Typical fluctuations and rare losses). For cost-optimal schedules, classify the possible central-limit and large-deviation behavior of the positive gain count and the total random loss. The normal limit derived here coexists with an exactly zero mean for the unnormalized surplus. Obtain quantitative estimates that separate typical fluctuations from the negative tail carrying the expectation.

Research question 68.10 (Average cost across players). The displayed construction has realized population average cost converging to $3/2$, while its maximum individual expected cost has supremum 2. Determine optimal population-average costs under specified excess growth and visibility constraints. Such averages are not governed by the same optimization as $\sup_i \mathbb{E}Q_i$; inserting balanced pairs can alter the former without changing the latter.

Research question 68.11 (Quantitative spectral escape). High confidence of positive excess forces its normalized Fourier mass out of every fixed degree. Determine how fast that transfer must occur for prescribed confidence and surplus, taking account of the own-coordinate restrictions on the individual score functions. The variance bound suggests that query influences, Fourier degree, and the tail of the query depth should be studied jointly.

Research question 68.12 (A kernel-checked cost classification). Implement the finite-tree semantics, the degree bound, a suitable hypercontractivity estimate, the finite-cylinder obstruction, and the geometric stopping-time comparison in a proof assistant. The finite trace construction offers a separate test of a certificate interface. The objective is to certify the quantified infinite statements, including the positive-probability conclusion and the exact supremum over individual costs, rather than only their finite examples.

[write] Status of the questions, 4 October 2026, batch 92. All twelve questions stay open; source 06 settles one value asked for in 68.2 (below). The write searched no literature for them. *Overlaps with source 06's questions of Section 57:* 68.1 is 57.3; 68.2 asks about the same quantity as 57.1, since $1 - p_k$ is source 06's α_k ; 68.6 and 57.2 overlap (randomized thresholds, public and private coins); 68.8 is 57.9; 68.9 is close to 57.11; 68.10 is 57.10; 68.11 is close to 57.8; 68.7 and 57.6 both separate query count from query distance. *Partial answers in this report:* for 68.2, source 06's Theorem 50.2 gives $p_1 = 3/4$, so the trace bound $p_{2^{r-1}-1} \geq 1 - 2^{-r}$ is an equality at $k = 1$ ($r = 2$), and its sparse-support bound (50.4) gives the lower bound $1 - p_k \geq c_k^{\text{ad}}$; for 68.6, source 06's Theorem 55.2 shows that with free *private* coins every uniform expected cap $C > 1$ admits almost-sure divergence, while the sharp randomized thresholds stay open. 68.5 retains the open part of Part I's Q9 and of Part II's Research questions 29.4–29.5 (see the notes after the questions of Sections 15 and 29).

69 What has been resolved

[write] Added 4 October 2026, batch 92. The “definitive negative answer” below is shared with source 06, which reached it independently against the same revision (Corollary 48.4, with category, and Section 54 for the other clauses of the question); what this section lists as this source’s own beyond that answer is the frontier for every nondecreasing cost, the fluctuation, variance and density results and the finite classification.

The uniform-query question in the preceding ProveIt manuscript has a definitive negative answer in the stated model. The proof also shows why a mere expectation calculation is insufficient: a centered score may be positive on almost every input, with rare losses carrying its expectation. Bounded decision depth prevents that concentration uniformly, and conditioning on finite hat assignments upgrades this obstruction to probability zero for divergent excess.

The positive direction has a different role. Eldredge’s construction already supplies divergent excess, and evaluating its exception by early stopping yields truncated geometric query costs. Combining that implementation with the lower bound determines all increasing worst-player costs simultaneously. This is the basis of the exact mean threshold 2, the tail envelope 2^{-r} , and the exponential-moment boundary.

For finite teams, first-moment optimality imposes a concrete amount of inspection at every all-wrong assignment. At the sharp depth, the failure set must be affine with replicated simplex parity-check columns. This gives a full parameter criterion, and the replicated trace strategy realizes every allowed case. In particular, strict majority for an odd team size at minimum depth forces an affine Hamming failure code.

These conclusions are conventional mathematical proofs with independent finite diagnostics. They do not replace the remaining formalization work or establish bibliographic priority.

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